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(54) **COMPOSITIONS AND METHODS FOR
TREATING ORNITHINE
TRANSCARBAMYLASE DEFICIENCY**

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CPC **C12N 9/1018** (2013.01); **A61P 9/00**
(2018.01); **C12Y 201/03003** (2013.01); **A61K**
48/00 (2013.01)

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CPC **C12Y 201/03003**
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(57)

ABSTRACT

The present disclosure provides a modified human OTC
protein having improved properties for the treatment of OTC
deficiency in a patient. Preferably, the protein of the disclo-
sure is produced from a codon optimized mRNA suitable for
administration to a patient suffering from OTC deficiency
wherein upon administration of the mRNA to the patient, the
protein of the disclosure is expressed in the patient in
therapeutically effective amounts to treat OTC deficiency.
The present disclosure also provides codon optimized
mRNA sequences encoding wild type human OTC compris-
ing a 5' UTR derived from a gene expressed by *Arabidopsis
thaliana* for use in treating OTC deficiency in a patient.

20 Claims, 15 Drawing Sheets

Specification includes a Sequence Listing.

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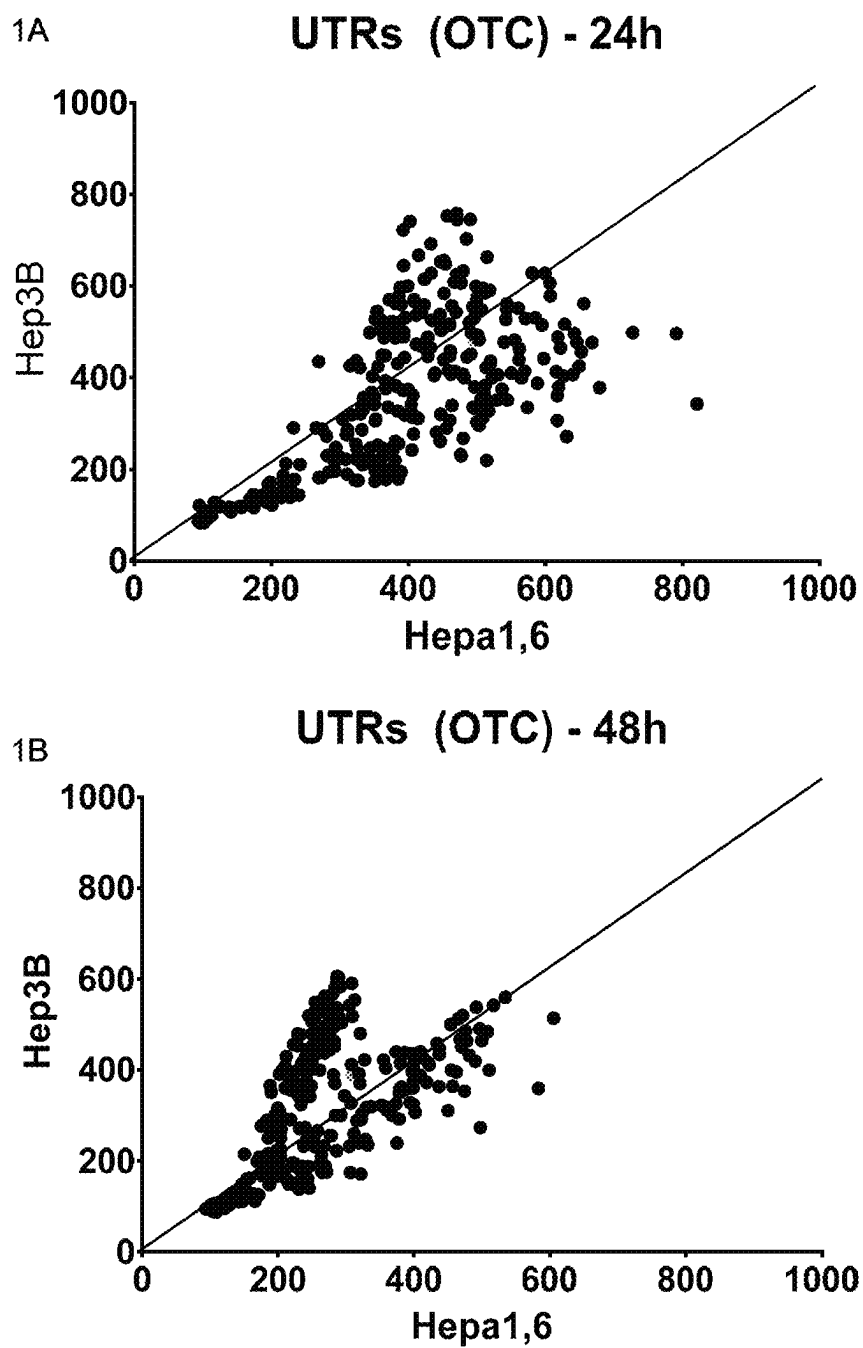


FIG. 1

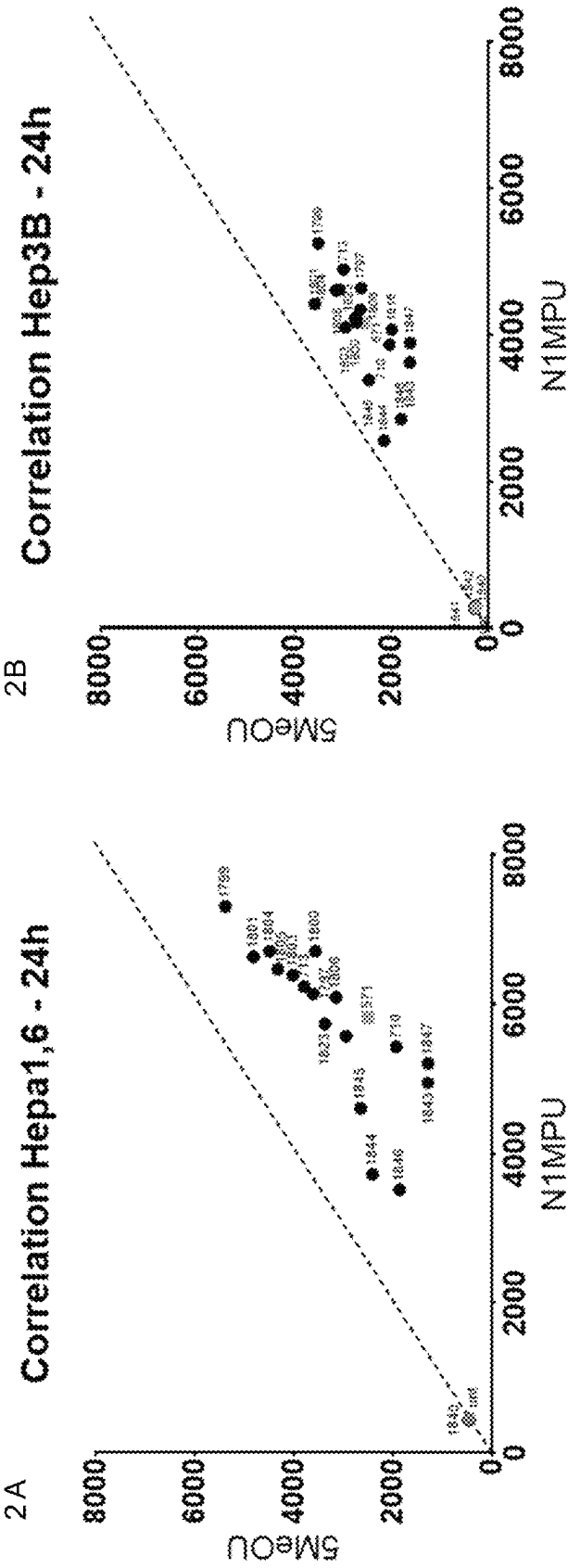


FIG. 2

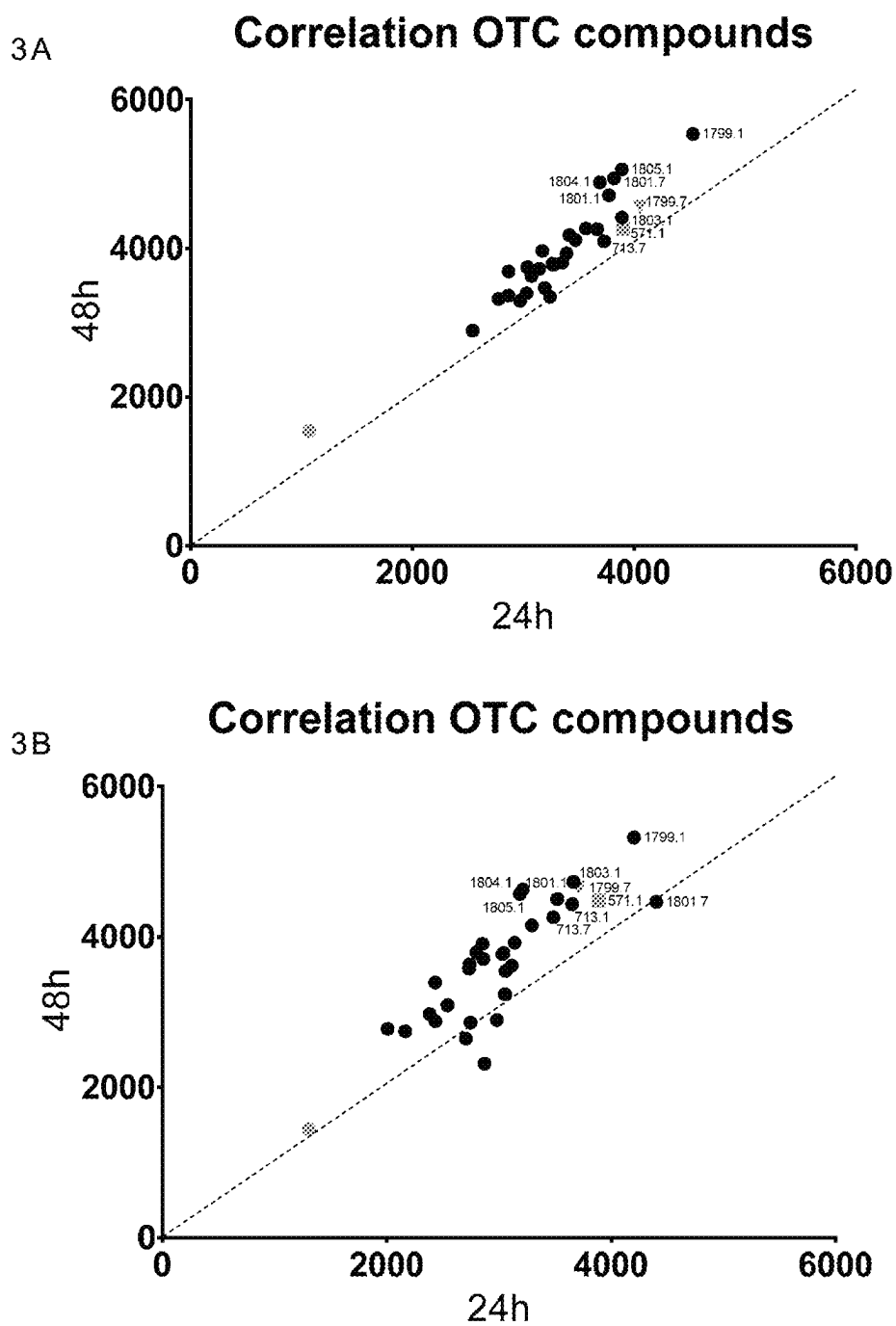


FIG. 3

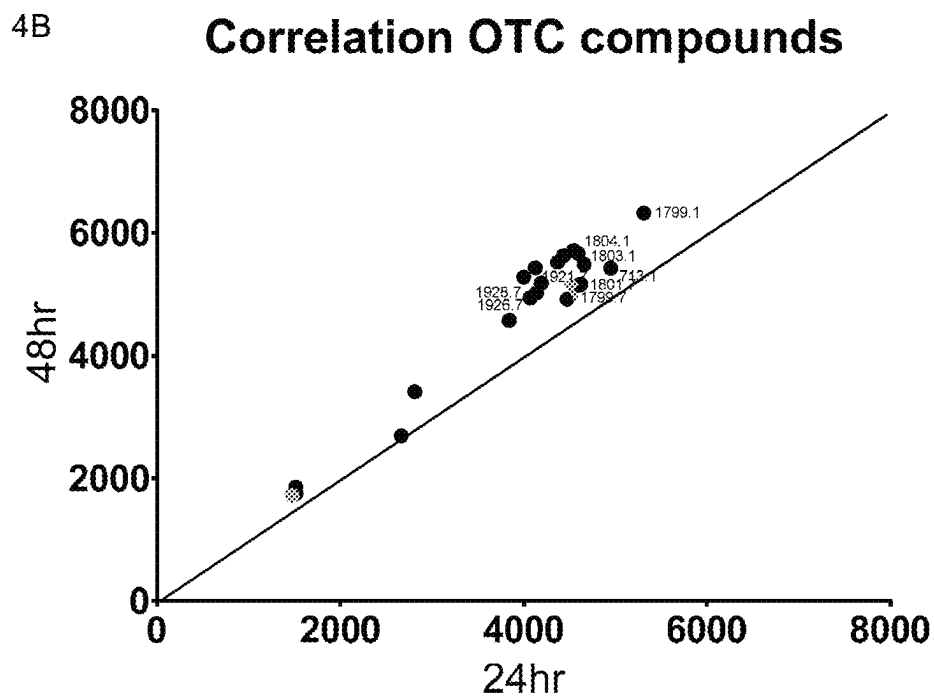
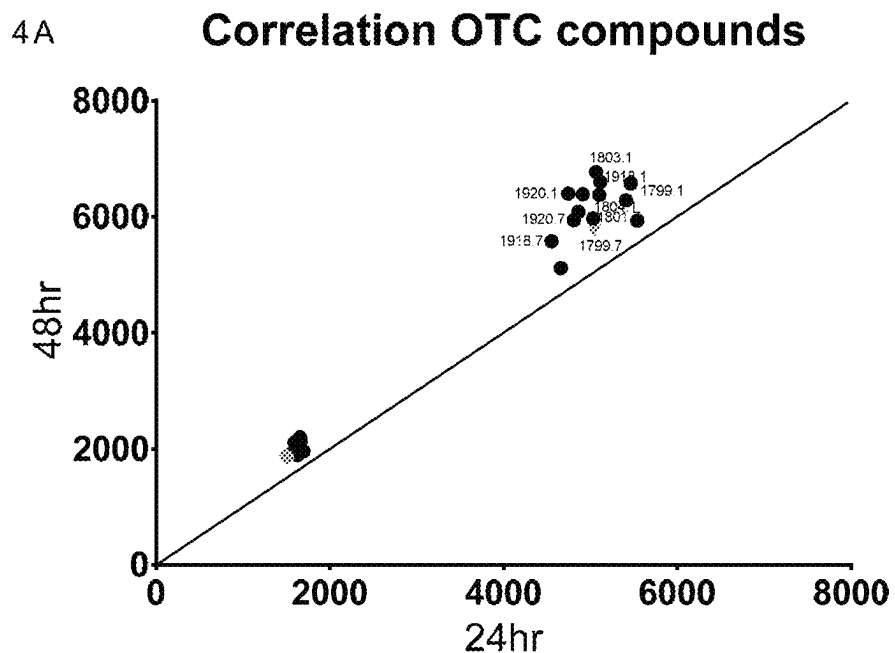


FIG. 4

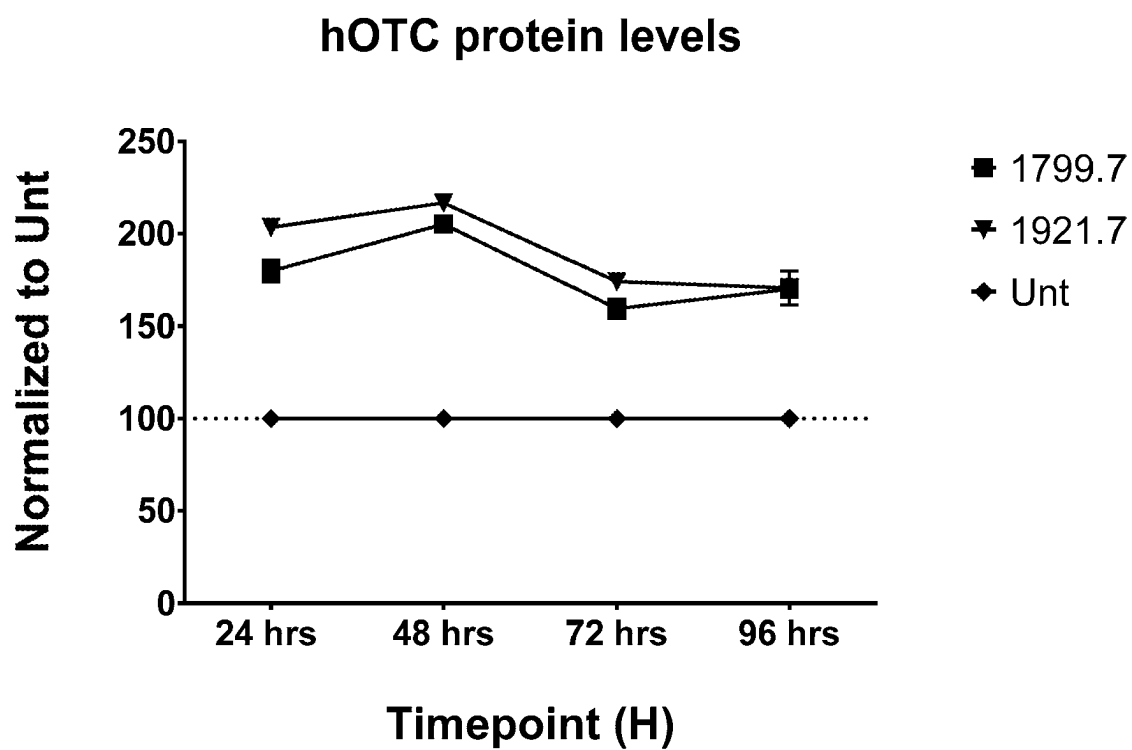


FIG. 5

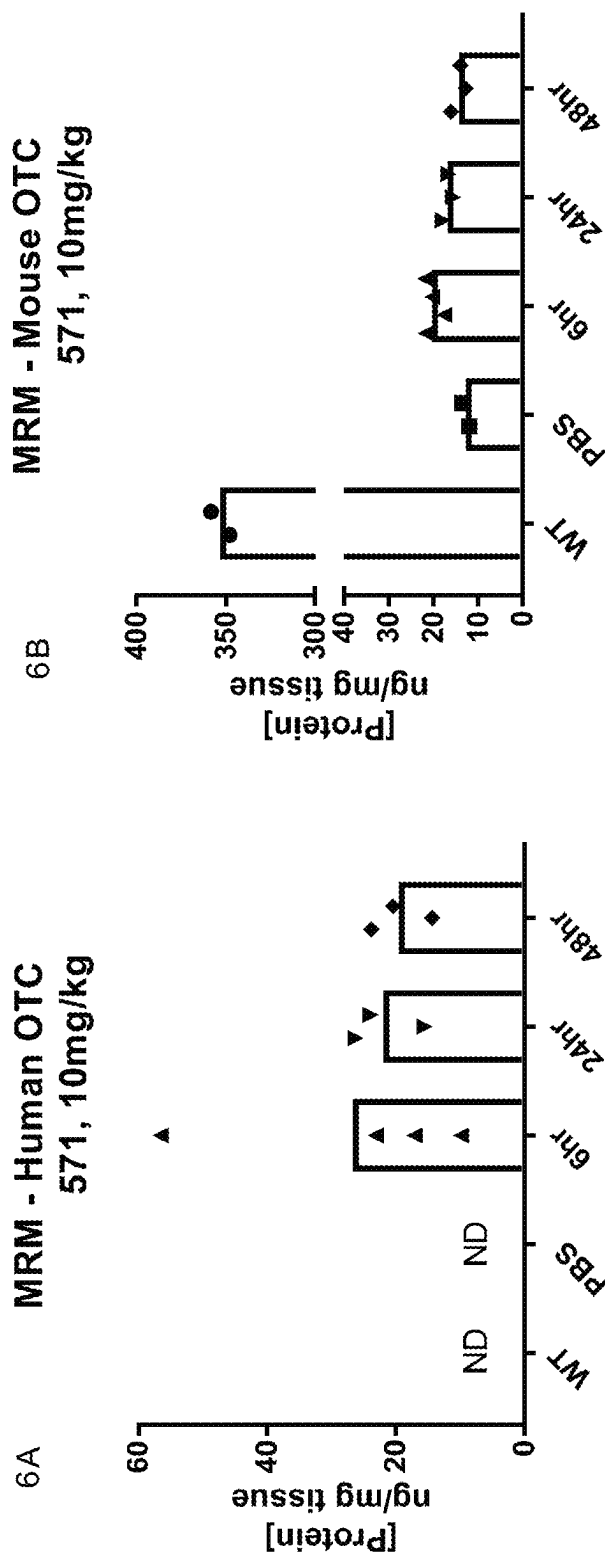


FIG. 6

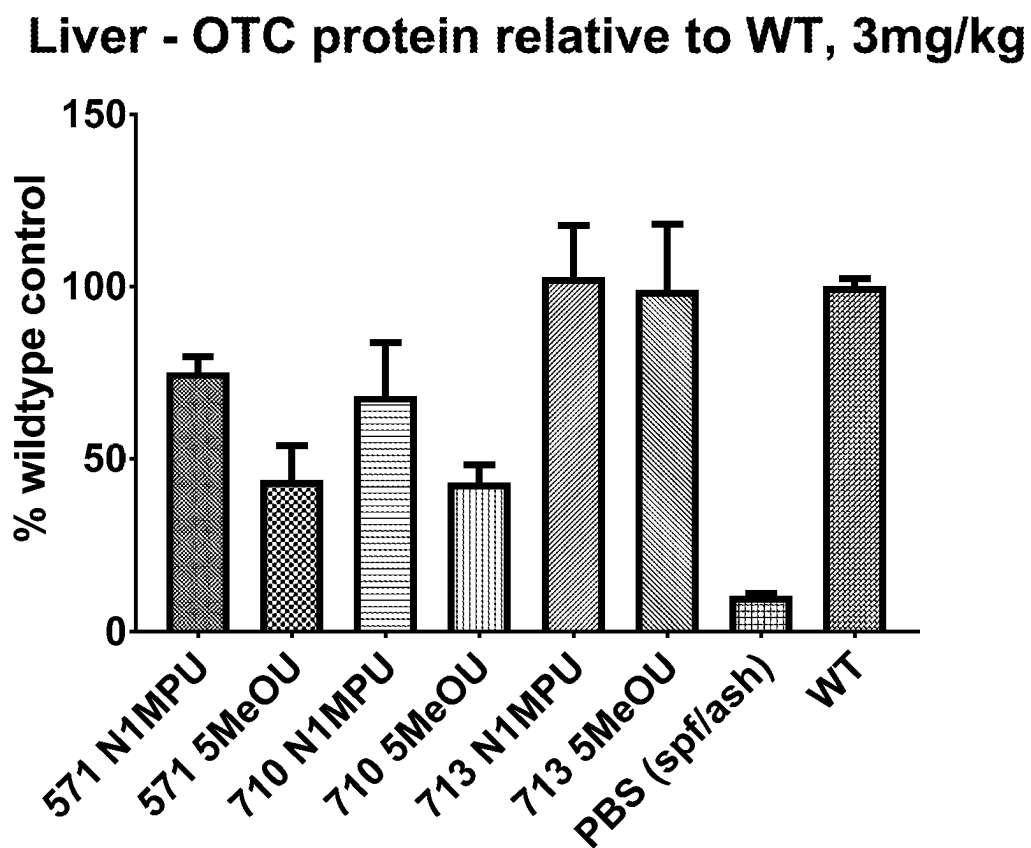


FIG. 7

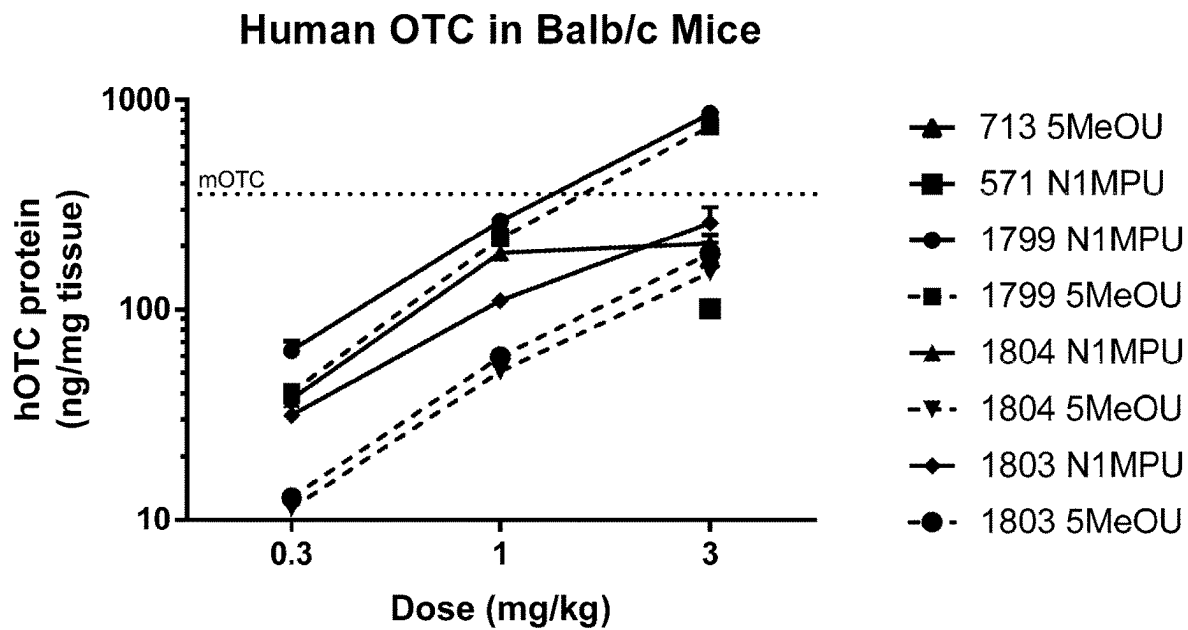


FIG. 8

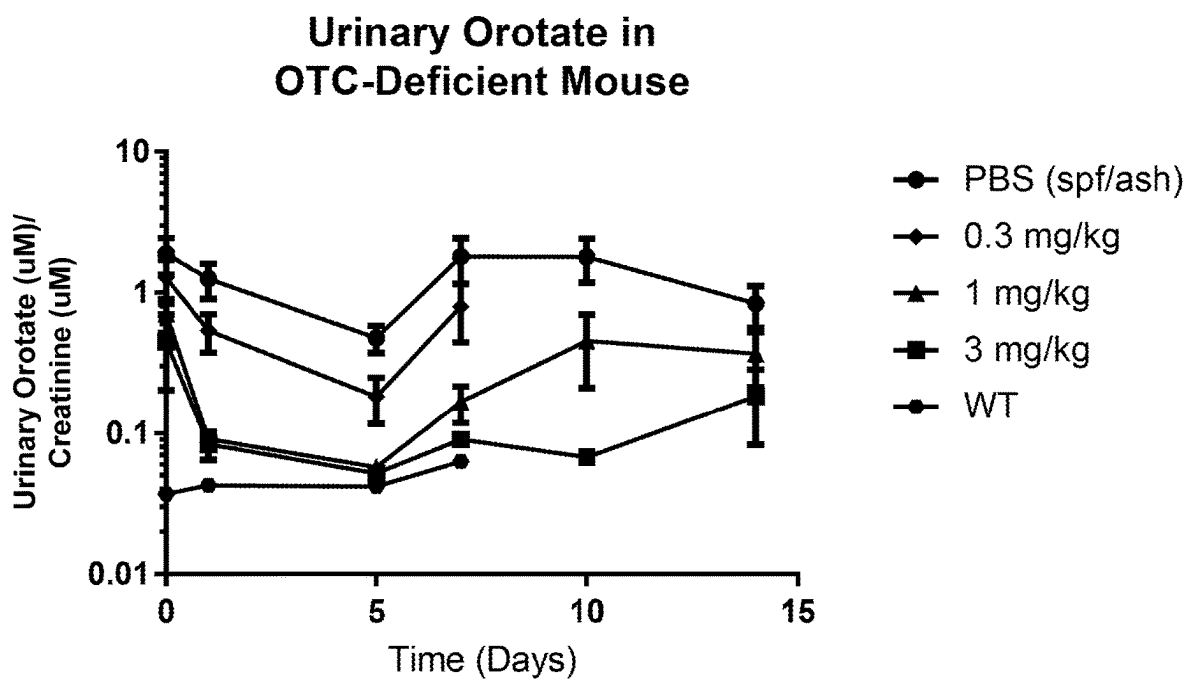


FIG. 9

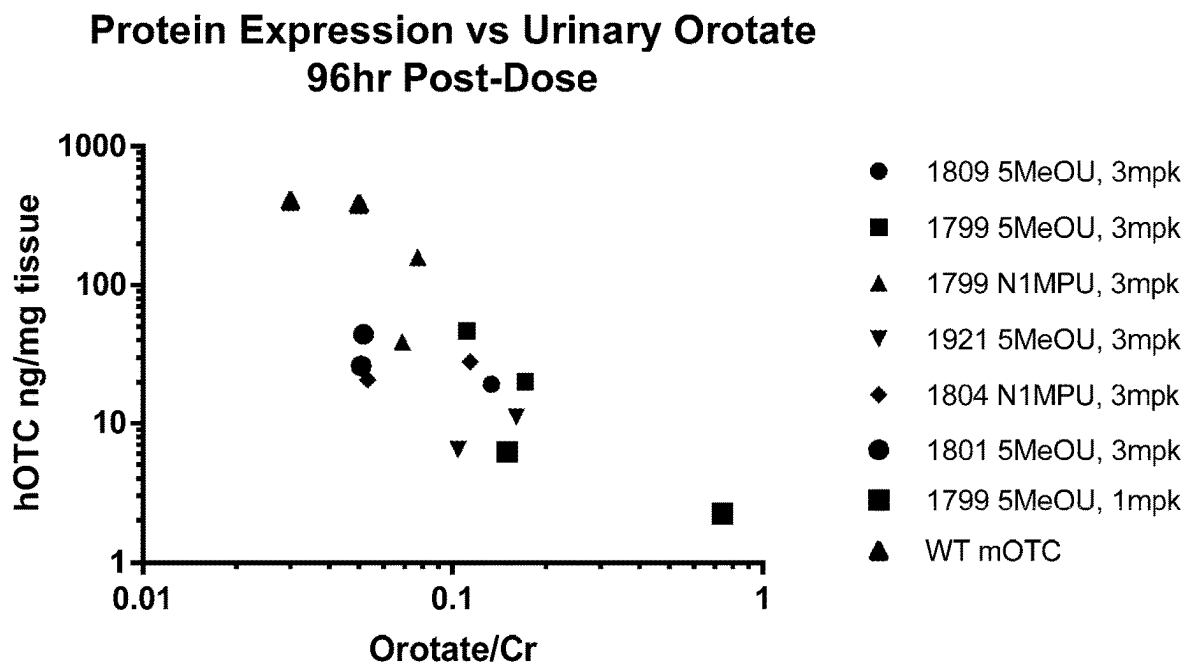


FIG. 10

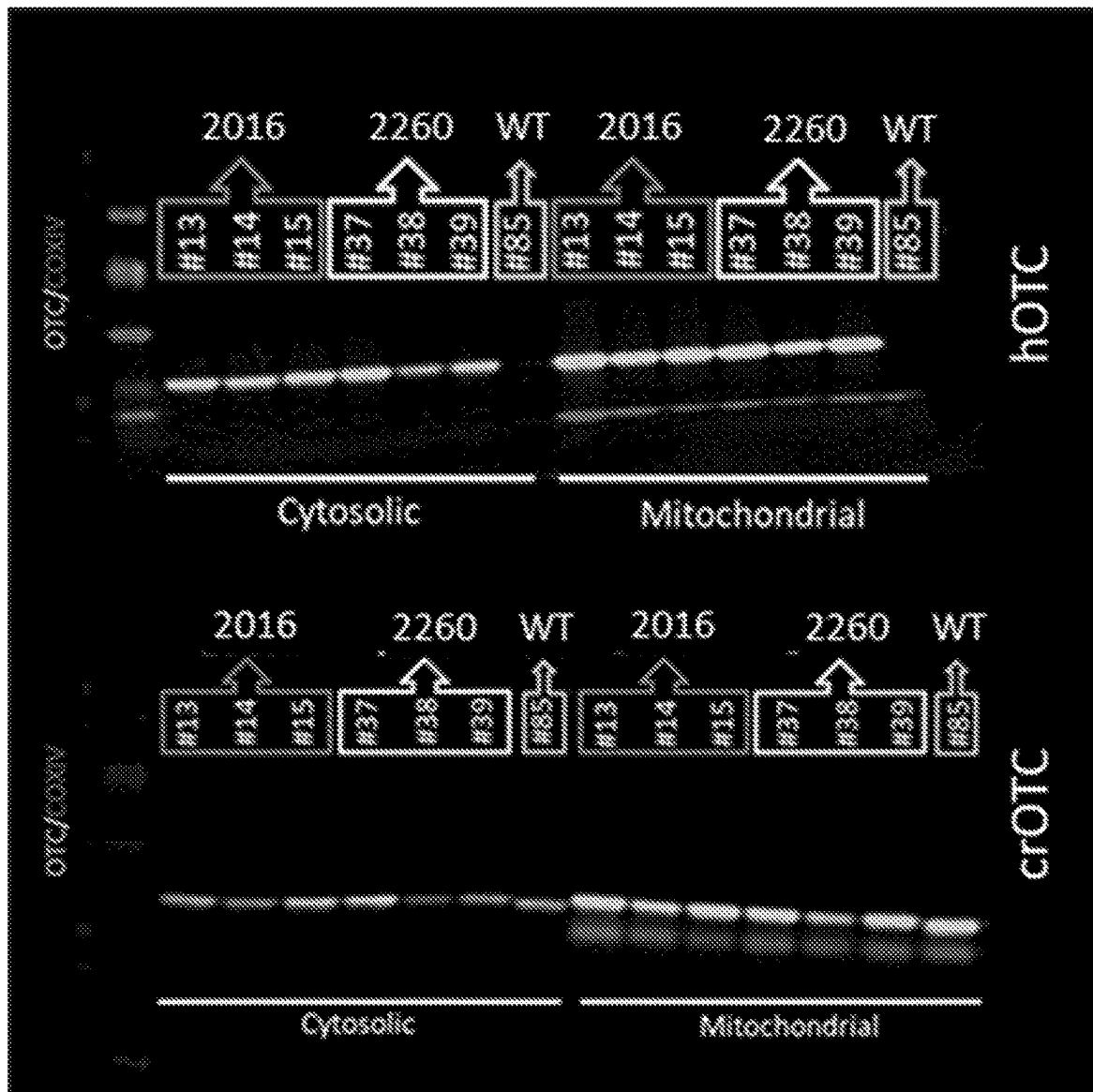


FIG. 11

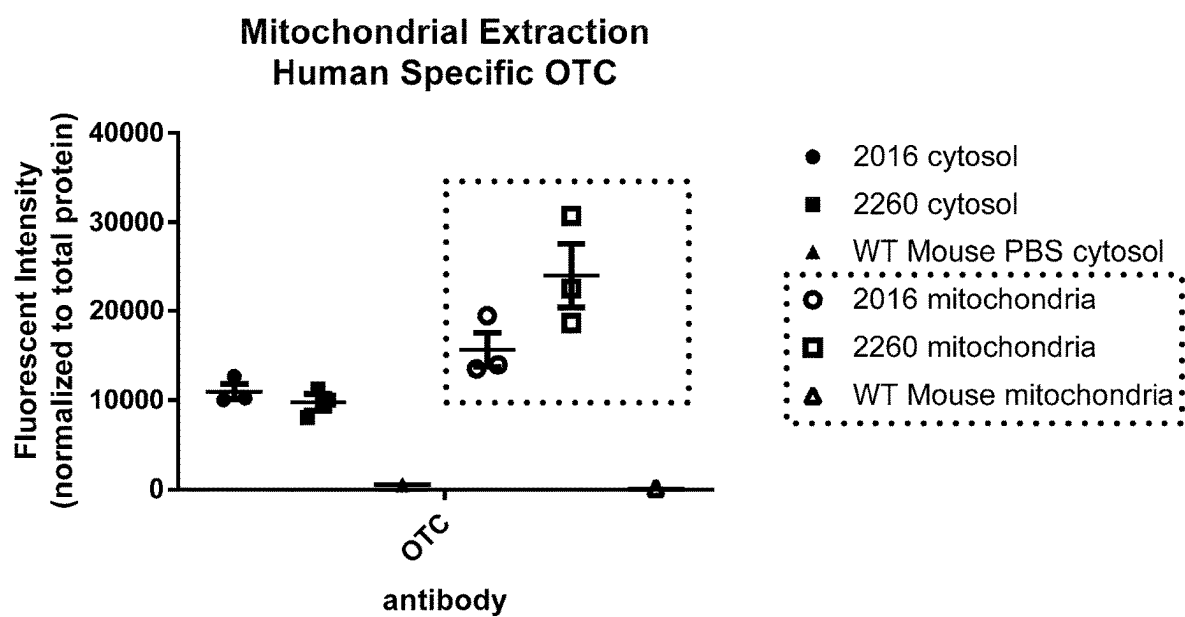


FIG. 12

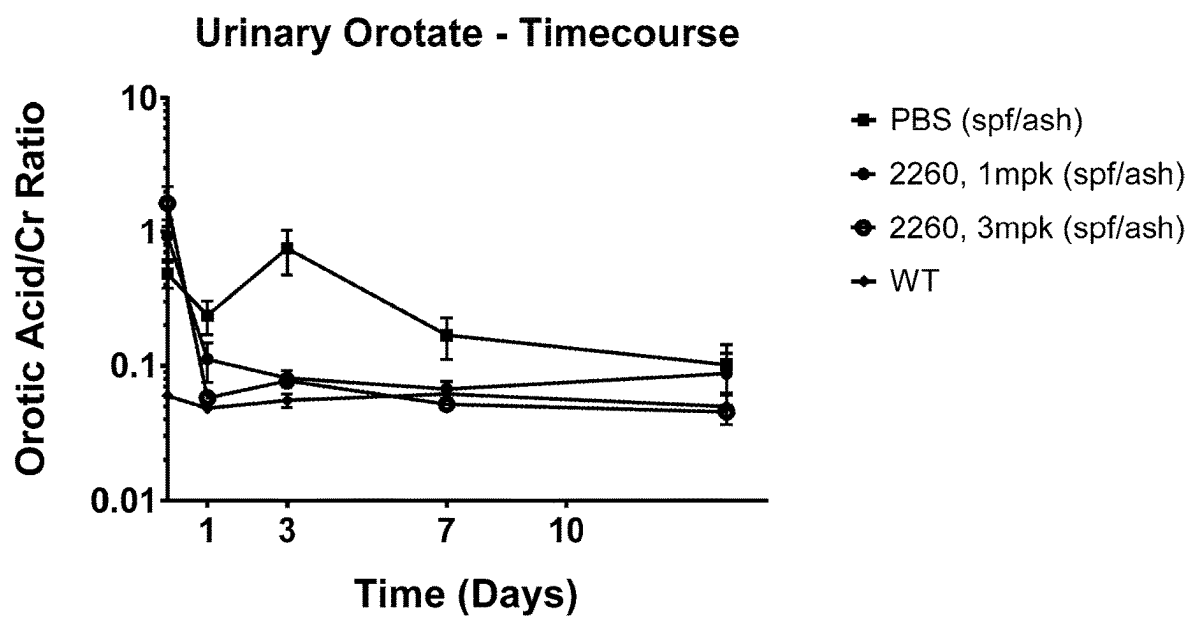


FIG. 13

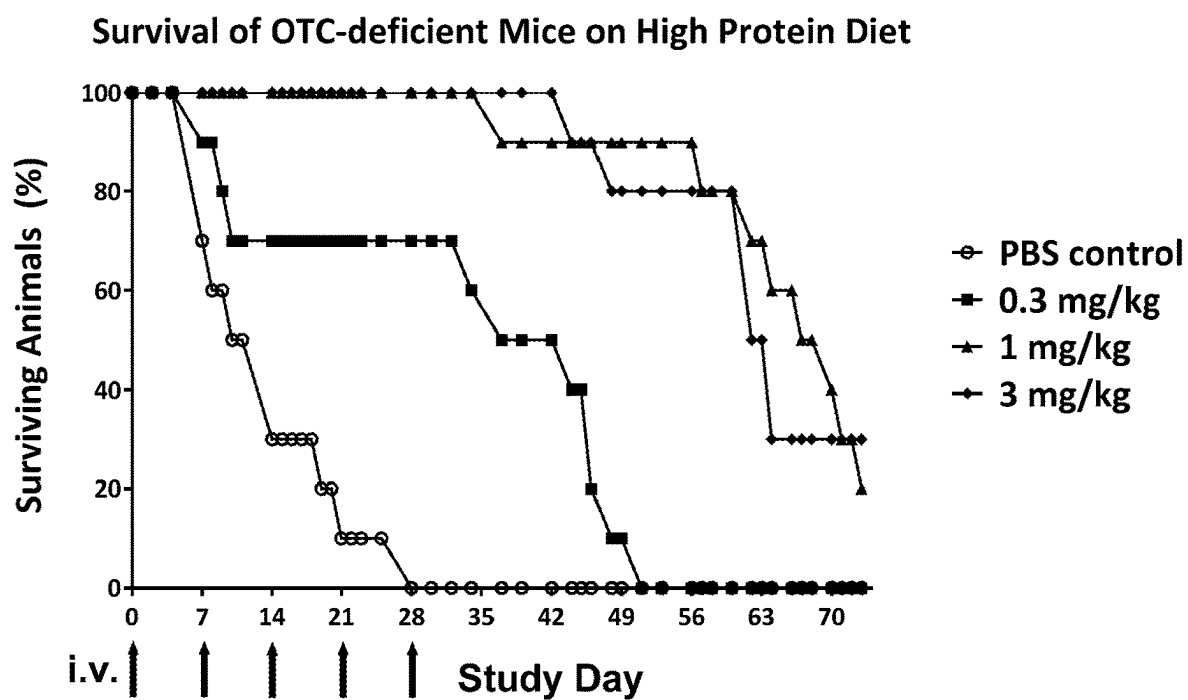


FIG. 14

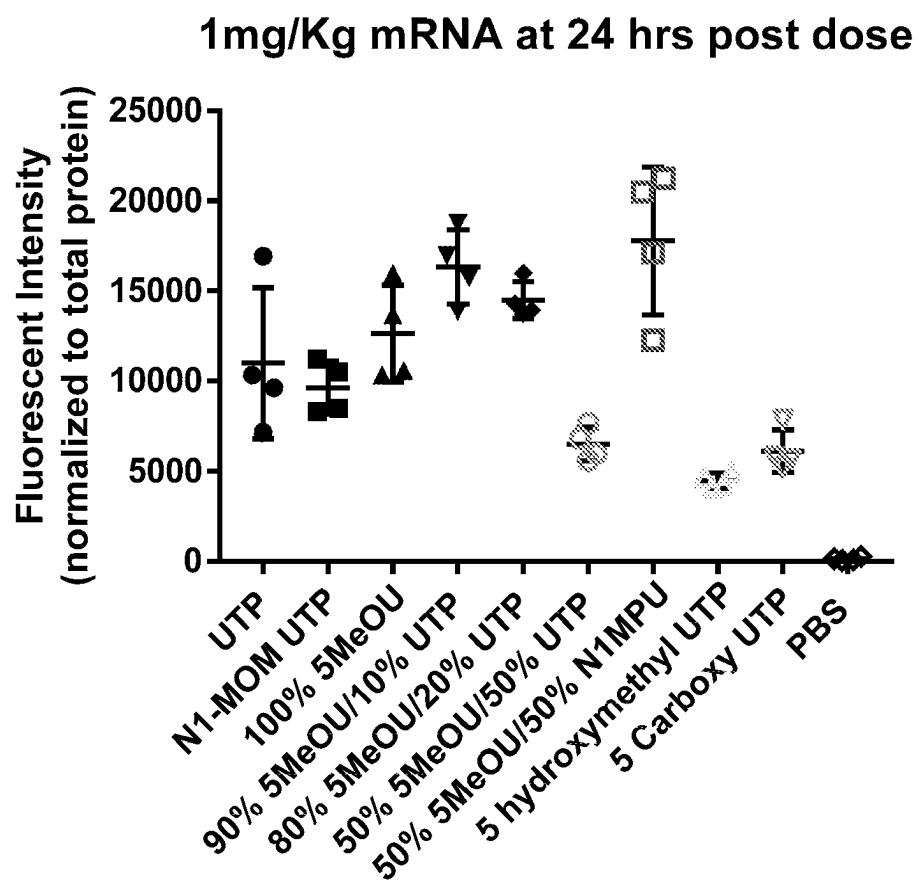


FIG. 15

1

COMPOSITIONS AND METHODS FOR TREATING ORNITHINE TRANSCARBAMYLASE DEFICIENCY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/705,102, filed Dec. 5, 2019, which claims the benefit of and priority to U.S. Provisional Application No. 62/776,302, filed Dec. 6, 2018. The contents of each application are incorporated herein by reference in their entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on May 11, 2023 is named 049386-523C01US_SL_ST26.xml and is 568,374 bytes in size.

BACKGROUND

Ornithine transcarbamylase (OTC) is a mitochondrial enzyme present in mammals which plays an essential role in detoxifying the organism from toxic ammonia. OTC mRNA has a mitochondrial signaling peptide (MSP) that is critical to redirect the nascent pre-peptide from the cytosol into the mitochondria. OTC protein exist as a precursor in the cytosol, the presence of the MSP redirects the pro-peptide into the mitochondria, where it undergoes excision of the signaling peptide and delivery of the functional protein into the mitochondrial matrix. OTC is one of six enzymes that play a role in the breakdown of proteins and removal of ammonia from the body, a process known as the urea cycle, a metabolic process that occurs in hepatocytes. OTC is responsible for converting carbamoyl phosphate and ornithine into citrulline.

Deficiency of the OTC enzyme results in excessive accumulation of nitrogen, in the form of ammonia (hyperammonemia), in the blood. Excess ammonia, which is a neurotoxin, travels to the central nervous system through the blood, resulting in the symptoms and physical findings associated with OTC deficiency. These symptoms can include vomiting, refusal to eat, progressive lethargy, or coma. If left untreated a hyperammonemic episode may progress to coma and life-threatening complications.

The severity and age of onset of OTC deficiency vary from person to person, even within the same family. A severe form of the disorder affects some infants, typically males, shortly after birth (neonatal period). A milder form of the disorder affects some children later in infancy. Both males and females may develop symptoms of OTC deficiency during childhood. Presently, the treatment of OTC deficiency is aimed at preventing excessive ammonia from being formed or from removing excessive ammonia during a hyperammonemic episode. Long-term therapy for OTC deficiency combines dietary restrictions and the stimulation of alternative methods of converting and excreting nitrogen from the body (alternative pathways therapy).

Dietary restrictions in individuals with OTC deficiency are aimed at limiting the amount of protein intake to avoid the development of excess ammonia. However, enough protein must be taken in by an affected infant to ensure proper growth. Infants with OTC deficiency are placed on a low protein, high calorie diet supplemented by essential amino acids.

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In addition to dietary restrictions, individuals with OTC deficiency are treated by medications that stimulate the removal of nitrogen from the body. These medications provide an alternative method to the urea cycle in converting and removing nitrogen waste. These medications are unpalatable to many patients and are often administered via a tube that is placed in the stomach through the abdominal wall (gastrostomy tube) or a narrow tube that reaches the stomach via the nose (nasogastric tube).

In cases where there is no improvement or in cases where hyperammonemic coma develops, the removal of wastes by filtering an affected individual's blood through a machine (hemodialysis) may be necessary. Hemodialysis is also used to treat infants, children, and adults who are first diagnosed with OTC deficiency during hyperammonemic coma.

In some cases, liver transplantation, may be an appropriate treatment option. Liver transplantation can cure the hyperammonemia in OTC deficiency. However, this operation is risky and may result in post-operative complications. Also, after liver transplantation, patients will need to follow a medication regimen throughout their lives for immunosuppression.

Novel approaches and therapies are still needed for the treatment of OTC enzyme deficiency. Strategies are needed which overcome the challenges and limitations associated with, for example, gene therapy. Poor stability, getting enough OTC in the mitochondria and efficient delivery to the target cells are still challenges.

SUMMARY

The present disclosure provides a modified human OTC protein of SEQ ID NO: 4 having improved properties for the treatment of OTC deficiency in a patient. SEQ ID NO: 4 has been modified from wild-type OTC to remove one or more predicted ubiquitination sites resulting in a protein that is less susceptible to ubiquitination and degradation by ubiquitin ligases. However, the modified OTC protein of SEQ ID NO: 4 maintains the catalytic enzyme activity of human wild type OTC. The removal of predicted ubiquitination sites preferably comprises replacing N-terminus residues that have been found to support ubiquitination such as asparagine, arginine, leucine, lysine or phenylalanine with N-terminus residues that have been found to be stabilizing against ubiquitination such as alanine, glycine, methionine, serine, threonine, valine and proline. Stabilization of the modified OTC protein of SEQ ID NO: 4 in this manner is particularly advantageous for preserving the stability of the modified OTC protein during its transport from the cytosol to the mitochondria wherein it exerts its enzymatic activity.

Preferably, the protein of SEQ ID NO: 4 described herein is produced from a nucleic acid encoding the protein of SEQ ID NO: 4. The nucleic acid may be RNA or DNA that encodes the protein of SEQ ID NO: 4. Preferably the nucleic acid is a heterologous mRNA construct comprising an open reading frame encoding for the modified protein of SEQ ID NO: 4. Preferably, the open reading frame is a codon-optimized open reading frame. Preferably, the open reading frame sequence is optimized to have a theoretical minimum of uridines possible to encode for the modified protein. Preferably, the heterologous mRNA construct comprises a 5' cap, a 5'UTR, a 3'UTR, an open reading frame encoding a modified protein of SEQ ID NO: 4 and a 3' poly A tail. Preferably, the 5'UTR derived from a gene expressed by *Arabidopsis thaliana*. Preferably the 5' UTR derived from a gene expressed by *Arabidopsis thaliana* is found in Table 2.

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The present disclosure also provides mRNA (also referred to herein as mRNA constructs or mRNA sequences) comprising an optimized coding region encoding wild type human ornithine transcarbamylase (OTC) protein of SEQ ID NO: 3 or an OTC protein sequence that is at least 95% identical over the full length of SEQ ID NO: 3 and having OTC protein enzymatic activity wherein the mRNA sequences comprise a 5'UTR derived from a gene expressed by *Arabidopsis thaliana*.

The mRNA constructs described herein provide high-efficiency expression of the OTC proteins described herein. The expression can be in vitro, ex vivo, or in vivo.

The present disclosure also provides pharmaceutical compositions comprising the mRNA sequences described herein and methods of treating ornithine transcarbamylase (OTC) deficiency by administering the pharmaceutical compositions comprising the mRNA sequences described herein to a patient in need thereof wherein the OTC protein of SEQ ID NO: 3 or SEQ ID NO: 4 is expressed in the patient.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1B show scatter plots illustrating ornithine transcarbamylase (OTC) protein expression in hepatocyte cell lines Hepa1,6 (mouse) and Hep3B (human) at 24 hours (FIG. 1A) and 48 hours (FIG. 1B) using In-Cell Western (ICW) assays.

FIGS. 2A-2B show scatter plots illustrating the correlation of protein stability compounds screened in Hepa1,6 cells (FIG. 2A) and Hep3B cells (FIG. 2B) at 24 h in Round 1.

FIGS. 3A-3B show scatter plots illustrating the correlation of protein stability compounds screened in human primary hepatocytes at 24 h and 48 h in Round 2 (newly optimized compounds based on Round 1) as shown with FIG. 3A and FIG. 3B.

FIGS. 4A-4B show scatter plots illustrating the correlation of protein stability compounds screened in human primary hepatocytes at 24 h and 48 h in Round 3 (newly optimized compounds based on rounds 1 and 2) as shown with FIG. 4A and FIG. 4B.

FIG. 5 is a plot illustrating OTC protein expression levels in human primary hepatocytes transfected with lead OTC mRNAs. 1799.7 is an mRNA having the sequence of SEQ ID NO: 175 wherein 100% of the uridines in SEQ ID NO: 175 are N¹-methylpseudouridine (N1MPU).

FIGS. 6A-6B show bar graphs depicting time course OTC expression levels in spf/ash mice dosed at 10 mg/kg with human-specific OTC-mRNA epitopes (FIG. 6A) or mouse-specific OTC-mRNA epitopes (FIG. 6B).

FIG. 7 is a bar graph depicting OTC expression levels in spf/ash mice dosed at 3 mg/kg with OTC-mRNAs using two different chemistries wherein 100% of the uridines are N¹-methylpseudouridine (N1MPU) and 100% of the uridines are 5-methoxyuridine (5MeOU).

FIG. 8 is a graph depicting OTC expression levels in Balb/c mice dosed with OTC-mRNAs at three different doses and using two different chemistries (N1MPU and 5MeOU).

FIG. 9 is a graph depicting urinary orotate levels measured in spf/ash mice dosed with OTC-mRNA 1799.7 at three different doses: 0.3 mg/kg, 1 mg/kg and 3 mg/kg. 1799.7 is an mRNA having the sequence of SEQ ID NO: 175 wherein 100% of the uridines in SEQ ID NO: 175 are 5MeOU.

FIG. 10 is a scatter plot comparing human OTC expression levels and Urinary Orotate at 96 h in spf/ash mice dosed

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with OTC-mRNAs at 1 mg/kg and 3 mg/kg using two different chemistries (N1MPU and 5MeOU).

FIG. 11 is a western blot illustrating the protein expression levels of OTC-mRNAs in spf/ash mice dosed at 1 mg/kg and 3 mg/kg with lead OTC-mRNAs.

FIG. 12 is a western blot illustrating the protein expression levels in mitochondrial vs cytosolic fractions of spf/ash mice treated with lead OTC-mRNAs.

FIG. 13 is a plot illustrating the time course of expression of urinary orotate levels in spf/ash mice treated with lead OTC-mRNA.

FIG. 14 is a plot illustrating the survival of OTC-deficient mice (spf/ash) on a high protein diet during treatment with three different doses of OTC-mRNA 1799.7.

FIG. 15 is a plot illustrating hOTC expression levels in male C57BL/6 mice dosed with OTC-mRNAs (2262) having different modifications.

DETAILED DESCRIPTION

Definitions

The term “ornithine transcarbamylase” as used interchangeably herein with “OTC” or “hOTC”, or “OTC_HUMAN” generally refers to the human protein associated with UniProtKB-P00480. The amino acid sequence for the wild type human OTC protein is represented herein by SEQ ID NO: 3.

The term “nucleic acid,” in its broadest sense, includes any compound and/or substance that comprise a polymer of nucleotides. These polymers are often referred to as polynucleotides. Exemplary nucleic acids or polynucleotides described herein include, but are not limited to, ribonucleic acids (RNAs), deoxyribonucleic acids (DNAs), threose nucleic acids (TNAs), glycol nucleic acids (GNAs), peptide nucleic acids (PNAs), locked nucleic acids (LNAs, including LNA having a β -D-ribo configuration, α -LNA having an α -L-ribo configuration (a diastereomer of LNA), 2'-amino-LNA having a 2'-amino functionalization, and 2'-amino- α -LNA having a 2'-amino functionalization) or hybrids thereof.

As used herein, the term “polynucleotide” is generally used to refer to a nucleic acid (e.g., DNA or RNA). When RNA, such as mRNA, is specifically being referred to, the term polyribonucleotide may be used. The terms polynucleotide, polyribonucleotide, nucleic acid, ribo nucleic acid, DNA, RNA, mRNA, and the like include such molecules that may be comprised of standard or unmodified residues; nonstandard or modified residues (e.g., analogs); and mixtures of standard and nonstandard (e.g., analogs) residues. In certain embodiments a polynucleotide or a polyribonucleotide is a modified polynucleotide or a polyribonucleotide. In the context of the present disclosure, for each RNA (polyribonucleotide) sequence listed herein, the corresponding DNA (polydeoxyribonucleotide or polynucleotide) sequence is contemplated and vice versa. “Polynucleotide” may be used interchangeably with the “oligomer”. Polynucleotide sequences shown herein are from left to right, 5' to 3', unless stated otherwise.

As used herein, the term “messenger RNA” (mRNA) refers to any polynucleotide which encodes a protein or polypeptide of interest and which is capable of being translated to produce the encoded protein or polypeptide of interest in vitro, in vivo, in situ or ex vivo.

As used herein, the term “translation” is the process in which ribosomes create polypeptides. In translation, messenger RNA (mRNA) is decoded by transfer RNAs (tRNAs)

in a ribosome complex to produce a specific amino acid chain, or polypeptide. The coding region of a polynucleotide sequence (DNA or RNA), also known as the coding sequence or CDS, is capable of being converted to a protein or a fragment thereof by the process of translation.

As used herein, the term “codon-optimized” means a natural (or purposefully designed variant of a natural) coding sequence which has been redesigned by choosing different codons without altering the encoded protein amino acid sequence. Codon optimized sequence can increase the protein expression levels (Gustafsson et al., *Codon bias and heterologous protein expression*. 2004, Trends Biotechnol 22: 346-53) of the encoded proteins amongst providing other advantages. Variables such as high codon adaptation index (CAI), LowU method, mRNA secondary structures, cis-regulatory sequences, GC content and many other similar variables have been shown to somewhat correlate with protein expression levels (Villalobos et al., *Gene Designer: a synthetic biology tool for constructing artificial DNA segments*. 2006, BMC Bioinformatics 7:285). High CAI (codon adaptation index) method picks a most frequently used synonymous codon for an entire protein coding sequence. The most frequently used codon for each amino acid is deduced from 74,218 protein-coding genes from a human genome. The Low U method targets only U-containing codons that can be replaced with a synonymous codon with fewer U moieties. If there are a few choices for the replacement, the more frequently used codon will be selected. The remaining codons in the sequence are not changed by the Low U method. This method may be used in conjunction with the disclosed mRNAs to design coding sequences that are to be synthesized with, for example, 5-methoxyuridine or N¹-methylpseudouridine.

As used herein, “modified” refers to a change in the state or structure of a molecule disclosed herein. The molecule may be changed in many ways including chemically, structurally or functionally. Preferably a polynucleotide or polypeptide of the disclosure are modified as compared to the native form of the polynucleotide or polypeptide or as compared to a reference polypeptide sequence or polynucleotide sequence. For example, mRNA disclosed herein may be modified by codon optimization, or by the insertion of non-natural nucleosides or nucleotides. Polypeptides may be modified, for example, by site specific amino acid deletions or substitutions to alter the properties of the polypeptide.

As used herein, the term “homology” refers to the overall relatedness between polymeric molecules, e.g. between nucleic acid molecules (e.g. DNA molecules and/or RNA molecules) and/or between polypeptide molecules. In some embodiments, polymeric molecules are considered to be “homologous” to one another if their sequences are at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% identical or similar. The term “homologous” necessarily refers to a comparison between at least two sequences (polynucleotide or polypeptide sequences). In accordance with the present disclosure, two polynucleotide sequences are considered to be homologous if the polypeptides they encode are at least about 50%, 60%, 70%, 80%, 90%, 95%, or even 99% for at least one stretch of at least about 20 amino acids. In some embodiments, homologous polynucleotide sequences are characterized by the ability to encode a stretch of at least 4-5 uniquely specified amino acids. For polynucleotide sequences less than 60 nucleotides in length, homology is determined by the ability to encode a stretch of at least 4-5 uniquely specified amino acids. In accordance with the present disclosure, two protein sequences are considered to be homolo-

gous if the proteins are at least about 50%, 60%, 70%, 80%, or 90% identical for at least one stretch of at least about 20 amino acids.

As used herein, the term “identity” refers to the overall relatedness between polymeric molecules, e.g., between oligonucleotide molecules (e.g. DNA molecules and/or RNA molecules) and/or between polypeptide molecules. Calculation of the percent identity of two polynucleotide sequences, for example, can be performed by aligning the two sequences for optimal comparison purposes. In certain embodiments, the length of a sequence aligned for comparison purposes is at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or 100% of the length of the reference sequence. The nucleotides at corresponding nucleotide positions are then compared. When a position in the first sequence is occupied by the same nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences, taking into account the number of gaps, and the length of each gap, which needs to be introduced for optimal alignment of the two sequences. The comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. Methods commonly employed to determine percent identity between sequences include, but are not limited to those disclosed in Carillo, H., and Lipman, D., *SIAM J Applied Math.*, 48:1073 (1988); incorporated herein by reference. Techniques for determining identity are codified in publicly available computer programs. Exemplary computer software to determine homology between two sequences include, but are not limited to, GCG program package, Devereux, J., et al., *Nucleic Acids Research*, 12(1), 387 (1984), BLASTP, BLASTN, and FASTA Altschul, S. F. et al., *J. Molec. Biol.*, 215, 403 (1990).

An “effective amount” of the mRNA sequence encoding an open reading frame (ORF) protein or a corresponding composition thereof is generally that amount of mRNA that provides efficient ORF protein production in a cell. Preferably protein production using an mRNA composition described herein is more efficient than a composition containing a corresponding wild type mRNA encoding an ORF protein. Increased efficiency may be demonstrated by increased cell transfection (i.e., the percentage of cells transfected with the nucleic acid), increased protein translation from the nucleic acid, decreased nucleic acid degradation (as demonstrated, e.g., by increased duration of protein translation from a modified nucleic acid), or reduced innate immune response of the host cell. When referring to an ORF protein described herein, an effective amount is that amount of ORF protein that overcomes an ORF protein deficiency in a cell.

As used herein, the term “in vitro” refers to events that occur in an artificial environment, e.g., in a test tube or reaction vessel, in cell culture, in a Petri dish, etc., rather than within an organism (e.g., animal, plant, or microbe).

As used herein, the term “in vivo” refers to events that occur within an organism (e.g., animal, plant, or microbe or cell or tissue thereof).

As used herein, the term “isolated” refers to a substance or entity that has been separated from at least some of the components with which it was associated (whether in nature or in an experimental setting). Isolated substances may have varying levels of purity in reference to the substances from which they have been associated. Isolated substances and/or entities may be separated from at least about 10%, about

20%, about 30%, about 40%, about 50%, about 60%, about 70%, about 80%, about 90%, or more of the other components with which they were initially associated. In some embodiments, isolated agents are more than about 80%, about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or more than about 99% pure. As used herein, a substance is "pure" if it is substantially free of other components. Substantially isolated: By "substantially isolated" is meant that the compound is substantially separated from the environment in which it was formed or detected. Partial separation can include, for example, a composition enriched in the compound described herein. Substantial separation can include compositions containing at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 95%, at least about 97%, or at least about 99% by weight of the compound described herein, or salt thereof. Methods for isolating compounds and their salts are routine in the art.

As used herein, the term "subject" or "patient" refers to any organism to which a composition in accordance with the present disclosure may be administered, e.g., for experimental, diagnostic, prophylactic, and/or therapeutic purposes. Typical subjects include animals (e.g., mammals such as mice, rats, rabbits, non-human primates, and humans) and/or plants. Preferably "patient" refers to a human subject who may seek or be in need of treatment, requires treatment, is receiving treatment, will receive treatment, or a subject who is under care by a trained professional for a particular disease or condition.

The phrase "pharmaceutically acceptable" is employed herein to refer to those compounds, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

As used herein, the term "preventing" refers to partially or completely delaying onset of an infection, disease, disorder and/or condition; partially or completely delaying onset of one or more symptoms, features, or clinical manifestations of a particular infection, disease, disorder, and/or condition; partially or completely delaying onset of one or more symptoms, features, or manifestations of a particular infection, disease, disorder, and/or condition; partially or completely delaying progression from an infection, a particular disease, disorder and/or condition; and/or decreasing the risk of developing pathology associated with the infection, the disease, disorder, and/or condition.

As used herein, the term "substantially" refers to the qualitative condition of exhibiting total or near-total extent or degree of a characteristic or property of interest. One of ordinary skill in the biological arts will understand that biological and chemical phenomena rarely, if ever, go to completion and/or proceed to completeness or achieve or avoid an absolute result. The term "substantially" is therefore used herein to capture the potential lack of completeness inherent in many biological and chemical phenomena.

As used herein, the term "therapeutically effective amount" means an amount of an agent to be delivered (e.g., nucleic acid, protein or peptide, drug, therapeutic agent, diagnostic agent, prophylactic agent, etc.) that is sufficient, when administered to a subject suffering from or susceptible to an infection, disease, disorder, and/or condition, to treat, improve symptoms of, diagnose, prevent, and/or delay the onset of the infection, disease, disorder, and/or condition.

As used herein, a "total daily dose" is an amount given or prescribed in a 24 hr period. It may be administered as a single unit dose.

As used herein, the term "treating" refers to partially or completely alleviating, ameliorating, improving, relieving, delaying onset of, inhibiting progression of, reducing severity of, and/or reducing incidence of one or more symptoms or features of OTC deficiency. Treatment may be administered to a subject who does not exhibit signs of OTC deficiency and/or to a subject who exhibits only early signs of OTC deficiency for the purpose of decreasing the risk of developing pathology associated with the disease, disorder, and/or condition.

As used herein, the terms "transfect" or "transfection" mean the intracellular introduction of a nucleic acid into a cell, or preferably into a target cell. The introduced nucleic acid may be stably or transiently maintained in the target cell. The term "transfection efficiency" refers to the relative amount of nucleic acid up-taken by the target cell which is subject to transfection. In practice, transfection efficiency is estimated by the amount of a reporter nucleic acid product expressed by the target cells following transfection. Preferred are compositions with high transfection efficacies and in particular those compositions that minimize adverse effects which are mediated by transfection of non-target cells and tissues.

As used herein, the term "target cell" refers to a cell or tissue to which a composition of the disclosure is to be directed or targeted. In some embodiments, the target cells are deficient in a protein or enzyme of interest. For example, where it is desired to deliver a nucleic acid to a hepatocyte, the hepatocyte represents the target cell. In some embodiments, the nucleic acids and compositions of the present disclosure transfect the target cells on a discriminatory basis (i.e., do not transfect non-target cells). The compositions and methods of the present disclosure may be prepared to preferentially target a variety of target cells, which include, but are not limited to, hepatocytes, epithelial cells, hematopoietic cells, epithelial cells, endothelial cells, lung cells, bone cells, stem cells, mesenchymal cells, neural cells (e.g., meninges, astrocytes, motor neurons, cells of the dorsal root ganglia and anterior horn motor neurons), photoreceptor cells (e.g., rods and cones), retinal pigmented epithelial cells, secretory cells, cardiac cells, adipocytes, vascular smooth muscle cells, cardiomyocytes, skeletal muscle cells, beta cells, pituitary cells, synovial lining cells, ovarian cells, testicular cells, fibroblasts, B cells, T cells, reticulocytes, leukocytes, granulocytes and tumor cells.

Following transfection of one or more target cells by the compositions and nucleic acids described herein, expression of the protein encoded by such nucleic acid may be preferably stimulated and the capability of such target cells to express the protein of interest is enhanced. For example, transfection of a target cell with an OTC mRNA will allow expression of the OTC protein product following translation of the nucleic acid. The nucleic acids of the compositions and/or methods provided herein preferably encode a product (e.g., a protein, enzyme, polypeptide, peptide, functional RNA, and/or antisense molecule), and preferably encode a product whose in vivo production is desired.

As used herein "an OTC protein enzymatic activity" refers to enzyme activity that catalyzes the reaction between carbamoyl phosphate and ornithine to form citrulline as part of the urea cycle in mammals.

As used herein, the term "about" or "approximately" as applied to one or more values of interest, refers to a value that is similar to a stated reference value. In certain embodi-

ments, the term “approximately” or “about” refers to a range of values that fall within 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, or less in either direction (greater than or less than) of the stated reference value unless otherwise stated or otherwise evident from the context (except where such number would exceed 100% of a possible value).

Polynucleotide Sequences

The present disclosure provides improved methods and compositions for the treatment of Ornithine transcarbamylase (OTC) deficiency using, for example, mRNA therapy. The present disclosure provides methods of treating ornithine transcarbamylase (OTC) deficiency, comprising administering to a subject in need of treatment a composition comprising an mRNA sequence described herein encoding a human ornithine transcarbamylase (OTC) protein, modified forms of human OTC protein or active fragments of OTC protein at an effective dose and an administration interval such that at least one symptom or feature of the OTC deficiency is reduced in intensity, severity, or frequency or has delayed onset. The present disclosure also provides modified OTC proteins encoded by the mRNA sequences wherein the modified OTC proteins have improved properties such as enhanced stability and resistance to protein degradation and increased half-life as compared to wild type human OTC proteins.

Preferably, the administration of an mRNA composition described herein results in an increased OTC protein expression or activity of the subject as compared to a control level. Preferably, the control level is a baseline serum OTC protein expression or activity level in the subject prior to the treatment and/or the control level is indicative of the average serum OTC protein expression or activity level in OTC patients without treatment.

Preferably, administration of a mRNA described herein composition results in a reduced urinary orotic acid level in the subject as compared to a control orotic acid level. Preferably, the control orotic acid level is a baseline urinary orotic acid level in the subject prior to the treatment and/or the control orotic acid level is a reference level indicative of the average urinary orotic acid level in OTC patients without treatment.

Preferably, the OTC proteins encoded by the mRNA described herein are produced from a heterologous mRNA construct comprising an open reading frame (ORF) also referred to herein as a “coding sequence” (CDS) encoding for an OTC protein. Preferably, the coding sequence is codon-optimized. Preferably, coding sequence is optimized to have a theoretical minimum of uridines possible to encode for an OTC protein. Preferably, the mRNA constructs described herein comprise one or more of the following features: a 5' cap; a 5'UTR, a 5'UTR enhancer sequence, a Kozak sequence or a partial Kozak sequence, a 3'UTR, an open reading frame encoding an OTC protein and a poly A tail. Preferably, the mRNA constructs described herein can provide high-efficiency expression of an OTC protein. The expression can be in vitro, ex vivo, or in vivo.

Preferably, a human OTC protein encoded by an mRNA described herein comprises a modified human OTC protein of SEQ ID NO: 4 shown in Table 1. SEQ ID NO: 4 has been modified from wild-type OTC of SEQ ID NO: 3 (Table 1) to remove one or more predicted ubiquitination sites resulting in a protein that is less susceptible to ubiquitination and degradation by ubiquitin ligases. The removal of predicted ubiquitination sites preferably comprises replacing N-terminus residues that have been found to support ubiquitination such as asparagine, arginine, leucine, lysine or phenylalanine with N-terminus residues that have been found to be

stabilizing against ubiquitination such as alanine, glycine, methionine, serine, threonine, valine and proline. Stabilization of the modified OTC protein of SEQ ID NO: 4 in this manner is particularly advantageous for preserving the stability of the modified OTC protein during its transport from the cytosol to the mitochondria wherein it exerts its enzymatic activity.

Preferably, an OTC protein encoded by an mRNA described herein comprises a protein sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 10000 identical to human wild type OTC protein of SEQ ID NO: 3 as shown in Table 1, while retaining the OTC protein activity of catalyzing the synthesis of citrulline (in the liver and small intestine) from carbamyl phosphate and ornithine.

TABLE 1

Selected OTC Nucleotide and Peptide Sequences

mRNA coding sequence for wild type human OTC	<u>AUGCUGUUUAAUCUGAGGAU</u> <u>CCUGUUAAACAAUCGAGCUU</u> <u>UUAGAAAUUGGUCACAAUCU</u> <u>UAGGUUCGAAAUUUUCGGUG</u> <u>UGGACAACCCACUACAAAU</u> AAGUGCAGCUGAAGGGCCGU GACCUUCUCACUCUAAAAA CUUUACCGGAGAAGAAUUA AAUAUAUGCUAUGGCUAUA GCAGAUUCGAAAUUUAGGAU AAAACAGAAAGGAGAGUAUU UGCCUUUAUUGCAAGGGAAG UCCUAGGCAGUAUUUUUGA GAAAAGAAGUACUCGAACAA GAUUGUCUACAGAAACAGGC UUUGCACUUCUGGGAGGACA UCCUUGUUUUCUACACAC AAGAUUAUCUUGGGUGUG AAUGAAAGUCUCACGGACAC GGCCCGUGUAUUGUCUAGCA UGGCAGAUAGCAGUAUUGGCU CGAGUGUAUAAACAAUCAGA UUUGGACACCCUGGCUAAG AAGCAUCCAUCCCAUUUUC AAUGGGCUGUCAGAUUUUGA CCAUCCUUAUCCAGAUCCUGG CUGAUUACCUACGCUCCAG GAACACUAUAGCUCUCUGAA AGGUCUUUACCCUAGCUGGA UCGGGGAUGGGAACAAUUC CUGCACUCCAUCAUGAUGAG CGCAGCGAAAUUCGGAUUC ACCUUCAGGCAGCUAUCUCA AAGGGUUUAUGAGCCGAGUC UAGUGUAACCAAGUUGGCAG AGCAGUAUGCCAAAGAGAAU GGUACCAAGCUGUUGCUGAC AAUGAUCCAUUGGAAGCAG CGCAUGGAGGCAAGUUAUA AUUACAGACACUUGGAUAAG CAUGGGACAAGAAGAGGAGA AGAAAAAGCGGCCUCCAGGCU UUCCAAGGUUACCAAGGUUAC AAUGAAGACUGCUAAGAUUG CUGCCUCUGACUGGACAUUU UUACACUGCUUGCCAGAAA GCCAGAAAGUUGGAUGAUG AAGUCUUUUUUUCCUCCGCA UCACUAGUGUUCCAGAGGC AGAAAAAGAAAGUGGACAA UCAUGGCUGUACUGGUGUCC CUGCUGACAGAUUACUCACC UCAGUCCAGAAAGCCUAAAU UUUGA (SEQ ID NO: 1)
DNA coding sequence for wild type human OTC	<u>ATGCTGTTTAATCTGAGGAT</u> <u>CCTGTTAAACAATGCAGCTT</u> <u>TTAGAAATGG</u>

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TABLE 1-continued

Selected OTC Nucleotide and Peptide Sequences	
	<u>TCACAACTTCATGGTTCGAA</u>
	<u>ATTTTCGGTGTGGACAACCA</u>
	<u>CTACAAATA</u>
	AAGTGCAGCTGAAGGGCCGT
	GACCTTCTCACTCTAAAAAA
	CTTTACCGG
	AGAAGAAATTAATATATGC
	TATGGCTATCAGCAGATCTG
	AAATTTAGG
	ATAAAACAGAAAGGAGAGTA
	TTTGCCTTTATTGCAAGGGA
	AGTCCTTAG
	GCATGATTTTGGAGAAAAGA
	AGTACTCGAACCAAGATTGTC
	TACAGAAAC
	AGGCTTTGCACTTCTGGGAG
	GACATCCTTGTCTTCTTACC
	ACACAAGATA
	TTCATTTGGGTGTGAATGAA
	AGTCTCACGGACACGGGCCG
	TGTATTGTCT
	AGCATGGCAGATGCAGTATT
	GGCTCGAGTGTATAACAAT
	CAGATTTGG
	ACACCCCTGGCTAAGAAGCA
	TCCATCCCAATTATCAATGG
	GCTGTCAGA
	TTTGTACCATCCTATCCAGA
	TCCTGGCTGATTACTCTACG
	CTCCAGGAAC
	ACTATAGCTCTCTGAAAGGT
	CTTACCCTCAGCTGGATCGG
	GGATGGGAA
	CAATATCCTGCACTCCATCA
	TGATGAGCGCAGCGAAATTC
	GGAATGCAC
	CTTCAGGCAGCTACTCCAAA
	GGGTTATGAGCCGATGCTA
	GTGTAACCA
	AGTTGGCAGAGCAGTATGCC
	AAAGAGAATGGTACCAAGCT
	GTTGCTGAC
	AAATGATCCATTGGAAGCAG
	CGCATGGAGGCAATGTATTA
	ATTACAGAC
	ACTTGGATAAGCATGGGACA
	AGAAAGAGGAGAAGAAAAGC
	GGCTCCAG
	GCTTTCCAAGGTTACCAAGT
	TACAATGAAGACTGCTAAAG
	TTGCTGCCTC
	TGACTGGACATTTTACACT
	GCTTGCCCGAGAAAGCCAGAA
	GAAGTGGAT
	GATGAAGTCTTTTATTCTCC
	TCGATCACTAGTGTCCAG
	AGGCAGAAAA
	CAGAAAGTGGACAATCATGG
	CTGTCATGGTGTCCCTGCTG
	ACAGATTACTCACCTCAGCT
	CCAGAAGCCTAAATTTTGA
	(SEQ ID NO: 2)
Human wild type OTC amino acid sequence (The signal peptide for mitochondrial import is underlined*)	<u>MLFNLRIILLNNAAFRNGHN</u>
	<u>FMVRNFRCGQLQNVQLKGR</u>
	DLTLTKNFTGEEIKYMLWLS
	ADLKFKRIKQKGEYLPQLQK
	SLGMIFEKRSTRRLSTETG
	FALLGGHPCFLTQDIHLGV
	NESLTD TARVLSSMADAVLA
	RVYKQSDLDLTAKEASIPF
	GLSDLYHPIQILADYLTQ
	EHYSSLKGLTLISWIGDGNNI
	LHSSIMMSAAKFGMHLQAATP
	KGYEPDASVTKLAEQYAKEN
	GTKLLLTNDPLEAAHGGNVL
	ITDWTISMGGQEEKKRLQA
	FQGYQVTMKTAKVAASDWT

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TABLE 1-continued

Selected OTC Nucleotide and Peptide Sequences	
	LHCLPRKPPEEVDDEVFYSR
	SLVFPPEAENRKWTIMAVMVS
	LLTDYSPQLQKPKF
	(SEQ ID NO: 3)
Modified OTC amino acid sequence (The signal peptide for mitochondrial import is underlined*)	<u>MLVFNLRIILLNNAAFRNGHN</u>
	<u>FMVRNFRCGQLQNVQLKKG</u>
	RDLLTLKNFTGEEIRYMLWL
	SADLKFKRIKQKGEYLPQLQK
	KSLGMIFEKRSTRRLSTET
	GFALLGGHPCFLTQDIHLG
	VNESLTD TARVLSSMADAVL
	ARVYKQSDLDLTAKEASIPF
	INGLSLDLYHPIQILADYLTQ
	QEHYSSLKGLTLISWIGDGNN
	ILHSSIMMSAAKFGMHLQAAT
	PKGYEPDASVTKLAEQYAKE
	NGTKLLLTNDPLEAAHGGNV
	LITDWTISMGGQEEKKRLQ
	AFQGYQVTMKTAKVAASDWT
	FLHCLPRKPPEEVDDEVFYSR
	RSLVFPPEAENRKWTIMAVMV
	SLLTDYSPQLQKPKF
	(SEQ ID NO: 4)
	*The OTC protein comprises a signal peptide which is translated and which is responsible for translocation to the mitochondria. This signal peptide is represented by the first 32 amino acids as underlined in SEQ ID NO: 3 and SEQ ID NO: 4. The signal sequence of SEQ ID NO: 4 has also been modified as compared to SEQ ID NO: 3. An amino acid, valine is inserted at position 3 of SEQ ID NO: 4. This modification provides better mitochondrial localization of the modified OTC of SEQ ID NO: 4 as compared to wild type human OTC of SEQ ID NO: 3.
Preferably, the open reading frame (ORF) or coding sequence (CDS) of an mRNA sequence described herein encodes an amino acid sequence that is substantially identical to the modified OTC protein of SEQ ID NO: 4.	
Preferably, the open reading frame (ORF) or coding sequence (CDS) of an mRNA sequence described herein encodes an amino acid sequence that is substantially identical to wild type human OTC protein of SEQ ID NO: 3.	
Preferably, an OTC protein encoded by an mRNA described herein comprises a protein sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to a modified human OTC protein of SEQ ID NO: 3 shown in Table 1 while retaining the OTC protein activity of catalyzing the synthesis of citrulline (in the liver and small intestine) from carbamyl phosphate and ornithine.	
Preferably, the ORF or CDS of an mRNA described herein encodes an amino acid sequence that is substantially identical to modified human OTC protein of SEQ ID NO: 4.	
Preferably, the ORF or CDS of an mRNA described herein encoding a human OTC protein comprises a codon optimized polynucleotide sequence at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to the mRNA coding sequence of SEQ ID NO: 1 of Table 1.	
Preferably an mRNA described herein further comprises a sequence immediately downstream (i.e., in the 3' direction from) of the CDS that creates a triple stop codon. The triple stop codon may be incorporated to enhance the efficiency of translation. In some embodiments, the translatable oligomer may comprise the sequence AUAAGUGAA (SEQ ID NO: 25) immediately downstream of an OTC CDS of an mRNA sequence described herein.	
Preferably, an mRNA described herein further comprises a 5' untranslated region (UTR) sequence. As is understood in the art, the 5' and/or 3' UTR may affect an mRNA's stability or efficiency of translation. The 5' UTR may be derived from an mRNA molecule known in the art to be relatively stable	

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(e.g., histone, tubulin, globin, glyceraldehyde 1-phosphate dehydrogenase (GAPDH), actin, or citric acid cycle enzymes) to increase the stability of the translatable oligomer. In other embodiments, a 5' UTR sequence may include a partial sequence of a cytomegalovirus (CMV) immediate-early 1 (IE1) gene.

Preferably, the 5' UTR comprises a sequence selected from the 5' UTRs of human IL-6, alanine aminotransferase 1, human apolipoprotein E, human fibrinogen alpha chain, human transthyretin, human haptoglobin, human alpha-1-antichymotrypsin, human antithrombin, human alpha-1-antitrypsin, human albumin, human beta globin, human

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complement C3, human complement C5, SynK (thylakoid potassium channel protein derived from the cyanobacteria, *Synechocystis* sp.), mouse beta globin, mouse albumin, and a tobacco etch virus, or fragments of any of the foregoing. Preferably, the 5' UTR is derived from a tobacco etch virus (TEV). Preferably, an mRNA described herein comprises a 5' UTR sequence that is derived from a gene expressed by *Arabidopsis thaliana*. Preferably, the 5' UTR sequence of a gene expressed by *Arabidopsis thaliana* is AT1G58420. Preferred 5' UTR sequences comprise SEQ ID NOS: 5-10, 125-127 and 230-250: as shown in Table 2.

TABLE 2

5'UTR sequences		
Name	Sequence	Seq ID No. :
EV	UCAACACAACAUUACAAAACAAACGAAUCUCAAGCAAUCAAGC AUUCUACUUUCUAUUGCAGCAAUUUAAUCAAUUUUUAAAGCA AAAGCAAUUUUUCUGAAAAUUUUCACCAUUUACGAACGAUAG	SEQ ID NO: 5
AT1G58420	AUUUUUACAUCAAAACAAAAGCCGCCA	SEQ ID NO: 6
ARC5-2	CUUAAAGGGGCGCGUCGCUACGGAGGUGGCAGCCAUCCUUCU CGGCAUCAAGCUUACCAUGGUGCCCCAGGCCUGCUCUUGGUC CGCUGCUGGUGUUCUUUUUCGCUUCGGCAAGUCCCCAUCUAC ACCAUCCCCGACAAGCUGGGGCGUGGAGCCCCAUCGACAUCCA CCACCUGUCCUGCCCCAACACCCUGGUGGAGGACGAGGGGCU GCACCAACCUGAGCGGGUUCUCCUAC	SEQ ID NO: 7
HCV	UGAGUGUCGU ACAGCCUCCA GGCCCCCCCC UCCCCGGAGA GCCAUAGUGG UCUGCGGAACCGUGAGUAC ACCGGAUUG CCGGGAAGAC UGGGUCCUUU CUUGGAUAAA CCCACUCUAUGCCCCGCCAU UUGGGCGUGC CCCCAGAGA CUGCUAGCCG AGUAGUGUUG GGUUGCG	SEQ ID NO: 8
HUMAN ALBUMIN	AAUUUUUGGUUAAAGAUUAUUUAGUGCUAAUUUCCUCCG UUUGUCCUAGCUUUUCUUCUGUCAACCCACACGCCUUUGG CACA	SEQ ID NO: 9
EMCV	CUCCUCCCC CCCCCUAAC GUUACUGGCC GAAGCCGCUU GGAAUAAGGC CGGUGUGCGU UUGUCUAUUA GUUAAUUUCC ACCAUAUUGC CGUCUUUUGG CAAUGUGAGG GCCCGGAAC CUGGCCUGU CUUCUUGACG AGCAUCCUA GGGGUCUUUC CCUCUCGCC AAAGGAAUGC AAGGUCUGUU GAAUGUCGUG AAGGAAGCAG UUCUCUGGA AGCUUCUUGA AGACAAACAA CGUCUGUAGC GACCCUUUGC AGGCAGCGGA ACCCCCACC UGGCGACAGG UGCCUCUGCG GCCAAAAGCC ACGUGUAUAA GAUACACCUG CAAAGGCGGC ACAACCCAG UGCCACGUUG UGAGUUGGAU AGUUGUGGAA AGAGUCAAAU GGCUCUCCUC AAGCGUAUUC AACAAAGGGC UGAAGGAUGC CCAGAAGGUA CCCCAUUGUA UGGGAUCUGA UCUGGGGCGU CGGUGCACA GCUUUACGUG UGUUUAGUCG AGGUUAAAAA ACGUCUAGGC CCCCCGAACC ACGGGGACGU GGUUUUCCUU UGAAAAACAC GAUGAUAAU	SEQ ID NO: 10
AT1G67090	CACAAAGAGUAAAGAAGACA	SEQ ID NO: 125
AT1G35720	AACACUAAAAGUAGAAGAAAA	SEQ ID NO: 126
AT5G45900	CUCAGAAAGAUAGAUCAGCC	SEQ ID NO: 127
AT5G61250	AACCAAUCGAAAGAAACAAA	SEQ ID NO: 230
AT5G46430	CUCUAUACACAGGAGUAAAA	SEQ ID NO: 231
AT5G47110	GAGAGAGAUUUACAAAAAA	SEQ ID NO: 232
AT1G03110	UGUGUAACAACAACAACAACA	SEQ ID NO: 233

TABLE 2-continued

5'UTR sequences		
Name	Sequence	Seq ID No.:
AT3G12380	CCGCAGUAGGAAGAGAAAGCC	SEQ ID NO: 234
AT5G45910	AAAAAAAAAAGAAUCAUAAA	SEQ ID NO: 235
AT1G07260	GAGAGAAGAAAGAAGACG	SEQ ID NO: 236
AT3G55500	CAAUUAAAAUACUACCAA	SEQ ID NO: 237
AT3G46230	GCAACAGAGUAAGCGAAACG	SEQ ID NO: 238
AT2G36170	GCGAAGAAGACGAACGCAAAG	SEQ ID NO: 239
AT1G10660	UUAGGACUGUAUUGACUGGCC	SEQ ID NO: 240
AT4G14340	AUCAUCGGAAUUCGAAAAAG	SEQ ID NO: 241
AT1G49310	AAAACAAAAGUUAAGCAGAC	SEQ ID NO: 242
AT4G14360	UUUAUCUCAAAUAGAAGGCA	SEQ ID NO: 243
AT1G28520	GGUGGGGAGGUGAGAUUUCUU	SEQ ID NO: 244
AT1G20160	UGAUUAGGAAACUACAAAGCC	SEQ ID NO: 245
AT5G37370	CAUUUUUCAAUUUCAUAAAAC	SEQ ID NO: 246
AT4G11320	UUACUUUUUAGCCCAACAAA	SEQ ID NO: 247
AT5G40850	GGCGUGUGUGUGUGUUGUUGA	SEQ ID NO: 248
AT1G06150	GUGUGAAGGGGAAGGUUUAG	SEQ ID NO: 249
AT2G26080	UUGUUUUUUUUGGUUUGGUU	SEQ ID NO: 250

Preferably the 5'UTR sequence comprises SEQ ID NO: 6 (AT1G58420).

Preferably, an mRNA described herein comprises a translation enhancer sequence. Translation enhancer sequences enhance the translation efficiency of a mRNA described herein and thereby provide increased production of the protein encoded by the mRNA. The translation enhancer region may be located in the 5' or 3' UTR of an mRNA sequence. Examples of translation enhancer regions include naturally-occurring enhancer regions from TEV 5'UTR and *Xenopus* beta-globin 3'UTR. Preferred 5' UTR enhancer sequences include but are not limited to those derived from mRNAs encoding human heat shock proteins (HSP) including HSP70-P2, HSP70-M1 HSP72-M2, HSP17.9 and HSP70-P1. Preferred translation enhancer sequences used in accordance with the embodiments of the present disclosure are represented by SEQ ID NOS 11-15 as shown in Table 3.

TABLE 3

5'UTR enhancers			
Name	Sequence	Seq ID No.:	
HSP70-P2	GUCAGCUUUCAAA CUCUUUGUUUCUU GUUUGUUGAUUGA GAAUA	SEQ ID NO: 11	
HSP70-M1	CUCUCGCCUGAGA AAAAAAAUCCACG AACCAUUUCUCA GCAACCAGCAGCA CG	SEQ ID NO: 12	
HSP72-M2	ACCUGUGAGGGUU CGAAGGAAGUAGC AGUGUUUUUGUU CCUAGAGGAAGAG	SEQ ID NO: 13	

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TABLE 3-continued

5'UTR enhancers		
Name	Sequence	Seq ID No.:
HSP17.9	ACACAGAAACAUU	SEQ ID NO: 14
	CGCAAAACAAAA	
	UCCCAGUAUCAA	
	AUUCUUCUUCUUU	
	UUUCAUAUUUCGC	
HSP70-P1	AAAGAC	
	CAGAAAAUUUGC	SEQ ID NO: 15
	UACAUUGUUUCAC	
	AAACUUCAAAUAU	
	UAUUCAUUUUUUU	

Preferably, an mRNA described herein comprises a Kozak sequence. As is understood in the art, a Kozak sequence is a short consensus sequence centered around the translational initiation site of eukaryotic mRNAs that allows for efficient initiation of translation of the mRNA. The ribosomal translation machinery recognizes the AUG initiation codon in the context of the Kozak sequence. A Kozak sequence, may be inserted upstream of the coding sequence for OTC, downstream of a 5' UTR or inserted upstream of the coding sequence for OTC and downstream of a 5' UTR. Preferably, an mRNA described herein comprises a Kozak sequence having the amino acid sequence GCCACC (SEQ ID NO: 23). Preferably an mRNA described herein comprises a partial Kozak sequence "p" having the amino acid sequence GCCA (SEQ ID NO: 24).

Preferably an mRNA described herein comprises a 3'UTR. Preferably, the 3' UTR comprises a sequence selected from the 3' UTRs of alanine aminotransferase 1, human apolipoprotein E, human fibrinogen alpha chain, human haptoglobin, human antithrombin, human alpha globin, human beta globin, human complement C3, human growth factor, human hepcidin, MALAT-1, mouse beta globin, mouse albumin, and *Xenopus* beta globin, or fragments of any of the foregoing. Preferably, the 3' UTR is derived from *Xenopus* beta globin. Preferred 3' UTR sequences include SEQ ID NOS 16-22 as shown in Table 4.

TABLE 4

3'UTR sequences		
Name	Sequence	Seq ID No.:
XBG	CUAGUGACUGACUAG	SEQ ID NO: 16
	GAUCUGGUUACCACU	
	AAACCAGCCUCAGAG	
	ACACCCGAAUGGAGU	
	CUCUAAGCUACAUA	
	UACCAACUUACACUU	
	ACAAAUGUUGUCCC	
	CCAAAUGUAGCCAU	
	UCGUUUCUGUCCUA	
	AUAAAAAGAAAGUUU	
	CUUCACAU	
	UGCAAGGCUGGCCGG	
	AAGCCCUUGCCUGAA	
	AGCAAGAUUUCAGCC	
HUMAN HAPTOGLOBIN	UGGAAGAGGGCAAAG	SEQ ID NO: 17
	UGGACGGGAGUGGAC	
	AGGAGUGGAUGCGAU	
	AAGAUGUGGUUGAA	
	GCUGAUGGGUGCCAG	
	CCCUGCAUUGCUGAG	

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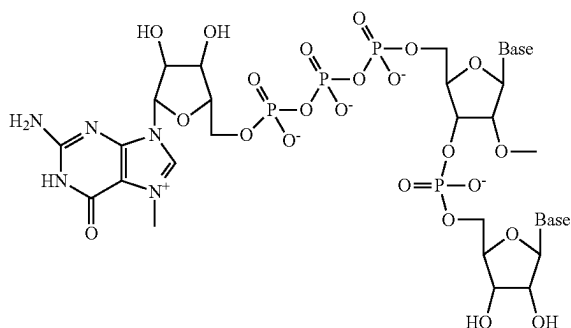
TABLE 4-continued

3'UTR sequences		
Name	Sequence	Seq ID No.:
HUMAN APOLIPOPROTEIN E	UCAAUCAUAUAAAGAG	SEQ ID NO: 18
	CUUUCUUUUGACCCA	
	U	
	ACGCCGAAGCCUGCA	
	GCCAUGCGACCCAC	
	GCCACCCGUGCCUC	
	CUGCCUCCGCGCAGC	
	CUGCAGCGGGAGACC	
	CUGUCCCCGCCCCAG	
	CCGUCCUCCUGGGGU	
HCV	GGACCCUAGUUUAAU	SEQ ID NO: 19
	AAAGAUAUACCAAGU	
	UUCACGCA	
	UAGAGCGGCAAAACC	
	UAGCUACACUCCAUA	
	GCUAGUUUCUUUUUU	
	UUUUGUUUUUUUUUU	
	UUUUUUUUUUUUUUU	
	UUUUUUUUUUUUUUC	
	CUUUUUUUUCCUUUU	
MOUSE ALBUMIN	UUUUCUUCUUUUUU	SEQ ID NO: 20
	GGUGGCUCUUAUUA	
	GCCUAGUCACGGCU	
	AGCUGUGAAAGGUCC	
	GUGAGCCGCAUGACU	
	GCAGAGAGUGCCGUA	
	ACUGGUCUCUCUGCA	
	GAUCAUGU	
	ACACAUCACAACCAC	
	AACCUUCUCAGGCUA	
HUMAN ALPHA GLOBIN	CCCUGAGAAAA	SEQ ID NO: 21
	AAAGACAUGAAGACU	
	CAGGACUCAUCCUUU	
	CUGUUGGUGU	
	AAAAUCAACACCCUA	
	AGGAACACAAAUUUC	
	UUUAAACAUUU	
	GACUUUCUGUCUCUG	
	UGCUGCAAUUAAUAA	
	AAAUGGAAA	
EMCV	GAUUCUAC	SEQ ID NO: 22
	GCUGGAGCCUCGGUA	
	GCCGUUCCUCCUGCC	
	CGCUGGGCCUCCCAA	
	CGGGCCUCCUCCCC	
	UCCUUGCACCCGGCCC	
	UCCUGGUCUUUGAA	
	UAAAGUCUGAGUGGG	
	CAGCA	
	UAGUGCAGUCAC	
	UGGCACAACG	
	CGUUGCCCGG	
	UAAGCCAAUC	
	GGGUAUACAC	
	GGUCGUCUA	
	CUGCAGACAG	
	GGUUCUUCUA	
	CUUUGCAAGA	
	UAGUCUAGAG	
	UAGUAAAAUA	
	AAUAGUAUAG	

Preferably, an mRNA described herein comprises a 3' tail region, which can serve to protect the mRNA from exonuclease degradation. The tail region may be a 3'poly(A) and/or 3'poly(C) region. Preferably, the tail region is a 3' poly(A) tail. As used herein a "3' poly(A) tail" is a polymer of sequential adenine nucleotides that can range in size from, for example: 10 to 250 sequential adenine nucleotides; 60-125 sequential adenine nucleotides, 90-125 sequential

adenine nucleotides, 95-125 sequential adenine nucleotides, 95-121 sequential adenine nucleotides, 100 to 121 sequential adenine nucleotides, 110-121 sequential adenine nucleotides; sequential adenine nucleotides, 112-121 sequential adenine nucleotides; 114-121 adenine sequential nucleotides; and 115 to 121 sequential adenine nucleotides. Preferably a 3' poly A tail as described herein comprise 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, or 125 sequential adenine nucleotides. 3' Poly(A) tails can be added using a variety of methods known in the art, e.g., using poly(A) polymerase to add tails to synthetic or in vitro transcribed RNA. Other methods include the use of a transcription vector to encode poly A tails or the use of a ligase (e.g., via splint ligation using a T4 RNA ligase and/or T4 DNA ligase), wherein poly(A) may be ligated to the 3' end of a sense RNA. Preferably, a combination of any of the above methods is utilized.

Preferably, an mRNA described herein comprises a 5' cap. 5'-ends capped with various groups and their analogues are known in the art. The 5' cap may be selected from m7GpppA, m7GpppC; unmethylated cap analogs (e.g., GpppG); dimethylated cap analog (e.g., m2,7GpppG), a trimethylated cap analog (e.g., m2,2,7GpppG), dimethylated symmetrical cap analogs (e.g., m7Gpppm7G), or anti reverse cap analogs (e.g., ARCA; m7, 2'OmeGpppG, m72'dGpppG, m7,3'OmeGpppG, m7,3'dGpppG and their tetraphosphate derivatives) (see, e.g., Jemielity, J. et al., RNA 9: 1108-1122 (2003)). The 5' cap may be an ARCA cap (3'-Ome-m7G(5')pppG). The 5' cap may be an mCAP (m7G(5')ppp(5')G, N⁷-Methyl-Guanosine-5'-Triphosphate-5'-Guanosine). The 5' cap may be resistant to hydrolysis. A preferred 5' cap is referred to herein as "m7GpppGm cap" also referred to herein as "Cap1" and has the following core structure:



Preferably an mRNA described herein comprises one or more chemically modified nucleotides. Examples of nucleic acid monomers include non-natural, modified, and chemically-modified nucleotides, including any such nucleotides known in the art. mRNA sequences comprising chemically modified nucleotides have been shown to improve mRNA expression, expression rates, half-life and/or expressed protein concentrations. mRNA sequences comprising chemically modified nucleotides have also been useful to optimize protein localization thereby avoiding deleterious bio-responses such as the immune response and/or degradation pathways.

Examples of modified or chemically-modified nucleotides include 5-hydroxycytidines, 5-alkylcytidines, 5-hydroxyalkylcytidines, 5-carboxycytidines, 5-formylcytidines,

5-alkoxycytidines, 5-alkynylcytidines, 5-halocytidines, 2-thiocytidines, N⁴-alkylcytidines, N⁴-aminocytidines, N⁴-acetylcytidines, and N⁴,N⁴-dialkylcytidines.

Examples of modified or chemically-modified nucleotides include 5-hydroxycytidine, 5-methylcytidine, 5-hydroxymethylcytidine, 5-carboxycytidine, 5-formylcytidine, 5-methoxycytidine, 5-propynylcytidine, 5-bromocytidine, 5-iodocytidine, 2-thiocytidine; N⁴-methylcytidine, N⁴-aminocytidine, N⁴-acetylcytidine, and N⁴,N⁴-dimethylcytidine.

Examples of modified or chemically-modified nucleotides include 5-hydroxyuridines, 5-alkyluridines, 5-hydroxyalkyluridines, 5-carboxyuridines, 5-carboxyalkylesteruridines, 5-formyluridines, 5-alkoxyuridines, 5-alkynyluridines, 5-halouridines, 2-thiouridines, and 6-alkyluridines.

Examples of modified or chemically-modified nucleotides include 5-hydroxyuridine, 5-methyluridine, 5-hydroxymethyluridine, 5-carboxyuridine, 5-carboxymethyluridine, 5-formyluridine, 5-methoxyuridine (also referred to herein as "5MeOU"), 5-propynyluridine, 5-bromouridine, 5-fluorouridine, 5-iodouridine, 2-thiouridine, and 6-methyluridine.

Examples of modified or chemically-modified nucleotides include 5-methoxycarbonylmethyl-2-thiouridine, 5-methylaminomethyl-2-thiouridine, 5-carbamoylmethyluridine, 5-carbamoylmethyl-2'-O-methyluridine, 1-methyl-3-(3-amino-3-carboxypropyl)pseudouridine, 5-methylaminomethyl-2-selenouridine, 5-carboxymethyluridine, 5-methylhydrouridine, 5-taurinomethyluridine, 5-taurinomethyl-2-thiouridine, 5-(isopentenylaminomethyl)uridine, 2'-O-methylpseudouridine, 2-thio-2'-O-methyluridine, and 3,2'-O-dimethyluridine.

Examples of modified or chemically-modified nucleotides include N⁶-methyladenosine, 2-aminoadenosine, 3-methyladenosine, 8-azaadenosine, 7-deazaadenosine, 8-oxoadenosine, 8-bromoadenosine, 2-methylthio-N⁶-methyladenosine, N⁶-isopentenyladenosine, 2-methylthio-N⁶-isopentenyladenosine, N⁶-(cis-hydroxyisopentenyl)adenosine, 2-methylthio-N⁶-(cis-hydroxyisopentenyl)adenosine, N⁶-glycinyllcarbamoyl-adenosine, N⁶-threonyllcarbamoyl-adenosine, N⁶-methyl-N⁶-threonyllcarbamoyl-adenosine, 2-methylthio-N⁶-threonyllcarbamoyl-adenosine, N⁶,N⁶-dimethyladenosine, N⁶-hydroxynorvalyllcarbamoyl-adenosine, 2-methylthio-N⁶-hydroxynorvalyllcarbamoyl-adenosine, N⁶-acetyl-adenosine, 7-methyl-adenine, 2-methylthio-adenine, 2-methoxy-adenine, alpha-thio-adenosine, 2'-O-methyl-adenosine, N⁶,2'-O-dimethyl-adenosine, N⁶,N⁶,2'-O-trimethyl-adenosine, 1,2'-O-dimethyl-adenosine, 2'-O-ribosyladenosine, 2-amino-N⁶-methyl-purine, 1-thio-adenosine, 2'-F-ara-adenosine, 2'-F-adenosine, 2'-OH-ara-adenosine, and N⁶-(19-amino-pentaoxonadecyl)-adenosine.

Examples of modified or chemically-modified nucleotides include N¹-alkylguanosines, N²-alkylguanosines, thienoguanosines, 7-deazaguanosines, 8-oxoguanosines, 8-bromoguanosines, O⁶-alkylguanosines, xanthosines, inosines, and N¹-alkylinosines.

Examples of modified or chemically-modified nucleotides include N¹-methylguanosine, N²-methylguanosine, thienoguanosine, 7-deazaguanosine, 8-oxoguanosine, 8-bromoguanosine, O⁶-methylguanosine, xanthosine, inosine, and N¹-methylinosine.

Examples of modified or chemically-modified nucleotides include pseudouridines. Examples of pseudouridines include N¹-alkylpseudouridines, N¹-cycloalkylpseudouridines, N¹-hydroxypseudouridines, N¹-hydroxyalkylpseudouridines, N¹-phenylpseudouridines, N¹-phenylalkylpseudouridines, N¹-aminoalkylpseudouridines,

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N³-alkylpseudouridines, N⁶-alkylpseudouridines, N⁶-alkoxypseudouridines, N⁶-hydroxypseudouridines, N⁶-hydroxyalkylpseudouridines, N⁶-morpholinopseudouridines, N⁶-phenylpseudouridines, and N⁶-halopseudouridines. Examples of pseudouridines include N¹-alkyl-N⁶-alkylpseudouridines, N¹-alkyl-N⁶-alkoxypseudouridines, N¹-alkyl-N⁶-hydroxypseudouridines, N¹-alkyl-N⁶-hydroxyalkylpseudouridines, N¹-alkyl-N⁶-morpholinopseudouridines, N¹-alkyl-N⁶-phenylpseudouridines, and N¹-alkyl-N⁶-halopseudouridines. In these examples, the alkyl, cycloalkyl, and phenyl substituents may be unsubstituted, or further substituted with alkyl, halo, haloalkyl, amino, or nitro substituents.

Examples of pseudouridines include N¹-methylpseudouridine (also referred to herein as "N1MPU"), N¹-ethylpseudouridine, N¹-propylpseudouridine, N¹-cyclopropylpseudouridine, N¹-phenylpseudouridine, N¹-aminomethylpseudouridine, N³-methylpseudouridine, N¹-hydroxypseudouridine, and N¹-hydroxymethylpseudouridine.

Examples of nucleic acid monomers include modified and chemically-modified nucleotides, including any such nucleotides known in the art.

Examples of modified and chemically-modified nucleotide monomers include any such nucleotides known in the art, for example, 2'-O-methyl ribonucleotides, 2'-O-methyl purine nucleotides, 2'-deoxy-2'-fluoro ribonucleotides, 2'-deoxy-2'-fluoro pyrimidine nucleotides, 2'-deoxy ribonucleotides, 2'-deoxy purine nucleotides, universal base nucleotides, 5-C-methyl-nucleotides, and inverted deoxybasic monomer residues.

Examples of modified and chemically-modified nucleotide monomers include 3'-end stabilized nucleotides, 3'-glyceryl nucleotides, 3'-inverted abasic nucleotides, and 3'-inverted thymidine.

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Examples of modified and chemically-modified nucleotide monomers include locked nucleic acid nucleotides (LNA), 2'-0,4'-C-methylene-(D-ribofuranosyl) nucleotides, 2'-methoxyethoxy (MOE) nucleotides, 2'-methyl-thio-ethyl, 2'-deoxy-2'-fluoro nucleotides, and 2'-O-methyl nucleotides. In an exemplary embodiment, the modified monomer is a locked nucleic acid nucleotide (LNA).

Examples of modified and chemically-modified nucleotide monomers include 2',4'-constrained 2'-O-methoxyethyl (cMOE) and 2'-O-Ethyl (cEt) modified DNAs.

Examples of modified and chemically-modified nucleotide monomers include 2'-amino nucleotides, 2'-O-amino nucleotides, 2'-C-allyl nucleotides, and 2'-O-allyl nucleotides.

Examples of modified and chemically-modified nucleotide monomers include N6-methyladenosine nucleotides.

Examples of modified and chemically-modified nucleotide monomers include nucleotide monomers with modified bases 5-(3-amino)propyluridine, 5-(2-mercapto)ethyluridine, 5-bromouridine, 8-bromoguanosine, or 7-deazaadenosine.

Examples of modified and chemically-modified nucleotide monomers include 2'-O-aminopropyl substituted nucleotides.

Examples of modified and chemically-modified nucleotide monomers include replacing the 2'-OH group of a nucleotide with a 2'-R, a 2'-OR, a 2'-halogen, a 2'-SR, or a 2'-amino, where R can be H, alkyl, alkenyl, or alkynyl.

Example of base modifications described above can be combined with additional modifications of nucleoside or nucleotide structure, including sugar modifications and linkage modifications. Certain modified or chemically-modified nucleotide monomers may be found in nature.

Preferred nucleotide modifications include N¹-methylpseudouridine and 5-methoxyuridine.

The constructs for preferred mRNA sequences are provided in Table 5.

TABLE 5

Exemplary mRNA Constructs							
mRNA Construct No.	Cap	5'UTR	Kozak*	OTC Protein Encoded	3'UTR	3' Poly A Tail	mRNA Construct SEQ ID NO:
563	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	26
564	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	27
565	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	28
566	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	29
567	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	30
568	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	31
569	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	32
570	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	33
571	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	34
572	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	35
573	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	36
574	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	37
575	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	38
708	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	39
709	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	40
710	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	41
711	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	42
712	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	43
713	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	44
714	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	45
715	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	46
716	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	47
717	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	48
718	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	49
719	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	50
720	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	51
721	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	52
722	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	53

TABLE 5-continued

Exemplary mRNA Constructs							
mRNA Construct No.	Cap	5'UTR	Kozak*	OTC Protein Encoded	3'UTR	3' Poly A Tail	mRNA Construct SEQ ID NO:
723	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	54
724	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	55
725	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	56
726	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	57
727	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	58
728	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	59
729	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	60
1787	Cap1	ARC5-2	No	SEQ ID NO: 3	Hu haptoglob	Yes	61
1788	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu ApoE	Yes	62
1789	Cap1	ARC5-2	No	SEQ ID NO: 3	Hu haptoglob	Yes	63
1790	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu ApoE	Yes	64
1791	Cap1	ARC5-2	No	SEQ ID NO: 3	Hu haptoglob	Yes	65
1792	Cap1	HCV5'	P	SEQ ID NO: 3	HCV3'	Yes	66
1793	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu a-glob	Yes	67
1794	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu a-glob	Yes	68
1795	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu a-glob	Yes	69
1796	Cap1	Hu alb	No	SEQ ID NO: 3	Ms Alb	Yes	70
1797	Cap1	Hu alb	No	SEQ ID NO: 3	Ms Alb	Yes	71
1798	Cap1	Hu alb	No	SEQ ID NO: 3	Ms Alb	Yes	72
1799	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	73
1800	Cap1	Hu alb	No	SEQ ID NO: 3	Ms Alb	Yes	74
1801	Cap1	AT1G58420	P	SEQ ID NO: 3	Hu ApoE	Yes	75
1802	Cap1	ARC5-2	No	SEQ ID NO: 3	Hu haptoglob	Yes	76
1803	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	77
1804	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	78
1805	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	79
1806	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	80
1808	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	81
1809	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	82
1816	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	83
1822	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	84
1823	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	85
1840	Cap1	EMCV	No	SEQ ID NO: 3	EMCV	Yes	86
1841	Cap1	EMCV	No	SEQ ID NO: 3	EMCV	Yes	87
1842	Cap1	EMCV	No	SEQ ID NO: 3	EMCV	Yes	88
1843	Cap1	HSP70-P2-TEV	Yes	SEQ ID NO: 3	XBG	Yes	89
1844	Cap1	HSP70-M1-TEV	Yes	SEQ ID NO: 3	XBG	Yes	90
1845	Cap1	HSP70-M2-TEV	Yes	SEQ ID NO: 3	XBG	Yes	91
1846	Cap1	HSP17.9-TEV	Yes	SEQ ID NO: 3	XBG	Yes	92
1847	Cap1	HSP70-P1-TEV	Yes	SEQ ID NO: 3	XBG	Yes	93
1882	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	94
1883	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	95
1884	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	96
1885	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	97
1886	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	98
1887	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	99
1888	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	100
1889	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	101
1890	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	102
1891	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	103
1898	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	104
1899	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	105
1900	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	106
1903	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	107
1904	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	108
1905	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	109
1906	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	110
1907	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	111
1908	Cap1	TEV	P	SEQ ID NO: 3	XBG	Yes	112
1915	Cap1	HSP70-P1-AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	113
1916	Cap1	HSP70-P1-AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	114
1917	Cap1	HSP70-P1-AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	115
1918	Cap1	HSP70-P1-AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	116
1919	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	117
1920	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	118

TABLE 5-continued

Exemplary mRNA Constructs							
mRNA Construct No.	Cap	5'UTR	Kozak*	OTC Protein Encoded	3'UTR	3' Poly A Tail	mRNA Construct SEQ ID NO:
1921	Cap1	AT1G58420	Yes	SEQ ID NO: 4**	Hu a-glob	Yes	119
1925	Cap1	TEV	Yes	SEQ ID NO: 3	XBG	Yes	120
1926	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	121
1927	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	122
1928	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	123
1929	Cap1	HSP70-P1-AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	124
2016	Cap1	AT1G58420	Yes	SEQ ID NO: 3	Hu a-glob	Yes	253
2260	Cap1	AT1G58420	Yes	SEQ ID NO: 4**	Hu a-glob	Yes	251
2262	Cap1	AT1G58420	Yes	SEQ IS NO: 4**	Hu a-glob	Yes	252

*Kozak sequence defined as GCCACC (SEQ ID NO: 23). Partial (P) Kozak defined as GCCA (SEQ ID NO: 24).

**Construct encodes modified human OTC protein of SEQ ID NO: 4.

Preferred mRNA sequences include all of the mRNA sequences listed in Table 5. Preferred mRNA sequences include all of the mRNA sequences listed wherein, 0% to 100%, preferably 1% to 100%, preferably 25% to 100%, preferably 50% to 100% and preferably 75% to 100% of the uracil nucleotides of the mRNA sequences are modified. Preferably, 1% to 100% of the uracil nucleotides are N¹-methylpseudouridine or 5-methoxyuridine. Preferably 100% of the uracil nucleotides are N¹-methylpseudouridine. Preferably 100% of the uracil nucleotides are 5-methoxyuridine.

Preferred mRNA sequences comprise a 5' cap, a 5'UTR that is derived from a gene expressed by *Arabidopsis thaliana*, an optional translation enhancer sequence, an optional Kozak sequence or partial Kozak sequence, a codon optimized coding sequence (CDS/ORF) coding for an OTC protein, a 3' UTR and a poly A tail. Preferably the codon optimized CDS encodes a protein of SEQ ID NO: 3 or SEQ ID NO: 4. Preferably, the 5' UTR that is derived from a gene expressed by *Arabidopsis thaliana* is selected from found in Table 5. Preferably, the 5' UTR that is derived from a gene expressed by *Arabidopsis thaliana* is selected from the group consisting of: SEQ ID NO: 6, SEQ ID NOS: 125-127 and SEQ ID NOS: 227-247. Preferably the 5' UTR sequence is AT1G58420 having the sequence of SEQ ID NO: 6. Preferably, the uracil content of the codon optimized sequence has been reduced with respect to the percentages of uracil content of SEQ ID NO: 1. Preferably, 0% to 100% of the uracil nucleotides of the mRNA sequences are modified. Preferably, 0% to 100% of the uracil nucleotides are N¹-methylpseudouridine or 5-methoxyuridine. Preferably 100% of the uracil nucleotides are N¹-methylpseudouridine. Preferably 100% of the uracil nucleotides are 5-methoxyuridine.

Preferred mRNA constructs comprise codon optimized coding sequences and a 5' UTR from a gene expressed by *Arabidopsis thaliana* and are selected from: SEQ ID NOS: 62, 67, 68, 69, 73, 113-119, 121-127.

A preferred mRNA construct of the disclosure comprises mRNA construct 1921 (SEQ ID NO: 119) having an optimized ORF encoding the modified human OTC protein of SEQ ID NO: 4 and comprising a 3' Poly A tail of 121 nucleotides. Another preferred mRNA construct comprises construct 2260 (SEQ ID NO: 251) encoding the modified human OTC protein of SEQ ID NO: 4 and comprising a 3' Poly A tail of 100 nucleotides. Another preferred mRNA construct comprises construct 2262 (SEQ ID NO: 252)

encoding the modified human OTC protein of SEQ ID NO: 4 and comprising a 3' Poly A tail of 100 nucleotides.

A preferred mRNA sequence of the disclosure includes the mRNA construct 1799 (SEQ ID NO: 73) having a codon optimized ORF encoding wild type human OTC of SEQ ID NO: 3 and having a 3' Poly A tail of 121 nucleotides. Another preferred mRNA construct of the disclosure includes the mRNA construct 2016 (SEQ ID NO: 253) having a codon optimized ORF encoding wild type human OTC of SEQ ID NO: 3 and comprising a 3' Poly A tail of 100 nucleotides.

Preferably 100% of the uridine nucleotides of mRNA constructs 1799, 2016, 1921, 2260 and 2262, are N1-methylpseudouridine. Preferably 100% of the uracil nucleotides of mRNA constructs 1799, 2016, 1921, 2260 and 2262, are 5-methoxyuridine.

The mRNA for use in accordance with this disclosure can exhibit increased translation efficiency. As used herein, translation efficiency refers to a measure of the production of a protein or polypeptide by translation of an mRNA in accordance with the disclosure. Preferably, an mRNA of the disclosure can exhibit at least 2-fold, 3-fold, 5-fold, or 10-fold increased translation efficiency in vivo as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure. Preferably an mRNA of the disclosure can provide at least a 2-fold, 3-fold, 5-fold, or 10-fold increased polypeptide or protein level in vivo as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized and/or does not comprise the preferred UTRs of the disclosure. Preferably, an mRNA of the disclosure can provide increased levels of a polypeptide or protein in vivo as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure. For example, the level of a polypeptide or protein can be increased by 10%, or 20%, or 30%, or 40%, or 50%, or more.

Preferably the mRNA of the disclosure can provide increased functional half-life in the cytoplasm of mammalian cells over mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure. The inventive translatable molecules can have increased half-life of activity as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure.

Preferably, the mRNA of the disclosure can reduce cellular innate immune response as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure.

Preferably, the mRNA of the disclosure can reduce the dose levels required for efficacious therapy as compared to mRNA encoding SEQ ID NO: 3 or SEQ ID NO: 4 that is not codon optimized in accordance with the disclosure and/or that does not comprise the preferred UTRs of the disclosure.

Design and Synthesis of the mRNA Sequences

mRNA for use in accordance with the disclosure may be prepared according to any available technique including, but not limited to chemical synthesis, in vitro transcription (IVT) or enzymatic or chemical cleavage of a longer precursor, etc. Methods of synthesizing RNAs are known in the art.

In some embodiments, mRNA is produced from a primary complementary DNA (cDNA) construct. The process of design and synthesis of the primary cDNA constructs described herein generally includes the steps of gene construction, mRNA production (either with or without modifications) and purification. In the IVT method, a target polynucleotide sequence encoding an OTC protein is first selected for incorporation into a vector which will be amplified to produce a cDNA template. Optionally, the target polynucleotide sequence and/or any flanking sequences may be codon optimized. The cDNA template is then used to produce mRNA through in vitro transcription (IVT). After production, the mRNA may undergo purification and clean-up processes. The steps of which are provided in more detail below.

The step of gene construction may include, but is not limited to gene synthesis, vector amplification, plasmid purification, plasmid linearization and clean-up, and cDNA template synthesis and clean-up. Once a human OTC protein (e.g. SEQ ID NO: 3 or SEQ ID NO: 4) is selected for production, a primary construct is designed. Within the primary construct, a first region of linked nucleosides encoding the polypeptide of interest may be constructed using an open reading frame (ORF) of a selected nucleic acid (DNA or RNA) transcript. The ORF may comprise the wild type ORF, an isoform, variant or a fragment thereof. As used herein, an "open reading frame" or "ORF" is meant to refer to a nucleic acid sequence (DNA or RNA) which is capable of encoding a polypeptide of interest. ORFs often begin with the start codon, ATG and end with a nonsense or termination codon or signal.

Further, nucleotide sequence of any region of the mRNA or DNA template may be codon optimized. Codon optimization methods are known in the art and may be useful in efforts to achieve one or more of several goals. These goals include to match codon frequencies in target and host organisms to ensure proper folding, to bias GC content to increase mRNA stability or reduce secondary structures, to minimize tandem repeat codons or base runs that may impair gene construction or expression, to customize transcriptional and translational control regions, to insert or remove protein trafficking sequences, to remove/add post translation modification sites in encoded protein (e.g. glycosylation sites), to add, remove or shuffle protein domains, to insert or delete restriction sites, to modify ribosome binding sites and mRNA degradation sites, to adjust translational rates to allow the various domains of the protein to fold properly, or to reduce or eliminate problematic secondary structures within the mRNA. Suitable codon optimization tools, algorithms and services are known in the art.

Preferably, the primary cDNA template may include reducing the occurrence or frequency of appearance of certain nucleotides in the template strand. For example, the occurrence of a nucleotide in a template may be reduced to a level below 25% of nucleotides in the template. In further examples, the occurrence of a nucleotide in a template may be reduced to a level below 20% of nucleotides in the template. In some examples, the occurrence of a nucleotide in a template may be reduced to a level below 16% of nucleotides in the template. Preferably, the occurrence of a nucleotide in a template may be reduced to a level below 15%, and preferably may be reduced to a level below 12% of nucleotides in the template.

For example, the present disclosure provides nucleic acids wherein with altered uracil content at least one codon in the wild-type sequence has been replaced with an alternative codon to generate a uracil-altered sequence. Altered uracil sequences can have at least one of the following properties:

- (i) an increase or decrease in global uracil content (i.e., the percentage of uracil of the total nucleotide content in the nucleic acid of a section of the nucleic acid, e.g., the open reading frame); or,
- (ii) an increase or decrease in local uracil content (i.e., changes in uracil content are limited to specific subsequences); or,
- (iii) a change in uracil distribution without a change in the global uracil content; or,
- (iv) a change in uracil clustering (e.g., number of clusters, location of clusters, or distance between clusters); or,
- (v) combinations thereof.

Preferably, the percentage of uracil nucleobases in the nucleic acid sequence is reduced with respect to the percentage of uracil nucleobases in the wild-type nucleic acid sequence. For example, 30% of nucleobases may be uracil in the wild-type sequence but the nucleobases that are uracil are preferably lower than 15%, preferably lower than 12% and preferably lower than 10% of the nucleobases in the nucleic acid sequences of the disclosure. The percentage uracil content can be determined by dividing the number of uracil in a sequence by the total number of nucleotides and multiplying by 100.

Preferably, the percentage of uracil nucleobases in a subsequence of the nucleic acid sequence is reduced with respect to the percentage of uracil nucleobases in the corresponding subsequence of the wild-type sequence. For example, the wild-type sequence may have a 5'-end region (e.g., 30 codons) with a local uracil content of 30%, and the uracil content in that same region could be reduced to preferably 15% or lower, preferably 12% or lower and preferably 10% or lower in the nucleic acid sequences of the disclosure.

Preferably, codons in the nucleic acid sequence of the invention reduce or modify, for example, the number, size, location, or distribution of uracil clusters that could have deleterious effects on protein translation. Although lower uracil content is desirable, in certain aspects, the uracil content, and in particular the local uracil content, of some subsequences of the wild-type sequence can be greater than the wild-type sequence and still maintain beneficial features (e.g., increased expression).

Preferably, the uracil-modified sequence induces a lower Toll-Like Receptor (TLR) response when compared to the wild-type sequence. Several TLRs recognize and respond to nucleic acids. Double-stranded (ds)RNA, a frequent viral constituent, has been shown to activate TLR3. Single-stranded (ss)RNA activates TLR7. RNA oligonucleotides, for example RNA with phosphorothioate internucleotide

linkages, are ligands of human TLR8. DNA containing unmethylated CpG motifs, characteristic of bacterial and viral DNA, activate TLR9.

As used herein, the term “TLR response” is defined as the recognition of single-stranded RNA by a TLR7 receptor, and preferably encompasses the degradation of the RNA and/or physiological responses caused by the recognition of the single-stranded RNA by the receptor. Methods to determine and quantify the binding of an RNA to a TLR7 are known in the art. Similarly, methods to determine whether an RNA has triggered a TLR7-mediated physiological response (e.g., cytokine secretion) are well known in the art. Preferably, a TLR response can be mediated by TLR3, TLR8, or TLR9 instead of TLR7. Suppression of TLR7-mediated response can be accomplished via nucleoside modification. RNA undergoes over a hundred different nucleoside modifications in nature. Human rRNA, for example, has ten times more pseudouracil ('P) and 25 times more 2'-O-methylated nucleosides than bacterial rRNA. Bacterial mRNA contains no nucleoside modifications, whereas mammalian mRNAs have modified nucleosides such as 5-methylcytidine (m5C), N6-methyladenosine (m6A), inosine and many 2'-O-methylated nucleosides in addition to N7-methylguanosine (m7G).

Preferably, the uracil content of polynucleotides disclosed herein and preferably polynucleotides encoding the modified OTC protein of SEQ ID NO: 4 is less than 50%, 49%, 48%, 47%, 46%, 45%, 44%, 43%, 42%, 41%, 40%, 39%, 38%, 37%, 36%, 35%, 34%, 33%, 32%, 31%, 30%, 29%, 28%, 27%, 26%, 25%, 24%, 23%, 22%, 21%, 20%, 19%, 18%, 17%, 16%, 15%, 14%, 13%, 12%, 11%, 10%, 90%, 80%, 70%, 60%, 5%, 4%, 3%, 2% or 1% of the total nucleobases in the sequence in the reference sequence. Preferably, the uracil content of polynucleotides disclosed herein and preferably polynucleotides encoding the modified OTC protein of SEQ ID NO: 4, is between about 5% and about 25%. Preferably, the uracil content of polynucleotides disclosed herein and preferably polynucleotides encoding the modified OTC protein of SEQ ID NO: 4 is about 15% and about 25%.

The cDNA templates may be transcribed to produce an mRNA sequence described herein using an in vitro transcription (IVT) system. The system typically comprises a transcription buffer, nucleotide triphosphates (NTPs), an RNase inhibitor and a polymerase. The NTPs may be selected from, but are not limited to, those described herein including natural and unnatural (modified) NTPs. The polymerase may be selected from, but is not limited to, T7 RNA polymerase, T3 RNA polymerase and mutant polymerases such as, but not limited to, polymerases able to incorporate modified nucleic acids.

The primary cDNA template or transcribed mRNA sequence may also undergo capping and/or tailing reactions. A capping reaction may be performed by methods known in the art to add a 5' cap to the 5' end of the primary construct. Methods for capping include, but are not limited to, using a Vaccinia Capping enzyme (New England Biolabs, Ipswich, Mass.) or CLEANCAP® technology (TriLink Biotechnologies). A poly-A tailing reaction may be performed by methods known in the art, such as, but not limited to, 2' O-methyltransferase and by methods as described herein. If the primary construct generated from cDNA does not include a poly-T, it may be beneficial to perform the poly-A-tailing reaction before the primary construct is cleaned.

Codon optimized cDNA constructs encoding an ornithine transcarbamylase (OTC) protein are particularly suitable for

generating mRNA sequences described herein. For example, such cDNA constructs may be used as the basis to transcribe, in vitro, a polyribonucleotide encoding an ornithine transcarbamylase (OTC) protein. Table 6 provides a listing of exemplary cDNA ORF templates used for in vitro transcription of the mRNA sequences listed in Table 5.

TABLE 6

Exemplary cDNA Templates		
DNA Construct No***:	SEQ ID NO:	Protein encoded by cDNA template SEQ ID NO:
p563	128	SEQ ID NO: 3*
p564	129	SEQ ID NO: 3*
p565	130	SEQ ID NO: 3*
p566	131	SEQ ID NO: 3*
p567	132	SEQ ID NO: 3*
p568	133	SEQ ID NO: 3*
p569	134	SEQ ID NO: 3*
p570	135	SEQ ID NO: 3*
p571	136	SEQ ID NO: 3*
p572	137	SEQ ID NO: 3*
p573	138	SEQ ID NO: 3*
p574	139	SEQ ID NO: 3*
p575	140	SEQ ID NO: 3*
p708	141	SEQ ID NO: 3*
p709	142	SEQ ID NO: 3*
p710	143	SEQ ID NO: 3*
p711	144	SEQ ID NO: 3*
p712	145	SEQ ID NO: 3*
p713	146	SEQ ID NO: 3*
p714	147	SEQ ID NO: 3*
p715	148	SEQ ID NO: 3*
p716	149	SEQ ID NO: 3*
p717	150	SEQ ID NO: 3*
p718	151	SEQ ID NO: 3*
p719	152	SEQ ID NO: 3*
p720	153	SEQ ID NO: 3*
p721	154	SEQ ID NO: 3*
p722	155	SEQ ID NO: 3*
p723	156	SEQ ID NO: 3*
p724	157	SEQ ID NO: 3*
p725	158	SEQ ID NO: 3*
p726	159	SEQ ID NO: 3*
p727	160	SEQ ID NO: 3*
p728	161	SEQ ID NO: 3*
p729	162	SEQ ID NO: 3*
p1787	163	SEQ ID NO: 3*
p1788	164	SEQ ID NO: 3*
p1789	165	SEQ ID NO: 3*
p1790	166	SEQ ID NO: 3*
p1791	167	SEQ ID NO: 3*
p1792	168	SEQ ID NO: 3*
p1793	169	SEQ ID NO: 3*
p1794	170	SEQ ID NO: 3*
p1795	171	SEQ ID NO: 3*
p1796	172	SEQ ID NO: 3*
p1797	173	SEQ ID NO: 3*
p1798	174	SEQ ID NO: 3*
p1799	175	SEQ ID NO: 3*
p1800	176	SEQ ID NO: 3*
p1801	177	SEQ ID NO: 3*
p1802	178	SEQ ID NO: 3*
p1803	179	SEQ ID NO: 3*
p1804	180	SEQ ID NO: 3*
p1805	181	SEQ ID NO: 3*
p1806	182	SEQ ID NO: 3*
p1808	183	SEQ ID NO: 3*
p1809	184	SEQ ID NO: 3*
p1816	185	SEQ ID NO: 3*
p1822	186	SEQ ID NO: 3*
p1823	187	SEQ ID NO: 3*
p1840	188	SEQ ID NO: 3*
p1841	189	SEQ ID NO: 3*
p1842	190	SEQ ID NO: 3*
p1843	191	SEQ ID NO: 3*
p1844	192	SEQ ID NO: 3*

TABLE 6-continued

Exemplary cDNA Templates		
DNA Construct No***:	SEQ ID NO:	Protein encoded by cDNA template SEQ ID NO:
p1845	193	SEQ ID NO: 3*
p1846	194	SEQ ID NO: 3*
p1847	195	SEQ ID NO: 3*
p1882	196	SEQ ID NO: 3*
p1883	197	SEQ ID NO: 3*
p1884	198	SEQ ID NO: 3*
p1885	199	SEQ ID NO: 3*
p1886	200	SEQ ID NO: 3*
p1887	201	SEQ ID NO: 3*
p1888	202	SEQ ID NO: 3*
p1889	203	SEQ ID NO: 3*
p1890	204	SEQ ID NO: 3*
p1891	205	SEQ ID NO: 3*
p1898	206	SEQ ID NO: 3*
p1899	207	SEQ ID NO: 3*
p1900	208	SEQ ID NO: 3*
p1903	209	SEQ ID NO: 3*
p1904	210	SEQ ID NO: 3*
p1905	211	SEQ ID NO: 3*
p1906	212	SEQ ID NO: 3*
p1907	213	SEQ ID NO: 3*
p1908	214	SEQ ID NO: 3*
p1915	215	SEQ ID NO: 3*
p1916	216	SEQ ID NO: 3*
p1917	217	SEQ ID NO: 3*
p1918	218	SEQ ID NO: 3*
p1919	219	SEQ ID NO: 3*
p1920	220	SEQ ID NO: 3*
p1921	221	SEQ ID NO: 4**
p1925	222	SEQ ID NO: 3*
p1926	223	SEQ ID NO: 3*
p1927	224	SEQ ID NO: 3*
p1928	225	SEQ ID NO: 3*
p1929	226	SEQ ID NO: 3*
p2016	175	SEQ ID NO: 3*
p2260	221	SEQ ID NO: 4**
p2262	221	SEQ ID NO: 4**

*SEQ ID NO: 3 is the amino acid sequence for wild type human OTC.

**SEQ ID NO: 4 is the amino acid sequence for modified human OTC.

***The entire plasmid sequence is not included.

Preferred cDNA template sequences include the DNA sequence of SEQ ID NO: 175 (p1779) having an optimized coding sequence encoding wild type human OTC of SEQ ID NO: 3. Preferred cDNA template sequences also include cDNA sequence of SEQ ID NO: 221 (p1921), having an optimized coding sequence encoding a modified OTC protein of SEQ ID NO: 4.

The present disclosure also provides polynucleotides (e.g. DNA, RNA, cDNA, mRNA) encoding a human OTC protein that may be operably linked to one or more regulatory nucleotide sequences in an expression construct, such as a vector or plasmid. In certain embodiments, such constructs are DNA constructs. Regulatory nucleotide sequences will generally be appropriate for a host cell used for expression. Numerous types of appropriate expression vectors and suitable regulatory sequences are known in the art for a variety of host cells.

Typically, said one or more regulatory nucleotide sequences may include, but are not limited to, promoter sequences, leader or signal sequences, ribosomal binding sites, transcriptional start and termination sequences, translational start and termination sequences, and enhancer or activator sequences. Constitutive or inducible promoters as known in the art are contemplated by the embodiments of the present disclosure. The promoters may be either naturally occurring promoters, or hybrid promoters that combine elements of more than one promoter.

An expression construct may be present in a cell on an episome, such as a plasmid, or the expression construct may be inserted in a chromosome. Preferably, the expression vector contains a selectable marker gene to allow the selection of transformed host cells. Selectable marker genes are well known in the art and will vary with the host cell used.

The present disclosure also provides expression vectors comprising a nucleotide sequence encoding an ornithine transcarbamylase (OTC) protein that is preferably operably linked to at least one regulatory sequence. Regulatory sequences are art-recognized and are selected to direct expression of the encoded polypeptide.

Accordingly, the term regulatory sequence includes promoters, enhancers, and other expression control elements. The design of the expression vector may depend on such factors as the choice of the host cell to be transformed and/or the type of protein desired to be expressed.

The present disclosure also provides a host cell transfected with an mRNA or DNA described herein which encodes an ornithine transcarbamylase (OTC) polypeptide described herein. Preferably, the human OTC polypeptide has the sequence of SEQ ID NO: 4. The host cell may be any prokaryotic or eukaryotic cell. For example, an ornithine transcarbamylase (OTC) polypeptide may be expressed in bacterial cells such as *E. coli*, insect cells (e.g., using a baculovirus expression system), yeast, or mammalian cells. Other suitable host cells are known to those skilled in the art.

The present disclosure also provides a host cell comprising a vector comprising a polynucleotide which encodes an mRNA sequence of any one of SEQ ID NOs: 26-229.

The present disclosure also provides methods of producing a human wild type OTC protein of SEQ ID NO: 3 or a modified human OTC protein SEQ ID NO: 4. Preferably, the OTC protein is SEQ ID NO: 4 and is encoded by mRNA of SEQ ID NO 119. For example, a host cell transfected with an expression vector encoding an OTC protein can be cultured under appropriate conditions to allow expression of the polypeptide to occur. The polypeptide may be secreted and isolated from a mixture of cells and medium containing the polypeptides. Alternatively, the polypeptides may be retained in the cytoplasm or in a membrane fraction and the cells harvested, lysed and the protein isolated. A cell culture includes host cells, media and other byproducts. Suitable media for cell culture are well known in the art.

The expressed OTC proteins described herein can be isolated from cell culture medium, host cells, or both using techniques known in the art for purifying proteins, including ion-exchange chromatography, gel filtration chromatography, ultrafiltration, electrophoresis, and immunoaffinity purification with antibodies specific for particular epitopes of the OTC polypeptide.

Lipid-Based Formulations

Lipid-based formulations have been increasingly recognized as one of the most promising delivery systems (also referred to herein as a delivery vehicle or carrier) for RNA due to their biocompatibility and their ease of large-scale production. Cationic lipids have been widely studied as synthetic materials for delivery of RNA. After mixing together, nucleic acids are condensed by cationic lipids to form lipid/nucleic acid complexes known as lipoplexes. These lipid complexes are able to protect genetic material from the action of nucleases and to deliver it into cells by interacting with the negatively charged cell membrane. Lipoplexes can be prepared by directly mixing positively charged lipids at physiological pH with negatively charged nucleic acids.

Conventional liposomes consist of a lipid bilayer that can be composed of cationic, anionic, or neutral (phospho)lipids and cholesterol, which encloses an aqueous core. Both the lipid bilayer and the aqueous space can incorporate hydrophobic or hydrophilic compounds, respectively. Liposome characteristics and behavior in vivo can be modified by addition of a hydrophilic polymer coating, e.g. polyethylene glycol (PEG), to the liposome surface to confer steric stabilization. Furthermore, liposomes can be used for specific targeting by attaching ligands (e.g., antibodies, peptides, and carbohydrates) to its surface or to the terminal end of the attached PEG chains (Front Pharmacol. 2015 Dec. 1; 6:286).

Liposomes are colloidal lipid-based and surfactant-based delivery systems composed of a phospholipid bilayer surrounding an aqueous compartment. They may present as spherical vesicles and can range in size from 20 nm to a few microns. Cationic lipid-based liposomes are able to complex with negatively charged nucleic acids via electrostatic interactions, resulting in complexes that offer biocompatibility, low toxicity, and the possibility of the large-scale production required for in vivo clinical applications. Liposomes can fuse with the plasma membrane for uptake; once inside the cell, the liposomes are processed via the endocytic pathway and the genetic material is then released from the endosome/carrier into the cytoplasm. Liposomes have long been perceived as drug delivery vehicles because of their superior biocompatibility, given that liposomes are basically analogs of biological membranes, and can be prepared from both natural and synthetic phospholipids (Int J Nanomedicine. 2014; 9: 1833-1843).

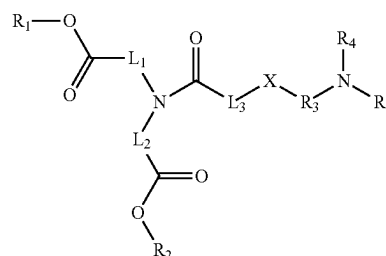
Cationic liposomes have been traditionally the most commonly used non-viral delivery systems for oligonucleotides, including plasmid DNA, antisense oligos, and siRNA/small hairpin R A-shRNA). Cationic lipids, such as DOTAP, (1,2-dioleoyl-3-trimethylammonium-propane) and DOTMA (N-[1-(2,3-dioleoyloxy)propyl]-N,N,N-trimethyl-ammonium methyl sulfate) can form complexes or lipoplexes with negatively charged nucleic acids to form nanoparticles by electrostatic interaction, providing high in vitro transfection efficiency. Furthermore, neutral lipid-based nanoliposomes for RNA delivery as e.g. neutral 1,2-dioleoyl-sn-glycero-3-phosphatidylcholine (DOPC)-based nanoliposomes were developed. (Adv Drug Deliv. Rev. 2014 February; 66: 110-116.)

Preferably, the mRNA constructs described herein are lipid formulated. The lipid formulation is preferably selected from, but not limited to, liposomes, lipoplexes, copolymers, such as PLGA, and lipid nanoparticles. Preferably a lipid nanoparticle (LNP) comprises:

- (a) a nucleic acid,
- (b) a cationic lipid,
- (c) an aggregation reducing agent (such as polyethylene glycol (PEG) lipid or PEG-modified lipid),
- (d) optionally a non-cationic lipid (such as a neutral lipid), and
- (e) optionally, a sterol.

Preferably, the lipid nanoparticle formulation consists of (i) at least one cationic lipid; (ii) a neutral lipid; (iii) a sterol, e.g., cholesterol; and (iv) a PEG-lipid, in a molar ratio of about 20-60% cationic lipid: 5-25% neutral lipid: 25-55% sterol; 0.5-15% PEG-lipid.

Some examples of lipids and lipid compositions for delivery of an active molecule of this disclosure are given in WO 2015/074085, U.S. 2018/0169268, WO 2018/119163, WO 20185/118102, U.S. 2018/0222863, WO 2016/081029, WO 2017/023817, WO 2017/117530, each of which is hereby incorporated by reference in its entirety. In certain embodiments, the lipid is a compound of the following Formula I:



Formula I

wherein

R₁ and R₂ both consist of a linear alkyl consisting of 1 to 14 carbons, or an alkenyl or alkynyl consisting of 2 to 14 carbons;

L₁ and L₂ both consist of a linear alkylene or alkenylene consisting of 5 to 18 carbons, or forming a heterocycle with N;

X is S;

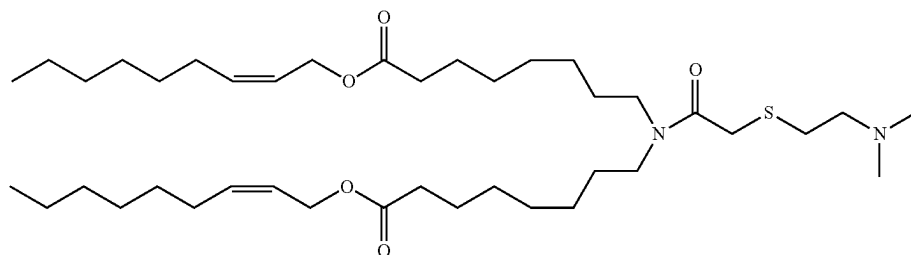
L₃ consists of a bond or a linear alkylene consisting of 1 to 6 carbons, or forming a heterocycle with N;

R₃ consists of a linear or branched alkylene consisting of 1 to 6 carbons; and

R₄ and R₅ are the same or different, each consisting of a hydrogen or a linear or branched alkyl consisting of 1 to 6 carbons;

or a pharmaceutically acceptable salt thereof.

The lipid formulation of may contain one or more ionizable cationic lipids selected from among the following (also referred to herein as "ATX lipids"):

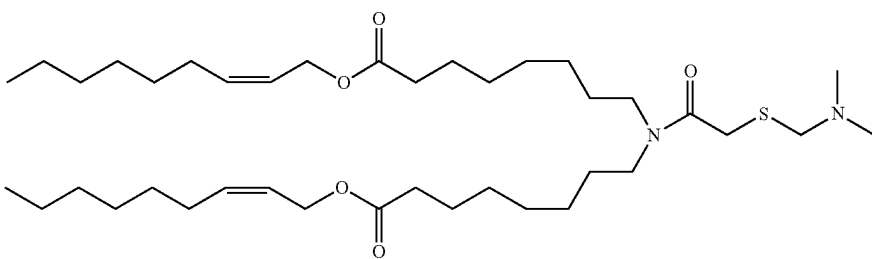


ATX-001

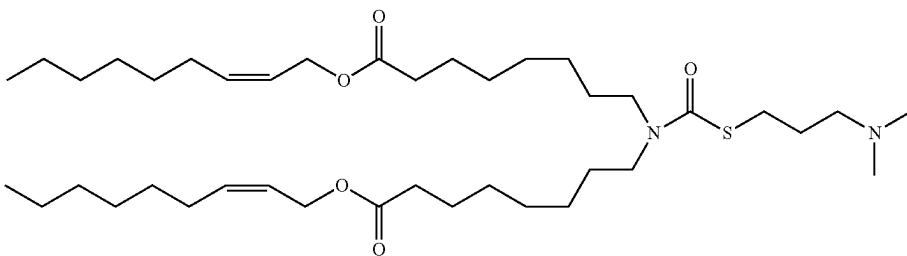
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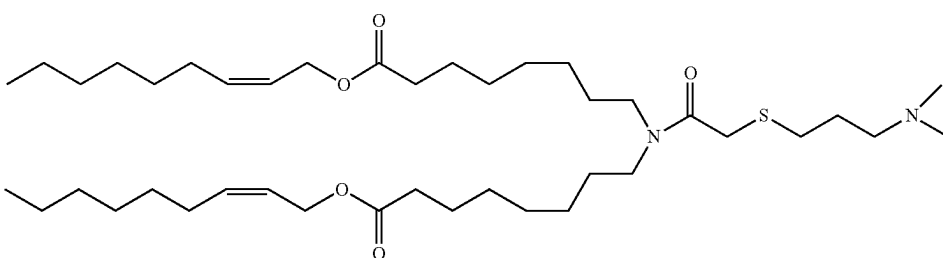
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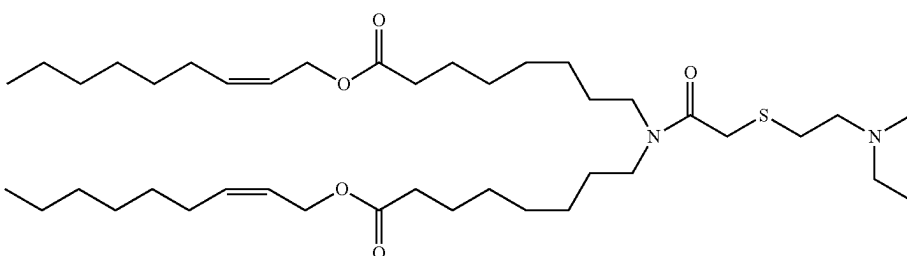
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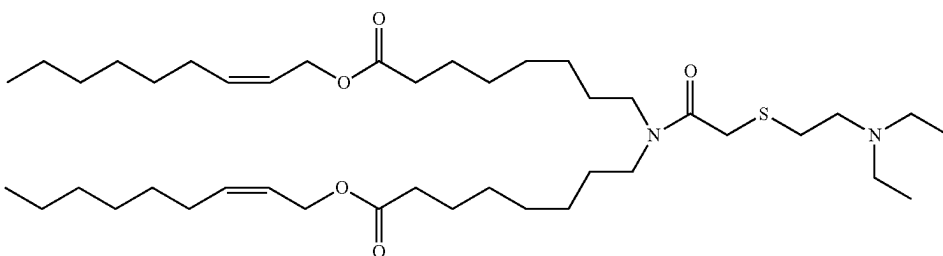
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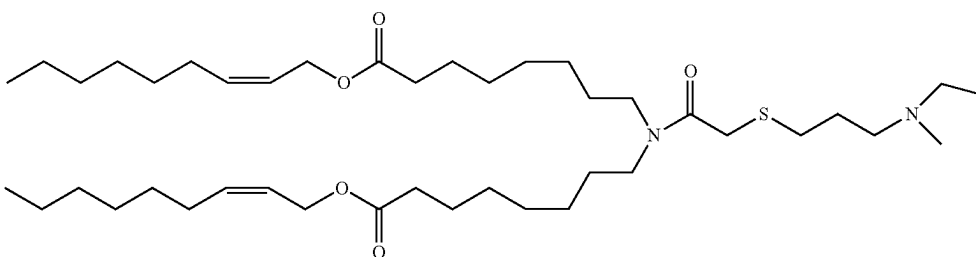
ATX-004



ATX-005



ATX-006



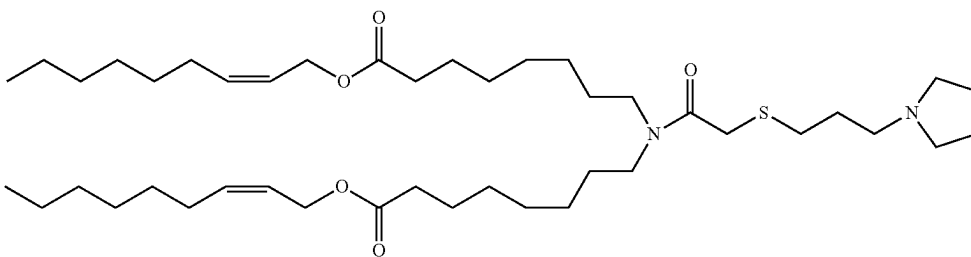
ATX-007

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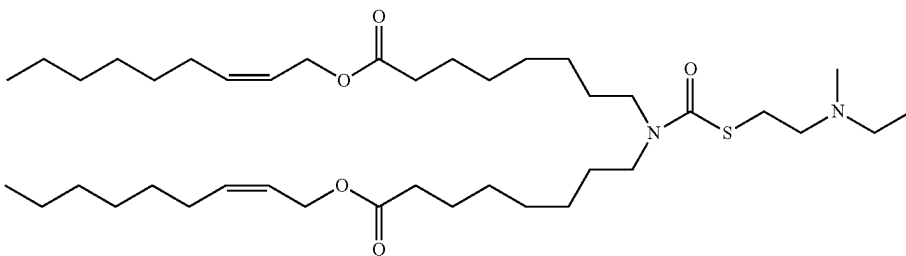
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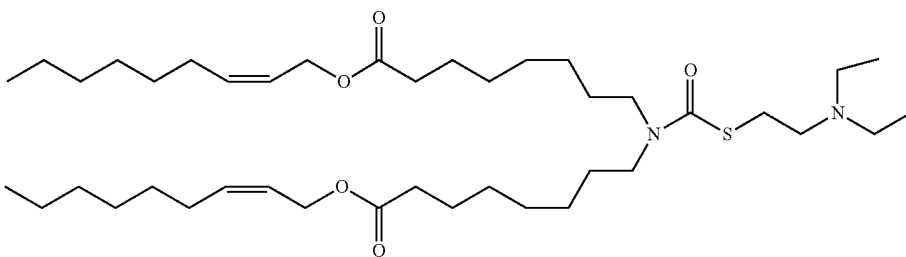
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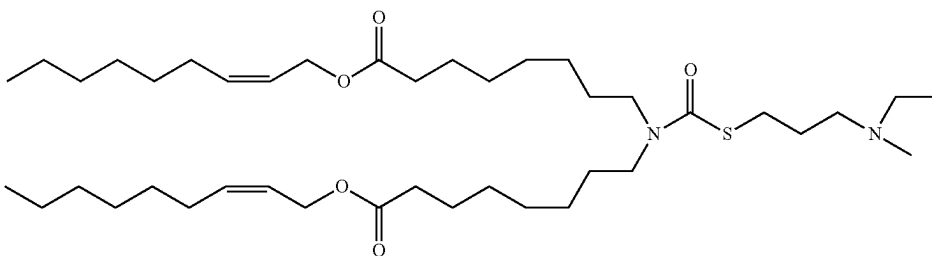
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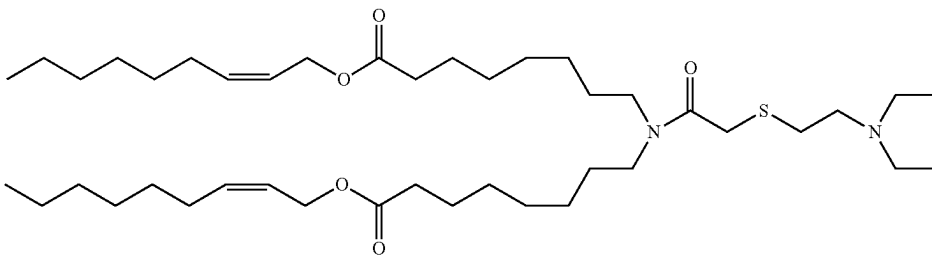
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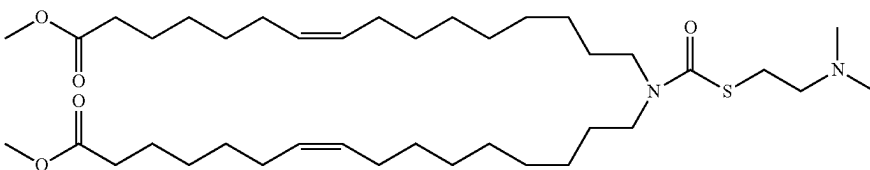
ATX-011



ATX-012



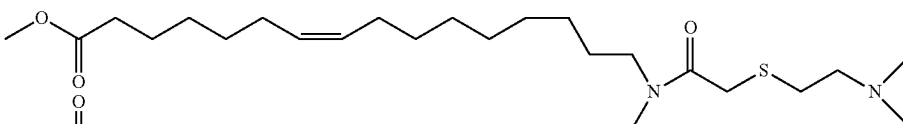
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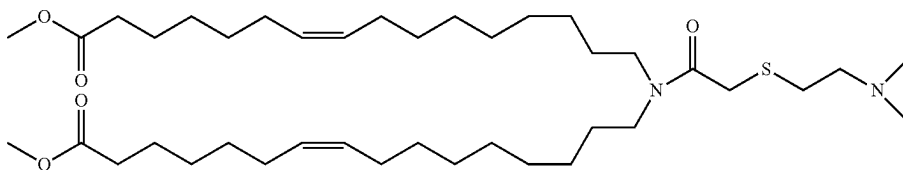
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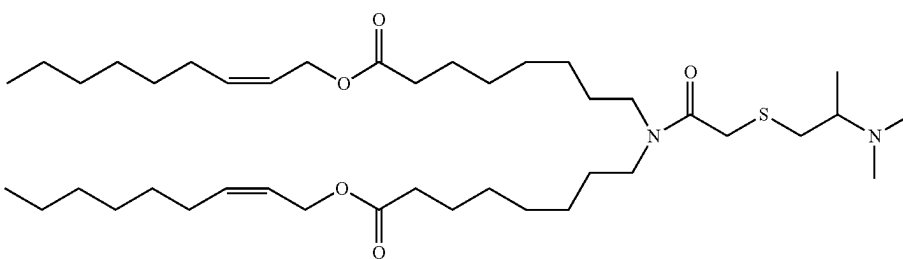
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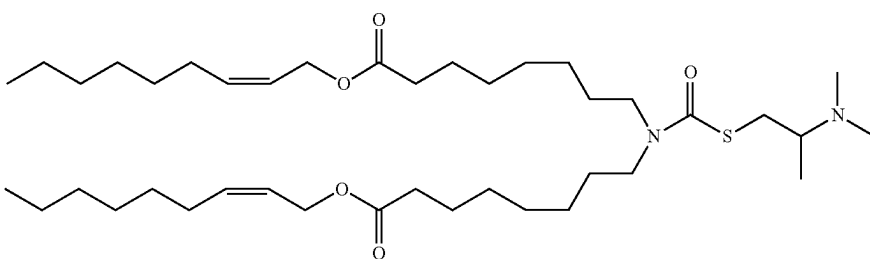
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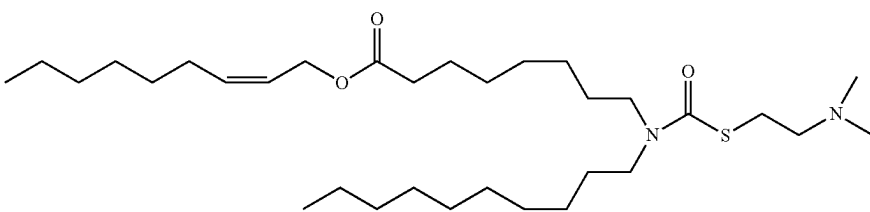
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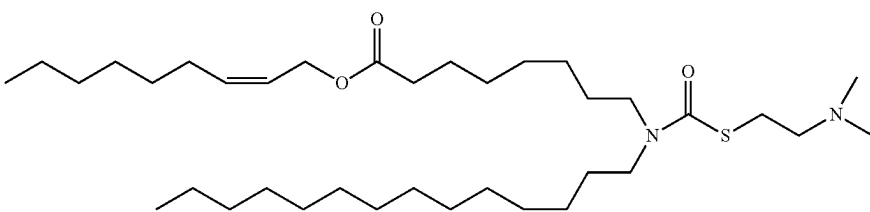
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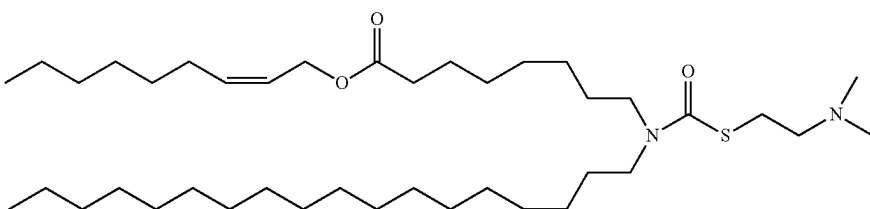
ATX-017



ATX-018



ATX-019



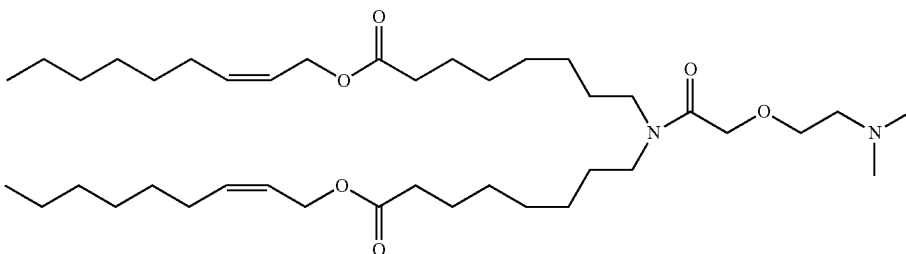
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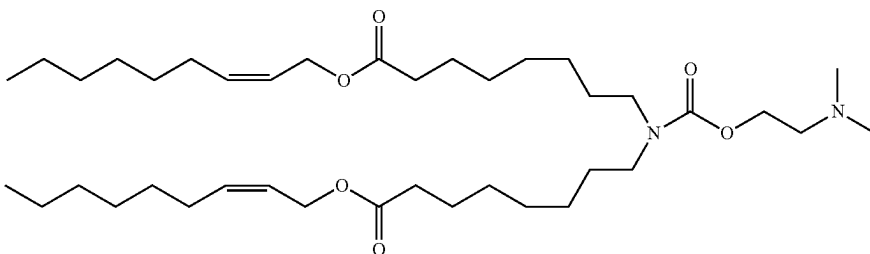
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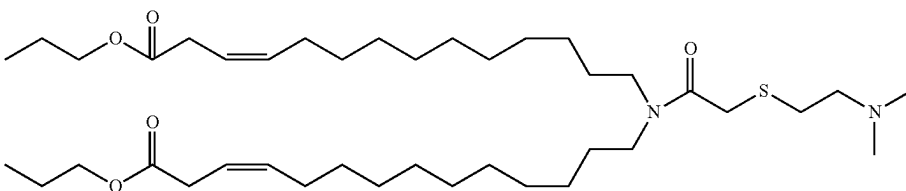
ATX-021



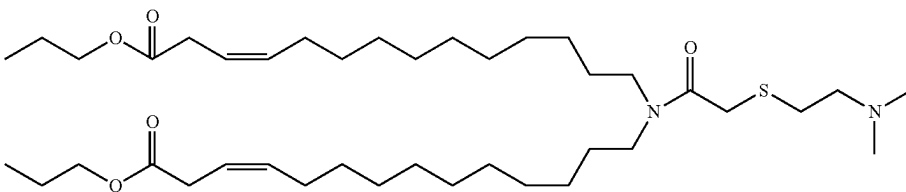
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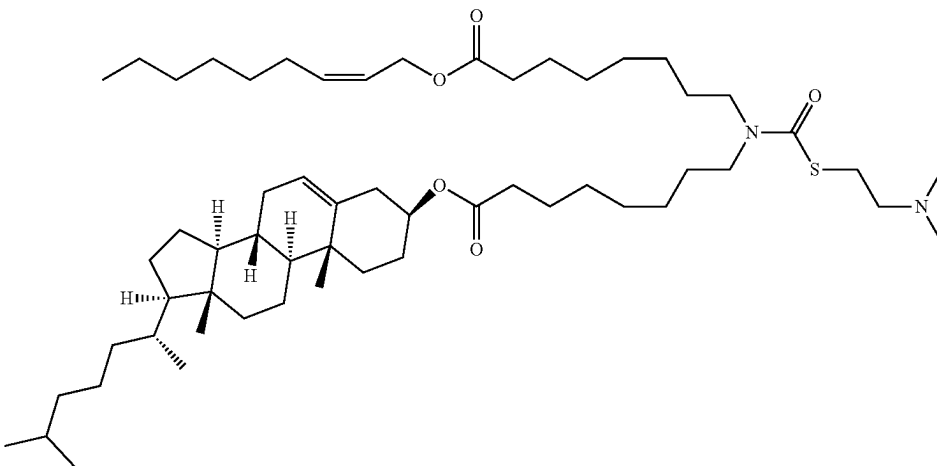
ATX-023



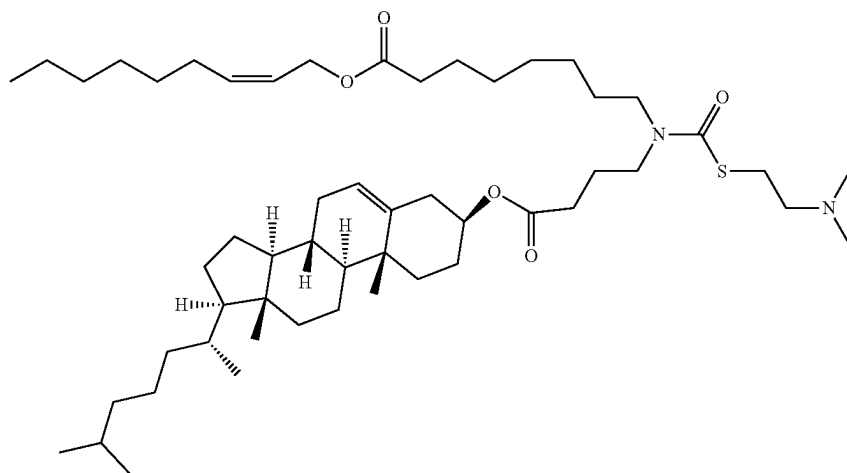
ATX-023



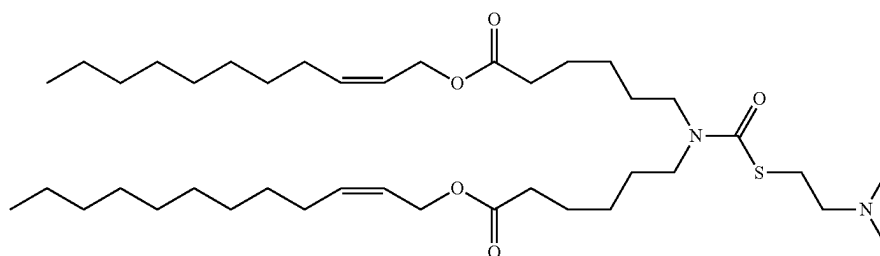
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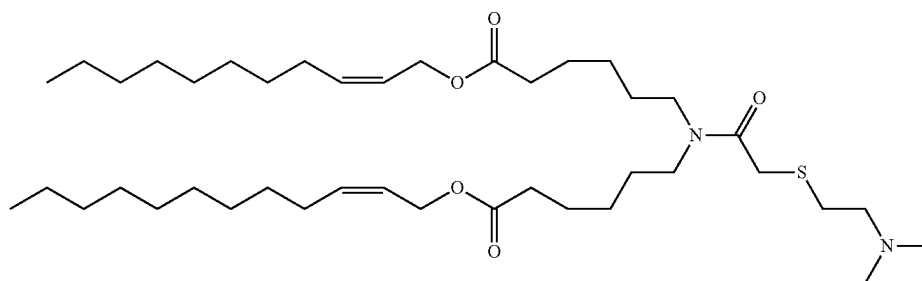
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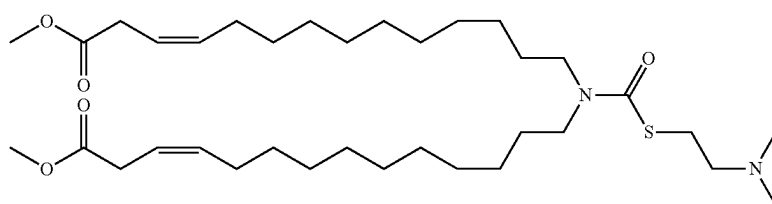
ATX-025



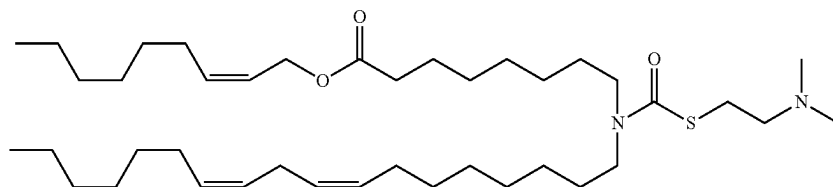
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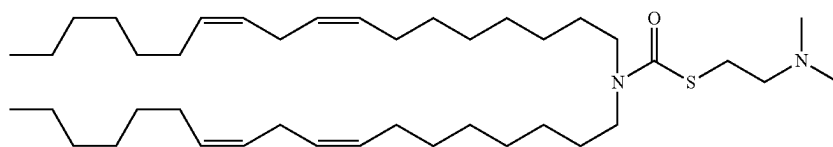
ATX-027



ATX-028

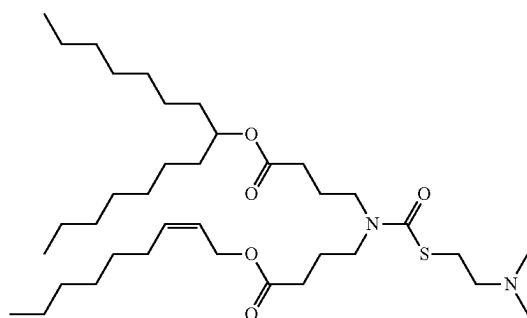


ATX-031

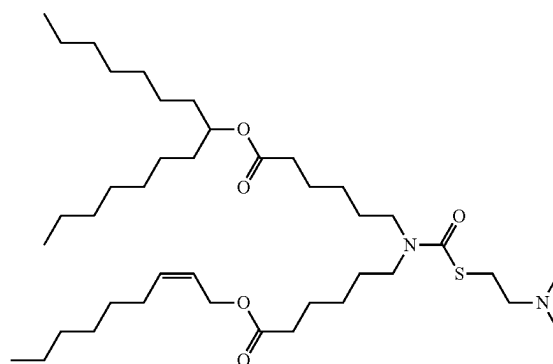


ATX-032

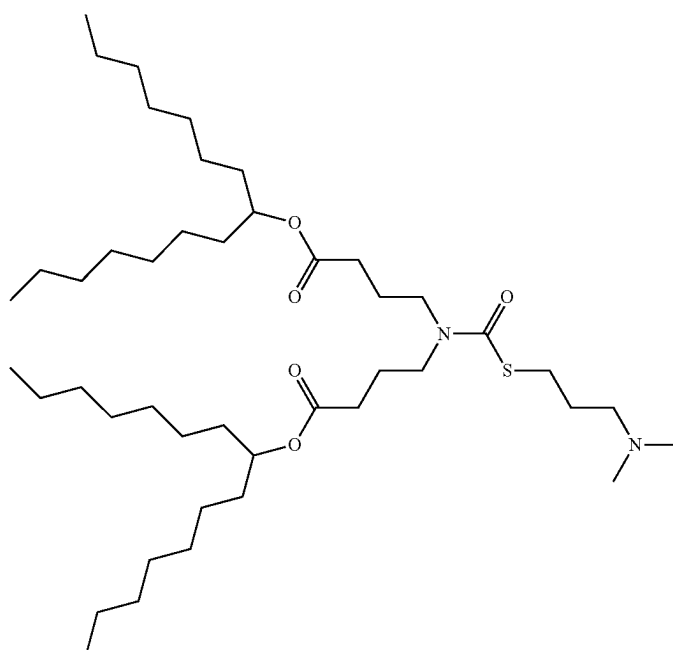
45

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ATX-081

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ATX-095



ATX-0126

The lipid nanoparticle preferably includes a cationic lipid suitable for forming a lipid nanoparticle. Preferably, the cationic lipid carries a net positive charge at about physiological pH. The cationic lipid may be, for example, N,N-dioleoyl-N,N-dimethylammonium chloride (DODAC), N,N-distearyl-N,N-dimethylammonium bromide (DDAB), 1,2-dioleoyltrimethylammoniumpropane chloride (DOTAP) (also known as N-(2,3-dioleoyloxy)propyl)-N,N,N-trimethylammonium chloride and 1,2-Dioleoyloxy-3-trimethylaminopropane chloride salt), N-(1-(2,3-dioleoyloxy)propyl)-N,N,N-trimethylammonium chloride (DOTMA), N,N-dimethyl-2,3-dioleoyloxypropylamine (DODMA), 1,2-Dilinoyleoxy-N,N-dimethylaminopropane (DLinDMA), 1,2-Dilinolenyloxy-N,N-dimethylaminopropane (DLenDMA), 1,2-di-γ-linolenyloxy-N,N-dimethylaminopropane (γ-DLenDMA), 1,2-Dilinoyleylcarbamoxyloxy-3-dimethylaminopropane (DLin-C-DAP), 1,2-Dilinoyleoxy-3-(dimethylamino)acetoxyp propane (DLin-DAC), 1,2-Dilinoyleoxy-3-morpholinopropane (DLin-MA), 1,2-Dilinoyleoxy-3-dimethylaminopropane (DLinDAP), 1,2-Dilinoylethio-3-dimethylaminopropane (DLin-S-DMA), 1-Linoyleyl-2-linoyleoxy-3-dimethylaminopropane (DLin-2-DMAP), 1,2-Dilinoyleoxy-3-trimethylaminopropane

chloride salt (DLin-TMA.Cl), 1,2-Dilinoyleoxy-3-trimethylaminopropane chloride salt (DLin-TAP.Cl), 1,2-Dilinoyleoxy-3-(N-methylpiperazino)propane (DLin-MPZ), or 3-(N,N-Dilinoylelamino)-1,2-propanediol (DLinAP), 3-(N,N-Dioylelamino)-1,2-propanedio (DOAP), 1,2-Dilinoyleoxy-3-(2-N,N-dimethylamino)ethoxypropane (DLin-EG-DMA), 2,2-Dilinoyleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-DMA) or analogs thereof, (3aR,5s,6aS)—N,N-dimethyl-2,2-di((9Z,12Z)-octadeca-9,12-dienyl)tetrahydro-3aH-cyclopenta[d][1,3]dioxol-5-amine, (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate (MC3), 1,1'-(2-(4-(2-((2-(bis(2-hydroxydodecyl)amino)ethyl)(2-hydroxydodecyl)amino)ethyl)piperazin-1-yl)ethylazanediy)didodecan-2-ol (C12-200), 2,2-dilinoyleyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLin-K-C2-DMA), 2,2-dilinoyleyl-4-dimethylaminomethyl-[1,3]-dioxolane (DLin-K-DMA), (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28 31-tetraen-19-yl 4-(dimethylamino)butanoate (DLin-M-C3-DMA), 3-((6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yloxy)-N,N-dimethylpropan-1-amine (MC3 Ether), 4-((6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yloxy)-N,N-dimethylbutan-1-amine (MC4 Ether), or any combination of any of the foregoing.

Other cationic lipids include, but are not limited to, N,N-distearyl-N,N-dimethylammonium bromide (DDAB), 3P—(N—(N',N'-dimethylaminoethane)-carbamoyl)cholesterol (DC-Choi), N-(1-(2,3-dioleoyloxy)propyl)-N-2-(spermin-ecarboxamido)ethyl)-N,N-dimethylammonium trifluoroacetate (DOSPA), dioctadecylamidoglycyl carboxyspermine (DOGS), 1,2-dioleoyl-sn-3-phosphoethanolamine (DOPE), 1,2-dioleoyl-3-dimethylammonium propane (DODAP), N-(1,2-dimyristyloxyprop-3-yl)-N,N-dimethyl-N-hydroxyethyl ammonium bromide (DMRIE), and 2,2-Dilinoleyl-4-dimethylaminoethyl-[1,3]-dioxolane (XTC). Additionally, commercial preparations of cationic lipids can be used, such as, e.g., LIPOFECTIN (including DOTMA and DOPE, available from GIBCO/BRL), and Lipofectamine (comprising DOSPA and DOPE, available from GIBCO/BRL).

Other suitable cationic lipids are disclosed in International Publication Nos. WO 09/086558, WO 09/127060, WO 10/048536, WO 10/054406, WO 10/088537, WO 10/129709, and WO 2011/153493; U.S. Patent Publication Nos. 2011/0256175, 2012/0128760, and 2012/0027803; U.S. Pat. No. 8,158,601; and Love et al., PNAS, 107(5), 1864-69, 2010.

Other suitable amino lipids include those having alternative fatty acid groups and other dialkylamino groups, including those, in which the alkyl substituents are different (e.g., N-ethyl-N-methylamino-, and N-propyl-N-ethylamino-). In general, amino lipids having less saturated acyl chains are more easily sized, particularly when the complexes must be sized below about 0.3 microns, for purposes of filter sterilization. Amino lipids containing unsaturated fatty acids with carbon chain lengths in the range of C14 to C22 may be used. Other scaffolds can also be used to separate the amino group and the fatty acid or fatty alkyl portion of the amino lipid.

Preferably, the LNP comprises the cationic lipid with formula (III) according to the patent application PCT/EP2017/064066. In this context, the disclosure of PCT/EP2017/064066 is also incorporated herein by reference.

Preferably, amino or cationic lipids of the disclosure have at least one protonatable or deprotonatable group, such that the lipid is positively charged at a pH at or below physiological pH (e.g. pH 7.4), and neutral at a second pH, preferably at or above physiological pH. It will, of course, be understood that the addition or removal of protons as a function of pH is an equilibrium process, and that the reference to a charged or a neutral lipid refers to the nature of the predominant species and does not require that all of the lipid be present in the charged or neutral form. Lipids that have more than one protonatable or deprotonatable group, or which are zwitterionic, are not excluded from use in the disclosure. In certain embodiments, the protonatable lipids have a pKa of the protonatable group in the range of about 4 to about 11, e.g., a pKa of about 5 to about 7.

The cationic lipid can comprise from about 20 mol % to about 70 or 75 mol % or from about 45 to about 65 mol % or about 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, or about 70 mol % of the total lipid present in the particle. In another embodiment, the lipid nanoparticles include from about 25% to about 75% on a molar basis of cationic lipid, e.g., from about 20 to about 70%, from about 35 to about 65%, from about 45 to about 65%, about 60%, about 57.5%, about 57.10%, about 50% or about 40% on a molar basis (based upon 100% total moles of lipid in the lipid nanoparticle). In one embodiment, the ratio of cationic lipid to nucleic acid is from about 3 to about 15, such as from about 5 to about 13 or from about 7 to about 11.

Pharmaceutical Compositions

Preferably, the disclosure provides pharmaceutical compositions containing a codon optimized mRNA encoding a human OTC protein of SEQ ID NO: 3 or SEQ ID NO: 4, preferably formulated in a lipid delivery system or lipid carrier and preferably comprising pharmaceutically acceptable excipients. Pharmaceutical compositions disclosed herein preferably facilitate expression of mRNA in vivo.

Suitable routes of administration include, for example, oral, rectal, vaginal, transmucosal, pulmonary including intratracheal or inhaled, or intestinal administration; parenteral delivery, including intradermal, transdermal (topical), intramuscular, subcutaneous, intramedullary injections, as well as intrathecal, direct intraventricular, intravenous, intraperitoneal, or intranasal.

Preferably, the intramuscular administration is to a muscle selected from the group consisting of skeletal muscle, smooth muscle and cardiac muscle. In some embodiments the administration results in delivery of the mRNA to a muscle cell. In some embodiments the administration results in delivery of the mRNA to a hepatocyte (i.e., liver cell). In a particular embodiment, the intramuscular administration results in delivery of the mRNA to a muscle cell.

Preferably, mRNAs and lipid formulations thereof may be administered in a local rather than systemic manner, for example, via injection of the pharmaceutical composition directly into a targeted tissue, preferably in a sustained release formulation. Local delivery can be affected in various ways, depending on the tissue to be targeted. For example, aerosols containing compositions of the present disclosure can be inhaled (for nasal, tracheal, or bronchial delivery); can be supplied in liquid, tablet or capsule form for administration to the stomach or intestines, can be supplied in suppository form for rectal or vaginal application; or can even be delivered to the eye by use of creams, drops, or even injection. Formulations containing provided compositions complexed with therapeutic molecules or ligands can even be surgically administered, for example in association with a polymer or other structure or substance that can allow the compositions to diffuse from the site of implantation to surrounding cells. Alternatively, they can be applied surgically without the use of polymers or supports.

Pharmaceutical compositions may be administered to any desired tissue. In some embodiments, the OTC mRNA delivered by provided liposomes or compositions is expressed in the tissue in which the liposomes and/or compositions were administered. In some embodiments, the mRNA delivered is expressed in a tissue different from the tissue in which the liposomes and/or compositions were administered. Exemplary tissues in which delivered mRNA may be delivered and/or expressed include, but are not limited to the liver, kidney, heart, spleen, serum, brain, skeletal muscle, lymph nodes, skin, and/or cerebrospinal fluid.

Preferably, a pharmaceutical composition can contain a polynucleotide described herein such as a primary DNA construct or mRNA described herein within a viral or bacterial vector.

Preferably, the primary DNA construct for an mRNA described herein or an mRNA described herein can be formulated using one or more excipients to: (1) increase stability; (2) increase cell transfection; (3) permit a sustained or delayed release (e.g., from a depot formulation of the polynucleotide, primary construct, or mRNA); (4) alter the biodistribution (e.g., target the polynucleotide, primary construct, or mRNA to specific tissues or cell types); (5) increase the translation of encoded protein in vivo; and/or (6) alter the release profile of encoded protein in vivo.

In addition to traditional excipients such as any and all solvents, dispersion media, diluents, or other liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, excipients of the present disclosure can include, without limitation, liposomes, lipid nanoparticles, polymers, lipoplexes, core-shell nanoparticles, peptides, proteins, cells transfected with primary DNA construct, or mRNA (e.g., for transplantation into a subject), hyaluronidase, nanoparticle mimics and combinations thereof.

Accordingly, the formulations described herein can include one or more excipients, each in an amount that together increases the stability of the primary DNA construct, or mRNA, increases cell transfection by the primary construct, or mRNA, increases the expression of polynucleotide, primary construct, or mRNA encoded protein, and/or alters the release profile of polynucleotide, primary construct, or mRNA encoded proteins. Further, the primary construct and mRNA of the present disclosure may be formulated using self-assembled nucleic acid nanoparticles.

Formulations of the pharmaceutical compositions described herein may be prepared by any method known or hereafter developed in the art of pharmacology. In general, such preparatory methods include the step of associating the active ingredient with an excipient and/or one or more other accessory ingredients.

A pharmaceutical composition in accordance with the present disclosure may be prepared, packaged, and/or sold in bulk, as a single unit dose, and/or as a plurality of single unit doses. As used herein, a "unit dose" refers to a discrete amount of the pharmaceutical composition comprising a predetermined amount of the active ingredient. The amount of the active ingredient may generally be equal to the dosage of the active ingredient which would be administered to a subject and/or a convenient fraction of such a dosage including, but not limited to, one-half or one-third of such a dosage.

Relative amounts of the active ingredient, the pharmaceutically acceptable excipient, and/or any additional ingredients in a pharmaceutical composition in accordance with the present disclosure may vary, depending upon the identity, size, and/or condition of the subject being treated and further depending upon the route by which the composition is to be administered. For example, the composition may comprise between 0.1% and 99% (w/w) of the active ingredient.

Pharmaceutical formulations may additionally comprise a pharmaceutically acceptable excipient, which, as used herein, includes, but is not limited to, any and all solvents, dispersion media, diluents, or other liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, and the like, as suited to the particular dosage form desired.

Various excipients for formulating pharmaceutical compositions and techniques for preparing the composition are known in the art (see Remington: The Science and Practice of Pharmacy, 21st Edition, A. R. Gennaro, Lippincott, Williams & Wilkins, Baltimore, Md., 2006; incorporated herein by reference in its entirety). The use of a conventional excipient medium may be contemplated within the scope of the embodiments of the present disclosure, except insofar as any conventional excipient medium may be incompatible with a substance or its derivatives, such as by producing any undesirable biological effect or otherwise interacting in a deleterious manner with any other component(s) of the pharmaceutical composition.

Therapeutic Uses

The mRNA sequences, primary DNA constructs that transcribe the mRNA sequences described herein and pharmaceutical compositions thereof provide numerous in vivo and in vitro methods and are useful to treat OTC deficiency. The treatment may comprise treating a human patient with OTC deficiency. Similarly, compositions described herein, may be used in vitro or ex vivo to study OTC deficiency in cell or animal-based models. For example, cells deficient for OTC expression can be used to analyze the ability to restore OTC expression and/or activity, as well as the time period over which expression and/or activity persists. Such cells and animal models are also suitable to identify other factors involved in the pathway, whether binding partners or factors in the same biochemical pathway. In other embodiments, compositions described herein can be used to study or track mitochondrial delivery.

Polynucleotides described herein, such as a DNA construct or template or an mRNA sequence described herein can be delivered to patients or cells experiencing OTC deficiency. Preferably the mRNA sequence comprises SEQ ID NO: 119 encoding a modified OTC protein of SEQ ID NO: 4. Preferably the DNA sequence comprises SEQ ID NO: 221 encoding a modified OTC protein of SEQ ID NO: 4.

Following administration, OTC is expressed in the cells or subject. Preferably, compositions described herein are delivered to mitochondria. Preferably compositions described herein are delivered to liver cells.

Preferably the therapeutic methods described herein decrease ammonia levels in plasma and/or urine in a subject in need thereof or in cells in culture, such as a subject having an OTC deficiency. Preferably, the therapeutic methods described herein decrease orotic acid levels in plasma and/or urine in a subject in need thereof or in cells in culture. Preferably, the therapeutic methods described herein increase citrulline in plasma and/or urine in a subject in need thereof or in cells in culture. Preferably, ammonia levels, orotic acid levels and/or citrulline levels are used as biomarkers to (i) identify subjects in need of treatment and/or (ii) to evaluate efficacy of treatment using the mRNA or DNA templates described herein.

Examples of mRNA sequences for use with these methods include those listed in Table 5. Preferably cDNA templates used to transcribe the mRNA sequences described herein are listed in Table 6. Preferred mRNA sequences for administering to patients for treatment of OTC deficiency are SEQ ID NOS: 1799 and SEQ ID NOS: 1921. Preferably the preferred mRNA sequences of SEQ ID NO: 1799 and SEQ ID NO: 1921.

Dosing

An effective dose of a mRNA, a protein or pharmaceutical formulations thereof of the present disclosure can be an amount that is sufficient to treat ORF protein deficiency in a cell and/or in a patient. A therapeutically effective dose can be an amount of an agent or formulation that is sufficient to cause a therapeutic effect. A therapeutically effective dose can be administered in one or more separate administrations, and by different routes. Generally, a therapeutically effective amount is sufficient to achieve a meaningful benefit to the subject (e.g., treating, modulating, curing, preventing and/or ameliorating phenylketonuria). For example, a therapeutically effective amount may be an amount sufficient to achieve a desired therapeutic and/or prophylactic effect. Generally, the amount of a therapeutic agent administered to a subject in need thereof will depend upon the characteristics of the subject. Such characteristics include the condition, disease severity, general health, age, sex and body weight of

the subject. One of ordinary skill in the art will be readily able to determine appropriate dosages depending on these and other related factors. In addition, both objective and subjective assays may optionally be employed to identify optimal dosage ranges.

Methods provided herein contemplate single as well as multiple administrations of a therapeutically effective amount of an mRNA sequence described herein. Pharmaceutical compositions comprising an mRNA sequence encoding an ORF protein described herein can be administered at regular intervals, depending on the nature, severity and extent of the subject's condition. Preferably, a therapeutically effective amount an mRNA sequence of the present disclosure may be administered periodically at regular intervals (e.g., once every year, once every six months, once every four months, once every three months, once every two months, once a month), biweekly, weekly, daily, twice a day, three times a day, four times a day, five times a day, six times a day, or continuously.

Preferably, the pharmaceutical compositions of the mRNA of the present disclosure are formulated such that they are suitable for extended-release of the translatable compound encoding a modified protein described herein contained therein. Such extended-release compositions may be conveniently administered to a subject at extended dosing intervals. For instance, in one embodiment, the pharmaceutical compositions of the present disclosure are administered to a subject twice a day, daily or every other day. In some embodiments, the pharmaceutical compositions of the present disclosure are administered to a subject twice a week, once a week, every 10 days, every two weeks, every 28 days, every month, every six weeks, every eight weeks, every other month, every three months, every four months, every six months, every nine months or once a year. Also contemplated herein are pharmaceutical compositions which are formulated for depot administration (e.g., subcutaneously, intramuscularly) to either deliver or release an mRNA sequence encoding an OTC protein described herein over extended periods of time. Preferably, the extended-release means employed are combined with modifications made to the translatable compound encoding an OTC protein described herein to enhance stability.

A therapeutically effective dose, upon administration, can result in serum or plasma levels of OTC of 1-1000 pg/ml, or 1-1000 ng/ml, or 1-1000 µg/ml, or more. In some embodiments, administering a therapeutically effective dose of a composition comprising an mRNA sequence described herein can result in increased liver modified protein levels in a treated subject. Preferably, administering a composition comprising a mRNA described herein results in a 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 95% increase in liver modified protein levels relative to a baseline modified protein level in the subject prior to treatment. Preferably, administering a therapeutically effective dose of a composition comprising an mRNA described herein will result an increase in liver OTC levels relative to baseline liver OTC levels in the subject prior to treatment. In some embodiments, the increase in liver OTC levels relative to baseline liver OTC levels will be at least 5%, 10%, 20%, 30%, 40%, 50%, 100%, 200%, or more.

Preferably, a therapeutically effective dose, when administered regularly, results in increased expression of OTC in the liver as compared to baseline levels prior to treatment. Preferably, administering a therapeutically effective dose of a composition comprising an mRNA sequence described herein results in the expression of a modified protein level at or above about 10 ng/mg, about 20 ng/mg, about 50 ng/mg,

about 100 ng/mg, about 150 ng/mg, about 200 ng/mg, about 250 ng/mg, about 300 ng/mg, about 350 ng/mg, about 400 ng/mg, about 450 ng/mg, about 500 ng/mg, about 600 ng/mg, about 700 ng/mg, about 800 ng/mg, about 900 ng/mg, about 1000 ng/mg, about 1200 ng/mg or about 1500 ng/mg of the total protein in the liver of a treated subject.

Preferably, a therapeutically effective dose, when administered regularly, results in a reduction of orotic acid levels in a biological sample. In some embodiments, administering a therapeutically effective dose of a composition comprising an mRNA described herein results in a reduction of orotic acid levels in a biological sample (e.g., urine, plasma or serum sample) by at least about 5%, at least about 10%, at least about 15%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, or at least about 95% as compared to baseline orotic acid levels before treatment. Preferably, the biological sample is selected from plasma, serum, whole blood, urine, or cerebrospinal fluid. Preferably, administering a therapeutically effective dose of a composition comprising an mRNA described herein results in reduction of orotic acid levels to about 1000 µmol/L or less, about 900 µmol/L or less, about 800 µmol/L or less, about 600 µmol/L or less, about 500 µmol/L or less, about 400 µmol/L or less, about 300 µmol/L or less, about 200 µmol/L or less, about 100 µmol/L or less or about 50 µmol/L or less in serum or plasma. Preferably, a therapeutically effective dose, when administered regularly results in reduction of orotic acid levels to about 600 µmol/L or less in serum or plasma. In another exemplary embodiment, a therapeutically effective dose, when administered regularly results in reduction of orotic acid levels to about 360 µmol/L or less in serum or plasma. Preferably, a therapeutically effective dose, when administered regularly results in reduction of orotic acid levels to about 120 µmol/L or less in serum or plasma.

A therapeutically effective dose of an mRNA described herein in vivo can be a dose of about 0.001 to about 500 mg/kg body weight. For instance, the therapeutically effective dose may be about 0.001-0.01 mg/kg body weight, or 0.01-0.1 mg/kg, or 0.1-1 mg/kg, or 1-10 mg/kg, or 10-100 mg/kg. Preferably, a Lipid-enabled and Unlocked Nucleomonomer Agent modified RNA (LUNAR)-mRNA (see WO 2015/074085 and U.S. 2018/0169268), encoding an OTC protein described herein, is provided at a dose ranging from about 0.1 to about 10 mg/kg body weight.

50 Combinations

The cDNA primary constructs, mRNA or encoded OTC proteins described herein may be used in combination with one or more other therapeutic, prophylactic, diagnostic, or imaging agents. By "in combination with," it is not intended to imply that the agents must be administered at the same time and/or formulated for delivery together, although these methods of delivery are within the scope of the present disclosure. Compositions can be administered concurrently with, prior to, or subsequent to, one or more other desired therapeutics or medical procedures. In general, each agent will be administered at a dose and/or on a time schedule determined for that agent. Preferably, the methods of treatment of the present disclosure encompass the delivery of pharmaceutical, prophylactic, diagnostic, or imaging compositions in combination with agents that may improve their bioavailability, reduce and/or modify their metabolism, inhibit their excretion, and/or modify their distribution

within the body. As a non-limiting example, mRNA disclosed herein and preferably an mRNA sequence comprising SEQ ID NO: 119, encoding a modified OTC protein of SEQ ID NO: 4 may be used in combination with a pharmaceutical agent for the treatment of OTC deficiency. The pharmaceutical agent includes, but is not limited to one or more of: sodium phenylbutyrate, glycerol phenylbutyrate, sodium phenylacetate, sodium benzoate, arginine, citrulline, Multiple vitamins, calcium supplements or combined with low protein/high caloric diet. In general, it is expected that agents utilized in combination with be utilized at levels that do not exceed the levels at which they are utilized individually. In some embodiments, the levels utilized in combination will be lower than those utilized individually. In one embodiment, the combinations, each or together may be administered according to the split dosing regimens as are known in the art.

EXAMPLES

Example 1: Material and Methods

In Vitro Transcription Protocol

The mRNAs were synthesized in vitro using T7RNA polymerase-mediated DNA-dependent RNA transcription where uridine triphosphate (UTP) was substituted and unsubstituted with modified UTPs such as 5 methoxy UTP (5MeOU), N1-methoxy methyl pseudo UTP (N1-MOM), 5-hydroxy methyl UTP, 5-carboxy UTP, and mixture of modifications using linearized template for each UTR combination. The mRNA was purified using column chromatography, the DNA and double stranded mRNA contamination of all mRNAs was removed using an enzymatic reaction, and the mRNA was concentrated, and buffer exchanged.

Preparation of Lipid Encapsulated mRNA

Lipid encapsulated mRNA particles were prepared by mixing lipids (ATX lipid: DSPC: Cholesterol: PEG-DMG) in ethanol with OTC mRNA dissolved in Citrate buffer. The mixed material was instantaneously diluted with Phosphate Buffer. Ethanol was removed by dialysis against phosphate buffer using regenerated cellulose membrane (100 kD MWCO) or by tangential flow filtration (TFF) using modified polyethersulfone (mPES) hollow fiber membranes (100 kD MWCO). Once the ethanol was completely removed, the buffer was exchanged with HEPES (4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid) buffer containing 50 mM NaCl and 9% sucrose, pH 7.3. The formulation was concentrated followed by 0.2 μ m filtration using PES filters. The mRNA concentration in the formulation was then measured by Ribogreen fluorimetric assay following which the concentration was adjusted to a final desired concentration by diluting with HEPES buffer containing 50 mM NaCl, 9% sucrose, pH 7.3 containing glycerol. The final formulation was then filtered through a 0.2 μ m filter and filled into glass vials, stoppered, capped and placed at $-70\pm 5^\circ$ C. The frozen formulations were characterized for their mRNA content and percent encapsulation by Ribogreen assay, mRNA integrity by fragment analyzer, lipid content by high performance liquid chromatography (HPLC), particle size by dynamic light scattering on a Malvern Zetasizer Nano ZS, pH and Osmolality.

In-Cell Western (ICW)

96-well collagen plates were used to seed the cells at the appropriate density in Dulbecco's Modified Eagle Media (DMEM)/Fetal Bovine Serum (FBS) culture media. At the optimal confluence, cells were transfected with the targeted

mRNAs diluted in the transfection reagent mix (Messenger-Max and Opti-MEM). Cells were placed in the CO₂ incubator and allowed to grow. At the desired timepoint, media was removed, and cells were fixed in 4% fresh paraformaldehyde (PFA) for 20 min. After that, fixative was removed, and cells were permeabilized in tris buffered saline with TWEEN (TBST) for 5 min several times. When permeabilization washes were complete, cells were incubated with a blocking buffer (ODYSSEY® Blocking Buffer (PBS) (Li-Cor, Lincoln, NE)) for 45 min. Primary antibody was then added and incubated for 1 h at room temperature. Cells were then washed several times in TBST and incubated for 1 h with a secondary antibody diluted in blocking buffer and containing a CellTag 700 stain. To finalize, cells were washed several times in TBST followed by a last wash in tris-buffered saline (TBS). The plate was imaged using the Licor detection system and data was normalized to the total number of cells labeled by the CellTag 700.

Example 2: UTRs Screening in Hepa1,6 and Hep3B—Correlation at 24 h and 48 h

A UTR library was screened in vitro using mRNA construct #571 comprising the sequence of SEQ ID NO: 34 as CDS (coding sequence). In-Cell Western assays as described in Example 1 were used to transfect the different mRNAs into Hepa1,6 and Hep3B using commercially available transfection reagents. OTC protein expression levels were measured by near-infrared fluorescent imaging systems. Commercially available antibodies for OTC were used for detection. Untransfected and reference sequences were used as internal controls. FIG. 1, Panel A is a scatter plot of OTC protein expression levels found in Hepa1,6 and Hep3B cells at 24 hours. FIG. 1, Panel B is a scatter plot of OTC protein expression levels found in Hepa1,6 and Hep3B cells at 48 hours. The aim of the screening was to determine an UTR-specific impact on OTC expression levels in a human (Hep3B) and a mouse (Hepa1,6) liver cell line that will be beneficial to determine which UTRs would work best in both models, in particular, its translatability from mouse-to-human. Top expressing UTRs were used in further profiling studies.

Example 3: Round 1 of Protein Stability Compounds Screened in Hepa1,6 and Hep3B at 24 h—Correlation

In vitro screening of certain mRNA constructs of Table 5 that were designed based on a protein-stability approach was performed. The mRNA constructs were tested in two different chemistries N¹-methylpseudouridine (N1MPU) and 5-methoxyuridine (5MeOU) meaning that 100% of the uridines in each mRNA were N1MPU only or 5MeOU only (not a combination of 5MeOU or N1MPU). In-Cell Western (ICW) assays as described in Example 1 were used to transfect the different mRNAs into Hepa1,6 and Hep3B using commercially available transfection reagents. OTC protein expression levels were measured by near-infrared fluorescent imaging systems. Commercially available antibodies for OTC were used for detection. Untransfected and reference sequences were used as internal controls. FIG. 2, panel A is a scatter plot showing the correlation of OTC protein expression levels in Hepa1,6 cells at 24 hours as a function of mRNAs tested in N1MPU and 5MeOU chemistries. FIG. 2, panel B is a scatter plot showing the correlation of OTC protein expression levels in Hep3B cells at 24 hours as a function of mRNAs tested in N1MPU and

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5MeOU chemistries. These figures exhibit the degree of variability in expression levels when mRNAs from two different chemistries are tested in a mouse and a human liver cell line. It can be seen that that in this experiment, most of the compounds express better when an N1MPU chemistry is used in the mRNA.

Example 4: Round 2 of Protein Stability
Compounds Screened in Human Primary
Hepatocytes at 24 h and 48 h—Correlation

In vitro screening of certain mRNA constructs of Table 5 that were designed based on a protein stability approach was performed. The mRNAs were tested in two different chemistries, 100% of the Uridines are N1MPU indicated by the name of mRNA constructs followed by “0.1” and 100% of the uridines are 5MeOU indicated by the name of mRNA constructs followed by “0.7”. In-Cell Western (ICW) assays as described in Example 1 were used to transfect the different mRNAs into human primary hepatocytes using commercially available transfection reagents. OTC protein expression levels were measured by near-infrared fluorescent imaging systems. Commercially available antibodies for OTC were used for detection. Untransfected and reference sequences were used as internal controls. (FIG. 3, Panels A and B). The results indicate that, in contrast to the experiments conducted in cancer cell lines (Hepa1.6; Hep3B; Example 3, both chemistries express similarly in human primary hepatocytes.

Example 5: Round 3 of Protein Stability
Compounds Screening in Human Primary
Hepatocytes at 24 h and 48 h—Correlation

In vitro screening of novel compounds designed based on a protein stability approach was performed. mRNAs were tested in two different chemistries, N1MPU indicated by the name of mRNA constructs followed by “0.1” and 5MeOU indicated by the name of mRNA constructs followed by “0.7”. In-Cell Western (ICW) assays as described in Example 1 were used to transfect the different mRNAs into human primary hepatocytes using commercially-available transfection reagents. OTC protein expression levels were measured by near-infrared fluorescent imaging systems. Commercially available antibodies for OTC protein were used for detection. Untransfected and reference sequences were used as internal controls. (FIG. 4, Panels A and B). The results indicate that, in contrast to the experiments conducted in cancer cell lines (Hepa1.6; Hep3B; Example 3), both chemistries express similarly in human primary hepatocyte.

Example 6: OTC Protein-Expression Levels in
Human Primary Cells Transfected with OTC
mRNA Constructs 1799.7 (5MeOU Chemistry)
Encoding the OTC Protein of SEQ ID NO: 3 and
1921.7 (5MeOU Chemistry) Encoding the Modified
OTC Protein of SEQ ID NO: 4

In-Cell Western (ICW) assays as described in Example 1 were used to transfect OTC mRNAs into human primary hepatocytes using commercially available transfection reagents. OTC protein expression levels were measured by near-infrared fluorescent imaging systems during a time course study up to 96 h. Commercially available OTC antibodies were used for detection. Untransfected cells were used as internal control. Plot shows OTC protein levels

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normalized to untransfected controls. (FIG. 5). The purpose of this study was to evaluate the half-life of the unmodified versus the modified protein sequence (encoded by constructs 1799.7, 1921.7, respectively) under in vitro conditions in transfected human primary hepatocytes. The results indicate that 1921.7 demonstrated more stable expression than 1799.7.

Example 7: OTC expression levels measured by multiple reaction monitoring (MRM) mass spectrometry in spf/ash mice dosed at 10 mg/kg. Spf/ash mice received an IV injection with either PBS or lipid-formulated (as described in Example 1) OTC-mRNAs at a 10 mg/kg dosing. WT mice were used as internal controls to determine endogenous levels. A time course (6 h, 24 h and 48 h) was performed, and expression levels were measured by MRM using human and mouse specific epitopes for OTC. Graphs were made that represent the amount of protein (ng/mg tissue) detected by MRM specific for human OTC (FIG. 6, Panel A) or mouse (FIG. 6, Panel B). Human- and mouse-specific heavy peptides were designed to measure total levels of OTC in both species. This data set shows quantitative levels of human-specific OTC derived from translation of delivered mRNAs are detected. This expression maintains a high level of stability up to 48 h.

Example 8: OTC Expression Levels Measured by
Western Blot in WT Mice Dosed at 3 mg/kg

Spf/ash mice received an IV injection with either phosphate buffered saline (PBS) or lipid-formulated OTC-mRNAs at a 3 mg/kg dosing using two different chemistries (N1MPU and 5MeOU). WT mice were used as internal controls to determine endogenous levels. Animals were sacrificed 24 h post-dose. OTC expression levels were measured by Western Blot (WB) using an OTC specific antibody. In the results provided in FIG. 7, the bars represent the percentage of expression relative to WT levels (100%). The data generated in this figure shows that WT levels of total OTC in a mouse background were achieved for several codon-optimized sequences.

Example 9: OTC Expression Levels Measured by
MRM in a Dose Range Finding Study

Balb/c mice received an IV injection with either PBS or lipid-formulated OTC-mRNAs at three different doses: 0.3 mg/kg, 1 mg/kg and 3 mg/kg and using two different chemistries (N1MPU and 5MeOU). Animals were sacrificed 24 h post-dose and expression levels were measured by MRM using human and mouse specific epitopes for OTC. The graph in FIG. 8 represents the percentage of expression of human OTC (ng) per mg of liver tissue in Balb/c mice. The horizontal dotted line represents the relative mouse OTC levels in Balb/c mice. (FIG. 8). Expression levels of hOTC protein for mRNA construct 713 5MeOU and mRNA construct 571 N1MPU are indicated by arrows in FIG. 8. In this figure, the MRM was used to quantitatively determine human-selective and mouse-selective OTC protein levels. The data generated in this figure shows, in a dose dependent manner, that WT levels of human OTC in a mouse background are achieved with the codon-optimized sequences disclosed herein.

Example 10: Urinary Orotate Levels Measured in
PBS and Treated Spf/Ash Mice

Spf/ash mice received an IV injection with either PBS or lipid-formulated OTC-mRNA construct 1799.7 (5MeOU

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chemistry) at three different doses: 0.3 mg/kg, 1 mg/kg and 3 mg/kg. WT and spf/ash mice were used to determine baseline and high urinary orotate levels, respectively. A spf/ash time course was determined, and urinary orotate levels were measured at each timepoint. The results can be seen in FIG. 9. Urinary orotate was normalized to creatinine, which is represented in the graph in the Y axis throughout the time course and serve as a proof-of-concept of functional restoration of OTC activity post-injection. At 3 mg/kg, a sustainable reduction of urinary orotate levels up to 14 days was observed.

Example 11: Pharmacokinetic/Pharmacodynamic
(PK/PD) Analysis Comparing Human OTC
Expression Levels and Urinary Orotate at 96 h

Spf/ash mice received an IV injection with either PBS or certain lipid-formulated OTC-mRNAs from Table 5 at 1 mg/kg and 3 mg/kg using two different chemistries (N1MPU and 5MeOU). WT mice were used as internal controls. Human-specific OTC levels were measured by MRM whereas urinary orotate was determined in each sample and normalized to creatinine. PK/PD is plotted in FIG. 10. PK/PD analysis shows the correlation between protein expression levels and reduction of urinary orotate in a compound-specific manner. Construct 1799.7 shows a high PK/PD correlation.

Example 12: Fractioning of Spf/Ash Mice In Vivo
Samples Treated with Selected mRNAs

Spf/ash mice received an IV injection with either PBS or lipid-formulated (as described in Example 1) OTC-mRNAs at 1 mg/kg and 3 mg/kg. WT mice were used as internal controls. Sample fractioning was performed on the liver samples, separating a cytosolic and a mitochondrial fraction. OTC levels were measured by WB using human specific (hOTC) and crossreactive (crOTC) antibodies (FIG. 11). Cyclooxygenase IV (CoxIV) was used as a mitochondrial control. OTC protein expression levels were measured by near-infrared fluorescent imaging systems and normalized to total protein. WB indicates differences in OTC expression levels within mitochondrial and cytosolic fractions when 2016 and 2260 mRNAs were delivered in the spf/ash mice. These results indicate that both compounds can efficiently target the mitochondria.

Example 13: Plot of the Mitochondrial vs Cytosolic
Fractions of Spf/Ash Mice Samples Treated with
mRNA Constructs 2016 and 2260

Spf/ash mice received an IV injection with either PBS or lipid-formulated OTC-mRNAs at 3 mg/kg. WT mice were used as internal controls. Sample fractioning was performed on the liver samples, separating a cytosolic and a mitochondrial fraction. OTC levels were measured by western blot using a human specific antibody. OTC protein expression levels were measured by near-infrared fluorescent imaging systems and both fractions normalized to total protein were plotted (FIG. 12). The plot of protein expression levels shown in FIG. 11 (Example 12), indicate that even though both compounds, 2016 and 2260, deliver similar protein levels in the cytosol, it is in the mitochondria where 2260 delivers more human OTC than 2016. The 2260 includes a modified mitochondrial signaling peptide sequence of the invention.

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Example 14: Urinary Orotate Levels in Spf/Ash
Mice Treated with mRNA Construct 2260

Spf/ash mice received an IV injection with either PBS or lipid-formulated (as described in Example 1) OTC-mRNA at 1 mg/kg and 3 mg/kg. WT mice were included as an internal control. Urinary orotate levels were measured at 0, 1, 3, 7 and 14 days, and levels were normalized to creatinine (FIG. 13). The functional read-out of this assay shows that urinary orotate levels are reduced for up to 14 days with compound 2260, in a dose dependent manner.

Example 15: Survival of Spf/Ash Mice on a High
Protein Diet During Treatment with OTC-mRNA
Construct 1799.7 (5MeOU Chemistry)

Spf/ash mice received an IV injection with either PBS or lipid-formulated OTC-mRNA (1799.7) at three doses: 0.3 mg/kg, 1 mg/kg and 3 mg/kg. Mice were under a high protein diet since day 0 to the end of the study. Treated animals were injected by IV on days 0, 7, 14, 21 and 28 (arrows). Survival rates were determined every week. The plot in FIG. 14 summarizes the entire study timeline and the survival rates observed for the different groups. The results show that animals treated with human OTC mRNAs described herein have a higher chance to survive during a hyperammonemic crisis, suggesting the protective role of OTC mRNAs described herein in detoxifying the animals from toxic ammonia. This survival rate was dose-dependent, and animals treated with a 3 mg/kg dose have a higher survival rate than animals treated at 1 mg/kg or 0.3 mg/kg.

Example 16: Comparison of hOTC Expression
Levels with Different Modifications

Lipid-formulated OTC-mRNA construct 2262 doses were injected by IV in 8 week-old male C57BL/6 mice at a dose of 1 mg/kg. Different chemistries were used in this study as indicated in the bottom axis of the chart provided in FIG. 15. The mice livers were harvested at 24 hours post IV-administration, and western blot was performed using a hOTC specific antibody. Levels were normalized to total protein (FIG. 15). Each construct was formulated as a lipid nanoparticle comprising an ATX lipid as described in Example 1. This dataset indicates the impact of different uridine chemistries on the expression levels of our codon-optimized mRNAs.

The patent and scientific literature referred to herein establishes the knowledge that is available to those with skill in the art. All United States patents and published or unpublished United States patent applications cited herein are incorporated by reference. All published foreign patents and patent applications cited herein are hereby incorporated by reference. All other published references, documents, manuscripts and scientific literature cited herein are hereby incorporated by reference.

While this disclosure has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the disclosure encompassed by the appended claims. It should also be understood that the embodiments described herein are not mutually exclusive and that features from the various embodiments may be combined in whole or in part in accordance with the disclosure.

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SEQUENCE LISTING

mRNA coding sequence for wild type human OTC

(SEQ ID NO: 1)

AUGCUGUUUAAUCUGAGGAUCCUGUUAAACAAUGCAGCUUUUAGAAAU

GGUCACACUUCUAGGUUUGGAAUUUUCGUGUGGACAAACCACUACAA

AAUAAAGUGCAGCUGAAGGGCCGUGACCUUCACACUAAAAACUUU

ACCGGAGAAGAAAUAAAUAAUAGCUAUGGCUAUCAGCAGAUUGGAA

UUUAGGAUAAACAGAAAGGAGAGUAUUUGCCUUUAUUGCAAGGGAAG

UCCUUAGGCAUGAUUUUUGAGAAAAGAAGUACUGAACAAAGAUUGUCU

ACAGAAACAGGCUUUGCACUUCUGGGAGGACAUCUUGUUUUUUUACC

ACACAAGAUUUCAUUGGGUGUGAUGAAAGUCACACGACACGGCC

CGUGUAUUUGUCUAGCAUGGCAGAUUGGCUUGAGUUAUAA

CAAUCAGAUUUGGACACCCUGGCUAAAGAAGCAUCCAUCCCAUUUAUC

AAUGGGCUGUCAGAUUUGUACAUCCAUCCAGAUCCUGGCUGAUUAC

CUCACGCUCACAGGAACACUAUAGCUCUCUGAAAGGUCUUAUCCUCAGC

UGGAUCGGGAUGGGAAACAAUUAUCCUGCACUCCAUCAUGAUGAGCGCA

GCGAAAUUCGGAUAGCACCUUACAGGCAGCUACUCCAAGGGUUAUGAG

CCGGAUGCUGUGUAACCAAGUUGGCAGAGCAGUAUGCCAAAGAGAAU

GGUACCAAGCUGUUGCUGACAAAUGAUCCAUGGAAGCAGCGCAUGGA

GGCAUGUAUUAAUUAACAGACACUUGGAUAGCAUGGGACAAGAAGAG

GAGAAGAAAAAGCGCUCACAGCUCUUAAGGUUACAGGUUACAAUG

AAGACUGCUAAAGUUGCUGCCUCUGACUGGACAUUUUACACUGCUUG

CCCAGAAAGCCAGAGAAGUGGAUGAUGAAGUCUUUUUUAUCCUCUGA

UCACUAGUGUUCACAGAGCAGAAAACAGAAAGUGGACAAUUAUGGCU

GUCAUGGUGUCCUGCUGACAGAUUACUACCUCAGCUCCAGAGCCU

AAAUUUUGA

DNA coding sequence for wild type human OTC

(SEQ ID NO: 2)

ATGCTGTTTAATCTGAGGATCCTGTTAAACAATGCAGCTTTTAGAAAT

GGTCACAACCTTATCGTTCGAAATTTTCGGTGTGGACAACCACTACAA

AATAAGTGCAGCTGAAGGGCGTGACCTTCTACTCTAAAAAATTTT

ACCGGAGAAGAAATTAATATATGCTATGGCTATCAGCAGATCTGAAA

TTTAGGATAAAACAGAAAGGAGAGTATTTCGCTTTATTGCAAGGGAAG

TCCTTAGGCATGATTTTGGAGAAAAGAGTACTCGAACAAGATTGTCT

ACAGAAACAGGCTTTGCACTTCTGGGAGGACATCCTTGTTTTCTTACC

ACACAAGATATTATTGGGTGTGAATGAAAGTCTCACGGACACGGCC

CGTGATTGTCTAGCATGGCAGATGCAGTATTGGCTCGAGTGATATAAA

CAATCAGATTGGACACCTTGGCTAAAGAAGCATCCATCCCAATTATC

AATGGGCTGTGAGATTGTACCATCCTATCCAGATCCTGGCTGATTAC

CTCACGCTCCAGGAACACTATAGCTCTCTGAAAGGTCTTACCTCAGC

TGGATCGGGGATGGGAACAATATCCTGCACTCCATCATGATGAGCGCA

GCGAAATTGGAATGCACCTTACGGCAGCTACTCCAAGGGTTATGAG

CCGGATGCTAGTGTAAACCAAGTTGGCAGAGCAGTATGCCAAAGAGAAT

60

-continued

GGTACCAAGCTGTTGCTGACAAATGATCCATTGGAAGCAGCGCATGGA

GGCAATGTATTAATTACAGACACTTGGATAAGCATGGGACAAGAAGAG

5 GAGAAGAAAAAGCGGCTCCAGGCTTCCAAGGTTACCAGGTTACAATG

AAGACTGCTAAAGTTGCTGCCTCTGACTGGACATTTTTTACTGCTTG

CCCAGAAAGCCAGAGAAGTGGATGATGAAGTCTTTTATTCTCTCGA

10 TCAC TAGTGTTCCAGAGGCAGAAAACAGAAAGTGGCAATCATGGCT

GTCATGGTGTCCCTGCTGACAGATTACTCACCTCAGCTCCAGAAGCCT

AAATTTTGA

15 Human wild type OTC amino acid sequence
(The signal peptide for mitochondrial
import is underlined*)

(SEQ ID NO: 3)

MLFNLRI LLNNAAFRNHNFVNRNFRCGQPLQKNKVQLKGRDLLTLKNF

20 TGEI KYMLWLSADLKFR IKQKGEYLP LLQGS LGMIFEKRSTRTRLS

TETGFALLGGHPCFLTQDIHLGVNESLTDARVLSSMA DAVLARVYK

QSDLDTLAKEASIP I FGLSDLYHP I QILADYLT LQEHYSS LKGLT LSW

25 IGDGNNILHSIMMSAAKFGMHLQAATPKGYEPDASVTKLAEQYAKENG

TKLLLTNDPLEAAHGGNVLITD TWISMGQEEKKRLQAFQGYQVTMK

TAKVAASDWTFLHCLPRKPEEVDDEVFYSRSLVFPEAENRKWTIMAV

MVSLLTDYSPQLQKPKF

30 Modified OTC amino acid sequence
(The signal peptide for mitochondrial
import is underlined*)

(SEQ ID NO: 4)

MLVFNLR ILLNNAAFRNHNFVNRNFRCGQPLQNRVQLKGRDLLTLKN

35 FTGEEIRYMLWLSADLKFR IKQKGEYLP LLQGS LGMIFEKRSTRTRL

STETGFALLGGHPCFLTQDIHLGVNESLTDARVLSSMA DAVLARVY

KQSDLDTLAKEASIP I I NGLSDLYHP I QILADYLT LQEHYSS LKGLT L

40 SWIGDNNILHSIMMSAAKFGMHLQAATPKGYEPDASVTKLAEQYAKE

NGTKLLLTNDPLEAAHGGNVLITD TWISMGQEEKKRLQAFQGYQVT

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45 AVMSLLTDYSPQLQKPKF

TEV

(SEQ ID NO: 5)

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AT1G58420

(SEQ ID NO: 6)

55 ATTATTACATCAAAACAAAAAGCCGCCA

ARC5-2

(SEQ ID NO: 7)

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CCTAC

61

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(SEQ ID NO: 9)
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EMCV
(SEQ ID NO: 10)
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(SEQ ID NO: 14)
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HUMAN HAPTOGLOBIN
(SEQ ID NO: 17)
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(SEQ ID NO: 18)
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>mARM563

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>mARM727

(SEQ ID NO: 58)

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91

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138

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139

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151

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152

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154

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157

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158

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161

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163

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164

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165

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167

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170

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15 >pARM1822

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173

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175

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177

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>pARM1846

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15 >pARM1882

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179

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182

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183

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>pARM1890

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 ctgctgacag attactcacc tcagctccag aagcctaaat tttga 1065

SEQ ID NO: 2 moltype = DNA length = 1065
 FEATURE Location/Qualifiers
 source 1..1065
 mol_type = genomic DNA
 organism = Homo sapiens

SEQUENCE: 2
 atgctgttta atctgaggat cctgttaaac aatgcagctt ttagaaatgg tcacaacttc 60
 atgggttcgaa attttcgggtg tggacaacca ctacaaaata aagtgcagct gaagggccgt 120
 gaccttctca ctctaaaaaa ctttaccgga gaagaaatta aatatatgct atggctatca 180
 gcagatctga aatttaggat aaaacagaaa ggagagtatt tgcctttatt gcaagggag 240
 tccttaggca tgatttttga gaaaagaagt actcgaacaa gattgtctac agaacacggc 300
 tttgcacttc tgggaggaca tccttggttt cttaccacac aagatattca tttgggtgtg 360
 aatgaaagtc tcacggacac ggcccgtgta ttgtctagca tggcagatgc agtattggct 420
 cgagtgtata aacaatcaga tttggacacc ctggctaaag aagcatccat cccaattatc 480
 aatgggctgt cagattttga ccatcctatc cagatcctgg ctgattacct cacgctccag 540
 gaacactata gctctctgaa aggtccttacc ctgagctgga tcggggatgg gaacaatc 600
 ctgcactcca tcatgatgag cgcagcgaaa ttcggaatgc accttcaggc agctactcca 660
 aagggttatg agccggatgc tagtgtaacc aagtggcag agcagtatgc caaagagaat 720
 ggtaccaagc tgttgctgac aaatgatcca ttggaagcag cgcattggagg caatgtatta 780
 attacagaca cttggataag catgggacaa gaagaggaga agaaaaagcg gctccaggct 840
 ttccaaggtt accaggttac aatgaagact gctaaagtgt ctgcctctga ctggacattt 900
 ttacactgct tcccagagaa gccagaagaa gtggatgatg aagtccttta ttctcctcga 960
 tcactagtgt tcccagagcg agaaaacaga aagtggacaa tcatggctgt catggtgtcc 1020
 ctgctgacag attactcacc tcagctccag aagcctaaat tttga 1065

SEQ ID NO: 3 moltype = AA length = 353
 FEATURE Location/Qualifiers
 source 1..353
 mol_type = protein
 organism = Homo sapiens

SEQUENCE: 3
 MLFNLRLILL NAAFRNGHNF MVRNFRCGQP LQNKVQLKGR DLLTLKNFTG EEIKYMLWLS 60
 ADLKFRIKQK GEYLPPLQGK SLGMIPEKRS TRTRLSTETG FALLGGHPCF LTTQDIHLGV 120

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NESLTD TARV LSSMADAVLA RVYKQSDLDL LAKEASIPF GLSDLYHPIQ ILADYLTQEQ 180
HYSSLKGLTL SWIGDGNLIL HSIIMMSAAKF GMHLQAATPK GYEPDASVTK LAEQYAKENG 240
TKLLLTNDPL EAAHGGNVLIT TDTWISMGQE EEKKKRLQAF QGYQVMTKTA KVAASDWTFL 300
HCLPRKPEEV DDEVFYSRPS LVPPEAENRK WTIMAVMVSL LTDYSPQLQK PKF 353

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SEQ ID NO: 4      moltype = AA length = 355
FEATURE          Location/Qualifiers
REGION           1..355
                 note = Synthetic construct
source           1..355
                 mol_type = protein
                 organism = synthetic construct

```

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SEQUENCE: 4
MLVFNLRLILL NNAAFRNGHN FMVRNFRCGQ PLQNRVQLKG RDLTLKNFT GEEIRYMLWL 60
SADLKFRKIQ KGEYLPPLQG KSLGMIFEKR STRTRLSTET GFALLGGHPC FLTTQDIHLG 120
VNESLTD TARV VLSSMADAVL ARVYKQSDLD TLAKEASIPF INGLSDLYHP IQILADYLT 180
QEHYSSLKGL TLSSWIGDGN ILHSIMMSAA KFGMHLQAAT PKGYEPDASV TKLAEQYAKE 240
NGTKLLLTND PLEAAHGGNV LITDTWISMG QEEKKKRLQ AFQGYQVMTK TAKVAASDWT 300
FLHCLPRKPE EVDDDEVFISP RSLVPPEAEN RKWTIMAVMV SLLTDYSPQL QKPKF 355

```

```

SEQ ID NO: 5      moltype = RNA length = 129
FEATURE          Location/Qualifiers
misc_feature      1..129
                 note = Synthetic construct
source           1..129
                 mol_type = other RNA
                 organism = synthetic construct

```

```

SEQUENCE: 5
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttct ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatag 129

```

```

SEQ ID NO: 6      moltype = RNA length = 28
FEATURE          Location/Qualifiers
misc_feature      1..28
                 note = Synthetic construct
source           1..28
                 mol_type = other RNA
                 organism = synthetic construct

```

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SEQUENCE: 6
attattacat caaaacaaaa agccgccca 28

```

```

SEQ ID NO: 7      moltype = RNA length = 245
FEATURE          Location/Qualifiers
misc_feature      1..245
                 note = Synthetic construct
source           1..245
                 mol_type = other RNA
                 organism = synthetic construct

```

```

SEQUENCE: 7
cttaaggggg cgctgcctac ggaggtggca gccatctcct tctcggcac aagcttacca 60
tgggtgccca ggccctgctc ttgggtcccgc tgctgggtgt cccctctgc ttcggcaagt 120
tccccatcta ccccatcccc gacaagctgg ggccgtggag ccccatcgac atccaccacc 180
tgtctgccc caacaacctc gtggtcgagg acgagggctg caccaacctg agcgggttct 240
cctac 245

```

```

SEQ ID NO: 8      moltype = RNA length = 177
FEATURE          Location/Qualifiers
misc_feature      1..177
                 note = Synthetic construct
source           1..177
                 mol_type = other RNA
                 organism = synthetic construct

```

```

SEQUENCE: 8
tgagtgtcgt acagctccca gggccccccc tcccgggaga gccatagtgg tctgcggaac 60
cggtgagtac accggaattg ccgggaagac tgggtccttt cttggataaa cccactctat 120
gccccggcat ttgggcgtgc ccccgcaaga ctgctagccg agtagtggtg ggttgcg 177

```

```

SEQ ID NO: 9      moltype = RNA length = 89
FEATURE          Location/Qualifiers
misc_feature      1..89
                 note = Synthetic construct
source           1..89
                 mol_type = other RNA
                 organism = synthetic construct

```

```

SEQUENCE: 9
aattattggt taaagaagta tattagtgtc aatttccctc cgtttgtcct agctttttctc 60
ttctgtcaac cccacacgcc ttgggcaca 89

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SEQ ID NO: 10      moltype = RNA  length = 569
FEATURE           Location/Qualifiers
misc_feature       1..569
                   note = Synthetic construct
source            1..569
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 10
ctccctcccc cccccctaac gttactggcc gaagccgctt ggaataaggc cgggtgtgcgt 60
ttgtctatat gttattttcc accatattgc cgtcttttgg caatgtgagg gcccggaac 120
ctggccctgt cttcttgacg agcattccta ggggtctttc ccctctcgcc aaaggaatgc 180
aaggctctgt gaatgtcgtg aaggaagcag ttctcttgga agcttcttga agacaaacaa 240
cgtctgtagc gaccctttgc aggcagcgga acccccacc tggcgacagg tgcctctgcg 300
gccaaaagcc acgtgtataa gatacacctg caaaggcgcc acaaccccag tgccacgttg 360
tgagttggat agttgtggaa agagtcaaat ggctctcttc aagcgtattc aacaaggggc 420
tgaaggatgc ccagaaggta cccattgta tgggatctga tctggggcct cgggtgcacat 480
gctttacgtg tgtttagtcg aggttaaaaa acgtctaggc ccccggaacc acggggacgt 540
ggttttcctt tgaaaaacac gatgataat 569

SEQ ID NO: 11      moltype = RNA  length = 44
FEATURE           Location/Qualifiers
misc_feature       1..44
                   note = Synthetic construct
source            1..44
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 11
gtcagctttc aaactctttg tttcttgttt gttgattgag aata 44

SEQ ID NO: 12      moltype = RNA  length = 54
FEATURE           Location/Qualifiers
misc_feature       1..54
                   note = Synthetic construct
source            1..54
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 12
ctctcgccctg agaaaaaaaa tccacgaacc aatttctcag caaccagcag cacg 54

SEQ ID NO: 13      moltype = RNA  length = 52
FEATURE           Location/Qualifiers
misc_feature       1..52
                   note = Synthetic construct
source            1..52
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 13
acctgtgagg gttcgaagga agtagcagtg tttttgttc ctagaggaag ag 52

SEQ ID NO: 14      moltype = RNA  length = 71
FEATURE           Location/Qualifiers
misc_feature       1..71
                   note = Synthetic construct
source            1..71
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 14
acacagaaac attcgcaaaa acaaaatccc agtatcaaaa ttcttctctt tttttcatat 60
ttcgcaaaga c 71

SEQ ID NO: 15      moltype = RNA  length = 52
FEATURE           Location/Qualifiers
misc_feature       1..52
                   note = Synthetic construct
source            1..52
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 15
cagaaaaatt tgctacattg tttcacaac ttcaaatatt attcatttat tt 52

SEQ ID NO: 16      moltype = RNA  length = 158
FEATURE           Location/Qualifiers
misc_feature       1..158
                   note = Synthetic construct
source            1..158
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 16
ctagtgcactg actaggatct ggttaccact aaaccagcct caagaacacc cgaatggagt 60
ctctaagcta cataatacca acttacactt acaaaatggt gtcccccaaa atgtagccat 120
tcgtatctgc tectaataaa aagaaagttt cttcacat 158

SEQ ID NO: 17 moltype = RNA length = 166
FEATURE Location/Qualifiers
misc_feature 1..166
 note = Synthetic construct
source 1..166
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 17
tgcaaggctg gccggaagcc cttgcctgaa agcaagattt cagcctggaa gagggcaaag 60
tggacgggag tggacaggag tggatgcgat aagatgtggt ttgaagctga tgggtgccag 120
ccctgcattg ctgagtcaat caataaagag ctttcttttg acccat 166

SEQ ID NO: 18 moltype = RNA length = 143
FEATURE Location/Qualifiers
misc_feature 1..143
 note = Synthetic construct
source 1..143
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 18
acgccgaagc ctgcagccat ggcacccac gccaccccggt gcctcctgcc tccgcgcagc 60
ctgcagcggg agaccctgtc cccgccccag ccgtcctcct ggggtggacc ctagtttaat 120
aaagattcac caagtttcac gca 143

SEQ ID NO: 19 moltype = RNA length = 220
FEATURE Location/Qualifiers
misc_feature 1..220
 note = Synthetic construct
source 1..220
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 19
tagagcggca aaccctagct aactccata gctagtttct ttttttttg ttttttttt 60
ttttttttt ttttttttt ttttttttc ctttcttttc cttcttttt tctcttttc 120
ttggtggctc catcttagcc ctagtacgg ctagtcttga aaggtccgtg agccgcata 180
ctgcagagag tgccgtaact ggtctctctg cagatcatgt 220

SEQ ID NO: 20 moltype = RNA length = 170
FEATURE Location/Qualifiers
misc_feature 1..170
 note = Synthetic construct
source 1..170
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 20
acacatcaca accacaacct tctcaggcta ccctgagaaa aaaagacatg aagactcagg 60
actcatcttt tctgttggtg taaaatcaac accctaagga acacaaattt ctttaaacat 120
tgacttctt gtctctgtgc tgcaattaat aaaaaatgga aagaatctac 170

SEQ ID NO: 21 moltype = RNA length = 110
FEATURE Location/Qualifiers
misc_feature 1..110
 note = Synthetic construct
source 1..110
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 21
gctggagcct cggttagcgt tctcctgcc cgctgggcct cccaacgggc cctcctcccc 60
tccttgacac ggccttcct ggtctttgaa taaagtctga gtgggcagca 110

SEQ ID NO: 22 moltype = RNA length = 123
FEATURE Location/Qualifiers
misc_feature 1..123
 note = Synthetic construct
source 1..123
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 22
tagtgcagtc actggcacia cgcgttgccc ggtaagccaa tcgggtatac acggtcgtea 60
tactgcagac agggttcttc tactttgcaa gatagtctag agtagtaaaa taaatagtat 120
aag 123

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SEQ ID NO: 23          moltype =   length =
SEQUENCE: 23
000

SEQ ID NO: 24          moltype =   length =
SEQUENCE: 24
000

SEQ ID NO: 25          moltype =   length =
SEQUENCE: 25
000

SEQ ID NO: 26          moltype = RNA length = 1368
FEATURE               Location/Qualifiers
misc_feature           1..1368
                        note = Synthetic construct
source                1..1368
                        mol_type = other RNA
                        organism = synthetic construct

SEQUENCE: 26
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctg tttaatctga ggaatcctgt aaacaatgca gcttttagaa 180
atggtcacaa cttcatgggt cgaaattttc ggtgtggaca accactacaa aataaagtgc 240
agctgaaggg ccgtgacctt ctcaactctaa aaaactttac cggagaagaa attaaatata 300
tgctatggct atcagcagat ctgaaattta ggataaaaca gaaaggagag tatttgccct 360
tattgcaagg gaagtcctta ggcattgatt ttgagaaaag aagtactcga acaagattgt 420
ctacagaaac aggcttttgc cttctgggag gacatccttg tttcttacc acacaagata 480
ttcatttggg tgtgaatgaa agtctcacgg acacggcccg tgtattgtct agcatggcag 540
atgcagtatt ggctcgagtg tataaacaat cagatttgga caccctggct aaagaagcat 600
ccatcccaat tatcaattgg ctgtcagatt tgtaccatcc tatccagatc ctggctgatt 660
acctcacgct ccagggaacac tatagctctc tgaaaaggct taccctcagc tggatcgggg 720
atgggaacaa tatcctgcac tccatcatga tgagcgcagc gaaattcgga atgcacctt 780
aggcagctac tccaaagggt tatgagccgg atgctagtgt aaccaagtgt gcagagcagt 840
atgccaaaag gaattggtag aagctgtgtg tgacaaatga tccattggaa gcagcgcagt 900
gaggcaatgt attaatata gacacttgga taagcatggg acaagaagag gagaagaaaa 960
agcggctcca ggctttccaa gggtaccagg ttacaatgaa gactgctaaa gttgctgct 1020
ctgactggag atttttacac tgcttgccca gaaagccaga agaagtgat gatgaagtct 1080
tttattctcc tcgatcacta gtgttcccag aggcagaaaa cagaagtggg acaatcatgg 1140
ctgtcatggt gtcctctgtg acagattact caccctcagc ccagaagcct aaattttgac 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccgcaat 1260
ggagttctta agctacataa taccacttta cacttacaaa atgttgtccc ccaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 27          moltype = RNA length = 1368
FEATURE               Location/Qualifiers
misc_feature           1..1368
                        note = Synthetic construct
source                1..1368
                        mol_type = other RNA
                        organism = synthetic construct

SEQUENCE: 27
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctc tttaatctgc gcatcttact gaacaacgac gcattccgga 180
acggtcacaa cttcatgggt cgcaatttcc gctgtggcca gccgcttcaa aacaagggtc 240
agctgaaggg acgggatctc ctgacactga agaacttcac cggagaagag atcaagtaca 300
tgctgtggct cagcgcagac ttgaagtctc ggatcaagca gaagggagaa tacttgcccc 360
tgctgcaagg aaagtctgtg ggaatgattt ttgagaagcg gtcaactcgc accagactct 420
ccaccgaaac tggtttcgca ctgcttggcg ggcacccttg cttcctgacg actcaggaca 480
tccacctcgg cgtgaacgaa tcgctaaccg ataccgccag agtgctttct tccatggcgg 540
acgcggtgct ggccagggtg tacaagcagt ccgacctcga taccttggca aaggagggtt 600
ccattcccat catcaaggcg ctgagcgacc tgtaccaccc aatccaaatc ctggctgact 660
acctgacctt gcaagagcac tacagcagcc tgaagggtct gacctgtca tggattggcg 720
atggaacaaa tattctgcac tccatcatga tgtccgcccg gaagtctcga atgcattctg 780
aagccgcgac tccaaaagga tacgaaccgg atgctccgtg gaccaagtgt gcggaacagt 840
acgcgaagga gaacggaacc aagcttctgc tgactaacga cccctcgag gctgctcatg 900
ggggcaacgt gctgattacc gacacctgga tctccatggg gcaggaggaa gagaagaaga 960
agagactgca ggcattccag gggtagcagg tcccatgaa aaccgcaaaa gtggcagctt 1020
cggactggac tttctgcat tgcctgccga ggaagccgga ggaagtgcag gacgaagtgt 1080
tctactcgcc tcggtccctg gtgttccccg aggcgaaaaa ccggaagtgg accatcatgg 1140
ccgtgatggt gtccttctgt actgactata gcccgacagt gcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccgcaat 1260
ggagttctta agctacataa taccacttta cacttacaaa atgttgtccc ccaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 28          moltype = RNA length = 1368
FEATURE               Location/Qualifiers

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misc_feature      1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 28
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctg tttaacctac gtattttgct caacaatgca gccttttagaa 180
acggacataa ctttatgggt cgaactttc gctgcgggca gccactgcag aacaagggtcc 240
agctgaaagg gagagatttg ctacagctga agaactttac tggcgaagaa atcaagtata 300
tgctgtgggt gtcgcgggac ctcaagtttc ggattaagca gaaaggggag tatctgccac 360
tgctgcaagg aaagagcctc ggcctgatct tcgagaagcg gagcactcgg accaggctga 420
gtaccgaaac tggcttcgca ttgttgggtg gacatccatg tttctgaca acgcaggaca 480
ttcatctggg cgtgaacgag agtctgacgg acacagctcg cgttctgtcc tctatggctg 540
atgctgggtt ggcctgggtc tataagcagt ccgatttgga cacttgggtt aaggaagcta 600
gcataccgat tatcaattgg ctgtccgacc tgtatcacc ttttcaaatc ctggccgact 660
acctcacact gcaagaacac tatagctcat tgaagggact gacctgagc tggatagggg 720
acggaaccaa gccctacat agcattatga tgtccgtgc caagtttggc atgcatcttc 780
aagccggcac cccaaagggt tatgagcccg acgctcagt gacaaagctg gccgagcagt 840
acgctaaggc gaattgggtc aaattactgc tgaactatga tccactggag gctgcacatg 900
gcggcaatgt actgatcacc gacacatgga tctcgatggg ccaggaggaa gaaaagaaga 960
agaggcttca ggccttccaa ggctaccagg tcaccatgaa aacagctaa gttgcagcat 1020
ctgattggac cttctgcgac tgtctgccc ggaagcccg agaggtggac gatgaagtat 1080
tctatagccc accgagtttg gtgttccctg aggtgaaaa taggaagtgg acaattatgg 1140
ccgtaattgt gtcctgttta accgactact ctccgcaact gcagaaacct aagttttagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccacttta cacttcaaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 29      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source             1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 29
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctg tttaaactta ggatcctgct gaacaacgac gcttttcgta 180
acggtcataa ctttatgggt cggaaactta gatgtggcca gccgctgcag aacaagggttc 240
agctgaaggg gagggatctg ctgaccttga agaactttac cggcgaagag atcaagtata 300
tgctgtgggt gagcggcgat ctgaagttta ggattaagca gaagggggag tatctgccac 360
tgctgcaagg aaaatccttg gggatgatct tcgagaagcg ctccactaga acccggttaa 420
gcacagaaac cggcttcgca cttctgggtg gacatccctg tttctgacg acgcaggata 480
tacacctggg cgtgaattgag agtctgacgg acacagctag ggtgttgagc agcatggcgg 540
atgcagtact ggcccgcgtt tataagcaga gcgacttgga cacactggcc aaggaagcgt 600
caattccgat tatcaattgg ctgtcagacc tgtatcatcc cattcaaatc ttggctgact 660
atctgacctc gcaagaacat tacagctccc tgaagggctt cactgtgtcc tggattggcg 720
acggaaccaa cattctgcat tcgatcatga tgagcgctgc taagtttggc atgcacctcc 780
aagccgctac acctaaaggga tatgagcctg atgccagcgt aaccaagctg gccgaacagt 840
acgcgaaggc gaattggcac aaactgctgt tgacaaatga cccactggag gcagctcacg 900
gtggcaacgt gctgatcacc gacacgtgga tatctatggg acaggaagaa gagaagaaga 960
agcggctgca ggccttccaa gggatcagg tcaccatgaa aacggccaag gttgctgcat 1020
ccgactggac atttctgcat tggcttcccc gcaaaccaga agaagtagac gacgaagtct 1080
tttattcccc acggtcgtcg gtgttcccc aggcgggagaa tcgaaagtgg acgattatgg 1140
ccgtgatggg gtcctatctg actgattact ctccccaaact gcaaaagcct aagttttagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccacttta cacttcaaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 30      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source             1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 30
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctt ttaacactga ggatcctcct gaacaacgac gcctttcgca 180
atggtcacaa ctttatgggt cggaaactta gatgcggcca gccgctgcag aacaagggtcc 240
agctgaaggg acgggatctg ctgactctga agaactttac cggagaagag atcaagtata 300
tgctgtgggt gtcggcgac ctgaagtcca ggatcaagca gaagggagaa tacctcccgc 360
tgctgcaagg aaagtccctg ggcctgattt tcgagaagcg ctcgaccaga actcggttgt 420
ccaccgaaac cgggtttgct ctgctgggag gacatccctg cttcctgacg actcaggata 480
ttcacctggg agtgaacgag tcgctgaccg acaccgccag agtgcctgagc tcgatggcgg 540

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acgccgtgtt ggcacgcgtg tacaagcagt ccgatctgga taccctggcc aaagaagctt 600
ccatcccgat cattaacggg ctgagcgacc tctaccaccc cattcaaatc ctggccgact 660
acctgactct gcaagaacac tacagctcgc tgaagggtt gactctgtcc tggatcggcg 720
acggaacaa catcctgcac tccatcatga tgtcggccgc aaagtctggc atgcatttgc 780
aagccgccc cccaaagggc tacgaaccag acgcgagcgt caccaagctg gccgaacagt 840
acgcgaagga aaatgggtact aagctgctgc tgaccaacga cccattggaa gctgcccatg 900
gtggaacgt gctgatcacc gacacctgga tctcgatggg ccaggaagag gagaagaaga 960
agcggctgca ggcgttccag gggatcagg tcaccatgaa aacagccaaa gtggcagcgt 1020
cagactggac cttctccac tgtctgcctc gcaagccaga ggaggtggac gacgaggtgt 1080
tctaactccc tcggtccctc gtgttccctg aggctgagaa ccggaagtgg accattatgg 1140
ccgtgatggt gtcactcctg actgattact ccccgcaact gcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 31      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature        1..1368
                    note = Synthetic construct
source              1..1368
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 31
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctg ttttaacctg ggatcctatt gaacaatgct gcttttcgta 180
atggccataa ctttatgggt cggaaacttta gatcggggca gccactgcag aacaagggtc 240
agttgaaggc ccgcgactct ttgacattga agaactttac cggcgaagag attaagtata 300
tgctgtggct gtcgtctgac ctcaagtttc gaatcaagca gaaggcgcaa tatctcccc 360
tgctgcaagg aaagtctctc ggcatgatct ttgagaagcg gagtaccga acacggctga 420
gcaccgaaac gggcttcgca ctgctggggg gccatccctg tttctgaca acgcaggaca 480
tcactttggg ggttaacgaa tcattgactg ataccgcccg cgtactgtca tccatggccg 540
acgctgtgct ggctaggggt tacaagcagt cagatctgga tacactggcc aaggaagcta 600
gcataccaat catcaatgga ctgagtgaac ttatcaccc gattcaataa ctagccgatt 660
atctgacctc gcaagagcat tactctctgc tgaaggcctc acgctgtcc tggatcggcg 720
acggcaacaa cattctgcat agtattatga tgtctgtcgc caaattcggc atgcattctc 780
aagctgtctc gccgaagggt tatgaacccg acgcgtcagt tacgaagctc gctgagcagt 840
atgcaaaagg gaatggcaca aagctgttgc ttaccaacga tccctggaa gctgctcatg 900
gcggcaatgt gctgattact gacacctgga tttcaatggg ccaggaggag gagaagaaga 960
agaggttaca ggcctttcaa ggttaccagg tcacgatgaa aaccgctaag gtcgcagcca 1020
gcgactggac attcctgcac tgtctgccaa gaaagccgga agaagtggac gacgaggtgt 1080
tctattcccc gcggtctttg gtgtttcccg aggcgcaaaa caggaaatgg accattatgg 1140
ccgtgatggt acggttctgt acggactaca gccctcagtt gcaaaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 32      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature        1..1368
                    note = Synthetic construct
source              1..1368
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 32
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctc ttttaacctc gcactcctct caacaacgac gccctccgga 180
atgggcataa cttcatgggt cggaaacttca gatcgggcca gccctgcga aacaagggtc 240
agttgaaggc acgggacctc cttacgctga agaactttac cggagaagag attaagtata 300
tgctgtgggt gtcgcgtgac ctcaagttcc gcattaagca gaaggagaa tatctgccg 360
tgctgcaagg aaagagcctg ggcatgatct tcgaaaagcg ctccactaga acccggtcgt 420
cgactgagac tggattcgcc ttgctcgggt gacacccgtg cttcctgacg acccaggaca 480
tcacacctgg agtgaacgag tcacttacgg ataccgcgag ggtgctgtcc tcaatggccg 540
acgcagtgtc gcgcgcgctg tacaagcagt cagatctgga taccctggcc aaggaagcca 600
gattcccat catcaacgga ctgagcgacc ttaccacccc aatccagatc ctgcgcgact 660
acttaacctc gcaagagcac tacagctccc tgaagggact gactctgtcc tggatcgggg 720
atggaacaa catcctgcac tccatcatga tgtctgcgcg taagtttggg atgcattctc 780
aagccgcaac cctaagggga tacgagcccg acgcctcggg gaccaaactc gcggaacagt 840
acgccaagga aaacgggtacc aagctgctgc tgaccaacga ccctctggaa gcggcccacg 900
gaggaaatgt gctgattacc gacacctgga ttctgatggg ccaggaggag gagaagaaga 960
agagactgca ggcgttccag ggatcagg tcaccatgaa aaccgccaag gtcgcgcca 1020
gcgactggac cttctgcac tgtctccctc ggaacccgga agaagtggtg gacgaggtgt 1080
tctaactccc gcgctcgtg gtgttcccg aggtgaaaaa caggaagtgg acaatcatgg 1140
ccgtgatggt gtccctgttg accgactact ccccaacaac gcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 33      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 33
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctgc gcatcctcct gaacaacgcc gccttcgcga 180
atggacacaa ctttatggtc cgaacttcc gctgtgggca gccgctgcag aacaagggtcc 240
agctcaaggg gagagatctc ctgaccctga agaacttcac tggagaggag atcaagtaca 300
tgctgtggct gtccgccgac ctgaaatttc ggattaagca gaaggcgcaa tacctccac 360
tgctgcaagg aaagtctttg ggcatgatct tcgaaaagag aagcaccgg acccggttga 420
gcaccgaaac tgggttcgag ctctcggtg gacaccctg cttcctgacc acccaagata 480
ttcatctggg tgtaacgaa agcctgacc acaccgccag ggtgctgtca tccatggctg 540
acgcagtgtc cgccgggtg tacaagcagt cagacctgga caccctcgcc aaggagcgt 600
cgatcctat catcaacgga ctttcgacc tgtaccacc catccaaatt ctggccgact 660
acctgactct gcaagaacac tatagctgc tgaaaggact tactctgtcc tggatcgggg 720
acggcaacaa cattctccat tccatcatga tgtccgctgc caagtccgga atgcacctc 780
aagcagcgac tccaaggga tacgaacctg atgctcctg gactaagctg gcagagcagt 840
acggcaaggga gaacgggtca aagctgctgc tcacgaacga cccctggag gcggccacg 900
gcggaaacgt gctgattacc gatacctgga tctcaatggg ccaggaagag gagaagaaga 960
agcgggtcca ggcgtttcaa ggctaccagg tcaccatgaa aaccgcgaag gtcgccgct 1020
ccgactggac tttcttgcac tgctgcccg ggaagcccga ggaagtggat gacgaagtgt 1080
tctactcgcc gagatcgttg gtgttccctg aggcggaaaa caggaagtgg accatcatgg 1140
ccgtgatggt gtcctgctg actgattaca gccacagct gcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 34      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 34
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatggtc cggaaactca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccgcgagc ttgaagtcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagctc ggcatgatct ttgagaagcg ctcaaccagg acccgcttct 420
ctactgaaac tgggttcgag ctgctcggtg gccaccctcg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tccctcacg ataccgccg ggtgttatcg agcatggcag 540
atgccgtgct ggccagggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caatctctat tatcaacggc cttagtgaac tctaccatcc gatcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcgcg 720
acggcaacaa cattctccat tccatcatga tgtccgccc aaaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagccc agcttccgt gactaagctc gccgagcagt 840
acgctaaggga gaacggaaac aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaa gtcgctgcct 1020
ccgactggac cttcctgcac tgcctgcctc gcaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc acggagcctc gtgttcccc aggcggagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 35      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 35
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaacctga gaatcctcct gaacaacgcc gccttcgcga 180
atggatcaaa cttcatggtc cgaactttc gctgcggaca gcctctccaa aacaagggtcc 240
agctcaaggg gccgcacctc ctcacactga agaacttcac tggagaagaa atcaagtaca 300
tgctgtggct gagcgccgat ctgaagtcc ggatcaagca gaaggggag taccttctc 360
tgctgcaagg gaagtcttg ggaatgattt tcgagaagcg gtccaccgg accaggctga 420

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gcaactgaaac tggcttcgcc ctgctgggag gccacccttg tttcctgacc actcaggaca 480
tccacctggg cgtgaacgag tccctgaccg atactgccag agtgctgtcc tccatggccg 540
acgcctgtct cgcccggtgt tacaagcagt cagacctcga tacgctggcc aaggaagcct 600
ccattcccat tatcaatggt ctgtcggacc tctaccatcc aatccaaatc ctgcgcgact 660
acctgactct gcaagaacac tacagctcac tcaagggcct caccctctcc tggatcggcg 720
acggaaacaa catccttcac tcgattatga tgtcggccgc gaagttcggg atgcacctcc 780
aagctgccac tccaaaaggc tacgagcccg atgctcagt gactaagttg gcggaacagt 840
atgcgaagga gaacggtagc aagctcctgc tgactaacga cccgctggag gccgccacg 900
ggggaacagt gctcatcacc gatacttggg ttccatggg acaggaggaa gagaagaaga 960
agcggttgca ggcatttcag ggctaccagg tcaccatgaa aactgccaaa gtcgccgcca 1020
gcgactggac cttcctgcac tgccctcccc gcaagcctga agaagtggac gacgaggtgt 1080
tctactctcc ccggtccctc gtgttccctg aggcggaaaa caggaaagtg accatcatgg 1140
ctgtgatggt gtcctcctg accgactaca gccctcagct ccaaaaacc aagttttagc 1200
tcgagctagt gctgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagctctta agctacataa taccaactta cacttacaaa atgttgctcc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 36      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 36
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaagcaaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaacctga gaatcctctt gaacaatgct gcttttcgga 180
atggccacaa ctttatgggt cggaacttcc gttgcggcca gccctttaca aacaagggtc 240
agctgaaggc ccgggatttg ctcacactga agaactttac tggggaggag attaagtata 300
tgctgtggct gtcgctgac ctgaagttaa ggatcaagca gaaggcgcaa tatctgccgc 360
tgctgcaagg gaaaagtctg ggcatgattt ttgaaaagcg ctctaccggg accagactgt 420
tcacggaaac aggttttgcc ctgctgggcg gccaccctcg tttctgaca acgcaggaca 480
tccatctggg cgtgaacgaa tcactgaccg atactgctcg ggtactcagt tctatggctg 540
acgcagtgtc ggctaggggt tacaagcaga gcgacttgga cacactggct aaggaggcca 600
gcatccccat tatcaatggc ctgtctgatt tgtaccatcc cattcaaatc ctggctgatt 660
atctgacact acaagagcat tactcaagtc tgaagggttt gactctctcc tggatcggcg 720
acggcaacaa cattttacat tccattatga tgagtgtgc taagtttggc atgcatttgc 780
aagctgtctc cccaaagggc tatgaacctg acgctagcgt aaccaagttg gccgaacagt 840
atgctaagaa gaattggcgc aagctgtctc tgacgaatga cccctggaa gctgctcatg 900
gggaaaacgt actataact gatacatgga ttagcatggg ccagggaagag gagaagaaga 960
agagactgca ggccttccaa ggctatcagg tcaccatgaa aactgccaaag gttgcagcta 1020
gcgactggac cttcctgcac tgcttgccga ggaaccgga ggaggtggac gatgaagtct 1080
tttattctcc ccgctccttg gtgtttcccg aggctgaaaa tcgaaagtgg acgataatgg 1140
cagtgatggt gtcctcactg accgactatt ctccacaact gcagaagcct aaattctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagctctta agctacataa taccaactta cacttacaaa atgttgctcc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

```

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SEQ ID NO: 37      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

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```

SEQUENCE: 37
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaagcaaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctga ggatcctgct gaacaacgct gcttttcgca 180
acggtcataa ctttatgggt cgcaattttc gttgtggcca gccgctgcag aacaagggtc 240
agctgaaggc cagagatctg ctgactctga agaacttcac tggggaagaa atcaagtata 300
tgttatggct gtcgctgcac ctgaaatttc gaatcaagca gaaggcgcaa tatctcccc 360
tgctgcaagg gaaatccttg ggcatgattt ttgagaagag gagcactagg actagattgt 420
caacagaaac aggttttgct ttgttgggcg gacatccctg cttctgacg acacaggata 480
tccacctcgg cgtaaacgag tccctcaccg acactgctag ggtactgagc agcatggccg 540
acgctgtgct agcccggttt tacaagcagt cagacctgga cacccttgcc aaggaagctt 600
ctattccaat tatcaacggc ctgagtgacc tgtatcacc cattcaaatc ctgcgcgact 660
atttgacgct tcaagaacat tacagcagcc tcaagggtct aaccttgagt tggataggcg 720
acggcaacaa tatcctgcat tccattatga tgtctgccgc taagtttggc atgcactac 780
aagccgcaac acccaagggc tatgaacctg acgctagcgt gaccaagctg gccgagcagt 840
atgctaagga aaatggcaca aagctccttc ttaccaacga tccctggag gctgctcacg 900
gcggcaacgt gctgattacc gatacatgga ttagcatggg ccaggaggag gagaagaaga 960
agcggctcca ggtttttcaa ggctatcagg tcaccatgaa aactgcaaaag gtcgctgcct 1020
ccgactggac tttcctgcat tgtctacccc gcaagcctga ggaagtggac gatgaggtgt 1080
tctactcccc acggagtctg gtgttcccgg aagcagagaa tcggaagtgg accatcatgg 1140
ctgtcatggt gtcgctcttg actgactatt ctccccaaact gcaaaaacc aagttttagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260

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ggagctctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgtccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 38      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 38
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgtta ttcaaccttc gtatcctgct aaacaatgct gcttttcgca 180
atggccataa ctttatgggt cgcaacttta gatcgggcca gccgctgcag aacaagggtc 240
agctgaaggg ccgggacttg ctgacgctga aaaactttac cggggaagag attaagtata 300
tgctgtggct aagcgctgat ctgaagttta ggatcaagca gaagggcgaa tatctgccac 360
tgctgcaagg gaagagctct gccatgattt ttgaaaagcg gtctaccaga acccgctgt 420
cgaccgagac aggttttgcct tcgctggggg gccatccctg tttctgaca actcaggaca 480
ttcacctggg cgtgaatgag tccctgaccg atactgctag ggtgttgagt agcatggcgg 540
acgctgtact cgtcagagtg tataagcagt ctgatctgga cactctggct aaggaagcgt 600
ccattcctat tatcaacggc ttgagcgacc tgtaccaccc cattcaaatc ctgctgatt 660
acttgacttt gcaagaacat tacagcagct tgaaggcctt aacactgagc tggataggcg 720
acggaacaaa catcttgcac tcataatga tgcctcgccg taagttcggg atgcacctcc 780
aagcagccac acctgaagggc tatgaaccgg atgcttccgt gacaaaactg gctgagcagt 840
atgctaagga aactgggcacg aaaactgctgc tcaccaacga ccatttgaa gctgcacatg 900
gtggcaacgt actgatcact gacacttgga tctcaatggg ccaggaggaa gagaagaaga 960
aaaggctgca ggcatttcag ggataccaag tcaatgaa aactgccaa gtcgctgcct 1020
cagactggac attcctgcac tgtctgccac ggaagcctga ggaagtcgat gacgaagtgt 1080
tctatagccc acgaagcttg gtgtttcccg aggtcgagaa taggaagtgg accattatgg 1140
ctgttatggt gtcctgctc accgactatt cccctcaact gcaaaaacc aagttttagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagctctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgtccta ataaaaagaa agtttcttca cattctag 1368

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```

SEQ ID NO: 39      moltype = RNA length = 1365
FEATURE           Location/Qualifiers
misc_feature       1..1365
                  note = Synthetic construct
source            1..1365
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 39
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc catgcttttt aatctccgca tctctcttaa caacgcccgg ttagaaacg 180
gccacaactt catggctcgg aacttcagat gtggccagcc gcttcaaaac aaggctccagc 240
tgaagggccg ggatcttctg accctgaaga acttactgg cgaagagatc aagtacatgc 300
tctggctctc cgcggacttg aagttccgca ttaagcagaa gggggaatac cttccgctgc 360
ttcaaggaaa gagctcggc atgatctttg agaagcgtc aaccaggacc cgcctttcta 420
ctgaaactgg gttcgcgctg ctcggtggcc acccctgctt cctgacgacc caggacatcc 480
acctcggagt gaacgaatcc ctacccgata ccgcccgggt gttatcgagc atggcagatg 540
ccgtgctggc caaggtgtac aaacagtccg atctggacac tctggccaag gaggcgtcaa 600
ttcccatcat caacggcctg agcgacctgt accacccaat ccaaatcctg gctgactacc 660
tgacctcgca agagcactac agcagcctga agggctgac cctgtcatgg attggcgatg 720
gaaacaatat tctgcactcc atcatgatgt ccgcccggaa gttcggaatg catctgcaag 780
ccgccacgcc aaaaggatac gaaccggatg cgtccgtgac gaagtggcg gaacagtacg 840
cgaaggagaa cggaaaccaag cttctgctga ctaacgaccc cctcagggt gcgcatgggg 900
gcaacgtgct gattaccgac acctggatct ccatggggca ggagggaag aagaagaaga 960
gactgcaggc attccagggg taccagggtc ccatgaaaac cgcaaaagtg gcagcttcgg 1020
actggacttt cctgcattgc ctgccaggga agccggagga agtcgacgac gaagtgttct 1080
actcgcctcg gtcctgggtg ttcccagagg ccgaaaaccg gaagtggacc atcatggcgg 1140
tgatgggtgc cttgctgact gactatagcc cgcagctgca gaagcctaag ttctagctcg 1200
agctagtga ctagctagat ctgggtacca ctaaacagc ctaagaaca cccgaatgga 1260
gtctctaagc tacataatac caacttacac ttacaaaatg ttgtccccc aaatgtagcc 1320
attcgtatct gctctaata aaaagaaagt ttcttcacat tctag 1365

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SEQ ID NO: 40      moltype = RNA length = 1365
FEATURE           Location/Qualifiers
misc_feature       1..1365
                  note = Synthetic construct
source            1..1365
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 40
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc catgcttttc aacctgagaa tctcttgaa caatgctgct ttcggaatg 180
gccacaactt tatggttcgg aacttccgtt gcggccagcc tttacaaaac aaggctccagc 240

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tgaagggccg ggatttggctc aactgaaga actttactgg agaagagatc aagtacatgc 300
tgtggctgtc ggccgacctg aagttcagga tcaagcagaa gggagaatac cttccgctgc 360
ttcaaggaaa gagcctcggc atgatctttg agaagcgctc aaccaggacc cgcctttcta 420
ctgaaactgg gttcgcgctg ctccgtggcc acccctgctt cctgacgacc caggacatcc 480
acctcggagt gaacgaatcc ctaccgata ccgcccggtt gttatcgagc atggcagatg 540
ccgtgctggc caggggtgtac aaacagtccg atctcgatac cttggcaaaag gaggcttcca 600
ttcccatcat caacggcctg agcgacctgt accaccaat ccaaatcctg gctgactacc 660
tgacctgca agagcactac agcagcctga agggctctgac cctgtcatgg attggcgatg 720
gaaacaatat tctgcactcc atcatgatgt ccgcccgcaa gttcggaatg catctgcaag 780
ccgccactcc aaaaggatac gaaccggatg cgtccgtgac caagttaggc gaacagtacg 840
cgaaggagaa cggaaccaag ctctctgtga ctaacgaccc cctcgaggct gcgcattggg 900
gcaacgtgct gattaccgac acctggatct ccatggggca ggaggaagag aagaagaaga 960
gactgcagggc attccagggg taccaggtca ccatgaaac cgcaaaagtg gcagcttcgg 1020
actggacttt cctgcagctg ctgccgagga agtcgagcac gaagtgttct 1080
actcgctcgc gtcctcggtg ttccccgagg ccgaaaaccg gaagtggacc atcatggccg 1140
tgatgggtgc ttctgtgact gactatagcc cgcagctgca gaagcctaag ttctagctcg 1200
agctagtgc tgactaggat ctggttacca cttaaccagc ctcaagaaca ccgaatgga 1260
gtctctaagc tacataatac caacttacac ttacaaaatg ttgtccccc aaatgtagcc 1320
attcgtatct gtcctaata aaaagaaagt ttcttcacat tctag 1365

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SEQ ID NO: 41      moltype = RNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = Synthetic construct
source              1..1365
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 41
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaagcaaaa agcaattttc tgaaaaattt caccattttac 120
gaacgatagc catgcttttc aacctgagaa tcctcttgaa caatgctgct tttcggaatg 180
gccacaactt tatggttcgg aacttcggtt gcggccagcc tttaaaaaac aagggtccagc 240
tgaagggccg ggatttggctc aactaaaga actttactgg agaagagatc aagtacatgc 300
tatggctgtc ggccgacctg aagttccgta tcaagcagaa gggagaatac cttccgctgc 360
ttcaaggaaa gagcctcggc atgatctttg agaagcgctc aaccaggacc cgcctttcta 420
ctgaaactgg gttcgcgctg ctccgtggcc acccctgctt cctgacgacc caggacatcc 480
acctcggagt gaacgaatcc ctaccgata ccgcccggtt gttatcgagc atggcagatg 540
ccgtgctggc caggggtgtac aaacagtccg atctcgatac cttggcaaaag gaggcttcca 600
ttcccatcat caacggcctg agcgacctgt accaccaat ccaaatcctg gctgactacc 660
tgacctgca agagcactac agcagcctga agggctctgac cctgtcatgg attggcgatg 720
gaaacaatat tctgcactcc atcatgatgt ccgcccgcaa gttcggaatg catctgcaag 780
ccgccactcc aaaaggatac gaaccggatg catccgtgac caagttaggc gaacagtacg 840
cgaaggagaa cggaaccaag ctctctgtga ctaacgaccc gctcgaggct gcgcattggg 900
gtaacgtgct gattaccggc acctggatct ccatggggca ggaggaagag aagaagaaga 960
gactgcagggc attccagggg taccaggtca ccatgaaac cgcaaaagtg gcagcttcgg 1020
actggacttt cctgcagctg ctgccgagga agtcgagcac gaagtgttct 1080
actcgctcgc gtcctcggtg ttccccgagg ccgaaaaccg gaagtggacc atcatggccg 1140
tgatgggtgc ttctgtgact gactatagcc cgcagctgca gaagcctaag ttctagctcg 1200
agctagtgc tgactaggat ctggttacca cttaaccagc ctcaagaaca ccgaatgga 1260
gtctctaagc tacataatac caacttacac ttacaaaatg ttgtccccc aaatgtagcc 1320
attcgtatct gtcctaata aaaagaaagt ttcttcacat tctag 1365

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SEQ ID NO: 42      moltype = RNA length = 1365
FEATURE            Location/Qualifiers
misc_feature        1..1365
                    note = Synthetic construct
source              1..1365
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 42
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaagcaaaa agcaattttc tgaaaaattt caccattttac 120
gaacgatagc catgcttttc aacctgagaa tcctcttgaa caatgctgct tttcggaatg 180
gccacaactt tatggttcgg aacttcggtt gcggccagcc tttaaaaaac aagggtccagc 240
tgaagggccg ggatttggctc aactaaaga actttactgg agaagagatc aagtacatgc 300
tatggctgtc ggccgacctg aagttccgta tcaagcagaa gggagaatac cttccgctgc 360
ttcaaggaaa gagcctcggc atgatctttg agaagcgctc aaccaggacc cgcctttcta 420
ctgaaactgg gttcgcgctg ctccgtggcc acccctgctt cctgacgacc caggacatcc 480
acctcggagt gaacgaatcc ctaccgata ccgcccggtt gttatcgagc atggcagatg 540
ccgtgctggc caggggtgtac aaacagtccg atctcgatac cttggcaaaag gaggcttcca 600
ttcccatcat caacggcctg agcgacctgt accaccaat ccaaatcctg gctgactacc 660
tgacctgca agagcactac agcagcctga agggctctgac cctgtcatgg attggcgatg 720
gaaacaatat tctgcactcc atcatgatgt ccgcccgcaa gttcggaatg catctgcaag 780
ccgccactcc aaaaggatac gaaccggatg cgtccgtgac caagttaggc gaacagtacg 840
cgaaggagaa cggaaccaag ctctctgtga ctaacgaccc cctcgaggct gcgcattggg 900
gcaacgtgct gattaccgac acctggatct ccatggggca ggaggaagag aagaagaaga 960
gactgcagggc attccagggg taccaggtca ccatgaaac cgcaaaagtg gcagcttcgg 1020
actggacttt cctgcagctg ctgccgagga agtcgagcac gaagtgttct 1080
actcgctcgc gtcctcggtg ttccccgagg ccgaaaaccg gaagtggacc atcatggccg 1140

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tgatgggtgc	cttctgtgact	gactatagcc	cgcagctgca	gaagcctaag	ttctagctcg	1200
agctagtgac	tgacttaggat	ctgggtacca	ctaaaccagc	ctcaagaaca	cccgaatgga	1260
gtctctaagc	tacataatac	caacttacac	ttacaaaatg	ttgtccccc	aaatgtagcc	1320
attcgtatct	gctcctaata	aaaagaaagt	ttcttcacat	tctag		1365

SEQ ID NO: 43 moltype = RNA length = 1371
 FEATURE Location/Qualifiers
 misc_feature 1..1371
 note = Synthetic construct
 source 1..1371
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 43

tcaacacaaac	atatacaaaa	caaacgaatc	tcaagcaatc	aagcattcta	cttctattgc	60
agcaatttaa	atcatttctt	ttaaagcaaa	agcaattttc	tgaaaatttt	caccatttac	120
gaacgatagc	caccatgctg	ttaaacctgc	gaatcctgct	gaacaacgcc	gcttttcgga	180
acgggcacaa	ctttatgggt	aggaactttc	gctgcggaca	gccccccag	aataagggtcc	240
agctgaaggc	cagggaacctg	ctgacctga	aaaattttac	aggggaggaa	atcaagtata	300
tgctgtggct	gtcagctgat	ctgaagttcc	ggatcaagca	gaaggcgcaa	tatctgctc	360
tgctccaggc	caaaagcctg	gggatgatct	tcgaaaagcg	cagtactcgg	accagactgt	420
caaccgagac	tggattcgct	ctgctgggag	gacacccttg	ttttctgacc	actcaggaca	480
ttcacctggg	agtgaacgag	tccctgaccg	acactgctcg	cgtcctgagc	tctatggccg	540
acgctgtgct	ggctcgagtc	tacaaacagt	ccgacctgga	taccctggcc	aaggaagcct	600
ctatcccaat	tattaacggc	ctgtcagacc	tgtatcacc	catccagatt	ctggccgatt	660
acctgacctc	ccaggagcac	tattctagtc	tgaaagggct	gacactgagt	tggattgggg	720
acggaaacaa	tatcctgcac	tctattatga	tgctagccgc	caagtttgga	atgcacctcc	780
aggctgcaac	cccaaaaggc	tacgaaccgc	atgcctcagt	gacaaagctg	gctgaacagt	840
acgccaaaga	gaacggcact	aagctgctgc	tgaccaacga	ccctctggag	gccgctcacg	900
gaggcaacgt	gctgatcacc	gatacctgga	ttagtattgg	acaggaggaa	gagaagaaga	960
agcggctcca	ggccttcag	ggctaccagg	tgacaatgaa	aaccgctaag	gtcgcagcca	1020
gcgattggac	ctttctgcac	tgctgcccc	gaaagccgca	agaggtggac	gacgaggtct	1080
tctactctcc	cagaagcctg	gtgtttcccg	aagctgagaa	taggaagtgg	acaattatgg	1140
cagtgatggt	cagcctgctg	actgattatt	cacctcagct	ccagaaacca	aagttctgat	1200
aaactcgact	agtgactgac	taggatctgg	ttaccactaa	accagcctca	agaacaccgc	1260
aatggagtct	ctaagctaca	taataccaac	ttacacttac	aaaatgttgt	cccccaaat	1320
gtagccattc	gtatctgctc	ctaataaaaa	gaaagtttct	tcacattcta	g	1371

SEQ ID NO: 44 moltype = RNA length = 1371
 FEATURE Location/Qualifiers
 misc_feature 1..1371
 note = Synthetic construct
 source 1..1371
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 44

tcaacacaaac	atatacaaaa	caaacgaatc	tcaagcaatc	aagcattcta	cttctattgc	60
agcaatttaa	atcatttctt	ttaaagcaaa	agcaattttc	tgaaaatttt	caccatttac	120
gaacgatagc	caccatgctg	ttaaacctgc	gcatcctgct	gaacaacgcc	gcttccgca	180
acggccacaa	cttcattggg	cgcaacttcc	gctgcggcca	gccccctgag	aacaagggtc	240
agctgaaggc	ccgcgacctg	ctgacctga	agaaattcac	cgccgaggag	atcaagtaca	300
tgctgtggct	gagcgcgacg	ctgaagttcc	gcatcaagca	gaaggcgag	tacctgccc	360
tgctgcaggc	caagagcctg	ggcatgatct	tcgagaagcg	cagcacccgc	accgcctga	420
gcaccgagac	aggcctggcc	ctgctgggag	gccacccctg	cttctgacc	accaggaca	480
tccacctggg	cgtgaacgag	agcctgaccg	acaccgccc	cgtgctgagc	agcatggccg	540
acgcctgctg	ggcccgctg	tacaagcaga	gcgacctgga	cacctggcc	aaggaggcca	600
gcatcccat	catcaacggc	ctgagcgacc	tgtaccaccc	catccagatc	ctggccgact	660
acctgacctc	gcaggagcac	tacagcagcc	tgaagggcct	gacctgagc	tggatcgccg	720
acggcaacaa	catcctgcac	agcatcatga	tgagcgcgc	caagttcggc	atgcacctgc	780
aggccgccc	ccccaaaggc	tacgagccc	acgcccagct	gaccaagctg	gccgagcagt	840
acgccaagga	gaacggcacc	aagctgctgc	tgaccaacga	ccccctggag	gccgcccacg	900
gcggcaacgt	gctgatcacc	gacacctgga	tcagcatggg	ccaggaggag	gagaagaaga	960
agcgcctgca	ggccttcag	ggctaccagg	tgacctgaa	gaccgccaag	gtggccgcca	1020
gcgactggac	cttctgcac	tgctgcccc	gcaagccgca	ggaggtggac	gacgaggtgt	1080
tctacagccc	ccgacgctg	gtgttcccc	agggcgagaa	ccgcaagtgg	accatcatgg	1140
ccgtgatggt	gagcctgctg	accgactaca	gccccagct	gcagaagccc	aagttctgat	1200
aaactcgact	agtgactgac	taggatctgg	ttaccactaa	accagcctca	agaacaccgc	1260
aatggagtct	ctaagctaca	taataccaac	ttacacttac	aaaatgttgt	cccccaaat	1320
gtagccattc	gtatctgctc	ctaataaaaa	gaaagtttct	tcacattcta	g	1371

SEQ ID NO: 45 moltype = RNA length = 1371
 FEATURE Location/Qualifiers
 misc_feature 1..1371
 note = Synthetic construct
 source 1..1371
 mol_type = other RNA
 organism = synthetic construct

SEQUENCE: 45

tcaacacaaac	atatacaaaa	caaacgaatc	tcaagcaatc	aagcattcta	cttctattgc	60
agcaatttaa	atcatttctt	ttaaagcaaa	agcaattttc	tgaaaatttt	caccatttac	120

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gaacgatagc caccatgctg ttcaacctgc gaatcctgct gaacaacgcc gcttttcgga 180
acgggcacaa ctttatgggt aggaactttc gctgcggaaca gcccctccag aataagggtcc 240
agctgaaggc cagggacctg ctgaccctga aaaattttcac aggggaggaa atcaagtata 300
tgctgtggct gtcagctgat ctgaagttcc ggatcaagca gaaggcgcaa tatctgcctc 360
tgctccaggc caaaagcctg gggatgatct tcgaaaagcg cagtactcgg accagactgt 420
caaccgagac tggattcgct ctgctgggag gacacccttg tttctgacc actcaggaca 480
ttcacctggg agtgaacgag tccttgaccg acactgctcg cgtcctgagc tctatggccg 540
acgctgtgct agctcgagtc tacaacacgt ccgacctgga taccctggcc aaggaagcct 600
ctatcccaat tattaacggc ctgtcagacc tgtatcacc ccacagatt ctggccgatt 660
acctgacctt ccaggagcac tattctagtc tgaaagggtc gacactgagt tggattgggg 720
acggaaacaa tatcctgcac tctattatga tgtcagccgc caagtttgga atgcacctcc 780
aggctgcaac cccaaaaggc tacgaaccgc atgacctagt gacaaagctg gctgaacagt 840
acgcccaga gaacggcact aagctgctgc tgaccaacga ccctctggag gccgctcacg 900
gaggcaacgt gctgatcacc gatacctgga ttagtatggg acaggaggaa gagaagaaga 960
agcggctcca ggccttcag ggtaccagg tgacaatgaa aaccgctaag gtcgcagcca 1020
gcgattggac cttctgcac tgcttgccca gaaagccga agaggtggac gacgaggtct 1080
tctactctcc cagagcctg gtgtttcccg aagctgagaa taggaagtgg acaattatgg 1140
cagtgatggt cagcctgctg actgattatt ccctcagct ccagaaacca aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacaccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 46      moltype = RNA length = 1371
FEATURE            Location/Qualifiers
misc_feature        1..1371
                    note = Synthetic construct
source              1..1371
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 46
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaaacaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctg ttcaacctgc gaatcctgct gaacaatgcc gcttttcgga 180
acgggcacaa cttcatgggt aggaactttc gctgcggaaca gcccctccag aacaagggtcc 240
agctgaaggc cagggacctg ctgaccctga aaaattttcac aggggaggaa atcaagtata 300
tgctgtggct gtcagccgat ctgaagttcc ggatcaagca gaaggcgcaa tatctgcctc 360
tgctccaggc caaaagcctg gggatgatct tcgaaaagcg cagtactcgg accagactgt 420
caacagagac tggattcgca ctgctgggag gacaccctatg tttctgacc acacaggaca 480
ttcatctggg agtgaacgag tccttgaccg acacagcacg cgtcctgagc tccatggctg 540
atgcagtgct ggctcgagtc tacaacacgt ctgacctgga taccctggcc aaggaagcct 600
ctatcccaat cattaatggc ctgagtgaac tgtatcacc ccacagatt ctggccgatt 660
acctgacctt ccaggagcac tattctagtc tgaaagggtc gacactgagc tggattgggg 720
acggaaacaa tatcctgcac tccattatga tgagcgccgc caagtttgga atgcacctcc 780
aggctgcaac cccaaaaggc tacgaaccgc atgacctcgt gacaaagctg gcagaacagt 840
atgcccaga gaacggcact aagctgctgc tgaccaatga ccctctggag gccgctcacg 900
gaggcaacgt gctgatcacc gatacctgga ttagtatggg acaggaggaa gagaagaaga 960
agcggctcca ggccttcag ggtaccagg tgacaatgaa aactgctaag gtcgcagcca 1020
cgactggag cttctgcat tgcttgccca gaaagcctga agaggtggac gatgaggtct 1080
tctactcacc cagaagcctg gtgtttcctg aagctgagaa taggaagtgg acaatcatgg 1140
cagtgatggt cagcctgctg actgattatt ccctcagct ccagaaacca aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacaccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 47      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature        1..1368
                    note = Synthetic construct
source              1..1368
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 47
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaaacaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaaccttc gcattctcct caacaacgcc gcgtttagaa 180
acggacacaa cttcatgggt cgcaacttcc gctgcggaaca gccgctcgag aacaagggtcc 240
agctcaaggc tcgggatctc ctgacgctga agaactttac cggcgaagag attaatgata 300
tgctgtggct gtcgcccagc ctttaagttc ggatcaagca gaaggcgcaa taccctccc 360
tgctgcaagg aaagtccctg ggcagatgat tcgagaagcg cagtaccaga accagactct 420
ccactgaaac cgggttcgag ctgcttgagg gccaccctgt tttcctcact acgcaagaca 480
tccatcttgg cgtgaacgag tcccttaccg acaccgccag ggtgctgtca agcatggccg 540
acgccgtcct tgccgcccgtg tacaagcagt cagaccttga tactctggcc aaggaagcct 600
ccatccctat tatcaacggc ctatccgacc tttaccaccc gatccagatc ctgctgact 660
acctgacctt gcaagaacac tacagcagcc tcaagggaact gactctgtcc tggatcggcg 720
acgggaacaa catcctgcac tcaatcatga tgagcgagc caagttcggc atgcactccc 780
aagccgctac acccaagggt tatgaaccgc acgctctgtg gaccaagtgt gcagaacagt 840
acgccaagga gaacggctact aagctccttt taaccaacga cccctcgaa gcagcccatg 900
cggggaatgt gctcattacc gatacctgga tttcgatggg ccaggaggag gagaagaaga 960
agcggctgca ggcgttcag ggtaccagg tcaccatgaa aactgccaaa gtggccgct 1020

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cggattggac ctttctccac tgcctgcctc ggaagcctga ggaggtggac gacgaagtgt 1080
tctactcccc acggtccctc gtgttccccg aggcgcaaaa taggaagtgg accatcatgg 1140
ccgtgatggt gtcctctctg accgattaca gcccgagct tcagaagcct aaattctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 48      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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```

SEQUENCE: 48
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatcttc gcactcctgtt gaacaacgcc gccttccgca 180
atggtcacaa cttcatggct cggaaactta gatgtggaca gcctctccaa aacaagggtcc 240
agctgaaggg aagggaacctc ttaacctca aaaactttac tggagaggag atcaagtaca 300
tgctgtggct tagcgccgac ctttaagttcc ggatcaagca gaagggagag tactccccg 360
tgctgcaagg aaagagtctt ggaatgatct tcgagaagcg gtccaccaga actcgctct 420
ccactgaaac cggattcgca ctctgggtg gacacccgtg cttctgacc acccaagaca 480
tcacactcgg agtgaacgag agcctcacgg acaccgagag agtgctgtca tccatggccg 540
acgccgtgct tgcacgggtc tacaagcagt ccgatctgga cactcttgcc aaggaagcct 600
ccattcctat cattaacggg ctgtcggatc tgtaccaccc gattcagatc cttgcccact 660
acctcacact tcaagaacac tatcaagcc taaagggtct gacctgttcc tggatcggag 720
atggaaacaa cttctcccat tccatcatga tgagcgctgc caagttcgga atgcattctc 780
aagcagcgac tcctaagggt tacgagccgg acgctcagt gactaagctg gccgagcagt 840
acgccaagga gaacgggtacc aaactgttgc ttactaacga cccgcttgaa gcggcccatg 900
gaggaacagt gctgattacc gacacctgga ttctgatggg acaggaagag gagaagaaga 960
agcggtcca gggttccag ggataccagg tcaccatgaa aacggccaaa gtggccgcta 1020
cgattggac cttctgcac tgctccccg cgaagcctga agaagtgagc gacgaagtgt 1080
tctactcccc tcgctctctt gtgttccccg aagccgaaaa caggaagtgg accatcatgg 1140
ccgtgatggt gtcctctctg accgattaca gcccgagct gcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

```

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SEQ ID NO: 49      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

```

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SEQUENCE: 49
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatcttc gcactcctct caacaacgcc gcgttttagaa 180
acggccacaa cttcatggct cggaaactta gatgtggcca gccgcttcag aacaagggtcc 240
agctcaaggg ccgggatctt ctgacctga agaactttac tggcgaagaa atcaagtaca 300
tgctctggct ctccgccgac ttgaagttcc gcattaagca gaagggggaa tacttccgc 360
tgctgcaagg aaagtgcctc ggcattgatc ttgagaagcg ctcaaccgcc accaggctgt 420
ccactgaaac cgggttcgag ctgcttggtg gccacccctg cttctgacc acccaagaca 480
ttcactcggg agtgaacgaa tcgctcactg ataactgccc ggtgctgtcg tcgatggccg 540
atgcagtgtc ggccagggtg tacaacacgt ccgatctgga cactctggcc aaggaggcgt 600
ccatccctat tatcaacggc ctttccgacc tctaccaccc gattcagatc cttgccgatt 660
acctcacctt gcaagaacac tactcgtcac tgaagggtct gacctgttcc tggatcggcg 720
acggcaacaa catctcccat tccattatga tgtccgccgc caaattcggc atgcattctc 780
aagccgcaac ccctaagggt tacgagccgg acgcttccgt gaccaagctc gccgagcagt 840
acgctaagga gaacgggaacc aagcttctgc tgactaacga ccccttagag gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg acaggaagaa gagaagaaga 960
agcggttaca ggcttccag ggctatcagg tcaccatgaa aaccgccaaag gtcgctgcct 1020
cggactggac cttctgcat tgctgcctc gcaagccgca agaagtgagc gacgaggtgt 1080
tctactcgcc acggtccctt gtgttccctg aggcgagaa tagaaagtgg accattatgg 1140
cgtgatggt gtcctctctc accgactact ccgcccaact gcagaaaccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

```

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SEQ ID NO: 50      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 50

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tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
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gaacgatagc caccatgctt ttaaatcttc gcactcctct caacaacgcc gctttccgga 180
acggtcacaa cttcatggtc cggaacttcc gctgcgggcca gccgtccaa aacaaagtgc 240
agcttaaggg ccgcatcttc ctgacctga agaacttcac cggagaggaa atcaagtaca 300
tgctgtggct ctccggcgac ctgaagtta ggattaagca gaagggggag tatctgccgc 360
tgctccaagg gaagtcctt ggcatgatct tcgaaaagag gtccaccgg actcggtca 420
gcaccgaaac aggttttgca cttctggggg gccacccgtg cttcctgacg acccaggaca 480
tccatctggg tgtcaacgag agtttgaccg acactgccag agtgctgtca tccatggcgg 540
acgcggtgct cgcgagatg tacaagcagt ccgatcttga caccctggca aaagaggcct 600
caatcccgat cattaacgga ctctcgatc tgtaccacc tatccaaatc ttggccgact 660
acctgacct gcaagaacac tacagctccc tgaaggcct gactcttcc tggattggcg 720
atggaaacaa cattctccat tctattatga tgtccggcg caagttcggc atgcaccttc 780
aagccgccac ccggaagggc tacgaacctg acgctccgt gactaagcta gccgaacagt 840
acgctaagga gaactggcact aagcttctcc ttaccaacga tccgctggag gcggcccatg 900
gcggaatgt gcttatcacc gacacctgga ttagcatggg gcaggagaa gagaagaaga 960
aacggtcca ggcattccag ggtaccagg tcacctgaa aactgccaa gtcccgcta 1020
gcgactggac cttcctccac gtgtgctc gcaagctga agaagtggac gacgaggtgt 1080
tctactcccc gactctccctc gtgtttctc aggcggagaa cagaaagtgg accatcatgg 1140
ccgtgatggt gtcattactt acggactaca gcccgagct gcagaagccg aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgtgtccc caaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 51

moltype = RNA length = 1368

FEATURE

Location/Qualifiers

misc_feature

1..1368

note = Synthetic construct

source

1..1368

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 51

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tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttaacttga gaactcctct gaacaacgcc gctttccgca 180
acggtcacaa cttcatggtc cggaacttca gatgtggcca gccctccaa aacaaagtgc 240
agctgaaggg ccgggacctt cttacgctga agaatttcac cggcgaagaa atcaagtaca 300
tgctctggct gtccgcatg cttaagttcc gcattaagca gaagggggaa tactccccg 360
tgctgcaagg gaagtcctg ggcatgatt ttgagaagcg gtcaactcgc acccgctgt 420
ccactgaaac tggattcgca ctgctcggg gccatccctg cttcctgacc acccaagaca 480
tccacctcgg cgtgaacgag tccctgactg acaccgcccg ggtcttatcc tcgatggcgg 540
atgctgtgct tgcgaggggt tacaagcagt ccgacctcga cacactcgcg aaggaggcct 600
ccatccccat catcaacggc ctgtccgacc tttaccacc aattcagatc ctgcgcgatt 660
acctgacct gcaagagcac tactcgtcgc tcaaggggct tacctctcgc tggattggcg 720
acggcaacaa catccttcac tccatcatga tgtcggcagc gaagttcggc atgcattcgc 780
aagccgccac gcctaagggt tatgaaccgg atgcctcagt gaccaagctc gccgaacagt 840
acgcaagaaga gaattggaacc aagctacttc tgaccaacga cccctggag gccgtccacg 900
gcggcaacgt cctcattacc gatacttga tttcgatggg acaggagag gaaaaagaaga 960
agagactgca ggcgttccag ggataccagg tcacctgaa aactgccaaa gtggcagcct 1020
ccgactggac cttccttcac tgcctgccga ggaagcctga agaggtggac gacgaggtgt 1080
tctactcccc ggcctccttg gtgtttctc aggcgaaaa ccggaagtgg actatcatgg 1140
ccgtgatggt gtcctcctc accgactact cggccgaact gcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgtgtccc caaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 52

moltype = RNA length = 1368

FEATURE

Location/Qualifiers

misc_feature

1..1368

note = Synthetic construct

source

1..1368

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 52

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tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgcta ttaaacctta gaattctcct taacaacgcc gctttccgga 180
atgggcataa ctttatggtc cgcaatttcc gctgtggaca gccctcgcaa aacaaagtgc 240
agctcaaggg ccgggactct ctgactctca agaacttcac tggggaagaa atcaagtaca 300
tgctctggct gagcgccgag ctcaagttcc gcactcaagca gaaggagag tactccccg 360
tgctccaagg gaagtcctg ggcatgatct tcgagaagag atccaccgcg accagacttt 420
ccactgagac tggcttcgcc ttgctgggag gccacccatg cttcctgacg acccaggaca 480
ttcacttggc cgtgaacgag tccctgactg acaccgcaag ggtgtgttc tcgatggcgg 540
acgccgtgct gtcgcccggg tacaagcaga gcgatcttga caccctggct aagggaagct 600
ccattcccat catcaacggc ctgagcgacc tgtaccacc gattcagatc ctggcggact 660
acctaacct gcaagagcac tatagctccc tgaaggcct cacacttca tggatcggcg 720
acggcaacaa catcctgcac tctattatga tgagcgctgc caaattcggc atgcacctc 780
aagccgccac gcctaaggc tacgagcccg acgctcgtg gaccaagctt gcggagcagt 840

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acgcgaagga aaacggcacc aagctgcttc tcaccaacga tectctggaa gcggcccatg 900
gtggcaacgt gctcattacc gacacttgga tctccatggg acaggaggag gaaaagaaga 960
agcggctcca ggcttttcag ggttaccagg tcaccatgaa aaccgccaa gtcgcagcct 1020
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tttactcacc tcggtcactc gtgttcccg aagcagagaa caggaaatgg accattatgg 1140
ccgtgatggt gtccctgctc accgattaca gtccgcaact gcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 53      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

```

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SEQUENCE: 53
tcaacacaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctcc gcactcctct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccgcgagc ttgaagttcc gcattaagca gaagggggaa taccttccgc 360
tgcttcaagg aaagagcctc ggcctgatct ttgagaagcg ctcaaccagg acccgctttt 420
ctactgaaac tgggttcgag ctgctcgggt gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tcctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcaggggtg tacaacagtc ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcaaatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgcccg aaaattcggc atgcatcttc 780
aagccgccac ccctaagggtt tacgaacccg acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacggaacc aagctgctgc tgactaacga cccgctagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaggaa gagaagaaga 960
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tctactcgcc ccggagcctc gtgttccccg aggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactcttc accgactaca gcccgagctc tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 54      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

```

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SEQUENCE: 54
tcaacacaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctcc gcactcctct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agcttaaggg ccgggatctc ctcaacctta aaaacttcac cggcgaagag atcaagtaca 300
tgctctggct ctccgcgagc cttaaagtcc gcattaagca gaagggggaa taccttccgc 360
tgcttcaagg aaagagcctc ggcctgatct ttgagaagcg ctcaaccagg accaggcttt 420
ctactgaaac tgggttcgag ctctcggcgt gtcactccctg cttcctcagc acccaagaca 480
tccacctcgg agtgaacgaa tcctcaccg atactgcccg cgtgctttcg agcatggcag 540
acgccgtgct cgcccggttg tacaacagtc ccgatctcga cactctcgcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac ttaccaccg gatccagatc ctgcgcgatt 660
acctcacact ccaagaacac tacagctccc ttaagggtct taccctctcc tggatcggcg 720
acggcaacaa cattctccac tccatcatga tgtccgcccg aaagtctggc atgcatcttc 780
aagccgccac cccgaagggtt tacgagcctg atgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacggaacc aagcttcttc tcaactaacga cccactcgaa gcagcccacg 900
ggggcaacgt gcttatcact gacacctgga tctccatggg ccagggaaga gagaagaaga 960
agcggctcca ggcgttccag ggatatcagg tcaccatgaa aaccgccaa gtcgctgcct 1020
ccgactggac cttctccac tgctcctc gcaaacctga agaagtggac gacgaggtgt 1080
tctactcgcc ccggagcctc gtgttccccg aggcgagaa tagaaagtgg accattatgg 1140
ccgtgatggt gtcactcttc accgactaca gcccgagctc tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 55      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct

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source          1..1368
                mol_type = other RNA
                organism = synthetic construct

SEQUENCE: 55
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctt ttcaatctcc gcattctcct taacaacgcc gcgttttagaa 180
acggacataa cttcatggtc cggaacttca gatgtggaca gccgcttcaa aacaagggtcc 240
agctgaaggg tcgggatctt ctgaccctga agaactttac cggagaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagtccc gcattaagca gaaggagaa tacctcccgc 360
tgcttcaagg aaagagcctc ggaatgattt ttgagaagcg ctcaaccagg acccgcttt 420
ctactgaaac tggattcgcg ctgctgggtg gacacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tccctcactg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggccagggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat catcaacgga cttagtgaac tctaccatcc gattcaaatc ctggccgact 660
acctcacctc gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcgag 720
atggaaacaa cattctccac tccatcatga tgtccgccgc aaaattcgga atgcattctc 780
aagccgccac gcctaagggt tacgaaccgc acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacgggtac aagcttctcc tgaccaacga cccactagaa gcagcccacg 900
gtggaaacgt gcttattact gacacttgga tctccatggg acaggaggaa gagaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaa gtcgtgcct 1020
ccgactggac cttcctgcac ttcctgccc gcaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc gcggagcctc gtgttccccg aggcgcgaga tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgagctc tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccacatta cacttacaaa atgttgtccc caaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 56      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature        1..1368
                    note = Synthetic construct
source              1..1368
                    mol_type = other RNA
                    organism = synthetic construct

```

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SEQUENCE: 56
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctt ttcaacctcc gcattctcct caacaacgct gccttccgga 180
atggacataa cttcatggtc cggaacttca gatgcggaca gccgcttcag aacaagggtcc 240
agcttaaggg gagagatctc cttacctca aaaacttcac tggcgaagaa atcaagtaca 300
tgctctggct tagtgccgat ctcaagttcc gcataagca gaaggagaa tacctcccgc 360
tccttcaagg aaagagcctc ggcatgattt ttgagaagag gtccaccaga actcgcttt 420
caaccgagac tgggttcgcc ctgctggcg gtcacccctg cttcctcact acccaagaca 480
tccacctcgg cgtgaacgag agccttaccg acaccgcccg cgtgctctcc tcaatggccg 540
acgctgtgct cggccgggtg tacaagcagt ccgacctga tactctcgcc aaggaggcct 600
ccatcccaat tatcaacggg ctctctgata tctaccaccc tatccaaatc ctgcggcact 660
acctcacctc ccaagagcac tatagctcgc tcaagggcct caccctttcc tggattggcg 720
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aagccgcgac ccccaagggc tacgagcctg acgcatccgt gaccaagctc gccgagcagt 840
acgcgaagga aaatggcacc aagcttcttc tcaccaacga ccccttgag gccgctcatg 900
gcggcaacgt gctcatcact gacacttgga tcagcatggg ccaggaggag gaaaaagaaga 960
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ccgactggac cttctctcac tgtctccgcg ggaagcctga agaagtggat gacgaagtgt 1080
tttactcccc tcggtcactc gtgttccccg aagcagaaaa caggaaagtgg accatcatgg 1140
cggctatggt gtcctcctc accgactaca gcccgagctc tcagaaaccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccacatta cacttacaaa atgttgtccc caaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 57      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature        1..1368
                    note = Synthetic construct
source              1..1368
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 57
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctt ttcaatctcc gcattctcct taacaacgca gcgttttagaa 180
acggtcacaa cttcatggtc cggaacttcc gctgtggaca gccgcttcaa aacaagggtcc 240
agctgaaggg tcgggacctt ctgaccctga agaactttac tggagaagag atcaagtaca 300
tgctttggct gtcgcggcgc ttgaagtccc gcattaagca gaaggagaa tacctcccgc 360
tgctccaagg aaagagcctg ggaatgatct ttgagaagcg ctcaaccagg acccgcttt 420
ctactgaaac tggattcgcg ctgctgggtg gtcacccctg cttcctgacg acccaggaca 480
ttcacctcgg agtgaacgag tccctcactg ataccgcccg agtgttatcg agcatggcag 540
atgccgtgct ggctagggtg tacaaacagt ccgatctgga caccctggcc aaggaggcat 600
caattcctat tatcaacgga cttagtgaac tctaccatcc gattcaaatc ctggccgatt 660

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acctcaccct gcaagaacac tacagctccc tgaagggctc gacattgtcc tggatcggag 720
atggaaacaa cattctccat tccatcatga tgtccgcggc caagttcggg atgcattctc 780
aagccgccac gccgaaagga tacgagccgg acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacggaacc aagcttctgc tgactaacga cccgctagaa gccgcccacg 900
gtggaaacgt gcttattact gacacctgga tctccatggg acaggaagaa gaaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaa gtcgccgcct 1020
ccgactggac cttccttcac tgccctgccc ggaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc gccgagcctc gtgttccctg aggccgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactcttc accgactaca gcccgagct tcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccacttta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 58      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 58
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttaaatctcc gcatctccct caacaacgca gccttttaga 180
acggccacaa cttcatggct cggaacttca gatgtggcca gccgctcag aacaagggtc 240
agctcaaggg ccgggacctc ctacacctca aaaactttac cggcgaagag atcaagtaca 300
tgctctggct ttccggcgac cttaaagttc gcatcaagca gaagggggaa taccttccgc 360
tgcttcaagg aaagtccctc ggcctgatct ttgaaaagcg ctgcaccagg accgccttt 420
ccactgaaac cgggttcgag cttctcgggt gccacctctg cttcctcacc acccaagaca 480
ttcacctcgg agtgaacgaa tcccttaccg ataccgcaag agtgctttcg tcgatggcgg 540
atgccgtgct tgcgcgggtg tacaagcagt cagatctcga cactctcgcc aaggaggcgt 600
ccattcctat tatcaacggc ctttccgacc tttaccaccc gattcagatc ctgcgcgatt 660
acctcaccct gcaagagcac tactcgtcac tcaaggggtc tacctctccc tggatcggcg 720
acggaaacaa catctcccat tcgatcatga tgtccgcgcg caaattcggc atgcacctcc 780
aagccgcgac cccgaagggt tacgagcccg acgcttccgt gaccaagctc gccgaacagt 840
acgctaagga aaacggcacc aagctcctcc tcaactaacga ccctctcgaa gcagcccatg 900
ggggcaacgt gctcattact gacacttggg tctcgatggg ccaggaagag gaaaaaaga 960
agcggcttca gccgttccag ggatatcagg tcaccatgaa aaccgccaa gtcgctgcct 1020
cggactggac cttccttcac tgcttccgcg gcaagcctga agaggtggag gatgagggtg 1080
tctactcccc acggttccct gtgttccccg aggccgagaa taggaagtgg accatcatgg 1140
ccgtgatggt gtcgctcttc actgactact ccccgcaact tcagaagcct aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccacttta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 59      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

```

```

SEQUENCE: 59
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttaaatctga gaatacttct aaacaacgca gccttccgga 180
atggccataa ctttatgggt cggaatttcc gctgcggcca gccgctcag aacaagggtc 240
agctgaaggg aagagacttg ctgacctca agaacttcac cggagaagaa atcaagtata 300
tgcttggtct gtcgcgcgac ctgaaattcc gcatcaagca gaagggcgaa tatctgccgc 360
tgttgcaagg gaagtccctg gggatgatct tcgagaagag gtccaccaga acacggcttt 420
caaccgaaac cgggtttgca ctgctgggtg gacacctctg tttctgacc actcaagata 480
tccacctggg cgtgaacgag tcccttaccg acactgctag ggtgttgtcc agcatggcgg 540
atgccgtcct ggctcgcgtg tacaagcagt ccgacctgga taccctggca aaggaagcgt 600
ccattcccat tatcaacggg ctgtccgacc tgtaccatcc gattcaaatc ctggcggact 660
acctgactct gcaagagcat tacagcagct tgaaggggct tactctctcg tggatcggcg 720
acgggaacaa catctgcac tccatcatga tgtccgcgcg caagttcggg atgcatttgc 780
aagctgcgac cccgaaaggt tacgagcccg atgctagcgt aactaagctt gccgaacagt 840
acgccaaaga gaatgggtaca aaactgcttc tgactaacga cccgctggaa gcagccacg 900
gcgggaacgt gctgataacc gacacctgga tttcaatggg gcaggaggaa gagaagaaga 960
agcgactgca ggcgttccaa ggctatcagg ttaccatgaa aaccgccaaa gtggcagcca 1020
gcgattggac tttcctgcac tgtctccgcg ggaagccgga ggaagtgtat gacgaagtat 1080
tctactcacc ccggagcctc gtgttccccg aggccgaaaa ccggaagtgg actattatgg 1140
ccgtgatggt gtcgctgttg accgactaca gcccgcaact gcagaagccg aagtttttagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccacttta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 60      moltype = RNA length = 1368
FEATURE           Location/Qualifiers

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misc_feature      1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 60
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaattttt caccatttac 120
gaacgatagc caccatgctt ttcaacctga ggatcctttt gaacaacgcc gcctttcgca 180
acggccacaa ctttatggct cgcaatttcc gctgcgggca gccgctgcag aacaaggctc 240
agctgaaggc ccgggatctg ctgacctga agaacttcac cggggaggaa atcaagtaca 300
tgctttggct ctccgccgat ctgaagttca gaatcaagca gaaggagag taacctccgt 360
tgctgcaagg aaagtcactc ggaatgattt tcgaaaagag aagcactagg acccgctct 420
caactgaaac cgggttcgct ctgctcgggg gccatccgtg ttctctgact acccaagaca 480
tccacctggg agtgaacgag tcgctgaccg acaccgcacg cgtgctgtca tccatggcgg 540
acgcagtgtc tgccggggtg tacaagcagt cggacctgga cactcttgcc aaggaggcat 600
caatcccatc cattaacgga ctgtccgatc tctaccacc gccatcagatc ctggctgact 660
acctaacctc gcaagagcac tactcaagcc tgaaggggct gacctgtctg tggatcgggg 720
acggcaacaa cattctgcac tccatcatga tgtcggcgcc taagtctggg atgcatttgc 780
aagcggcaac ctcgaagggt tatgaacccg acgctccgtg gaccaagctg gccgaacagt 840
acgccaagga aaacgggaac aagttgtctg tgaactatga tccctggag gcggccacg 900
gggggaacgt gctgataacc gatacctgga tctccatggg gcaggaagaa gagaagaaaa 960
agcgggtgca ggcattccag ggataccagg tcaccatgaa aaccgcaaaa gtggcagcca 1020
gcgactggac tttctccat tgctcggcgc gaaagccgga ggaggtcgat gacgaggtgt 1080
tctactcccc gcggtcgtg gtgttccccg aggcggaaaa ccggaagtgg accattatgg 1140
ccgtgatggt gctactcttg actgactaca gcccgcacct gcagaagccg aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cactacaaa atgtgtgcc ccaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 61      moltype = RNA length = 1496
FEATURE            Location/Qualifiers
misc_feature       1..1496
                  note = Synthetic construct
source             1..1496
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 61
cttaaggggg cgctgcctac ggaggtggca gccatctcct tctcggcatc aagcttacca 60
tggtgcccca ggcctcgtc ttggtcccg tgctgggtgt cccctctgct tcggcaagt 120
tcccactcta caccatcccc gacaagctgg ggcgtggag ccccatcgac atccaccacc 180
tgtctgccc caacaacctc gtggtcgagg acgagggctg caccaacctg agcgggttct 240
cctacatgct tttcaatctc cgatccctcc ttaacaacgc cgcgtttaga aacggccaca 300
acttcatggt ccggaacttc agatgtggcc agccgcttca aaacaaggct cagctgaagg 360
gccgggatct tctgacctg aagaacttta ctggcgaaga gatcaagatc atgctctggc 420
tctccgcgga cttgaagttc cgcattaagc agaaggggga ataccttcg ctgcttcaag 480
gaaagagcct cggeatgatc tttgagaagc gctcaaccag gaccgcctt tctactgaaa 540
ctgggttcgc gctgctcggt ggcacccctc gcttctgac gaccaggac atccacctcg 600
gagtgaacga atccctcacc gataccgccc ggggtgtatc gagcatggca gatgcgtg 660
tgccagggtg gtacaaaacg tccgatctgg acactctggc caaggaggcg tcaattccta 720
ttatcaacgg cttagttagc ctctaccatc cgatccagat cctggccgat tactcacc 780
tgcaagaaca ctacagctcc ctgaagggtc tgacattgtc ctggatcgcc gacggcaaca 840
acattctcca ttccatcatg atgtccgccc caaaattcgg catgcatctt caagccgcca 900
cgccgaaggg ttacgagccc gacgcttccc tgactaagct cgccgagcag tacgctaagg 960
agaaacggaac caagcttctc ctgactaacg accactaga agcagccac gggggcaacg 1020
tgcttattac tgacacctgg atctccatgg gccaggaaga agagaaaaag aagcggctgc 1080
aggggttcca gggatatcag gtcaccatga aaaccgccaa ggtcgctgcc tccgactgga 1140
ccttctgca ctgcctgccc gcgaagcctg aagaagtgga cgacgaggtg ttctactcgc 1200
cacggagcct cgtgttcccc gaggcggaga atagaaaagt gaccatcatg gccgtgatgg 1260
tgtcactgct caccgactac agcccgcagc ttcagaagcc caagtcttag ataagtgaat 1320
gcaaggctgg ccggaagccc ttgcctgaaa gcaagatttc agcctggaag agggcaag 1380
ggacgggagt ggacaggagt ggatgcgata agatgtggtt tgaagctgat ggggtgccag 1440
cctgatttgc tgagtcaatc aataaagagc tttcttttga cccattctag atctag 1496

SEQ ID NO: 62      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source             1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 62
aggaaactta agattattac atcaaaacaa aaagccgcca atgctgttca acctgcgcat 60
cctgctgaac aacgcccgtc tccgcaacgg ccacaacttc atggtgcgca acttccgctg 120
cggccagccc ctgcagaaca aggtgcagct gaaggggccc gacctgctga cctgaagaa 180
cttcaccggc gaggagatca agtacatgct gtggctgagc gccgacctga agttccgcat 240
caagcagaag ggcgagtacc tgccccctgt gcagggcaag agcctgggca tgatcttcca 300
gaagcgagc acccgacccc gcttgagcac cgagacagcg ctggccctgc tggcgggcca 360
ccctgcttcc ctgaccaccc aggacatcca cctgggcgtg aacgagagcc tgaccgacac 420

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cgcccgctg ctgagcagca tggccgacgc cgtgctggcc cgcgtgtaca agcagagcga 480
cctggacacc ctggccaagg aggccagcat ccccatcadc aacggcctga ggcacctgta 540
ccaccccatc cagatcctgg ccgactacct gaccttgacg gagcactaca gcagcctgaa 600
gggctgacc ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag 660
cgccgccaag ttcggcatgc acctgcaggc cgccaccccc aagggtacg agcccgacgc 720
cagcgtgacc aagctggcgg agcagtacgc caaggagaac ggcaccaagc tgcctgtgac 780
caacgacccc ctggaggccg cccacggcgg caacgtgctg atcacggaca cctggatcag 840
catggggccg gaggaggaga agaagaagcg cctgcaggcc ttcagggt accaggtgac 900
catgaagacc gccaaagtgg ccgccagcga ctggaccttc ctgcaactgc tgcgccgcaa 960
gcccaggagg gtggacgacg aggtgttcta cagcccccgc agcctggtgt tccccgaggc 1020
cgagaaccgc aagtggacca tcatggccgt gatggtgagc ctgctgaccc actacagccc 1080
ccagctgcag aagcccaagt tctgaacgcc gaagcctgca gccatgcgac cccacgccac 1140
cccgctgcct ctgcctccgc gcagcctgca gcgggagacc ctgtcccccgc cccagccgtc 1200
ctcctgggggt ggaccctagt ttaataaaga ttcaccaagt ttcacgcaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaa 1368

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SEQ ID NO: 63      moltype = RNA length = 1495
FEATURE           Location/Qualifiers
misc_feature       1..1495
                   note = Synthetic construct
source            1..1495
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 63
cttaaggggg cgtgcctac ggaggtggca gccatctcct tctcggcadc aagcttacca 60
tgggtcccca ggcctgtctc ttgggtccgc tgtggtgttt cccctctgcg ttcggcaagt 120
tcccatctca caccatcccc gacaagctgg ggccgtggag ccccatcgac atccaccacc 180
tgtctcgccc caacaacctc gtggtcgagg acgagggtcg caccacactg agcgggttct 240
cctacatgct gttcaacctg cgcactcctg tgaacaacgc cgcttctcgc aacggccaca 300
acttcatggt gcgcaacttc cgtgcggccc agccctgca gaacaaggtg cagctgaagg 360
gcccgcagct gctgacctg aagaacttca cggcgaggga gatcaagtac atgctgtggc 420
tgagcgccga cctgaagttc cgcatacagc agaaggcgga gtacctgccc ctgctgcagg 480
gcaagagcct gggcatgac ttcgagaagc gcagcaccgc caccgcctg agcaccgaga 540
caggcctggc cctgctgggc ggcacccctt gcttctgac caccaggac atccacctg 600
gctgaacga gagcctgacc gacaccgccc gcgtgctgag cagcatggcc gacgccgtg 660
tggcccgctg gtacaagcag agcgacctgg acacctggc caaggaggcc agcatcccca 720
tcatcaacgg cctgagcgac ctgtaccacc ccatccagat cctggccgac tactgaccc 780
tgaggagaca ctacagcagc ctgaaggggc tgacctgag ctggatcgcc gacggcaaca 840
acatcctgca cagcatactg atgagcgccc ccaagttcgg catgcacctg caggccgcca 900
cccccaaggg ctacgagccc gacgcagcg tgaccaagct ggccgagcag tacgccaagg 960
agaacggcac caagctgtct ctgaccaacg acccctgga ggccgccacc ggccggcaacg 1020
tgtgatcac ggacacctg atcagcatgg gccaggagga ggagaagaag aagcgcctgc 1080
aggccttcca gggctaccag gtgacctga agaccgcaa ggtggcggc agcgactgga 1140
ccttctgca ctgcctgccc cgcaagcccc aggaggtgga cgacgaggtg tctacagcc 1200
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tgagcctgct gacgactac agccccagc tgcagaagcc caagttctga ataagtgat 1320
caaggctggc cggaagccct tgctgaaag caagatttca gcctggaaga gggcaaaagt 1380
gacgggagtg gcaggagtg gatgcgataa gatgtggttt gaagctgatg ggtgccagcc 1440
ctgcattgct gagtcaatca ataaagagct ttcttttgac ccattctaga tctag 1495

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SEQ ID NO: 64      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 64
aggaaactta agattattac atcaaaacaa aaagccgcca atgcttttca atctccgcat 60
cctccttaac aacgcccgct ttagaaacgg ccacaacttc atggtccgga acttcagatg 120
tggccagccg cttcaaaaca aggtccagct gaaggggcgg gatcttctga cctgaagaa 180
ctttactggc gaagagatca agtacctgct ctggctctcc gcggacttga agttccgcat 240
taagcagaag ggggaatacc ttccgctgct tcaaggaaag agcctcgga tgatctttga 300
gaagcgctca accaggaccc gcctttctac tgaaactggg ttccgctgct tcgggtggca 360
cccctgcttc ctgacgaccc aggacatcca cctcgagtg aacgaatccc tcaccgatac 420
cgcccggtgt ttatcgagca tggcagatgc cgtgctggcc aggtgttaca aacagtccga 480
tctggacact ctggccaagg aggcgtcaat tctattatc aacggcctta gtgacctcta 540
ccatccgatt cagatcctgg ccgattacct caccctgcaa gaacactaca gctccctgaa 600
gggtctgaca ttgtcctgga tcggcgacgg caacaacatt ctccattcca tcatgatgct 660
cgccgcaaaa ttccgcatgc atcttcaagc cgccacggcg aagggttacg agcccgacgc 720
ttccgtgact aagctcgccg agcagtacgc taaggagaac ggaaccaagc ttctgtgac 780
taacgaccca ctagaagcag cccacggggg caacgtgctt attactgaca cctggatctc 840
catggggccg gaagaagaga aaagaagcgg gctgcaggcg ttccagggat atcaggtcac 900
catgaaaaac ccgaaggtcg ctgcctccga ctggaccttc ctgcaactgc tgcctcgcaa 960
gcctgaagaa gtggacgacg aggtgttcta ctgccacgg agcctcggtt tccccgaggc 1020
cgagaataga aagtggacca tcatggccgt gatggtgtca ctgctcaccg actacagccc 1080
gcagcttcag aagcccaagt tctagacgcc gaagcctgca gccatgcgac cccacgccac 1140
cccgctgcct ctgcctccgc gcagcctgca gcgggagacc ctgtcccccgc cccagccgtc 1200

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ctcctgggggt ggaccctagt ttaataaaga ttcaccaagt ttcacgcaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1368

```

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SEQ ID NO: 65      moltype = RNA length = 1495
FEATURE           Location/Qualifiers
misc_feature       1..1495
                   note = Synthetic construct
source            1..1495
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 65
cttaaggggg cgctgcctac ggaggtggca gccatctcct tctcggcac aagcttacca 60
tggtgcccca ggccctgctc ttggtcccg tgcgtggtgt cccctctgc ttcggcaagt 120
tccccatcta caccatcccc gacaagctgg ggccgtggag ccccatcgac atccaccacc 180
tgtctgcccc caacaacctc gtggtcgagg acgagggctg caccaacctg agcgggttct 240
cctacatgct tttaacctcg agaatcctct tgaacaatgc tgcttttcgg aatggccaca 300
actttatggt tcggaacttc cgttgcggcc agcctttaca aaacaaggct cagctgaagg 360
gccgggattt gctcacacta aagaacttta ctggagaaga gatcaagtac atgctatggc 420
tgtcggccga cctgaagtgc cgtatcaagc agaagggaga atacctccg ctgcttcaag 480
gaaagagcct cggcatgata tttgagaagc gctcaaccag gaccgcctt tctactgaaa 540
ctgggttcgc gctgctcggt ggccaccctc gcttcctgac gaccaggac atccactcg 600
gagtgaacga atccctcacc gataccgccc ggggtttatc gagcatggca gatgccgtgc 660
tggccagggt gtacaaacag tccgatctcg atacctggc aaaggaggct tccattccca 720
tcatcaacgg cctgagcgac ctgtaccacc caatccaaat cctggctgac tacctgaccc 780
tgcaagagca ctacagcagc ctgaagggtc tgacctgtc atggattggc gatggaaaca 840
atatctctga ctccatcatg atgtccgccc cgaagtctcg aatgcactcg caagccgcca 900
ctccaaaagg atacgaaccg gatgcattcg tgaccaagt ggccgaacag tacgcgaagg 960
agaacggaac aagctcctcg ctgactaacg acccgctcga ggtcgcgcat gggggtaacg 1020
tgctgattac ggacacctgg atctccatgg ggcaggagga agagaagaag aagagactgc 1080
aggcattcca ggggtaccag gtcaccatga aaaccgcaaa agtggcagct tcggactgga 1140
ctttcctgca ttgctgccc aggaagcccg aggaagtcca cgacgaagtg ttctactcgc 1200
ctcggtccct ggtgttcccc gagggcgaaa accggaagtg gaccatcatg gccgtgatgg 1260
tgtccttgct gactgactat agcccgacg tgcagaagcc taagtcttag ataagtgatg 1320
caaggctggc gctgaagcct tgccctgaaa caagatttca gcctggaaga gggcgaagtg 1380
gacgggagtg gacaggagtg gatgcgataa gatgtggttt gaagctgatg ggtgccagcc 1440
ctgcattgct gagtcaatca ataaagagct ttcttttgac ccattctaga tctag 1495

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SEQ ID NO: 66      moltype = RNA length = 1475
FEATURE           Location/Qualifiers
misc_feature       1..1475
                   note = Synthetic construct
source            1..1475
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 66
tgagtgtcgt acagcctcca gggccccccc tcccgggaga gccatagtgg tctgcggaac 60
cggtgagtag accggaattg ccgggaagac tgggtccttt cttggataaa cccactctat 120
gcccgcccat ttgggctgtc gcccgcaaga ctgctagccg agtagtgttg ggttgcgatg 180
ctgttcaacc tgccatcctc gctgaacaac gccgccttcc gcaacggcca caacttcatt 240
gtgcgcaact tccgtcgcgg ccagccctcg cagaacaagg tgcagctgaa gggccgcgac 300
ctgctgaccc tcaagaactt caccggcgag gagatcaagt acatgctgtg gctgagcgcc 360
gacctgaagt tccgcatcaa gcagaaggcc gagtacctgc ccctgctgca gggcaagagc 420
ctgggcattg tottcgagaa gcgcagcacc cgcacccgcc tgagcaccca gacaggcctg 480
gccctgctgg gcggccacc ctgcttctcg accaccagg acatccacct gggcgtgaac 540
gagagcctga ccgacaccgc ccgcgtgctg agcagcatgg ccgacgccgt gctggcccgc 600
gtgtacaagc agagcgacct ggacaccctg gccaaaggag ccagcatccc catcatcaac 660
ggcctgagcg acctgtacca cccatccag atcctggccc actacctgac cctgcaggag 720
cactacagca gccctgaagg cctgaccctg agctggatcg gcgacggcaa caacatcctg 780
cacagcatca tgatgagcgc cgccaagtgc ggcattgcac tgcaggccgc ccccccaag 840
ggctacgagc ccgacgccc cgtgaccaag ctggccgagc agtacgcca ggagaacggc 900
accaagctgc tgcagacca cgaacccctg gagggccgcc acggcggcaa cgtgctgatc 960
accgacacct ggatcagcat gggccaggag gaggagaaga agaagcgctt gcaggccttc 1020
cagggtctacc aggtgacct gaagaccgcc aagggtggcc ccagcgactg gaccttctg 1080
cactgcctgc cccgcaagcc cgaggaggtg gacgacgagg tgttctacag ccccccgcgc 1140
ctgggtgttc ccgaggccga gaaccgcaag tggaccatca tggccgtgat ggtgagcctg 1200
ctgaccgact acagccccc gctgcagaag cccaagtctt gaataagtga tagagcggca 1260
aacctagctc aactccata gctagtttct tttttttttt tttttttttt tttttttttt 1320
tttttttttt tttttttttt ctttcttttt ctttcttttt tctctttttt ttggtggctc 1380
catcttagcc ctagtcaagg ctagtgtgta aaggctccgt agccgcatga ctgcagagag 1440
tgccgtaact ggcctctctg cagatcatgt tctag 1475

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SEQ ID NO: 67      moltype = RNA length = 1335
FEATURE           Location/Qualifiers
misc_feature       1..1335
                   note = Synthetic construct
source            1..1335
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 67

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aggaactta agattattac atcaaaacaa aaagccgcca atgcttttca atctccgcat 60
cctccttaac aacgcccgct ttagaacagg ccacaacttc atggtccgga acttcagatg 120
tggccagccg cttcaaaaca aggtccagct gaaggcccg gactttctga cctgaagaa 180
ctttactggc gaagagatca agtacatgct ctggctctcc gcggacttga agttccgcat 240
taagcagaag ggggaatacc ttccgctgct tcaaggaaa agcctcgga tgatctttga 300
gaagcgctca accaggaccc gccctttctac tgaactggg ttccgctgctc tcggtggcca 360
ccctgcttc ctgacgaccc aggacatcca cctcggagtg aacgaatccc tcaccgatac 420
cgcccggtgt ttatcgagca tggcagatgc cgtgctggcc aggggtgtaca aacagtccga 480
tctggacact ctggccaagg aggcgtcaat tcctattatc aacggcctta gtgacctcta 540
ccatccgatt cagatccctg cggattacct caccctgcaa gaacactaca gctccctgaa 600
gggtctgaca ttgtcctgga tcggcgacgg caacaacatt ctccattcca tcatgatgtc 660
cgccgcaaaa ttccgcatgc atcttcaagc cgccacggcg aagggttacg agcccgacgc 720
ttccgtgact aagctcgccg agcagtacgc taaggagaa ggaaccaagc ttctgctgac 780
taacgaccca ctagaagcag ccacggggg caactgtgctt attactgaca cctggatctc 840
catgggcccag gaagaagaga aaaagaagcg gctgcaggcg ttccagggat atcaggtcac 900
catgaaaacc gccaaaggtcg ctgcctccga ctggaccttc ctgcaactgcc tgccctcgaa 960
gcctgaagaa gtggacgacg aggtgttcta ctgccacgg agcctcgtgt tccccgaggc 1020
cgagaataga aagtggacca tcatggccgt gatggtgtca ctgctcaccg actacagccc 1080
gcagcttcag aagcccaagt tctaggctgg agcctcggtta gccgttcttc ctgcccctg 1140
ggcctcccaa cgggccctcc tccctcctt gcaccggccc ttctgtgtct ttgaataaag 1200
tctgagtggg cagcaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaa 1335

```

SEQ ID NO: 68

moltype = RNA length = 1335

FEATURE

Location/Qualifiers

misc_feature

1..1335

note = Synthetic construct

source

1..1335

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 68

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aggaactta agattattac atcaaaacaa aaagccgcca atgctgttca acctgcgcat 60
cctgctgaac aacgcccgct tccgcaacgg ccacaacttc atggtgcgca acttccgctg 120
cgccagcccc ctgcagaaca aggtgcagct gaaggcccg gacctgctga cctgaagaa 180
cttcaccggc gaggagatca agtacatgct gtggctgagc gccgacctga agttccgcat 240
caagcagaag ggcgagtacc tgccctgctt gcagggcaag agcctgggca tgatcttctga 300
gaagcgcgac acccgacccc gccctgagcac cgagacaggc ctggccctgc tgggcccga 360
ccctgcttc ctgacacccc aggacatcca cctgggctgt aacgagagcc tgaccgacac 420
cgcccgctgt ctgacgagca tggccgacgc cgtgctggcc cgcgtgtaca agcagagcga 480
cctggacacc ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta 540
ccaccctatc cagatccctg ccgactacct gacctgtcag gagcaactaca gcagcctgaa 600
gggctgacc ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag 660
cgccgccaa ttccgcatgc acctgcaggc cgccaccccc aagggttacg agcccgacgc 720
cagctgacc aagctggccg ccacggggcg caactgtctg atcaccgaca cctggatcag 780
caacgacccc ctggaggccg ccacggggcg caactgtctg atcaccgaca cctggatcag 840
catgggcccag gaggaggaga agaagaagcg cctgcaggcc ttccagggtt accagggtgac 900
catgaagacc gccaaaggtgg ccgccaagca ctggaccttc ctgcaactgcc tgccccgcaa 960
gcccaggagg gtggacgacg aggtgttcta cagcccccg agcctgtgtt tccccgaggc 1020
cgagaaccgc aagtggacca tcatggccgt gatggtgagc ctgctgacgg actacagccc 1080
ccagctcgag aagcccaagt tctgagctgg agcctcggtta gccgttcttc ctgcccctg 1140
ggcctcccaa cgggccctcc tccctcctt gcaccggccc ttctgtgtct ttgaataaag 1200
tctgagtggg cagcaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaa 1335

```

SEQ ID NO: 69

moltype = RNA length = 1335

FEATURE

Location/Qualifiers

misc_feature

1..1335

note = Synthetic construct

source

1..1335

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 69

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aggaactta agattattac atcaaaacaa aaagccgcca atgcttttca acctgagaat 60
cctcttgaac aatgctgctt ttccgaatgg ccacaacttt atggttcgga acttccgttg 120
cgccagacct ttacaaaaca aggtccagct gaaggcccg gatttgctca cactaaagaa 180
ctttactgga gaagagatca agtacatgct atggctgtcg gccgacctga agttccgtat 240
caagcagaag ggagaatacc ttccgctgct tcaaggaaa agcctcgga tgatctttga 300
gaagcgctca accaggaccc gccctttctac tgaactggg ttccgctgctc tcggtggcca 360
ccctgcttc ctgacgaccc aggacatcca cctcggagtg aacgaatccc tcaccgatac 420
cgcccggtgt ttatcgagca tggcagatgc cgtgctggcc aggggtgtaca aacagtccga 480
tctcgatacc ttggcaaaag aggttccat tcccatcatc aacggcctga gcgacctgta 540
ccaccctatc caaatcctgg ctgactacct gacctgcaa gagcaactaca gcagcctgaa 600
gggtctgacc ctgtcatgga ttggcgatgg aaacaatatt ctgcaactcca tcatgatgtc 660
cgccgcgaag ttccgaatgc atctgcaagc cgccactcca aaaggatagc aaccggatgc 720
atccgtgacc aagttggcgg aacagtacgc gaaggagaa ggaaccaagc tctgctgac 780
taacgacccg ctcgaggctg cgcattgggg taactgtctg attacggaca cctggatctc 840

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catggggcag gaggaagaga agaagaagag actgcaggca ttccaggggt accaggtcac 900
catgaaaacc gcaaaagtgg cagcttcgga ctggactttc ctgcattgcc tgccgaggaa 960
gccggaggaa gtcgacgacg aagtgttcta ctgcctcggg tccctgggtg tccccgaggc 1020
cgaaaaccgg aagtggacca tcatggccgt gatggtgtcc ttgctgactg actatagccc 1080
gcagctgcag aagcctaagt tctaggctgg agcctcggtg gccgttcctc ctgcccgctg 1140
ggcctcccaa cgggccctcc tcccctcctt gcaccggccc ttctcgtgtc ttgaataaag 1200
tctgagtggg cagcaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaa 1335

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SEQ ID NO: 70      moltype = RNA length = 1346
FEATURE           Location/Qualifiers
misc_feature       1..1346
                   note = Synthetic construct
source            1..1346
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 70
aggaaactta agaattattg gttaaagaag tatattagtg ctaattttcc tccgtttgtc 60
ctagcttttc tcttctgtca accccacacg cctttggcac aatgcttttc aatctccgca 120
tctcctttaa caacgcgcgg tttagaaacg gccacaactt catgggtccg aacttcagat 180
gtggccagcc gcttcaaaac aaggtccagc tgaaggggcg ggatcttctg accctgaaga 240
actttactgg cgaagagatc aagtacatgc tctggctctc cgcggacttg aagttccgca 300
ttaagcagaa gggggaatac ctcccgctgc ttcaaggaaa gagcctcggc atgatctttg 360
agaagcgctc aaccaggacc cgcttttcta ctgaaactgg gttcgcgctg ctcggtggcc 420
acccctgctt cctgacgacc caggacatcc accctggagt gaacgaatcc ctccaccgata 480
cgcgccgggt gttatcgagc atggcagatg ccgtgctggc cagggtgtac aaacagtcgc 540
atctggacac tctggccaag gaggcgtaaa ttctattat caacggcctt agtgacctct 600
accatccgat tcagatcctg gccgattacc tcaccctgca agaacactac agctccctga 660
agggctctgac attgtctctg atcggcgagc gcaacaacat tctccattcc atcatgatgt 720
ccgccgcaaa attcggcaat catcttcaag ccgccacgcc gaagggttac gagcccgacg 780
cttcctgtac taagctcgcc gagcagtagc ctaaggagaa cggaaccaag cttctcgtga 840
ctaaccgacc actagaagca gcccccgggg gcaacgtgct tattactgac acctggatct 900
ccatgggcca ggaagaagag aaaaagaagc ggctgcaggc gttccaggga tatcaggta 960
ccatgaaaac gcccaagggt gctgcctccg actggacctt cctgcactgc tcgcctcgca 1020
agcctgaaga agtggagcag gaggtgttct actgccacg gagcctcggt tccccgagg 1080
ccgagaatag aaagtggcag atcatggcg tgatggtgtc actgctcacc gactacagcc 1140
cgcagcttca gaagcccaag ttctagctcg agacacatca caaccacaac cttctcaggc 1200
taccctgaga aaaaaagaca tgaagactca ggactcatct tttctgttgg tgtaaaaatca 1260
acaccctaag gaacacaaat ttctttaaac atttgacttc ttgtctctgt gctgcaatta 1320
ataaaaaatg gaaagaatct atctag 1346

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SEQ ID NO: 71      moltype = RNA length = 1346
FEATURE           Location/Qualifiers
misc_feature       1..1346
                   note = Synthetic construct
source            1..1346
                   mol_type = other RNA
                   organism = synthetic construct

```

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SEQUENCE: 71
aggaaactta agaattattg gttaaagaag tatattagtg ctaattttcc tccgtttgtc 60
ctagcttttc tcttctgtca accccacacg cctttggcac aatgctgttc aacctgcgca 120
tctcgtgtaa caacgcgcgc ttccgcaacg gccacaactt catggtgcgc aacttccgct 180
gccggccagcc cctgcagaac aaggtgcagc tgaaggggcg gcacctgctg accctgaaga 240
acttcaccgg cgaggagatc aagtacatgc tgtggctgag cgcgcgacct aagttccgca 300
tcaagcagaa gggcgagtag ctgccctcgc tgcagggcaa gagcctgggc atgatcttcg 360
agaagcgtag caccgcacc cgctgagca ccgagacagg cctggccctg ctggggcgcc 420
acccctgctt cctgaccacc caggacatcc accctggcgt gaacgagagc ctgaccgaca 480
cgcgcccggt gctgagcagc atggccgagc ccgtgctggc ccgcgtgtac aagcagagcg 540
acctggacac cctggccaag gaggccagca tcccacatca caacggcctg agcgacctgt 600
accaccccat ccagatcctg gccgactacc tgaccctgca ggagcactac agcagcctga 660
agggcctgac cctgagctgg atcggcgagc gcaacaacat cctgcacagc atcatgatga 720
gcgcgcgcaa gttcggcaat cacctgcagg ccgccacccc caagggttac gagcccgacg 780
ccagcgtgac caagctggcc gagcagtagc ccaaggagaa cggcaccagg ctgctgctga 840
ccaacgaccc cctggaggcc gcccccgggc gcaacgtgct gatcaccgac acctggatca 900
gcatgggcca ggaggaggag aagaagaagc gcctgcaggc cttccagggc taccaggtga 960
ccatgaagac gcccaagggt gccgcagcg actggacctt cctgcactgc ctgcccgcga 1020
agcccgaggg ggtggagcag gaggtgttct acagcccccg cagcctgggt tccccgagg 1080
ccgagaacgg caagtggacc atcatggcg tgatggtgag cctgctgacc gactacagcc 1140
ccagctgca gaagcccaag ttctgactcg agacacatca caaccacaac cttctcaggc 1200
taccctgaga aaaaaagaca tgaagactca ggactcatct tttctgttgg tgtaaaaatca 1260
acaccctaag gaacacaaat ttctttaaac atttgacttc ttgtctctgt gctgcaatta 1320
ataaaaaatg gaaagaatct atctag 1346

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SEQ ID NO: 72      moltype = RNA length = 1346
FEATURE           Location/Qualifiers
misc_feature       1..1346
                   note = Synthetic construct

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source          1..1346
                mol_type = other RNA
                organism = synthetic construct

SEQUENCE: 72
aggaaactta agaattattg gttaaagaag tatattagtg ctaatttccc tccgtttgtc 60
ctagcttttc tcttctgtca accccacacg cctttggcac aatgcttttc aacctgagaa 120
tcctcttgaa caatgctgct ttccggaatg gccacaactt tatgggtcgg aacttccgtt 180
gcggccagcc tttacaaaac aaggtccagc tgaaggggcg ggattttgtc acactaaaga 240
actttactgg agaagagatc aagtacatgc tatggctgtc ggccgacctg aagttccgta 300
tcaagcagaa gggagaatac ctcccgctgc ttcaaggaaa gagcctcgcc atgatcttg 360
agaagcgctc aaccaggacc cgcttttcta ctgaaactgg gttcgcgtcg ctccgtggcc 420
acccctgctt cctgacgacc caggacatcc acctcgagtg gaacgaatcc ctaccgata 480
ccgcccggtt gttatcgagc atggcagatg ccgtgctggc cagggtgtac aaacagtcg 540
atctcgatac cttggcaaaag gaggttcca ttcccatcat caacggcctg agcgacctgt 600
accacccaat ccaaatcctg gctgactacc tgacctgca agagcactac agcagcctga 660
agggctctgac cctgtcatgg attggcgatg gaaacaatat tctgcaactc atcatgatgt 720
ccgcccgcaa gttcggaaag catctgcaag ccgccactcc aaaaggatag gaaccggatg 780
catccgtgac caagtgtggc gaacagtacg cgaaggagaa cggaaccaag ctctgctga 840
ctaaccagcc gctcgaggct gcgcattggg gtaacgtgct gattacggac acctggatct 900
ccatggggca ggaggagag aagaagaaga gactgcaggc attccagggg taccaggta 960
ccatgaaac cgcaaaagtg gcagcttcgg actggacttt cctgcattgc ctgccagga 1020
agccggagga agtcgacgac gaagtgttct actcgctcgt gtcctgtgtt ttcccagag 1080
ccgaaaaccg gaagtggacc atcatggccg tgatgggtgc cttgctgact gactatagcc 1140
cgcagctgca gaagcctaag ttctagctcg agacacatca caaccacaac ctctcaggc 1200
taccctgaga aaaaaagaca tgaagactca ggactcatct tttctgttgg tgtaaatca 1260
acaccctaag gaacacaaat ttctttaaac atttgacttc ttgtctctgt gctgcaatta 1320
ataaaaaatg gaaagaatct atctag 1346

SEQ ID NO: 73      moltype = RNA length = 1222
FEATURE           Location/Qualifiers
misc_feature       1..1222
                  note = Synthetic construct
source            1..1222
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 73
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcac cctgctgaac 60
aacgccgcct tccgcaacgg ccacaacttc atgggtgcga acttccgctg cggccagccc 120
ctgcagaaca aggtgcagct gaaggggccg gacctgctga ccctgaagaa cttcaccggc 180
gaggagatca agtacatgct gtggctgagc gccgacctga agttccgcat caagcagaag 240
ggcgagtacc tgccccctgt gcaggggcaag agcctgggca tgatcttcga gaagcgcagc 300
acccgcaccc gcctgagcac cgagacaggc ttgcctctgc tggcgggcca cccctgcttc 360
ctgaccaccc aggaatcca cttggcgctg aacgagagcc tgaccgacac cgcccgctg 420
ctgagcagca tggccgacgc cgtgctggcc ccgctgtaca agcagagcga cctggacacc 480
ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta ccacccatc 540
cagatctctg ccgactacct gacctgcag gagcactaca gcagcctgaa gggcctgacc 600
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgccgccaag 660
ttcgcatgac acctgcaggc cgccaccccc aagggtctac agcccgacgc cagcgtgacc 720
aagctggccg agcagtacgc caaggagaac ggcaccaagc tgcgtgtgac caacgacccc 780
ctggaggccg cccacggcgg caacgtgtg atcacgcaca cctggatcag catggggcag 840
gaggaggaga agaagaagcg cctgcaggcc ttccagggtc accaggtgac catgaagacc 900
gccaaaggtg ccgccagcga ctggaccttc ctgcaactgc tgcctcgcaa gcccgaggag 960
gtggacgagc aggtgttcta tagccccccg agcctggtgt tccccgaggg cgagaaccgc 1020
aagtggaaca tcattggcgt gatggtagc ctgctgaccg actacagccc ccagctgcag 1080
aagcccaagt tctgaggtct ctgtaatga gctggagcct cggtagccgt tctctctgcc 1140
cgctgggcct cccaacgggc cctcctcccc tctctgcacc ggcccttctt ggtctttgaa 1200
taaaagtctga gtgggcatct ag 1222

SEQ ID NO: 74      moltype = RNA length = 1346
FEATURE           Location/Qualifiers
misc_feature       1..1346
                  note = Synthetic construct
source            1..1346
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 74
aggaaactta agaattattg gttaaagaag tatattagtg ctaatttccc tccgtttgtc 60
ctagcttttc tcttctgtca accccacacg cctttggcac aatgcttttc aacctgcga 120
tcctgtctga caacgcgcgc ttccgcaacg gccacaactt catggtgcgc aacttccgct 180
gcgccagacc cctgcagaac aaggtgcagc tgaaggggcg cgacctgctg acctgaaga 240
acttcaccgg cgaggagatc aagtacatgc tgtggctgag cgccgacctg aagttccgca 300
tcaagcagaa gggcgagtac ctgccctgct tgcagggcaa gagcctgggc atgatcttcg 360
agaagcgcac caccgcacc cgctgagca ccgagacagg cttcgccctg ctggcgggcc 420
acccctgctt acctgaccac caggacatcc acctgggctg gaacgagagc ctgaccgaca 480
ccgcccgctt gctgagcagc atggccgacg ccgtgctggc ccgctgttac aagcagagcg 540
acctggacac cctggccaag gagggcagca tcccatcat caacggcctg agcgacctgt 600
accaccccat ccagatctct gcgactacc tgacctgca ggagcactac agcagcctga 660
agggcctgac cctgagctgg atcgcgagc gcaacaacat cctgcacagc atcatgatga 720
gcgccgcaa gttcggcatg cacctgcagg ccgccacccc caagggtctac gagcccgagc 780

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ccagcgtgac caagctggcc gagcagtacg ccaaggagaa cggcaccag ctgctgctga 840
ccaacgagcc cctggaggcc gccccaggcg gcaacgtgct gatcacgac acctggatca 900
gcatgggcca ggaggaggag aagaagaagc gcctgcagcg cttccaggcg taccagggtga 960
ccatgaagac cgccaaggtg gccgccagcg actggacett cctgcaactgc ctgccccgca 1020
agcccaggga ggtggagcgc gaggtgttct acagcccccg cagcctgggtg ttccccgagg 1080
ccgagaaccg caagtggacc atcatggccg tgatgggtgag cctgctgacc gactacagcc 1140
ccagcgtgca gaagcccaag ttctgactcg agacacatca caaccacaac cttctcaggc 1200
taccctgaga aaaaaagaca tgaagactca ggactcatct tttctgttgg tgtaaaatca 1260
acaccctaag gaacacaaat ttctttaaac atttgacttc ttgtctctgt gctgcaatta 1320
ataaaaaatg gaaagaatct atctag 1346

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SEQ ID NO: 75      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 75
aggaacctta agattattac atcaaaacaa aaagccgcca atgctgttca acctgcgcat 60
cctgctgaac aacgccgctt tccgcaacgg ccacaacttc atggtgcgca acttccgctg 120
cggccagccc ctgcagaaca aggtgcagct gaagggcccg gacctgctga cctgaagaa 180
cttcaccggc gaggagatca agtacctgct gtggctgagc gccgacctga agttccgcat 240
caagcagaag ggcgagtacc tgccccctgt gcagggcaag agcctgggca tgatcttcga 300
gaagcgcgag acccgacccc gcctgagcac cgagacaggg ttccgccctgc tgggcggcca 360
ccccctgctt ctgaccaccc agggacatcca cctgggctgtg aacgagagcc tgaccgacac 420
cgcccgctgt ctgagcagca tggccgacgc cgtgctggcc cgctgtgaca agcagagcga 480
cctggacacc ctggccaagg agggcagcat ccccatcacc aacggcctga gcgacctgta 540
ccaccccatc cagatcctgg ccgactacct gacctgcag gagcactaca gcagcctgaa 600
gggctgacc ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag 660
cgccgccaaag ttccgctgac acctgcaggg cgccaccccc aagggtctac agcccgcgc 720
cagcgtgacc aagctggccg agcagtagcc caaggagaaac ggcaccaaagc tgctgctgac 780
caacgacccc ctggaggccg ccacggcgcg caacgtgctg atcacggaca cctggatcag 840
catggggcag gaggaggaga agaagaagcg cctgcaggcc ttccagggt accaggtgac 900
catgaagacc gccaaagtggt ccgccagcga ctggaccttc ctgcaactgcc tgcgccgcaa 960
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cgagaaccgc aagtggacca tcatggccgt gatggtgagc ctgctgacgg actacagccc 1080
ccagctgcag aagcccaagt tctgaacgcc gaagcctgca gccatgcgac cccacgccac 1140
cccgtgcctc ctgcctccgc gcagcctgca cggggagacc ctgtccccgc cccagccgtc 1200
ctcctggggt ggaccctagt ttaataaaga ttcaccaagt ttcacgcaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1368

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SEQ ID NO: 76      moltype = RNA length = 1496
FEATURE           Location/Qualifiers
misc_feature       1..1496
                   note = Synthetic construct
source            1..1496
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 76
cttaaggggg cgctgcctac ggagggtggca gccatctcct tctcgccatc aagcttacc 60
tggtgcccc aagcctgctc ttggtcccg tgctggtgtt cccctctgc ttggcaagt 120
tccccatcta caccatcccc gacaagctgg ggcctgtag ccccatcgac atccaccac 180
tgtctctccc caacaacctc gtggtcgagg acgaggcgct caccacactg agcgggttct 240
ctacatgctt gttcaacctg cgcactctgc tgaacaacgc cgccttccgc aacggccaca 300
acttcatggt gcgcaacttc cgctgcggcc agccccctga gaacaagggt cagctgaagg 360
gccgcgacct cgtgacctgc aagaacttca ccggcgagga gatcaagtac atgctgtggc 420
tgagcgccga cctgaagttc cgcactcaagc agaaggggcg gtacctgccc ctgctgcagg 480
gcaagagcct gggcatgac ttcgagaagc gcagaccccg caccgcctg agcaccgaga 540
caggcttcgc cctgctgggg gccacccct gcttctgac caccaggac atccacctg 600
gcgtgaacga gagcctgacc gacaccgccc gcgtgctgag cagcatggcc gacgcctg 660
tgggcccgct gtacaagcag agcgacctgg acacctggc caaggaggcc agcatcccc 720
tcataacagg cctgagcgac ctgtaccacc ccatccagat cctggcgac tactgacc 780
tgaggaggca ctacagcagc ctgaaggggc tgacctgag ctggatcgcc gacggcaaca 840
acatcctgca cagcatcatg atgagcgccg ccaagtctcg catgcacctg caggccgcca 900
cccccaaggg ctacgagccc gacgccagcg tgaccaagct ggccgagcag tacgccaagg 960
agaacggcac caagctgctg ctgaccaacg acccctggga ggcgcgccac ggcggcaacg 1020
tgctgatcac cgacacctgg atcagcatgg gccaggagga ggagaagaag aagcgctgc 1080
aggccttcca gggctaccag gtgacctga agaccgcaa ggtggccgc agcgactgga 1140
ccttctgca ctgcctgccc gcgaagcccg aggaggtgga cgacgaggtg ttctacagcc 1200
cccgagcctt ggtgttcccc gagggcgaga accgcaagtg gaccatcatg gccgtgatgg 1260
tgagcctgct gaccgactac agccccagc tcgagaagcc caagtcttga ataagtgaat 1320
gcaaggctgg ccgaagccc ttgctgaaa gcaagattc agcctggaag agggcaaggt 1380
ggacgggagt ggacaggagt ggatgcgata agatgtggtg tgaagctgat gggtgccagc 1440
cctgcattgc tagtcaatc aataaagac tttcttttga ccatctctag atctag 1496

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SEQ ID NO: 77      moltype = RNA length = 1380
FEATURE           Location/Qualifiers

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misc_feature      1..1380
                  note = Synthetic construct
source            1..1380
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 77
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgggc gtcttcaacc tgcggatcct gctgaacaac gccgccttcc 180
ggaacggcca caacttcagt gtcgcaact tcagatgcgg ccagcccctg cagaacaagg 240
tgcagctgaa gggcggggac ctgctgacct tgaagaactt caccggcgaa gagatcaagt 300
acatgctgtg gctgagcgcc gacctgaagt tccggatcaa gcagaagggc gactacctgc 360
ccctgctgca aggcaagagc ctgggcatga tcttcgagaa gcggagcacc cggacccggc 420
tgagcaccga gacaggcttt gccctgctgg gaggccaccc ctgctttctg accaccagg 480
acatccacct gggcgtgaac gagagcctga ccgacaccgc cagagtgcgt agcagcatgg 540
ccgacgcctg gctggccggg gtgtacaagc agagcgacct ggacaccctg gccaaaggag 600
ccagcatccc catcatcaac ggctgagcgc acctgtacca cccatccag atcctggcgg 660
actacctgac cctgcaggaa cactacagct cctgaagggt cctgaccctg agctggatcg 720
gcgacggcaa caacatcctg cacagcatca tgatgagcgc cgccaagtgc ggcatgcac 780
tgacggcgcc caccoccaa ggctacgagc ctgatgccag cgtgaccaag ctggccgagc 840
agtacgccaa agagaacggc accaagctgc tgctgaccaa cgaccccctg gaagccggcc 900
acggcgggcaa cgtgctgat accgacacct ggatcagcat gggccaggaa gaggaaga 960
agaagcgggt cagggcttcc cagggctacc aggtcacaaat gaagaccgcc aaggtggcgg 1020
ccagcgactg gaccttctct cactgcctgc cccggaagcc cgaagagggt gacgacgagg 1080
tggtctacag ccccggctcc ctggtgttcc ccgaggccga gaaccggaag tggaccatta 1140
tggcctgatg ggtgtccctg ctgaccgact actccccca gctgcagaag cccaagttct 1200
agataagtga actcgagcta gtgactgact aggatctggt taccactaaa ccagcctcaa 1260
gaacaccoga atggagtctc taagctacat aataccaact tacacttaca aaatgttgtc 1320
cccaaaaatg tagccattcg tatctgctcc taataaaaag aaagtttctt cacattctag 1380

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SEQ ID NO: 78      moltype = RNA length = 1380
FEATURE            Location/Qualifiers
misc_feature        1..1380
                    note = Synthetic construct
source              1..1380
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 78
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgggc gtcttcaacc tgcggatcct gctgaacaac gccgccttcc 180
ggaacggcca caacttcagt gtcgcaact tcagatgcgg ccagcccctg cagaacaagg 240
tgcagctgaa gggcggggac ctgctgacct tgaagaactt caccggcgaa gagatcaggt 300
acatgctgtg gctgagcgcc gacctgaagt tccggatcaa gcagaagggc gactacctgc 360
ccctgctgca aggcaagagc ctgggcatga tcttcgagaa gcggagcacc cggacccggc 420
tgagcaccga gacaggcttt gccctgctgg gaggccaccc ctgctttctg accaccagg 480
acatccacct gggcgtgaac gagagcctga ccgacaccgc cagagtgcgt agcagcatgg 540
ccgacgcctg gctggccggg gtgtacaagc agagcgacct ggacaccctg gccaaaggag 600
ccagcatccc catcatcaac ggctgagcgc acctgtacca cccatccag atcctggcgg 660
actacctgac cctgcaggaa cactacagct cctgaagggt cctgaccctg agctggatcg 720
gcgacggcaa caacatcctg cacagcatca tgatgagcgc cgccaagtgc ggcatgcac 780
tgacggcgcc caccoccaa ggctacgagc ctgatgccag cgtgaccaag ctggccgagc 840
agtacgccaa agagaacggc accaagctgc tgctgaccaa cgaccccctg gaagccggcc 900
acggcgggcaa cgtgctgat accgacacct ggatcagcat gggccaggaa gaggaaga 960
agaagcgggt gcaggcttcc cagggctacc aggtcacaaat gaagaccgcc aaggtggcgg 1020
ccagcgactg gaccttctct cactgcctgc cccggaagcc cgaagagggt gacgacgagg 1080
tggtctacag ccccggctcc ctggtgttcc ccgaggccga gaaccggaag tggaccatta 1140
tggcctgatg ggtgtccctg ctgaccgact actccccca gctgcagaag cccaagttct 1200
agataagtga actcgagcta gtgactgact aggatctggt taccactaaa ccagcctcaa 1260
gaacaccoga atggagtctc taagctacat aataccaact tacacttaca aaatgttgtc 1320
cccaaaaatg tagccattcg tatctgctcc taataaaaag aaagtttctt cacattctag 1380

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SEQ ID NO: 79      moltype = RNA length = 1380
FEATURE            Location/Qualifiers
misc_feature        1..1380
                    note = Synthetic construct
source              1..1380
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 79
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctg gtcttcaacc tgcggatcct gctgaacaac gccgccttcc 180
ggaacggcca caacttcagt gtcgcaact tcagatgcgg ccagcccctg cagaacaagg 240
tgcagctgaa gggcggggac ctgctgacct tgaagaactt caccggcgaa gagatcaggt 300
acatgctgtg gctgagcgcc gacctgaagt tccggatcaa gcagaagggc gactacctgc 360
ccctgctgca aggcaagagc ctgggcatga tcttcgagaa gcggagcacc cggacccggc 420
tgagcaccga gacaggcttt gccctgctgg gaggccaccc ctgctttctg accaccagg 480
acatccacct gggcgtgaac gagagcctga ccgacaccgc cagagtgcgt agcagcatgg 540

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ccgacgcgct gctggcccg gtgtacaagc agagcgacct ggacacctg gccaaagagg 600
ccagcatccc catcatcaac ggctgagcg acctgtacca cccatccag atcctggccg 660
actacctgac cctgcaggaa cactacagct cctgaaggg cctgacctg agctggatcg 720
gcgacggcaa caacatcctg cacagcatca tgatgagcgc cgccaagttc ggcatgcac 780
tgcaggccgc caccoccaa ggctacgagc ctgatgccag cgtgaccaag ctggccgagc 840
agtacgccaa agagaacggc accaagctgc tgcgtacca cgaacccctg gaagccgccc 900
acggcggcaa cgtgctgac accgacacct ggatcagcat gggccaggaa gaggaaaaga 960
agaagcggct gcaggccttc cagggctacc aggtcacat gaagaccgcc aaggtggccg 1020
ccagcgactg gaccttctg cactgcctgc ccggaagcc cgaagaggtg gacgacgagg 1080
tggtctacag ccccggtcc ctgggtgttc ccgaggccga gaaccggaag tggaccatta 1140
tggccgtgat ggtgtccctg ctgaccgact actcccccga gctgcagaag cccaagtctt 1200
agataagtga actcgagcta gtgactgact aggatctggt taccactaaa ccagcctcaa 1260
gaacacccga atggagctct taagctacat aataccaact tacacttaca aaatgttgtc 1320
ccccaaaatg tagcattctg tatctgtctc taataaaaag aaagtcttct cacattctag 1380

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SEQ ID NO: 80      moltype = RNA length = 1377
FEATURE           Location/Qualifiers
misc_feature       1..1377
                   note = Synthetic construct
source            1..1377
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 80
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctg ttcaacctga ggatcctgct gaacaacgca gctttcagga 180
acggccacaa cttcatggtg aggaacttcc ggtgcggcca gccctgcag aacaagggtc 240
agctgaaggc caggagcctg ctgacctga agaacttcac cggagaggag atcaagtaca 300
tgctgtggct gagcgagac ctgaagtcca ggatcaagca gaaggagag tacctgcccc 360
tgctgcaggg gaagtccctg ggcattgatc tcgagaagag gagtaccagg accaggctga 420
gcaccgaaac cggcttcgac ctgctgggag gacacccctg cttcctgacg acccaggaca 480
tcacccctgg cgtgaacgag agtctgaccg acaccgccag ggtgctgtct agcatggccg 540
acggcgtgct ggccagggtg tacaagcagt cagacctgga caccctggct aaggaggcca 600
gcatcccat catcaacggc ctgagcgacc tgtaccaccc catccagatc ctggctgact 660
acctgacctc gcaggagcac tcagctctc tgaaggcctc gacctgagc tggatcggcg 720
acgggaacaa catcctgcac agcatcatga tgagcgccgc caagtctggc atgcacctgc 780
aggccgctac cccaagggtg tacgagcccg acgacagcgt gaccaagctg gcagagcagt 840
acgccaagga gaacgggacc aagctgctgc tgaccaacga cccctggag gccgcccacg 900
gaggcaacgt gctgatcacc gacacctgga tcagcatggg acaggaggag gagaagaaga 960
agcggctgca ggtttccag ggttaccagg tgacctgaa gaccgccaag gtggtgcca 1020
gcgactggac cttcctgcac tgctgcccga ggaagcccga ggagggtggc gacgaggtgt 1080
tctactctcc caggagcctg gtgttccccg agggcgagaa caggaaagtg accatcatgg 1140
ctgtgatggt gtcctgctg accgactaca gccccagct gcagaagccc aagttctgaa 1200
taagtgaact cgagctagt actgactagg atctggttac cactaaacca gcctcaagaa 1260
caccgcaatg gagtctctaa gctacataat accaacttac acttacaata tgttgtcccc 1320
caaatgtag ccattcgat ctgctcctaa taaaagaaa gtttcttcac attctag 1377

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SEQ ID NO: 81      moltype = RNA length = 1377
FEATURE           Location/Qualifiers
misc_feature       1..1377
                   note = Synthetic construct
source            1..1377
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 81
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctg ttcaacctga ggatcctgct gaacaacgca gctttcagga 180
acggccacaa cttcatggtg aggaacttcc ggtgcggcca gccctgcag aacaagggtc 240
agctgaaggc caggagcctg ctgacctga agaacttcac cggagaggag atcaagtaca 300
tgctgtggct gagcgagac ctgaagtcca ggatcaagca gaaggagag tacctgcccc 360
tgctgcaggg gaagtccctg ggcattgatc tcgagaagag gagtaccagg accaggctga 420
gcaccgaaac cggcttcgac ctgctgggag gacacccctg cttcctgacg acccaggaca 480
tcacccctgg agtgaacgaa tcctcaccg ataccgcccg ggtgttatca agcatcgag 540
atgccgtgct ggccagggtg taaaaacagt ccgatctgga cactctggct aaggaggcca 600
gcatcccat catcaacggc ctgagcgacc tgtaccaccc catccagatc ctggctgact 660
acctgacctc gcaggagcac tacagctctc tgaaggcctc gacctgagc tggatcggcg 720
acgggaacaa catcctgcac tcacatgga tgctcgccgc gaagtctgga atgcacctgc 780
aagccgccc gccaaaagga tacgaaccgg atgcgcccgt gacaaaagtg gcggaacagt 840
acgctaagga gaacgggacc aagctgctgc tgaccaacga cccctggag gccgcccacg 900
gaggcaacgt gctgatcacc gacacctgga tcagcatggg acaggaggag gagaagaaga 960
agcggctgca ggtttccag ggttaccagg tgacctgaa gaccgccaag gtggtgcca 1020
gcgactggac cttcctgcac tgctgcccga ggaagcccga ggagggtggc gacgaggtgt 1080
tctactctcc caggagcctg gtgttccccg agggcgagaa caggaaagtg accatcatgg 1140
ctgtgatggt gtcctgctg accgactaca gccccagct gcagaagccc aagttctgaa 1200
taagtgaact cgagctagt actgactagg atctggttac cactaaacca gcctcaagaa 1260
caccgcaatg gagtctctaa gctacataat accaacttac acttacaata tgttgtcccc 1320
caaatgtag ccattcgat ctgctcctaa taaaagaaa gtttcttcac attctag 1377

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SEQ ID NO: 82      moltype = RNA length = 1477
FEATURE           Location/Qualifiers
misc_feature       1..1477
                   note = Synthetic construct
source            1..1477
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 82
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctg ttaaacctgc gcatcctgct gaacaacgcc gccttcgcga 180
acggccacaa cttcatgggt cgaacttcc gctgcggcca gccctgcag aacaagggtc 240
agctgaaggg ccgcgacctg ctgacctga agaacttcac cggcgaggag atcaagtaca 300
tgctgtggct gagcgccgac ctgaagttcc gcatcaagca gaaggcgag tactgcccc 360
tgctgcaggg caagagcctg ggcattgatc tcgagaagcg cagcaccgc acccgctga 420
gcaccgagac aggtctcgcc ctgctgggcg gccaccctg cttcctgacc acccaggaca 480
tcacactggg cgtgaacgag agcctgaccg acaccgccg cgtgctgagc agcatggcgg 540
acgcctgtgt ggcccgctgt tacaagcaga gcgacctgga caccctggcc aaggaggcca 600
gcatcccat catcaacggc ctgagcgacc tgtaccacc catccagatc ctggccgact 660
acctgacct gcaggagcac tacagcagcc tgaaggcgct gacctgagc tggatggcgg 720
acggcaacaa catcctgcac agcatcatga tgagcgccg caagttcgcc atgcacctgc 780
aggccgccc acccaaggcc tacgagcccg acgcagcgt gaccaagctg gccgagcagt 840
acggcaagga gaacgggcac aagctgtctg tgaccaacga cccctggag gccgcccacg 900
ggcgcaacgt gctgatcacc gacactgga tcagcatggg ccaggaggag gagaagaaga 960
agcgctgca ggccttcacg ggctaccagg tgacctgaa gaccgccaa gtggccgcca 1020
gcgactggac cttcctgcac tgctgcccc gcaagcccga ggagggtggc gacgaggtgt 1080
ttacagccc ccgcagcctg gtgttcccc aggcgcgaga ccgcaagtgg accatcatgg 1140
ccgtgatggt gagcctgctg accgactaca gcccacgct gcagaagccc aagttctgaa 1200
taagtgaact cgagctagt accgactagg atctgggtac cactaaacca gcctcaagaa 1260
caccgcaatg gagtctctaa gctacataat accaacttac acttacaana tgttgtcccc 1320
caaatgtag ccattcgat ctgctcctaa taaaagaaa gtttcttcac attctagaaa 1380
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaa aaaaaaaaaa aaaaaaaaaa 1440
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaa aaaaaaaaaa aaaaaaaaaa 1477

SEQ ID NO: 83      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                   note = Synthetic construct
source            1..1371
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 83
aggaactta agtcaacaca acatatacaa aacaaacgaa tctcaagcaa tcaagcattc 60
tacttctatt gcagcaattt aaatcatttc ttttaaagca aaagcaattt tctgaaaatt 120
ttcaccattt acgaacgata gccatgctgt tcaacctgcg catcctgctg aacaacgccg 180
ccttcgcgaa cggccacaac ttcatgggtg gcaacttccg ctgcggccag cccctgcaga 240
acaagggtga gctgaagggc ccgcacctgc tgacctgaa gaacttcacc ggcgaggaga 300
tcaagtacat gctgtggctg agcgccgacc tgaagtccg catcaagcag aaggcgaggt 360
acctgcccc gctgcagggc aagagcctgg gcatgatctt cgagaagcgc agcaccgcga 420
ccgcctgag caccgagaca ggettcgccc tgctgggccc ccacctgc ttcctgacca 480
cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgccgcg gtgctgagca 540
gcatggccga ccgcgtgctg cccgcgctgt acaagcagag cgacctggac acctgggcca 600
aggaggccag catccccatc atcaacggcc tgagcgacct gtaccacccc atccagatcc 660
tgcccgacta cctgacctg caggagcact acagcagcct gaaggccctg acctgagct 720
ggatcggcga cggcaacaac atcctgcaca gcatcatgat gagcgccgca aagttcggca 780
tgcacctgca ggcgcgccac cccaagggt acgagcccga cgcagcgtg accaagctgg 840
ccgagcagta cggcaaggag aacggcacca agctgctgct gaccaacgac cccctggagg 900
ccgcccacgg cggcagcgtg ctgatcaccg acacctggat cagcatgggc caggaggagg 960
agaagaagaa gcgcctcgag gccttcagg gctaccaggt gacctgaag accgccaagg 1020
tgcccgccag cgactggacc ttcctgact gcctgcccc caagcccag gaggtggacg 1080
acgaggtgtt ctacagcccc cgcagcctgg tgttccccga ggccgagaac cgcaagtga 1140
ccatcatggc cgtgatgggt agcctgctga ccgactacag ccccagctg cagaagccca 1200
agttctgact agtgactgac taggatctgg ttaccactaa accagcctca agaacaccgg 1260
aatggagttc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

SEQ ID NO: 84      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                   note = Synthetic construct
source            1..1371
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 84
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttaacttga gaatcctgct gaacaacgcc gcctttcgca 180
acggtcaciaa ttttatgggt agaaacttca gatcggaaca gccctccaa aacaagggtc 240
agctgaaggg ccgcgatctc ctcacctga agaacttcac gggggaggag atcaagtaca 300

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tgetgtggct ctccgctgac ctgaagttca ggatcaagca gaagggagaa tatctgccc 360
tgetgcaagg gaagtcctct gggatgattt tcgagaagcg gagcaccgg actcggtct 420
ccactgaaac tggtttcgcc cttctgggcy gtcacccctg cttcctgacc actcaagaca 480
ttcacctcgg agtgaacgag tccctgactg acaccgccc ggtgctgtcg agcatggcag 540
acgccgtgct agcccgctg tacaagcagt cagacctcga taccctggcc aaggaggctt 600
cgatcccgat catcaacggg ttgtccgacc tgtaccacc gattcagatt ctgcgcgact 660
acctcacctt gcaagagcat tacagctccc tgaaggggct taccctgtcc tggattggcg 720
acggaaacaa catcctgcac tccattatga tgtcggcgcc caagttcggc atgcacctcc 780
aagccgcgac ccctaagggt tacgaaccag acgcgtcagt gactaagctg gccgaacagt 840
acgcaaaagg aaatggcacg aagctgctcc tgaccaacga tccgttggaa gccgcccatt 900
gcggaatagt gctcatcacc gacacctgga tctcgatggg acaggaggaa gagaagaaga 960
agcggctgca ggcgttcacg ggtaccagg taccatgaa aactgccaa gtggccgcca 1020
gcgactggac cttcctgcac tgcctccgc gcaagcctga ggaggtggac gatgaagtgt 1080
tctactctcc accgttcctg gtgttcccc aggcggagaa ccgcaaatg accatcatgg 1140
ctgtgatggt cagcctgctg accgattaca gccctcagtt gcaaaagccg aagttttgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacaccg 1260
aatggagtct ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 85      moltype = RNA length = 1471
FEATURE           Location/Qualifiers
misc_feature       1..1471
                  note = Synthetic construct
source            1..1471
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 85
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttct ttaaagcaaa agcaattttc tgaattttt caccatttac 120
gaacgatagc caccatgctg ttcaacctcc gcactcctct caacaacgcc gcattcagaa 180
acgggcacaa cttcatggct agaaacttcc gctcggggca accctacaa aacaaggctc 240
agctcaaggg cggggacctc ctgacctga agaacttcac cggcgaagag atcaagtaca 300
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aagccgcaac cccgaaggcg tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
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aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa a a 1471

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SEQ ID NO: 86      moltype = RNA length = 1771
FEATURE           Location/Qualifiers
misc_feature       1..1771
                  note = Synthetic construct
source            1..1771
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 86
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ctggccctgt cttcttgacg agcattccta ggggtcttcc cctctcgcgc aaaggaaatg 180
aaggctctgt gaatgtcgtg aaggaagcag ttcctctgga agcttcttga agacaaacaa 240
cgtctgtagc gaccttttgc aggcagcgga acccccacc tggcgacagg tgcctctgcy 300
gccaaggccc acgtgtataa gatacacctg caaaggcgcc acaaccccag tgccacgttg 360
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tgaaggatgc ccagaaggta cccatttga tgggatctga tctggggcct cgggtgacat 480
gctttacgtg tgtttagtgc aggttaaaaa acgtctaggc ccccggaacc acggggagct 540
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tatcgagcat ggcagatgcc gtgctggcca aggtgtacaa acagtcagat ctggacactc 1020
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agatcctggc cgattacctc accctgcaag aacactacag ctccctgaag ggtctgacat 1140
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tcggcatgca tcttcaagcc gccacgccga aggggttacga gcccgacgct tccgtgacta 1260
agctcgccga gcagtacgct aaggagaacg gaaccaagct tctgctgact aacgacccac 1320
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aagaagagaa aaagaagcgg ctgcaggcgt tccagggata tcagggtcacc atgaaaaccg 1440
ccaaggtcgc tgcctccgac tggaccttcc tgcactgcct gcctcgcaag cctgaagaag 1500
tggacgcaga ggtgttctac tcgccacgga gcctcggtgt ccccgaggcc gagaatagaa 1560
agtggaccat catggcgtg atgggtgtcac tgcaccgga ctacagcccg cagcttcaga 1620
agcccaagtt ctgaataagt agatagtgc gtcactggca caacgcgttg cccggtaagc 1680
caatcgggta tacacggtcg tcatactgca gacagggttc ttctactttg caagatagtc 1740
tagagtagta aaataaatag tataagtcta g 1771

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SEQ ID NO: 87      moltype = RNA length = 1771
FEATURE           Location/Qualifiers
misc_feature       1..1771
                  note = Synthetic construct
source            1..1771
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 87
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ctggccctgt cttcttgacg agcatctcta ggggtctttc ccctctcgcc aaagggaatgc 180
aagggtctgt gaattgctgt aaggagcag ttccctctgga agcttcttga agacaacaa 240
cgtctgtagc gaccctttgc aggcagcgga acccccacc tggcgacagg tgcctctgcg 300
gccccaaagc acgtgtataa gatacacctg caaaggcgcc acaaccgccg tgcacggttg 360
tgagttggat agttgtggaa agagtcaaat ggctctcctc aagcgtattc aacaaggggc 420
tgaaggatgc ccagaaggta ccccatgtga tctggggcct cggtgacacat 480
gctttacgtg tgtttagtcg aggttaaaaa acgtctaggc ccccgaacc acggggacgt 540
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tacaaaacaa ggtccagctg aagggccggg atttgcctac actaaagaac ttactggag 720
aagagatcaa gtacatgcta tgctgtcgg cgcacctgaa gttccgtatc aagcagaagg 780
gagaatacct tccgctgctt caaggaaaga gcctcgcat gatctttgag aagcgcctca 840
ccaggaccgc cctttctact gaaactgggt tcgcgctgct cgggtggccac cctgcttcc 900
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tatcgagcat ggcagatgcc gtgctggcca ggggtgtaca acagtcgat ctcgatacct 1020
tggcaaaagg ggtctccatt cccatcatca acggcctgag cgacctgtac caccatctc 1080
aaatcctggc agcttcggac accctgcaag agcactacag cagcctgaag ggtctgacct 1140
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agttggcgga acagtacggc aaggagaacg gaaccaagct cctgctgact aacgacccgc 1320
tcgagggtgc gcatgggggt aacgtgctga ttacggacac ctggatctcc atggggcagg 1380
aggaagagaa gaagaagaga ctgcaggcat tccagggata ccagggtcacc atgaaaaccg 1440
caaaagtggc agcttcggac tggactttcc tgcattgcct gccgaggaa cggagggaag 1500
tcgacgcaga agtgttctac tcgcctcggt cccctgggtt ccccgaggcc gaaaaaccga 1560
agtggaccat catggcgtg atgggtgtcct tgcctgactga ctatagcccg cagctgcaga 1620
agcctaagtt ctgaataagt agatagtgc gtcactggca caacgcgttg cccggtaagc 1680
caatcgggta tacacggtcg tcatactgca gacagggttc ttctactttg caagatagtc 1740
tagagtagta aaataaatag tataagtcta g 1771

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SEQ ID NO: 88      moltype = RNA length = 1771
FEATURE           Location/Qualifiers
misc_feature       1..1771
                  note = Synthetic construct
source            1..1771
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 88
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ctggccctgt cttcttgacg agcatctcta ggggtctttc ccctctcgcc aaagggaatgc 180
aagggtctgt gaattgctgt aaggagcag ttccctctgga agcttcttga agacaacaa 240
cgtctgtagc gaccctttgc aggcagcgga acccccacc tggcgacagg tgcctctgcg 300
gccccaaagc acgtgtataa gatacacctg caaaggcgcc acaaccgccg tgcacggttg 360
tgagttggat agttgtggaa agagtcaaat ggctctcctc aagcgtattc aacaaggggc 420
tgaaggatgc ccagaaggta ccccatgtga tctggggcct cggtgacacat 480
gctttacgtg tgtttagtcg aggttaaaaa acgtctaggc ccccgaacc acggggacgt 540
ggttttcctt tgaaaaaacac gatgataata tgctgttcaa cctgcgcacat ctgctgaaca 600
acgcgcctt cgcgaacggc cacaacttta tgggtcgcga cttccgttgc ggccagcccc 660
tgcaaacaa ggtgcagctg aagggccggc acctgctgac cctgaagaac ttcaccggcg 720
aggagatcaa gtacatgctg tggctgagcg ccgacctgaa gttccgcacat aagcagaagg 780
gcgagtagct gccctgctg cagggaaga gcctgggcat gatcttcgag aagcgcagca 840
cccgaccccg cctgagcacc gagacaggcc tggccctgct gggcgggccac cctgcttcc 900
tgaccaccca ggacatccac ctggcggtga acgagagcct gaccgacacc gcccgctg 960
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tggcaaaagg gccagacatc cccatcatca acggcctgag cgacctgtac caccatctc 1080
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tgagctggat cggcgacggc aacaacatcc tgcacagcat catgatgagc gccgccaagt 1200
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agcccaagtt ctgaataagt agatagtgc gtcactggca caacgcgctt cccggaagc 1680
caatcggtga tacacggctc tcatactgca gacagggttc ttctactttg caagatagtc 1740
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SEQ ID NO: 89      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 89
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agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctcc gcactcctct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttccgc 360
tgcttcaagg aaagagcctc ggcatgatct ttgagaagcg ctcaaccagg acccgcttt 420
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tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggccagggtg tacaacacgt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctc gcaagaacac tacagctccc tgaagggctc gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgcccg aaaattcggc atgcattctc 780
aagccgcccac gccgaagggt tacgagcccc acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacgggaac aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
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tctactcgcc acggagcctc gtgttccccg aggcggagaa tagaaagtgg accatcatgg 1140
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ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaatgta 1320
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SEQ ID NO: 90      moltype = RNA length = 1368
FEATURE           Location/Qualifiers
misc_feature       1..1368
                   note = Synthetic construct
source            1..1368
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 90
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gaacgatagc caccatgctt ttcaatctcc gcactcctct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttccgc 360
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tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
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caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctc gcaagaacac tacagctccc tgaagggctc gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgcccg aaaattcggc atgcattctc 780
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acgctaagga gaacgggaac aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaag gtcgctgcct 1020
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ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaatgta 1320
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SEQ ID NO: 91      moltype = RNA length = 1368
FEATURE           Location/Qualifiers

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misc_feature      1..1368
                  note = Synthetic construct
source            1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 91
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gaacgatagc caccatgctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatgggc cggaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
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tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgcg 360
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caattcctat tatcaacggc cttagtgcac tctaccatcc gattcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccggcgc aaaattcggc atgcattctc 780
aagccgcac ccggaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaagga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaaag gtcgctgcct 1020
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tctactcgcc acggagcctc gtgttccccg aggcggagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgtct accgactaca gcccgcagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 92      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source             1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 92
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gaacgatagc caccatgctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatgggc cggaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgcg 360
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caattcctat tatcaacggc cttagtgcac tctaccatcc gattcagatc ctggccgatt 660
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aagccgcac ccggaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
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agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaaag gtcgctgcct 1020
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tctactcgcc acggagcctc gtgttccccg aggcggagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgtct accgactaca gcccgcagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccactaaacc agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccaactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

SEQ ID NO: 93      moltype = RNA length = 1368
FEATURE            Location/Qualifiers
misc_feature       1..1368
                  note = Synthetic construct
source             1..1368
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 93
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agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatgggc cggaacttca gatgtggcca gccgcttcaa aacaagggtcc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgcg 360
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ctactgaaac tgggttcgcg ctgctcgggt gccacccctg cttcctgacg acccaggaca 480
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atgccgtgct ggccagggtg tacaacagct ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtacc tctaccatcc gattcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgccgc aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttccgt gactaagctc gccgagcagt 840
acgctaaggga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttccag ggatatcagg tcaccatgaa aaccgccaaag gtcgctgcct 1020
ccgactggac cttctgcac tgcctgcctc gcaagcctga agaagtggac gacgaggtgt 1080
tctaactcgc acggagcctc gtgttccccg aggcggagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gccgcagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaactaaac agcctcaaga acaccggaat 1260
ggagtctcta agctacataa taccactta cacttacaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 94      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                  note = Synthetic construct
source            1..1371
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 94
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgggtg ttcaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcacaa cttcatgggtc agaaacttcc gctgcgggca acccctacaa aacagggtcc 240
agctcaaggg gcgggacctc ctgaccctga agaacttcac cggcgaagag atcaagtaca 300
tgctgtggct ctcgcgcgac ctgaagttcc gcatcaagca gaaggagag tagctccgc 360
tgctgcaagg gaagtgcgtg gggatgatct tcgagaagcg gtcaaccaga acccggtgt 420
caaccgaaac cgggttcgca ctgctggggg gacaccctgt cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgcgctgct ggcccgctg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tacagctccc tgaagggcct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataatga tgtcagccgc caagttcggga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aaacggcacc aagctcctgc tgaccaacga cccgctggag gccgcacag 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttccag gggtagcagg tcaccatgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttctgcac tgcctgcccc ggaagccgga agaggtggac gacgaggtgt 1080
tctaactcgc gcgctgcgtg gtgttccccg aggcggagaa caggaagtgg accatcatgg 1140
cgggtgatggt agtgcactac accgactact ccgcgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacaccgg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 95      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                  note = Synthetic construct
source            1..1371
                  mol_type = other RNA
                  organism = synthetic construct

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SEQUENCE: 95
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgggtg ttcaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcacaa cttcatgggtc agaaacttcc gctgcgggca acccctacaa aacagggtcc 240
agctcaaggg gcgggacctc ctgaccctga agaacttcac cggcgaagag atcaagtaca 300
tgctgtggct ctcgcgcgac ctgaagttcc gcatcaagca gaaggagag tagctccgc 360
tgctgcaagg gaagtgcgtg gggatgatct tcgagaagcg gtcaaccaga acccggtgt 420
caaccgaaac cgggttcgca ctgctggggg gacaccctgt cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgcgctgct ggcccgctg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tacagctccc tgaagggcct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataatga tgtcagccgc caagttcggga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aaacggcacc aagctcctgc tgaccaacga cccgctggag gccgcacag 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttccag gggtagcagg tcaccatgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttctgcac tgcctgcccc ggaagccgga agaggtggac gacgaggtgt 1080
tctaactcgc gcgctgcgtg gtgttccccg aggcggagaa caggaagtgg accatcatgg 1140
cgggtgatggt cagcctcctg accgactact ccgcgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacaccgg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 96      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                   note = Synthetic construct
source            1..1371
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 96
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatggtg ttaaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcacaa cttcatggtc agaaacttcc gctgcgggca acccctacaa aaccgggtcc 240
agctcaaggg gcgggacctc ctgaccctga agaacttcac cggcgaagag atccggtaca 300
tgctgtggct ctccgccgac ctgaagtacc gcatcaagca gaaggagag tacctcccgc 360
tgctgcaagg gaagtgcgtg gggatgatct tcgagaagcg gtcaaccaga acccggtcgt 420
caaccgaaac cgggttcgca ctgctggggg gacacccgtg cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgcctgtct ggcccgctg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tcagctccc tgaaggccct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataatga tgtcagccgc caagttcgga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aacggcgacc aagctcctgc tgaccaacga cccgctggag gccgcacacg 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttcagg gggtagcagg tcacctgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttcctgcac tgcctgcccc ggaagccgga agagggtggac gacgaggtgt 1080
tctaactcgc gcgctcgtg gtgttcccc aggcggagaa caggaagtgg accatcatgg 1140
cggatgatgt cagctcctg accgactact cgcgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacacccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 97      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                   note = Synthetic construct
source            1..1371
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 97
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatggtg ttaaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcacaa cttcatggtc agaaacttcc gctgcgggca acccctacaa aacaagggtcc 240
agctcaaggg gcgggacctc ctgaccctga agaacttcac cggcgaagag atcaagtaca 300
tgctgtggct ctccgccgac ctgaagtacc gcatcaagca gaaggagag tacctcccgc 360
tgctgcaagg gaagtgcgtg gggatgatct tcgagaagcg gtcaaccaga acccggtcgt 420
caaccgaaac cgggttcgca ctgctggggg gacacccgtg cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgcctgtct ggcccgctg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tcagctccc tgaaggccct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataatga tgtcagccgc caagttcgga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aacggcgacc aagctcctgc tgaccaacga cccgctggag gccgcacacg 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttcagg gggtagcagg tcacctgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttcctgcac tgcctgcccc ggaagccgga agagggtggac gacgaggtgt 1080
tctaactcgc gcgctcgtg gtgttcccc aggcggagaa caggaagtgg accatcatgg 1140
cggatgatgt cagctcctg accgactact cgcgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacacccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 98      moltype = RNA length = 1471
FEATURE           Location/Qualifiers
misc_feature       1..1471
                   note = Synthetic construct
source            1..1471
                   mol_type = other RNA
                   organism = synthetic construct

SEQUENCE: 98
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatggtg gtaaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcacaa cttcatggtc agaaacttcc gctgcgggca acccctacaa aaccgggtcc 240
agctcaaggg gcgggacctc ctgaccctga agaacttcac cggcgaagag atcaagtaca 300
tgctgtggct ctccgccgac ctgaagtacc gcatcaagca gaaggagag tacctcccgc 360
tgctgcaagg gaagtgcgtg gggatgatct tcgagaagcg gtcaaccaga acccggtcgt 420

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caaccgaaac cgggttcgca ctgctggggg gacacccgtg cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgccgtgct ggcccgcgtg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tacagctccc tgaaggccct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataaatga tgtcagccgc caagttcgga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aaacggcacc aagctcctgc tgaccaacga cccgctggag gccgcacacg 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttcagg ggggtaccagg tcaccatgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttcctgcac tgctgcccc ggaagccgga agagggtgag gacgaggtgt 1080
tctactcgcc gcgctcgctg gtgttcccc aggcggagaa caggaagtgg accatcatgg 1140
cggatgatgt cagcctcctg accgactact cgccgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaaccaccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtgttct tcacattcta gaaaaaaaaa 1380
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1440
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa a 1471

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SEQ ID NO: 99      moltype = RNA length = 1371
FEATURE           Location/Qualifiers
misc_feature       1..1371
                   note = Synthetic construct
source            1..1371
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 99
tcaacacaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaattttt caccatttac 120
gaacgatagc caccatgctg gtcaacctcc gcatcctcct caacaacgcc gcattcagaa 180
acgggcaacaa cttcatggct agaaacttcc gctgcgggca accctacaa aaccgggtcc 240
agctcaaggg cggggacctc ctgacctga agaacttcac cggcgaagag atccggtaca 300
tgctgtggct ctcgcccagc ctgaagttcc gcatcaagca gaaggagag tacctcccgc 360
tgctgcaagg gaagtgcctg gggatgatct tcgagaagcg gtcaaccaga acccggtgtg 420
caaccgaaac cgggttcgca ctgctggggg gacacccgtg cttcctgacc acccaagaca 480
tccacctggg agtgaacgaa tcgctgaccg acaccgcccg cgtgctgagc tcaatggcgg 540
acgccgtgct ggcccgcgtg tacaagcagt ccgacctgga caccctggcc aaggaagcgt 600
ccatcccgat catcaacgga ctgtccgacc tgtaccaccc gatccagatc ctggcagact 660
acctgacctt gcaagaacac tacagctccc tgaaggccct gacctgtca tggatcgggg 720
acgggaacaa catctgcac tcataaatga tgtcagccgc caagttcgga atgcacctcc 780
aagccgcaac cccgaagggc tacgaaccgg acgcatcagt gaccaaactg gccgagcagt 840
acgccaagga aaacggcacc aagctcctgc tgaccaacga cccgctggag gccgcacacg 900
gggggaacgt gctgatcacc gacacctgga tctccatggg acaggaggag gaaaagaaga 960
agcggctgca ggcgttcagg ggggtaccagg tcaccatgaa aaccgcgaag gtcgcggcat 1020
cagactggac cttcctgcac tgctgcccc ggaagccgga agagggtgag gacgaggtgt 1080
tctactcgcc gcgctcgctg gtgttcccc aggcggagaa caggaagtgg accatcatgg 1140
cggatgatgt cagcctcctg accgactact cgccgcagct gcagaagccg aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaaccaccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtgttct tcacattcta g 1371

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SEQ ID NO: 100    moltype = RNA length = 1471
FEATURE           Location/Qualifiers
misc_feature       1..1471
                   note = Synthetic construct
source            1..1471
                   mol_type = other RNA
                   organism = synthetic construct

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SEQUENCE: 100
tcaacacaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaattttt caccatttac 120
gaacgatagc caccatgctg gtcaacctgc gcatcctcct gaacaacgcc gccttccgca 180
acggccacaa cttcatggct cgcaacttcc gctgcggcca gccctgcag aacaagggtc 240
agctgaaggg ccgagacctg ctgacctga agaacttcac cggcgaagag atcaagtaca 300
tgctgtggct gagcgcggac ctgaagttcc gcatcaagca gaaggagag tacctgcccc 360
tgctgcaggg caagagcctg ggcgatgatct tcgagaagcg cagcaccgcc acccgctga 420
gcaccgagac aggtcttcgcc ctgctggggc gccacccttg cttcctgacc acccaggaca 480
tccacctggg cgtgaacgag acgctgaccg acaccgcccg cgtgctgagc agcatggcgg 540
acgccgtgct ggcccgcgtg tacaagcaga gcgacctgga caccctggcc aaggaggcca 600
gcatccccat catcaacggc ctgagcgacc tgtaccaccc catccagatc ctggcagact 660
acctgacctt cgaggagcac tacagcagc tgaaggccct gacctgagc tggatcgggc 720
acggcaacaa catctgcac agcatcatga tgagcgccgc caagttcggc atgcacctgc 780
agggcgccac ccccaagggc tacgagcccg acgcccagct gaccaagctg gccgagcagt 840
acgccaagga gaaacggcacc aagctgctgc tgaccaacga cccctggag gccgcccacg 900
ggggcaacgt gctgatcacc gacacctgga tcagcatggg ccaggaggag gagaagaaga 960
agcgcctgca ggccttcagg ggctaccagg tgaccatgaa gaccgccaag gtggccgcca 1020
gcgactggac cttcctgcac tgctgcccc gcaagccgga ggagggtgag gacgaggtgt 1080
tctacagccc ccgagcctg gtgttcccc aggcggagaa ccgcaagtgg accatcatgg 1140
ccgtgatgtg gagcctgctg accgactaca gccccagct gcagaagccc aagttctgat 1200

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aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacacccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt cccccaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta gaaaaaaaaa 1380
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1440
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa a a 1471

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SEQ ID NO: 101      moltype = RNA length = 1371
FEATURE             Location/Qualifiers
misc_feature        1..1371
                     note = Synthetic construct
source              1..1371
                     mol_type = other RNA
                     organism = synthetic construct

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SEQUENCE: 101
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaagcaaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctg gtaaacctgc gcatcctgct gaacaacgcc gccttcgcga 180
acggccacaa cttcatgggt cgcaacttcc gctgcggcca gcccttcgag aaccgggtgc 240
agctgaaggg ccgagcctgc ctgaccctga agaacttcac cggcgaggag atcaagtaca 300
tgctgtggct gagcgccgac ctgaagttcc gcatcaagca gaaggcgag tacctgcccc 360
tgctgcaggg caagagcctg ggcattgatc tcgagaagcg cagcaccgcc acccgctga 420
gcaccgagac aggtttcgcc ctgctgggcg gccaccctcg cttctgacc acccaggaca 480
tccacctggg cgtgaacgag agcctgaccg acaccgcccg cgtgctgagc agcatggccg 540
acggcgtgct ggcccgctg tacaagcaga gcgacctgga caccctggcc aaggaggcca 600
gcattcccat catcaacggc ctgagcgacc tgtaccaccc catccagatc ctggccgact 660
acctgacctt gcaggagcac tacagcagcc tgaaggccct gacctgagc tggatcgccg 720
acggcaacaa catctgcac agcatcatga tgagcgccgc caagttcggc atgcacctgc 780
agggcgccac cccaaggggc tacgagcccg acgacagcgt gaccaagctg gccgagcagt 840
acggcaagga gaacgggacc aagctgtctg tgaccaacga ccccttgagg gccgcccacg 900
ggcgcaacgt gctgatcacc gacacctgga tcagcatggg ccaggaggag gagaagaaga 960
agcgctgca ggccttcagg ggctaccagg tgacctgaa gaccgccaag gtggccgcca 1020
gcgactggac tgcctgcacc tgctgcctcc gcaagcccgga ggagggtgac gacgagggtg 1080
tctacagccc ccgagcctg gtgttccccg agggcgagaa ccgcaagtgg accatcatgg 1140
ccgtgatggt gagcctgctg accgactaca gccccagct gcagaagccc aagttctgat 1200
aactcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaacacccg 1260
aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt cccccaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 102      moltype = RNA length = 1368
FEATURE             Location/Qualifiers
misc_feature        1..1368
                     note = Synthetic construct
source              1..1368
                     mol_type = other RNA
                     organism = synthetic construct

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SEQUENCE: 102
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaagcaaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgctt gtaaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatgggt cggaacttca gatgtggcca gccgcttcaa aacaagggtc 240
agctgaaggg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccgaggac ttgaagttcc gcattaagca gaagggggaa taccttcgcg 360
tgcttcaagg aaagagcctc ggcatgatct ttgagaagcg ctcaaccagg acccgctttt 420
ctactgaaac tgggttcgag ctgctcggtg gccaccctcg cttctgacc acccaggaca 480
tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgcctgtct ggccagggtg taaaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctt gcaagaacac tacagctccc tgaagggtct gacattgttc tggatcgccg 720
acggcaacaa cattctccat tccatcatga tgtccgcccg aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccagggaagaa gagaanaaga 960
agcggtctga ggcgttcagg ggatatcagg tcacctgaa aaccgccaaag gtcgtgctct 1020
ccgactggac cttctgcac tgctgcctc gcaagcctga agaagtgagc gacgagggtg 1080
tctactgcc acggagcctc gtgttccccg agggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgac accgactaca gcccgagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactag gatctgggta ccaataaacc agcctcaaga acaccgaat 1260
ggagtctcta agctacataa taccactta cacttaaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 103      moltype = RNA length = 1368
FEATURE             Location/Qualifiers
misc_feature        1..1368
                     note = Synthetic construct
source              1..1368
                     mol_type = other RNA
                     organism = synthetic construct

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SEQUENCE: 103

```

tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgctt gtcattctcc gcattcctct taacaacgcc gcgttttagaa 180
acggccacaaa cttcatgggc cggaacttca gatgtggcca gccgcttcaa aaccgggtcc 240
agctgaaggcg ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagcctc ggcatgatct ttgagaagcg ctcaaccagg acccgcttt 420
ctactgaaac tgggttcgcg ctgctcgggt gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggccagggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctc gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa catttccat tccatcatga tgtccgcccg aaaattcggc atgcatcttc 780
aagccgccac gccgaagggt tacgagcccg acgcttccgt gactaagctc gccgagcagt 840
acgctaagga gaacggaac aagcttctgc tgactaagca cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggggttcag ggatatcagg tccacctgaa aaccgccaa gtcgtgcct 1020
ccgactggac cttcctgcac gtgctgctc gcaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc accgagcctc tgtttcccg aggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgagct tcagaagccc aagttctagc 1200
tcgagctagt gactgactga gatctgggta ccactaaacc agcctcaaga accccgaat 1260
ggagtctcta agctacataa taccacatta cacttcaaaa atgttgtccc ccaaaatgta 1320
gccattcgta tctgctccta ataaaaagaa agtttcttca cattctag 1368

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SEQ ID NO: 104

FEATURE

misc_feature

source

moltype = RNA length = 1371

Location/Qualifiers

1..1371

note = Synthetic construct

1..1371

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 104

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tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgggc cttgtcaatc tccgcatcct ccttaacaac gccgctttaa 180
gaaacggcca caacttcatg gtcgggaact tcagatgtgg ccagccgctt caaaaacaagg 240
tccagctgaa gggccgggat cttctgaccc tgaagaactt tactggcgaa gagatcaagt 300
acatgctctg gctctccgcg gacttgaagt tccgcattaa gcagaagggg gaataccttc 360
cgctgcttca aggaagagac ctcgccatga tctttgagaa gcgctcaacc aggacccgcc 420
ttttactga aactgggttc gcgctgctcg gtggccaccc ctgcttctcg acgacccagg 480
acatccacct cggagtgaac gaatccctca ccgataccgc ccgggtgtta tcgagcatgg 540
cagatgccgt tctggccagg gtgtacaaac agtccgatct ggacactctg gccaggagg 600
cgtcaattcc tattatcaac ggcttagtg acctctacca tccgattcag atcctggcg 660
attacctcac cctgcaagaa cactacagct ccctgaaggg tctgacattg tcttgatcg 720
gcgacggcaa caactcttc cattccatca tgatgtccgc cgcaaaattc ggcatgcac 780
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agtacgctaa ggagaacgga accaagcttc tgcgtactaa cgacccacta gaagcagccc 900
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tggttctact gccacggagc ctgctgttcc ccgagggcca gaatagaaag tggacatca 1140
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aatggagctc ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta g 1371

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SEQ ID NO: 105

FEATURE

misc_feature

source

moltype = RNA length = 1485

Location/Qualifiers

1..1485

note = Synthetic construct

1..1485

mol_type = other RNA

organism = synthetic construct

SEQUENCE: 105

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tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgggc cttgtcaatc tccgcatcct ccttaacaac gccgctttaa 180
gaaacggcca caacttcatg gtcgggaact tcagatgtgg ccagccgctt caaaaaccggg 240
tccagctgaa gggccgggat cttctgaccc tgaagaactt tactggcgaa gagatcaagt 300
acatgctctg gctctccgcg gacttgaagt tccgcattaa gcagaagggg gaataccttc 360
cgctgcttca aggaagagac ctcgccatga tctttgagaa gcgctcaacc aggacccgcc 420
ttttactga aactgggttc gcgctgctcg gtggccaccc ctgcttctcg acgacccagg 480
acatccacct cggagtgaac gaatccctca ccgataccgc ccgggtgtta tcgagcatgg 540
cagatgccgt tctggccagg gtgtacaaac agtccgatct ggacactctg gccaggagg 600
cgtcaattcc tattatcaac ggcttagtg acctctacca tccgattcag atcctggcg 660
attacctcac cctgcaagaa cactacagct ccctgaaggg tctgacattg tcttgatcg 720
gcgacggcaa caactcttc cattccatca tgatgtccgc cgcaaaattc ggcatgcac 780
ttcaagccgc cagccgaag ggttacgagc ccgacgcttc cgtgactaag ctgcgcgagc 840

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agtagcgctaa ggagaacgga accaagcttc tgctgactaa cgaccacta gaagcagccc 900
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agaagcggct gcaggcgctt cagggatatc aggtcaccat gaaaaccgcc aaggtcgctg 1020
cctccgactg gaccttctcg cactgcctgc ctgcgaagcc tgaagaagtg gacgacgagg 1080
tggttctact gccacggagc ctggtgttcc ccgaggccga gaatagaaag tggaccatca 1140
tggcgtgat ggtgtcactg ctcaccgact acagcccgca gcttcagaag cccaagtctt 1200
agctcgagct agtgactgac taggatctgg ttaccactaa accagcctca agaaccaccg 1260
aatggagtct ctaagctaca taataccaac ttacacttac aaaatgttgt ccccaaaaat 1320
gtagccattc gtatctgctc ctaataaaaa gaaagtttct tcacattcta gaaaaaaaaa 1380
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1440
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaa 1485

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SEQ ID NO: 106      moltype = RNA length = 1374
FEATURE            Location/Qualifiers
misc_feature        1..1374
                    note = Synthetic construct
source              1..1374
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 106
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaattttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc caccatgggc ggacttgtca atctccgcat cctccttaac aacgcgcgct 180
tagaaaacgg ccacaacttc atgggtccgga acttcagatg tggccagccg cttcaaaaaca 240
aggtccagct gaagggccgg gatcttctga ccctgaagaa ctttactggc gaagagatca 300
agtacatgct ctggctctcc gcggacttga agttccgcat taagcagaag ggggaatacc 360
ttccgctgct tcaaggaag agcctcgga tgatctttga gaagcgctca accaggaccc 420
gcctttctac tgaactggg ttccgctgct cgggtggcca cccctgttc ctgacgaccc 480
aggacatcca cctcgagtg aacgaatccc tcaccgatac cgcccggtg ttatcgagca 540
tggcagatgc cgtgctggcc aggggtgtaca aacagtccga tctggacact ctggccaagg 600
agggctcaat tcctattatc aacggcctta gtgacctcta ccacgcgatt cagatcctgg 660
ccgattacct caccctgcaa gaacactaca gctccctgaa ggggtgtgaca ttgtcctgga 720
tcggcgacgc caacaacatt ctccattcca tcatgatgtc cgccgcaaaa ttcggcatgc 780
atcttcaagc cgccacgcgc aaggggttac agcccgacgc ttccgtgact aagctcgccg 840
agcagtacgc taaggagaac ggaaccaagc ttctgctgac taacgaccca ctagaagcag 900
cccacggggg caacgtgctt attactgaca cctggatctc catgggccag gaagaagaga 960
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aggtgttcta ctgccacgg agcctcggtg tcccgaggc cgagaataga aagtggacca 1140
tcatggccgt gatgggtgtc ctgctcaccg actacagccc gcagcttcag aagcccaagt 1200
tctagctcga gctagtgaat gactaggatc tggttaccac taaccagccc tcaagaacac 1260
ccgaatggag tctctaagct acataatacc aacttacact tacaaaatgt tgtcccccaa 1320
aatgtagcca ttctgtatct ctcttaataa aaagaaagtt tcttcacatt ctag 1374

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SEQ ID NO: 107      moltype = RNA length = 1377
FEATURE            Location/Qualifiers
misc_feature        1..1377
                    note = Synthetic construct
source              1..1377
                    mol_type = other RNA
                    organism = synthetic construct

```

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SEQUENCE: 107
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaattttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc catggccctt ttcaatctcc gcatctctct taacaacgcc gcgtttagaa 180
acggccacaa cttcatggtc cggaacttca gatgtggcca gccgcttcaa ggcaaggctc 240
agctgaaggc ccgggatctt ctgaccctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccgggac ttgaagttcc gcattaagca gaagggggaa taccttccgc 360
tgcttcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgccctt 420
ctactgaaac tactgttcgc ctgctcggtg gccaccctcg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tcctcaccg ataccgcccc ggtgttatcg agcatggcag 540
atgcctgctc ggcagggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgcccg aaaattcggc atgcattctc 780
aagccgcac gccgaagggt tacgagcccc acgcttccgt gactaagctc gccgagcagt 840
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ggggcaacgt gcttattact gacacctgga tctccatggg ccagggaagaa gagaaaaaga 960
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ccgactggac cttcctgcac tgccctgcct gcaagcctga agaagtggac gacgaggtgt 1080
tctactgcgc acggagcctc gtgttccccg agggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgacgct tcagaagccc aagttctaga 1200
taagtgaact cgagctagtg actgactagg atctggttac cactaaacca gcctcaagaa 1260
caccgaatg gagtctctaa gctacataat accaacttac acttacaata tgttgtcccc 1320
caaatgttag ccattcgtat ctgctcctaa taaaagaaa gtttcttcac attctag 1377

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SEQ ID NO: 108      moltype = RNA length = 1377
FEATURE            Location/Qualifiers
misc_feature        1..1377
                    note = Synthetic construct
source              1..1377
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 108
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc catggccctt ttcaatctcc gcatcctcct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaaattca gatgtggcca gccgcttcaa ggccgggtcc 240
agctgaaggc cgggatctct ctgacctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgcttt 420
ctactgaaac tgggttcgcg ctgctcggtg gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcagggtg tacaacacgt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccggcgc aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaagga gaacgggaac aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttcag ggatatcagg tcacatgaa aaccgccaaag gtcgctgcct 1020
cggactggac cttcctgcac tgctgcctc gcaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc acggagcctc gtgttccccg aggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgagctc tcagaagccc aagttctaga 1200
taagtgaact cgagctagtg actgactagg atctggttac cactaaacca gcctcaagaa 1260
caccgcaatg gagtctctaa gctacataat accaacttac acttacaaaa tgttgtcccc 1320
caaaatgtag ccattcgtat ctgctcctaa taaaagaaa gtttcttcac attctag 1377

SEQ ID NO: 109      moltype = RNA length = 1377
FEATURE            Location/Qualifiers
misc_feature        1..1377
                    note = Synthetic construct
source              1..1377
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 109
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc catggccctt ttcaatctcc gcatcctcct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaaattca gatgtggcca gccgcttcaa ggccgggtcc 240
agctgaaggc cgggatctct ctgacctga agaactttac tggcgaagag atcaggtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgcttt 420
ctactgaaac tgggttcgcg ctgctcggtg gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tccctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcagggtg tacaacacgt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcaccct gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccggcgc aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaagga gaacgggaac aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttcag ggatatcagg tcacatgaa aaccgccaaag gtcgctgcct 1020
cggactggac cttcctgcac tgctgcctc gcaagcctga agaagtggac gacgaggtgt 1080
tctactcgcc acggagcctc gtgttccccg aggcgagaa tagaaagtgg accatcatgg 1140
ccgtgatggt gtcactgctc accgactaca gcccgagctc tcagaagccc aagttctaga 1200
taagtgaact cgagctagtg actgactagg atctggttac cactaaacca gcctcaagaa 1260
caccgcaatg gagtctctaa gctacataat accaacttac acttacaaaa tgttgtcccc 1320
caaaatgtag ccattcgtat ctgctcctaa taaaagaaa gtttcttcac attctag 1377

SEQ ID NO: 110      moltype = RNA length = 1377
FEATURE            Location/Qualifiers
misc_feature        1..1377
                    note = Synthetic construct
source              1..1377
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 110
tcaacacaaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcattttctt ttaaagcaaa agcaattttc tgaaaatttt caccattttac 120
gaacgatagc catggccctt gtcaatctcc gcatcctcct taacaacgcc gcgttttagaa 180
acggccacaa cttcatggtc cggaaattca gatgtggcca gccgcttcaa ggccgggtcc 240
agctgaaggc cgggatctct ctgacctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360

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tgcctcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgccctt 420
ctactgaaac tgggttcgcg ctgctcggtg gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tcctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcaggggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctt gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgccg aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaaggga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttcag ggatatcagg tcaccatgaa aaccgccaaag gtcgtgcct 1020
cggactggac cttcctgcac tgctcgctc gcaagcctga agaagtggac gacgaggtgt 1080
tctaactgcc acggagcctc gtgttcccc aggccgagaa tagaaagtgg accatcatgg 1140
cgtgatgggt cgtactgtc accgactaca gcccgagct tcagaagccc aagttctaga 1200
tgaatagact cgaactagtg actgactagg atctgggtac cactaaacca gcctcaagaa 1260
caccgcaatg gactctctaa gctacataat accaacttac acttacaaaa tgttgtcccc 1320
caaaatgtag ccattcgtat ctgctcctaa taaaaagaaa gtttcttcac attctag 1377

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SEQ ID NO: 111      moltype = RNA length = 1496
FEATURE            Location/Qualifiers
misc_feature        1..1496
                    note = Synthetic construct
source              1..1496
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 111
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc catggccctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatggtc cggaaacttca gatgtggcca gccgcttcaa gtcaagggtc 240
agctgaaggg ccgggatctt ctgacctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgccctt 420
ctactgaaac tgggttcgcg ctgctcggtg gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tcctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcaggggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctt gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgccg aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaaggga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttcag ggatatcagg tcaccatgaa aaccgccaaag gtcgtgcct 1020
cggactggac cttcctgcac tgctcgctc gcaagcctga agaagtggac gacgaggtgt 1080
tctaactgcc acggagcctc gtgttcccc aggccgagaa tagaaagtgg accatcatgg 1140
cgtgatgggt gtcactgtc accgactaca gcccgagct tcagaagccc aagttctaga 1200
taagtgaact cgaactagtg actgactagg atctgggtac cactaaacca gcctcaagaa 1260
caccgcaatg gactctctaa gctacataat accaacttac acttacaaaa tgttgtcccc 1320
caaaatgtag ccattcgtat ctgctcctaa taaaaagaaa gtttcttcac attctagaaa 1380
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1440
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaa 1496

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SEQ ID NO: 112      moltype = RNA length = 1377
FEATURE            Location/Qualifiers
misc_feature        1..1377
                    note = Synthetic construct
source              1..1377
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 112
tcaacacaa atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc catggccctt ttcaatctcc gcatcctcct taacaacgcc gcgtttagaa 180
acggccacaa cttcatggtc cggaaacttca gatgtggcca gccgcttcaa gtcaagggtc 240
agctgaaggg ccgggatctt ctgacctga agaactttac tggcgaagag atcaagtaca 300
tgctctggct ctccggcgac ttgaagttcc gcattaagca gaagggggaa taccttcgc 360
tgcttcaagg aaagagcctc ggcgatgatc ttgagaagcg ctcaaccagg acccgccctt 420
ctactgaaac tgggttcgcg ctgctcggtg gccacccctg cttcctgacg acccaggaca 480
tccacctcgg agtgaacgaa tcctcaccg ataccgcccg ggtgttatcg agcatggcag 540
atgccgtgct ggcaggggtg tacaaacagt ccgatctgga cactctggcc aaggaggcgt 600
caattcctat tatcaacggc cttagtgaac tctaccatcc gattcagatc ctggccgatt 660
acctcacctt gcaagaacac tacagctccc tgaagggtct gacattgtcc tggatcggcg 720
acggcaacaa cattctccat tccatcatga tgtccgccg aaaattcggc atgcattctc 780
aagccgccac gccgaagggt tacgagcccg acgcttcctg gactaagctc gccgagcagt 840
acgctaaggga gaacgggaacc aagcttctgc tgactaacga cccactagaa gcagcccacg 900
ggggcaacgt gcttattact gacacctgga tctccatggg ccaggaagaa gagaaaaaga 960
agcggctgca ggcgttcag ggatatcagg tcaccatgaa aaccgccaaag gtcgtgcct 1020
cggactggac cttcctgcac tgctcgctc gcaagcctga agaagtggac gacgaggtgt 1080
tctaactgcc acggagcctc gtgttcccc aggccgagaa tagaaagtgg accatcatgg 1140

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ccgtgatggt gtcactgtct accgactaca gcccgagct tcagaagccc aagttctaga 1200
taagtgaact cgagctagtg actgactagg atctggttac cactaaacca gctcaagaa 1260
caccogaatg gagtctctaa gctacataat accaacttac acttacaana tgttgtcccc 1320
caaatgtag ccattcgtag ctgctcctaa taaaagaaa gtttcttcac attctag 1377

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SEQ ID NO: 113      moltype = RNA length = 1282
FEATURE
misc_feature        Location/Qualifiers
                    1..1282
                    note = Synthetic construct
source              1..1282
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 113
ggcagaaaaa tttgtacat tgtttcacia acttcaaata ttattcattt atttagatct 60
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcat cctgctgaac 120
aacgccgctt cccgcaacgg ccacaacttc atggtgcgca acttcgctg cgccagccc 180
ctgcagaaac aggtgcagct gaaggggcgc gacctgtga cctgaagaa cttcaccggc 240
gaggagatca agtactgtct gtggtgagc gccgacctga agttccgcat caagcagaag 300
ggcgagtacc tgccccgtct gcagggcaag agcctgggca tgatcttcga gaagcgcagc 360
acccgcaccc gctgagcac cgagacaggc ttgcctctgc tgggcggcca cccctgcttc 420
ctgaccaccc aggcacatcca cctgggcgtg aacgagagcc tgaccgacac cgcccgctg 480
ctgagcagca tggccgacgc cgtgctggcc cgctgtgaca agcagagcga cctggacacc 540
ctggccaagg agggcagcat ccccatcatc aacggcctga gcgacctgta ccccccac 600
cagatcctgg ccgactacct gacctgcag gagcactaca gcagcctgaa gggcctgacc 660
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgccgccaag 720
ttcgcatcgc acctgcagcg cgccaccccc aagggtctac agcccgacgc cagcgtgacc 780
aagctggccc agcagtagc caaggagaa ggccaccaag tgctgctgac caacgacccc 840
ctggaggccg ccaacggcgg caacgtgctg atcaccgaca cctggatcag catgggccag 900
gaggaggaga acaagaagcg cctgcaggcc ttccagggtc accaggtgac catgaagacc 960
gccaaggttg ccgcccagca ctggaccttc ctgcactgcc tgccccgcaa gcccgaggag 1020
gtggacgacg aggtgttcta cagccccccc agcctggtgt tccccgaggg cgagaaccgc 1080
aagtggacca tcatggccgt gatggtgagc actacagccc ccagctgcag 1140
aagcccaagt tctgaggtct ctgaaatga gctggagcct cggtagccgt tctctctgcc 1200
cgctgggcct cccaacgggc cctcctcccc tccttgaccc ggcccttctt ggtctttgaa 1260
taaaagtctga gtgggcatct ag 1282

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SEQ ID NO: 114      moltype = RNA length = 1285
FEATURE
misc_feature        Location/Qualifiers
                    1..1285
                    note = Synthetic construct
source              1..1285
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 114
ggcagaaaaa tttgtacat tgtttcacia acttcaaata ttattcattt atttagatct 60
attattacat caaaacaaaa agccgccacc atgggagtag tcaacctgcg catcctgctg 120
aacaacgccc ccttcgcaa cgccacacac ttcattggtg gcaacttcgg ctgcggccag 180
cccctgcaga acaaggtgca gctgaagggc cgcgacctgc tgacctgaa gaacttcacc 240
ggcgaggaga tcaagtacat gctgtggctg agcgccgacc tgaagttccg catcaagcag 300
aaggcgagtg acctgcccc gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 360
agcaccgcga cccgctgag caccgagaca ggcttcgccc tgctggcgcg ccacccctgc 420
ttcctgacca cccaggacat ccacctgggc gtgaacgaga gacctgacga caccgcccgc 480
gtgctgagca gcatggccga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 540
accctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 600
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 660
accctgagct ggttcggcga cggcaacaac atcctgcaca gcatcatgat gagcgcggcc 720
aagttcgcca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 780
accaagctgg ccgaccagta cggcaaggag aacggcacca agctgctgct gaccaacgac 840
cccctggagg ccgcccacgg cggcaacgtg ctgacaccg acacctggat cagcatgggc 900
caggaggagg agaagaagaa gcgcctgcag gccttcagg gctaccaggt gacctgaag 960
accgccaagg tggccgccag cgactggacc ttctgacct gcctgcccc caagcccgag 1020
gaggtggacg acgaggtgtt ctacagcccc cgcagcctgg tgttccccga ggccgagaa 1080
cgcaagtgga ccatcatggc cgtgatggtg agcctgtgta ccgactacag ccccgagctg 1140
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttctctct 1200
gcccgtggg cctcccaacg ggccctctc cctccttgc accggccctt cctggtcttt 1260
gaataaagtc tgagtgggca tctag 1285

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SEQ ID NO: 115      moltype = RNA length = 1285
FEATURE
misc_feature        Location/Qualifiers
                    1..1285
                    note = Synthetic construct
source              1..1285
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 115
ggcagaaaaa tttgtacat tgtttcacia acttcaaata ttattcattt atttagatct 60
attattacat caaaacaaaa agccgccacc atgggagtag tcaacctgcg catcctgctg 120
aacaacgccc ccttcgcaa cgccacacac ttcattggtg gcaacttcgg ctgcggccag 180
cccctgcaga accgggtgca gctgaagggc cgcgacctgc tgacctgaa gaacttcacc 240

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ggcgaggaga tccggtacat gctgtggtg agcgccgacc tgaagttccg catcaagcag 300
aagggcgagt acctgcccct gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 360
agcaccgcga cccgctgtag caccgagaca ggcttcgccc tgctgggcgg ccacccttgc 420
ttctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 480
gtgctgagca gcatggccga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 540
acctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 600
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 660
acctgagct ggatcggcga cggcaacaac atcctgcaca gcatcatgat gagcgccgcc 720
aagttcggca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 780
accaagctgg ccgagcagta cggcaaggag aacggcacca agctgctgct gaccaacgac 840
cccctggagg ccgcccacgg cggcaacgtg ctgatcaccg acacctggat cagcatgggc 900
caggaggagg agaagaagaa gcgcctgcag gccttcagg gctaccaggt gacctgaag 960
accgccaagg tggccgccag cgaactggacc ttctgcaact gcctgccccg caagcccag 1020
gaggtggacg acgaggtgtt ctacagcccc cgcagcctgg tgttccccga ggccgagaa 1080
cgcaagtgga ccatcatggc cgtgatgggt agcctgctga ccgactacag cccccagctg 1140
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttctctct 1200
gcccgtggg cctcccaacg ggccctcctc cctccttgc accggccctt cctgggtctt 1260
gaataaagtc tgagtgggca tctag 1285

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SEQ ID NO: 116      moltype = RNA length = 1285
FEATURE            Location/Qualifiers
misc_feature        1..1285
                    note = Synthetic construct
source              1..1285
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 116
ggcagaaaaa ttgtctacat tgtttcacia acttcaaata ttattcattt atttagatct 60
attattacat caaaacaaaa agccgccacc atgctgggtat tcaacctgcg catcctgctg 120
aacaacgcgc ccttcgcgaa cggccacaac ttcattgggtc gcaacttccg ctgcccgcag 180
ccctgcaga accgggtgca gctgaagggc cgcgacctgc tgacctgaa gaacttcacc 240
ggcgaggaga tccggtacat gctgtggctg agcgccgacc tgaagttccg catcaagcag 300
aagggcgagt acctgcccct gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 360
agcaccgcga cccgctgtag caccgagaca ggcttcgccc tgctgggcgg ccacccttgc 420
ttctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 480
gtgctgagca gcatggccga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 540
acctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 600
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 660
acctgagct ggatcggcga cggcaacaac atcctgcaca gcatcatgat gagcgccgcc 720
aagttcggca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 780
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caggaggagg tgaagaagaa gcgcctgcag gccttcagg gctaccaggt gacctgaag 960
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cgcaagtgga ccatcatggc cgtgatgggt agcctgctga ccgactacag cccccagctg 1140
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttctctct 1200
gcccgtggg cctcccaacg ggccctcctc cctccttgc accggccctt cctgggtctt 1260
gaataaagtc tgagtgggca tctag 1285

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SEQ ID NO: 117      moltype = RNA length = 1225
FEATURE            Location/Qualifiers
misc_feature        1..1225
                    note = Synthetic construct
source              1..1225
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 117
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aacaacgcgc ccttcgcgaa cggccacaac ttcattgggtc gcaacttccg ctgcccgcag 120
ccctgcaga acaaggtgca gctgaagggc cgcgacctgc tgacctgaa gaacttcacc 180
ggcgaggaga tcaagtacat gctgtggctg agcgccgacc tgaagttccg catcaagcag 240
aagggcgagt acctgcccct gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 300
agcaccgcga cccgctgtag caccgagaca ggcttcgccc tgctgggcgg ccacccttgc 360
ttctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 420
gtgctgagca gcatggccga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 480
acctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 540
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 600
acctgagct ggatcggcga cggcaacaac atcctgcaca gcatcatgat gagcgccgcc 660
aagttcggca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 720
accaagctgg ccgagcagta cggcaaggag aacggcacca agctgctgct gaccaacgac 780
cccctggagg ccgcccacgg cggcaacgtg ctgatcaccg acacctggat cagcatgggc 840
caggaggagg agaagaagaa gcgcctgcag gccttcagg gctaccaggt gacctgaag 900
accgccaagg tggccgccag cgaactggacc ttctgcaact gcctgccccg caagcccag 960
gaggtggacg acgaggtgtt ctacagcccc cgcagcctgg tgttccccga ggccgagaa 1020
cgcaagtgga ccatcatggc cgtgatgggt agcctgctga ccgactacag cccccagctg 1080
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttctctct 1140
gcccgtggg cctcccaacg ggccctcctc cctccttgc accggccctt cctgggtctt 1200
gaataaagtc tgagtgggca tctag 1225

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SEQ ID NO: 118          moltype = RNA length = 1225
FEATURE                Location/Qualifiers
misc_feature            1..1225
                        note = Synthetic construct
source                 1..1225
                        mol_type = other RNA
                        organism = synthetic construct

SEQUENCE: 118
attattacat caaaacaaaa agccgccacc atgggagtat tcaacctgcg catcctgctg 60
aacaacgcgc ccttcgcgaa cgccacacaac ttcattgggtgc gcaacttcgc ctgcggccag 120
cccttcgaga accgggtgca gctgaagggc cgcgacctgc tgacctcgaa gaacttcacc 180
ggcgaggaga tccggtacat gctgtggctg agcgccgacc tgaagttccg catcaagcag 240
aagggcgagt acctgccctt gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 300
agcaccgcga ccgcctgag caccgagaca ggcttcgccc tgctgggcgg ccaccctgc 360
ttcctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 420
gtgctgagca gcatggcgga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 480
accctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 540
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 600
accctgagct ggatcgcgga cggaacacaac atcctgcaca gcatcatgat gagcgccgcc 660
aagttcggca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 720
accaagctgg ccgagcagta cgccaaggag aacggcacca agctgctgct gaccaacgac 780
cccttggaag ccgcccacgg cggaacgtg ctgatacccg acacctggat cagcatgggc 840
caggaggagg agaagaagaa ggcctgcag gccttcagg gctaccagg gaccatgaag 900
accgccaagg tggccgcgac cgactggacc ttcctgact gcctgcccgc caagcccgag 960
gaggtggacg acgaggtgtt ctacagcccc cgcagcctgg tgttcccga ggccgagaac 1020
cgcaagtggg ccatcatggc cgtgatgggt agcctgtgta ccgactacag cccccagctg 1080
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttcctcct 1140
gcccgtggg gccccaacg ggcctcctc cctccttgc accggccctt cctgggtctt 1200
gaataaagtc tgagtgggca tctag 1225

SEQ ID NO: 119          moltype = RNA length = 1228
FEATURE                Location/Qualifiers
misc_feature            1..1228
                        note = Synthetic construct
source                 1..1228
                        mol_type = other RNA
                        organism = synthetic construct

SEQUENCE: 119
attattacat caaaacaaaa agccgccacc atgctgggat tcaacctgcg catcctgctg 60
aacaacgcgc ccttcgcgaa cgccacacaac ttcattgggtgc gcaacttcgc ctgcggccag 120
cccttcgaga accgggtgca gctgaagggc cgcgacctgc tgacctcgaa gaacttcacc 180
ggcgaggaga tccggtacat gctgtggctg agcgccgacc tgaagttccg catcaagcag 240
aagggcgagt acctgccctt gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 300
agcaccgcga ccgcctgag caccgagaca ggcttcgccc tgctgggcgg ccaccctgc 360
ttcctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 420
gtgctgagca gcatggcgga cgccgtgctg gcccgctgtg acaagcagag cgacctggac 480
accctggcca aggaggccag catcccatc atcaacggcc tgagcgacct gtaccacccc 540
atccagatcc tggccgacta cctgacctg caggagcact acagcagcct gaagggcctg 600
accctgagct ggatcgcgga cggaacacaac atcctgcaca gcatcatgat gagcgccgcc 660
aagttcggca tgcacctgca ggccgccacc cccaagggct acgagccga cgccagcgtg 720
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cccttggaag ccgcccacgg cggaacgtg ctgatacccg acacctggat cagcatgggc 840
caggaggagg agaagaagaa ggcctgcag gccttcagg gctaccagg gaccatgaag 900
accgccaagg tggccgcgac cgactggacc ttcctgact gcctgcccgc caagcccgag 960
gaggtggacg acgaggtgtt ctacagcccc cgcagcctgg tgttcccga ggccgagaac 1020
cgcaagtggg ccatcatggc cgtgatgggt agcctgtgta ccgactacag cccccagctg 1080
cagaagccca agttctgagg tctctagtaa tgagctggag cctcggtagc cgttcctcct 1140
gcccgtggg gccccaacg ggcctcctc cctccttgc accggccctt cctgggtctt 1200
gaataaagtc tgagtgggca gcatctag 1228

SEQ ID NO: 120          moltype = RNA length = 1376
FEATURE                Location/Qualifiers
misc_feature            1..1376
                        note = Synthetic construct
source                 1..1376
                        mol_type = other RNA
                        organism = synthetic construct

SEQUENCE: 120
tcaacacaac atatacaaaa caaacgaatc tcaagcaatc aagcattcta cttctattgc 60
agcaatttaa atcatttctt ttaaaagcaaa agcaattttc tgaaaatttt caccatttac 120
gaacgatagc caccatgttg ttcaacttga ggatcttggt gaacaacgcc gccttcagga 180
acggacacaa cttcatggta aggaacttca ggtgcggaca gcccttgtag aacaaagtac 240
agttgaaagg aagggaactg ttgacattga aaaacttcac aggagaagaa atcaaatata 300
tgttgtgggt gtcggccgac ttgaaattca ggatcaaaa gaaaggagaa tacttgccct 360
tgttgcaggg aaaatcgttg ggaatgatct tcgaaaaaag gtcgacaagg acaaggttgt 420
cgacagaaac aggattcgcc ttgttgggag gacaccctgt cttcttgaca acacaggaca 480
tccacttggg agtaaacgaa tcgttgacag acacagccag ggtattgtcg tcgatggcgc 540
acgccgtatt ggccagggta tacaacacgt cggacttgga cacattggcc aaagaagcct 600

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cgatccccat catcaacgga ttgtcggact tgtaccaccc catccagatc ttggccgact 660
acctgacatt gcaggaacac tactcgtcgt tgaaaggatt gacattgtcg tggatcggag 720
acggaaacaa catcttcac tcgatcatga tgtcggccgc caaattcggga atgcacttgc 780
aggccgcac acccaagga tacgaaccgc acgctcgtt acaaaaattg gccgaacagt 840
acgcaaaaga aaacggaaca aaattgttgt tgacaaacga ccccttgga gccgccacg 900
gaggaaacgt attgatcaca gacacatgga tctcgatggg acaggaagaa gaaaaaaa 960
aaaggttgca ggccttcag ggataccagg taacaatgaa aacagccaaa gtagccgcct 1020
cggactggac attcttcgac tgcttgcga ggaaaccga agaagtagac gacgaagtat 1080
tctactcgcc caggctcgtt gtattcccc aagccgaaaa caggaaatgg acaatcatgg 1140
ccgtaatggt atcgttgttg acagactact cgccccagtt gcagaaacc aaattctgaa 1200
tagtgaactc gagctagtga ctgactagga tctggttacc actaaaccag cctcaagaac 1260
acccgaatgg agtctctaag ctacataata ccaacttaca cttacaaaat gttgtcccc 1320
aaaatgtagc cattcgtatc tgtcctaata aaaaagaaag tttcttcaca ttctag 1376

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SEQ ID NO: 121      moltype = RNA length = 1222
FEATURE
misc_feature       1..1222
                    note = Synthetic construct
source             1..1222
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 121
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcac cctgctgaac 60
aacgccgcct tccgcaacgg ccacaacttc atggtgcgca acttccgctg cggccagccc 120
ctgcagggca aggtgcagct gaagggccgc gacctgtga ccctgaagaa cttaccggc 180
gaggagatca agtacatgct gtggctgagc gccgacctga agttccgcat caagcagaag 240
ggcgagtacc tgccccgtct gcagggcaag agcctgggca tgatcttcga gaagcgcagc 300
acccgcaccc gccctgagcac cgagacaggc ttgcctctgc tgggcggcca cccctgcttc 360
ctgaccaccc aggacatcca cctgggcgtg aacgagagcc tgaccgacac cgcccgcgtg 420
ctgagcagca tggccgacgc cgtgctggcc cgcgtgtaca agcagagcga cctggacacc 480
ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta ccaccccatc 540
cagatcctgg ccgactacct gacctgcag gagcactaca gcagcctgaa gggcctgacc 600
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgcgcgaag 660
ttcgccatgc acctgcaggc cgccaccccc aagggtctac agcccgacgc cagcgtgacc 720
aagctggccg agcagtacgc caaggagaac ggcaccaagc tgcgtgtgac caacgacccc 780
ctggaggccg cccacggcgg caacgtgctg atcaccgaca cctggatcag catggggcag 840
gaggaggaga agaagaagcg cctgcaggcc ttccagggct accaggtgac catgaagacc 900
gccaaggtgg ccgcccagcg ctggaccttc ctgcaactgc tgccccgcaa gcccgaggag 960
gtggacgacg aggtgttcta cagccccccg agcctggtgt tccccgaggc cgagaaccgc 1020
aagtggacca tctggccgt gatggtgagc ctgctgaccg actacagccc ccagctgcag 1080
aagcccaagt tctgaggtct ctagtaatga gctggagcct cggtagccgt tctctctgcc 1140
cgctgggcct cccaacgggc cctcctcccc tccttgcaac ggcccttctt ggtctttgaa 1200
taaagtctga gtgggcatct ag 1222

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SEQ ID NO: 122      moltype = RNA length = 1222
FEATURE
misc_feature       1..1222
                    note = Synthetic construct
source             1..1222
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 122
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcac cctgctgaac 60
aacgccgcct tccgcaacgg ccacaacttc atggtgcgca acttccgctg cggccagccc 120
ctgcagggca ggtgcagct gaagggccgc gacctgtga ccctgaagaa cttaccggc 180
gaggagatca agtacatgct gtggctgagc gccgacctga agttccgcat caagcagaag 240
ggcgagtacc tgccccgtct gcagggcaag agcctgggca tgatcttcga gaagcgcagc 300
acccgcaccc gccctgagcac cgagacaggc ttgcctctgc tgggcggcca cccctgcttc 360
ctgaccaccc aggacatcca cctgggcgtg aacgagagcc tgaccgacac cgcccgcgtg 420
ctgagcagca tggccgacgc cgtgctggcc cgcgtgtaca agcagagcga cctggacacc 480
ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta ccaccccatc 540
cagatcctgg ccgactacct gacctgcag gagcactaca gcagcctgaa gggcctgacc 600
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgcgcgaag 660
ttcgccatgc acctgcaggc cgccaccccc aagggtctac agcccgacgc cagcgtgacc 720
aagctggccg agcagtacgc caaggagaac ggcaccaagc tgcgtgtgac caacgacccc 780
ctggaggccg cccacggcgg caacgtgctg atcaccgaca cctggatcag catggggcag 840
gaggaggaga agaagaagcg cctgcaggcc ttccagggct accaggtgac catgaagacc 900
gccaaggtgg ccgcccagcg ctggaccttc ctgcaactgc tgccccgcaa gcccgaggag 960
gtggacgacg aggtgttcta cagccccccg agcctggtgt tccccgaggc cgagaaccgc 1020
aagtggacca tctggccgt gatggtgagc ctgctgaccg actacagccc ccagctgcag 1080
aagcccaagt tctgaggtct ctagtaatga gctggagcct cggtagccgt tctctctgcc 1140
cgctgggcct cccaacgggc cctcctcccc tccttgcaac ggcccttctt ggtctttgaa 1200
taaagtctga gtgggcatct ag 1222

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SEQ ID NO: 123      moltype = RNA length = 1222
FEATURE
misc_feature       1..1222
                    note = Synthetic construct

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source          1..1222
                mol_type = other RNA
                organism = synthetic construct

SEQUENCE: 123
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcat cctgctgaac 60
aacgccgcct tccgcaacgg ccacaacttc atggtgcgca acttccgctg cggccagccc 120
ctgcagggcc ggggtgcagct gaagggccgc gacctgtga ccctgaagaa cttcaccggc 180
gaggagatcc ggtacatgct gtggctgagc gccgacctga agttccgcat caagcagaag 240
ggcgagtacc tgccctgtct gcagggcaag agcctgggca tgatcttcga gaagcgacgc 300
acccgcaccc gcttgagcac cgagacaggg ttccgcttgc tgggcggcca cccctgcttc 360
ctgaccaccc aggacatcca cctgggctgt aacgagagcc tgaccgacac cggccgctgt 420
ctgagcagca tggccgacgc cgtgctggcc cgctgtgaca agcagagcga cctggacacc 480
ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta ccacccatc 540
cagatcctgg ccgactacct gacctgtcag gagcactaca gcagcctgaa gggcctgacc 600
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgccgccaag 660
ttcggcctgc acctgcaggg cgccaccccc aagggtctac agcccgacgc cagcgtgacc 720
aagctggccg ccgacagcgc caaggagaac ggcaccaagc tgctgctgac caacgacccc 780
ctggaggccg cccacggcgg caacgtgtgt atcacgcaca cctggatcag catgggcccag 840
gaggaggaga agaagaagcg cctgcaggcc ttccagggtc accaggtgac catgaagacc 900
gccaaggttg ccgccagcga ctggaccttc ctgcactgcc tgccccgcaa gcccgaggag 960
gtggacgacg aggtgttcta cagccccccg agcctggtgt tccccgaggc cgagaaccgc 1020
aagtggacca tcatggccgt gatggtgagc ctgctgacgc actacagccc ccagctgcag 1080
aagcccaagt tctgaggtct ctagtaatga gctggagcct cggtagccgt tctctctgcc 1140
cgctgggcct cccaacgggc cctcctcccc tccttgaccc ggcccttctt ggtctttgaa 1200
taaagtctga gtgggcatct ag 1222

SEQ ID NO: 124      moltype = RNA length = 1282
FEATURE            Location/Qualifiers
misc_feature        1..1282
                    note = Synthetic construct
source              1..1282
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 124
ggcagaaaaa tttgtacat tgtttcacaa acttcaaata ttattcattt atttagatct 60
attattacat caaaacaaaa agccgccacc atgctgttca acctgcgcat cctgctgaac 120
aacgccgcct tccgcaacgg ccacaacttc atggtgcgca acttccgctg cggccagccc 180
ctgcagggca aggtgcagct gaagggccgc gacctgtga ccctgaagaa cttcaccggc 240
gaggagatca agtacctgct gtggctgagc gccgacctga agttccgcat caagcagaag 300
ggcgagtacc tgccctgtct gcagggcaag agcctgggca tgatcttcga gaagcgacgc 360
acccgcaccc gcttgagcac cgagacaggg ttccgcttgc tgggcggcca cccctgcttc 420
ctgaccaccc aggacatcca cctgggctgt aacgagagcc tgaccgacac cggccgctgt 480
ctgagcagca tggccgacgc cgtgctggcc cgctgtgaca agcagagcga cctggacacc 540
ctggccaagg aggccagcat ccccatcatc aacggcctga gcgacctgta ccacccatc 600
cagatcctgg ccgactacct gacctgtcag gagcactaca gcagcctgaa gggcctgacc 660
ctgagctgga tcggcgacgg caacaacatc ctgcacagca tcatgatgag cgccgccaag 720
ttcggcctgc acctgcaggg cgccaccccc aagggtctac agcccgacgc cagcgtgacc 780
aagctggccg agcagtagcg caaggagaac ggcaccaagc tgctgctgac caacgacccc 840
ctggaggccg cccacggcgg caacgtgtgt atcacgcaca cctggatcag catgggcccag 900
gaggaggaga agaagaagcg cctgcaggcc ttccagggtc accaggtgac catgaagacc 960
gccaaggttg ccgccagcga ctggaccttc ctgcactgcc tgccccgcaa gcccgaggag 1020
gtggacgacg aggtgttcta cagccccccg agcctggtgt tccccgaggc cgagaaccgc 1080
aagtggacca tcatggccgt gatggtgagc ctgctgacgc actacagccc ccagctgcag 1140
aagcccaagt tctgaggtct ctagtaatga gctggagcct cggtagccgt tctctctgcc 1200
cgctgggcct cccaacgggc cctcctcccc tccttgaccc ggcccttctt ggtctttgaa 1260
taaagtctga gtgggcatct ag 1282

SEQ ID NO: 125      moltype = RNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Synthetic construct
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 125
cacaaagagt aaagaagaac a 21

SEQ ID NO: 126      moltype = RNA length = 21
FEATURE            Location/Qualifiers
misc_feature        1..21
                    note = Synthetic construct
source              1..21
                    mol_type = other RNA
                    organism = synthetic construct

SEQUENCE: 126
aacactaaaa gtagaagaaa a 21

SEQ ID NO: 127      moltype = RNA length = 21
FEATURE            Location/Qualifiers

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misc_feature      1..21
                  note = Synthetic construct
source            1..21
                  mol_type = other RNA
                  organism = synthetic construct

SEQUENCE: 127
ctcagaaaaga taagatcagc c                                     21

SEQ ID NO: 128      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                  note = Synthetic construct
source              1..1065
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 128
atgctgttta atctgaggat cctgttaaac aatgcagctt ttagaaatgg tcacaacttc 60
atggttcgaa attttcgggtg tggacaacca ctacaaaata aagtgcagct gaagggccgt 120
gaccttctca ctctaaaaaa ctttaccgga gaagaaatta aatatatgct atggctatca 180
gcagatctga aatttaggat aaaacagaaa ggagagtatt tgcctttatt gcaaggggaag 240
tccttaggca tgatttttga gaaaagaagt actcgaaaca gattgtctac agaaacaggc 300
tttgcaactc tgggaggaca tccttgtttt cttaccacac aagatattca ttgggtgtg 360
aatgaaagtc tcacgggacac gggccgtgta ttgtctagca tggcagatgc agtatggct 420
cgagtgtata aacaatcaga ttgggacacc ctggctaaag aagcatccat cccaattatc 480
aatgggtgtg cagatttgta ccatcctatc cagatcctgg ctgattacct cacgctccag 540
gaacactata gctctctgaa aggtcttacc ctcagctgga tcggggatgg gaacaatatc 600
ctgcactcca tcctgatgag cgcagcgaaa ttcggaatgc accttcaggc agctactcca 660
aaggggtatg agcgggatgc tagtctaacc aagttggcag agcagtatgc caaagagaat 720
ggtaccaagc tgttgcgtgac aaatgatcca ttggaagcag cgcattggagg caatgtatta 780
attacagaca cttggataag catgggacaa gaagaggaga agaaaaagcg gctccaggct 840
ttccaaggtt accaggttac aatgaagact gctaaagttg ctgcctctga ctggacattt 900
ttacactgct tcccagaaaa gccagaagaa gtggatgatg aagctcttta ttctctcga 960
tcactagtgt tcccagaggc agaaaaacaga aagtggacaa tcatggctgt catggtgtcc 1020
ctgctgacag attactcacc tcagctccag aagcctaaat tttaga                                     1065

SEQ ID NO: 129      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                  note = Synthetic construct
source              1..1065
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 129
atgctcttta atctgcgcat cttactgaac aacgcgcgat tccggaacgg tcacaacttc 60
atggtcgcga atttcgcgtg tggccagccg cttcaaaaaca aggtccagct gaagggacgg 120
gatctgctga cactgaagaa cttcaccgga gaagagatca agtacatgct gtggctcagc 180
gcagacttga agttccggat caagcagaag ggagaatact tgcccctgct gcaaggaaaag 240
tcgctgggaa tgatttttga gaagcgggtc actcgaccca gactctccac cgaactgtgt 300
ttcgactgct ttggcgggga ccttgctctc ctgacgactc agacatcca cctcggcgtg 360
aacgaatcgc taaccgatc cgccagagtg cttttctcca tggccgacgc ggtgctggcc 420
aggggtgtata agcagtccca cctcgatacc ttggcaaaag aggtttccat tcccatcctc 480
aacggcctga ggcacctgta ccaccaatc caaatcctgg ctgactacct gaccctgcaa 540
gagcactaca gcagcctgaa gggctctgacc ctgtcatgga ttggcgatgg aaacaatatt 600
ctgcactcca tcctgatgtc cgcgcggaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aacccgatgc gtccgtgacc aagttggcgg aacagtagcg gaaggagaac 720
ggaaccaagc ttctgctgac taacgacccc ctcgaggtcg cgcattgggg caacgtgtcg 780
attaccgaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccaggggt accaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattggc tgccgaggaa gccggaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tcccagaggc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctag                                     1065

SEQ ID NO: 130      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                  note = Synthetic construct
source              1..1065
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 130
atgctgttta acctacgtat tttgtcaac aatgcagcct ttagaaacgg acataacttt 60
atggttcgaa actttcgtcg cgggcagcca ctgcagaaca aggtccagct gaaagggaga 120
gatttgctca cgctgaagaa ctttactggc gaagaaatca agtatatgct gtggtgtgct 180
gcggacctca agtttcggat taagcagaaa ggggagatgc tgccactgct gcaaggaaaag 240
agcctcggca tgtatctcga gaagcggagc actcggacca ggctgagtac cgaactgtgc 300
ttcgatttgt tgggttgaca tccatgtttt ctgacaaacg aggacattca tctgggcgtg 360
aacgagagtc tgacggacac agctcgcgtt ctgtcctcta tggctgatgc ggtgttgccc 420
cggtctata agcagtccca ttggacacc ttggctaagg aagctagcat accgattatc 480
aatgggtgtg ccgacctgta tcacctatt caaatcctgg ccgactacct cacactgcaa 540

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gaacactata gctcattgaa gggactgacc ctgagctgga taggggacgg aaacaacatc 600
ctacatagca ttatgatgtc cgtgccaag tttggcatgc atcttcaagc cgccacgcc 660
aagggttatg agcccgacgc gtcagtgaaca aagctggccg agcagtacgc taaggagaat 720
ggtaccaaata tactgtgac taatgatcca ctggaggctg cacatggcgg caatgtactg 780
atcaccgaca catggatctc gatgggccag gaggaagaaa agaagaagag gcttcaggcc 840
ttccaaggct accaggctac catgaaaaca gctaagggtg cagcatctga ttggaccttt 900
ctgcactgtc tgccaaggaa gccgaagag gtggacgatg aagtattcta tagcccacgg 960
agtttggtgt tccctgaggc tgaaaatagg aagtggacaa ttatggccgt aatgggtgtc 1020
ctgtaaacgc actactctcc gcaactgcag aaacctaaat tttag 1065

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```

SEQ ID NO: 131      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 131
atgctgttta acttaaggat cctgctgaac aacgccgctt ttcgtaacgg tcataacttt 60
atgggtccgga actttagatg tggccagccg ctgcagaaca aggttcagct gaaggggagg 120
gatctgctga ccttgaagaa ctttaccggc gaagagatca agtacatgtt gtggctgagc 180
gccgatctga agtttaggat taagcagaag ggggagattt tgccactgct gcaaggaaaa 240
tccctgggga tgatcttcca gaagcgtccc actagaaccc ggctaagcac agaaaccggc 300
tcgcactctc tgggttgaca tcctgtttt ctgacgacgc aggatataca cctgggctgt 360
aatgagagtc tgacggacac agctagggtg ttgacgacga tggccgatgc agtactggcc 420
cgcttttata agcagagcga ctgggacaca ctggccaagg aagcgtcaat tccgattatc 480
aatgggctgt cagacctgta tcattccatt caaatcttgg ctgactatct gacctgtcaa 540
gaacattaca gctcctgaa gggcctcacg ttgtcctgga ttggcgacgg aaacaacatt 600
ctgcattcca tcatgatgag cgtgctaa gcttggcatg acctccaagg cgtacacct 660
aagggatatg agcctgatgc cagcgttaac aagctggccg aacagtacgc gaaggagaat 720
ggcacgaaac tgtgttgac aatgaccca ctggaggcag ctacgggtgg caacgtgtctg 780
atcaccgaca cgtgggatgc tatgggacag gaagaagaga agaagaagcg gctgcaggca 840
ttccaagggt atcaggctac catgaaaacg gccaaagggt ctgcatccga ctggacattt 900
ctgcattgct tcccgcgcaa accagaagaa gtagacgacg aagtccttta tccccacgg 960
tcgctgggtg tccccgaggc ggagaatcga aagtggacga ttatggccgt gatgggtgtc 1020
ctgctgactg attactctcc ccaactgcaa aagcctaagt tttag 1065

```

```

SEQ ID NO: 132      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 132
atgcttttca acctgaggat cctcctgaac aacgccgcct ttcgcaatgg tcacaacttt 60
atgggtccgga acttcagatg cggccagccg ctgcagaaca aggtccagct gaaggggacg 120
gatctgctga ctctgaagaa cttcaccgga gaagagatca agtacatgct gtggctgtcg 180
gccgacctga agttcaggat caagcagaag ggagaatacc tcccgtgctg gcaaggaaa 240
tccctgggca tgattttcca gaagcgtctg accagaactc ggttgtccac cgaaccggg 300
tttgctgtgc tgggcccaga tccttgcttc ctgacgactc aggatattca cctgggagtg 360
aacgagtcgc tgaccgacac gccacagagt ctgagctcga tggccgacgc cgtgttggca 420
cgctgtgata agcagtcgga tctggatacc ctggccaagg aagcttccat cccgatcatt 480
aacgggctga gcgacctcta ccccccatt caaatcctgg ccgactacct gactctgcaa 540
gaacactaca gctcgtgact ggggttgact ctgtcctgga tcggcgacgg aaacaacatc 600
ctgcactcca tcatgatgtc ggccgcaaa gttcgcatgc atttgaagc cgccacccca 660
aagggtctac aaccagacgc gacgctcacc aagctggccg aacagtacgc gaaggaaaat 720
ggtactaagc tgcgtctgac caacgaccca ttggaagctg cccatgggtg aaacgtgtctg 780
atcaccgaca cctggatctc gatgggccag gaagaggaga agaagaagcg gctgcaggcg 840
ttcagggggt atcaggctac catgaaaaca gccaaagtgg cagcgtcaga ctggaccttc 900
ctccactgtc tgcctcgcaa gccagaggag gtggacgacg aggtgttcta ctcccctcg 960
tccctcgtgt tccctgaggc tgagaaccgg aagtggacca ttatggccgt gatgggtgtc 1020
ctcctgactg attactcccc gcaactgcag aagcccaagt tctag 1065

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```

SEQ ID NO: 133      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 133
atgctgttta acctgaggat cctattgaac aatgctgctt ttcgtaatgg ccataacttt 60
atgggtccgga actttagatg cgggcagcca ctgcagaaca aggtccagtt gaaaggccgc 120
gatctgttga cattgaagaa ctttaccggc gaagagatta agtatatgct gtggctgtct 180
gctgacctca agtttcgaat caagcagaag ggccaatata tcccctgctg gcaaggaaa 240
tctctcgga tgacttttga gaagcggagt acccgaaacac ggctgagcac cgaaccgggc 300
tcgcactgct tggggggcca tcctgtttt ctgacaacgc aggcacatcca cttgggggtt 360
aacgaatcat tgactgatac cgcccgcgta ctgtcatcca tggccgacgc tgtgctggct 420

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aggggtgtaca agcagtcaga tctggatata ctggccaagg aagctagcat accaatcatc 480
aatggactga gtgaccttta tcacccgatt caaatactag ccgattatct gaccctgcaa 540
gagcattact cctcgctgaa aggcctcacg ctgtcctgga tcggcgacgg caacaacatt 600
ctgcatagta ttatgatgtc tctgcccata ttcggcatgc atctgcaagc tgctacgccg 660
aaggggttatg aaccogacgc gtcagttacg aagctcgctg agcagtatgc aaaggagaat 720
ggcacaaggc tgttgcttac caacgatccc ctggaagctg ctcatggcgg caatgtgctg 780
attactgaca cctggatttc aatgggccag gaggaggaga agaagaagag gttacaggct 840
tttcaagggt accaagtcac gatgaaaacc gctaaggctg cagccagcga ctggacattc 900
ctgcaactgtc tgccaagaaa gccggaagaa gtggacgacg aggtgttcta tccccgcgg 960
tctttggtgt ttccggaggc cgaacaacag aaatggacca ttatggccgt gatggtatcg 1020
tgctgacgg actacagccc tcagtgcga aagcccaagt tctag 1065

```

```

SEQ ID NO: 134      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 134
atgctcttta acctccgcat cctcctcaac aacgcgcgct tccggaatgg gcataacttc 60
atggtccgga acttcagatg cggccagccc ctgcaaaaca aggtccagtt gaaggagcgg 120
gacctcctta cgctgaagaa ctttaccgga gaagagatta agtacatgct gtggtgtgcc 180
gtgcaactca agttccgcat taagcagaag ggagaatata tgccgctgct gcaaggaaaag 240
agcctgggca tgatcttcga aaagcgctcc actagaaccc ggctgtcgac tgagactgga 300
ttcgcttgcc tgggtggaca cccgtgcttc ctgacgaccc aggacatcca cctgggagtg 360
aacgagtcac ttacggatgc cgcgaggggtg ctgtcctcaa tggccgacgc agtgctcgccg 420
cgcggtgtaca agcagtcaga tctggatacc ctggccaagg aagccagcat tccatcatc 480
aacggactga gcgaccttta ccaccaatc cagatcctcg ccgactactt aacctctgca 540
gagcaactaca gctccctgaa gggactgact ctgtcctgga tcggggatgg aaacaacatc 600
ctgcaactcca tcatgatgtc tgcgctaaag ttgggatgc atctgcaagc cgcaaccctc 660
aagggatacg agcccgacgc ctcggtgacc aaacttgccg aacagtacgc caaggaaaac 720
ggtaccaagc tgetgctgac caacgaccct ctggaagcgg cccacggagg aaatgtgctg 780
attaccgaca cctggatttc gatgggccag gaggaggaga agaagaagag actgcaggcg 840
ttccagggat atcaggtcac catgaaaacc gccaaaggctg ccgccagcga ctggacattc 900
ctgcaactgtc tccctcgcaa accggaagaa gtggatgacg aggtgttcta ctcgccgcg 960
tcgctggtgt tcccggaggc tgaacaacag aagtggacca tcatggccgt gatggtgtcc 1020
ctgtgacgg actactcccc acaactgcag aagcccaagt tctag 1065

```

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SEQ ID NO: 135      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 135
atgcttttca atctgcgcat cctcctgaac aacgcgcgct tccgcaatgg acacaacttt 60
atggtccgca acttcgcgtg tgggcagccg ctgcagaaca aggtccagct caaggggaga 120
gatctcctga ccttgaagaa cttcactgga gaggagatca agtacatgct gtggtgtgcc 180
gccgacctga aatttcggat taagcagaag ggcgaatacc tccactgct gcaaggaaaag 240
tctttgggca tgatcttcga aaagagaagc acccgacccc ggttgagcac cgaaactggg 300
ttcgcgctcc tgggtggaca cccgtgcttc ctgaccaccc aagatattca tctgggtgtc 360
aacgaaagcc tgaccgacac cgcgaggggtg ctgtcatcca tggctgacgc agtgctcgcc 420
cgggtgtaca agcagtcaga cctggacacc ctgcgaaccc aagcttcgat ccctatcatc 480
aacggacttt ccgacctgta ccaccccatc caaattctgg ccgactacct gactctgcaa 540
gaacactata gctcgctgaa aggacttact ctgtcctgga tcggggacgg caacaacatt 600
ctccattcca tcatgtcaga cgctgccaag ttcggaatgc acctcaagc agcgactccc 660
aagggatacg aacctgatgc ctccgtgact aagctggcag agcagtacgc caaggagaa 720
ggtacaaagc tgetgctcac gaacgacccc ctggaggcgg cccacggcgg aaacgtgctg 780
attaccgata cctggatctc aatgggccag gaagaggaga agaagaagcg gctccaggcg 840
tttcaaggct accaggtcac catgaaaacc gcgaaggctg ccgctccga ctggactttc 900
ttgcaactgc tgccgcggaa gcccgaggaa gtggatgacg aagtgttcta ctcgccgaga 960
tcgctggtgt tccctgaggc cgaacaacag aagtggacca tcatggccgt gatggtgtcc 1020
ctgctgactg attacagccc acagctgcag aagcetaagt tctag 1065

```

```

SEQ ID NO: 136      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 136
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaaagg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaaag 240
agcctcgga tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300

```

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ttcgcgctgc tegggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagtg 360
aacgaatccc tcacogatac cggccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgaccteta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcgatgtgtc cggcgcaaaa ttccgcatgc atcttcaagc cggccgcggc 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtagcg taaggagaa 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catggggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggtcg ctgcctccga ctggaccttc 900
ctgcaactgcc tgccctcgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggcgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 137      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

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SEQUENCE: 137
atgcttttca acctgagaat cctcctgaac aacgcccgtt tccgcaatgg tcataacttc 60
atggtccgca actttcgtcg cggacagcct ctccaaaaca aggtccagct caaggggccc 120
gacctctca cactgaagaa cttcactgga gaagaaatca agtacatgct gtggctgagc 180
gccgatctga agttccggat caagcagaag ggagagtacc ttctctgctt gcaaggggag 240
tccttgggaa tgatttttga gaagcgggtc acccgagcca ggctgagcac tgaactgtgc 300
ttcgccctgc tgggaggcca ccttgttttc ctgaccactc aggacatcca cctgggctgt 360
aacgagtcct tgaccgatac tgccagagtg ctgtcctcca tggccgacgc cgtgctcgcc 420
cgggtgtaca agcagtcaga cctcgatacg ctggccaagg aagcctccat tccattatc 480
aatggtctgt cggacctcta ccatccaatc caaatcctcg ccgactacct gactctgcaa 540
gaacactaca gctcactcaa gggcctcacc ctctcctgga tcggcgacgg aaacaacatc 600
cttactctga ttatgatgtc ggcgcggaag ttccggatgc acctccaaag tgcactcca 660
aaaggctacg agccggatgc ctcagtgact aagttggcgg aacagtatgc gaaggagaa 720
ggtaccaagc tctctctgac taacgacccc ctggaggccc cccacggggg aaacgtgctc 780
atcacccgata cttggatttc catgggacag gaggaagaga agaagaagcg gttgcaggca 840
tttcagggct accaggtcac catgaaaact gccaaagtgc ccgacagcga ctggaccttc 900
ctgcaactgcc tgccctcgcaa gctgaagaa gtggacgacg aggtgttcta ctctccccgg 960
tcctcgtgtg tccctgaggc cgaaaacagg aagtggacca tcatggctgt gatggtgtcc 1020
ctcctgaccg actacagccc tcagctccaa aaacccaagt ttag 1065

```

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SEQ ID NO: 138      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 138
atgcttttca acctgagaat cctcttgaac aatgctgctt ttccgaatgg ccacaacttt 60
atggttcgga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgctca cactgaagaa ctttactggg gaggagatta agtatatgct gtggctgtcc 180
gtcgacctga agtttaggat caagcagaag ggcgaatata tgccgctgct gcaaggggaa 240
agtctgggca tgatttttga aaagcgtctc acccgagcca gactgtctac ggaacagggc 300
tttgccctgc tgggaggcca cccctgtttt ctgacaacgc aggacatcca tctgggctgt 360
aacgaatcac tgaccgatac tgctcgggta ctcagtctta tggctgacgc agtgctggct 420
aggggtgtaca agcagagcga cttggacaca ctggctaagg aggcagcat cccattatc 480
aatggcctgt ctgatttcta ccatccatt caaatcctgg ctgattatct gacactacaa 540
gagcattact caagtctgaa gggtttgact ctctcctgga tcggcgacgg caacaacatt 600
ttacattcca ttatgatgag tgctgctaag ttggcatgc atttgaagc tgctaccca 660
aagggtctatg aacctgacgc tagcgtaacc aagttggccc aacagtatgc taaagagaa 720
ggcaccgaagc tgctctgac gaatgacccc ctggaagctg ctcatggcgg aaacgtactt 780
ataactgata catggattag catgggcccag gaagaggaga agaagaagag actgcaggcc 840
ttcaaaggct atcaggtcac catgaaaact gccaaaggttg cagctagcga ctggaccttc 900
ctgcaactgtt tgccgaggaa acccgaggag gtggacgatg aagtctttta ttctccccgc 960
tccttggtgt ttcccgaggc tgaaaatcga aagtggacga taatggcagt gatggtgtcc 1020
ctactgaccg actattctcc acaactgcag aagcctaaat tctag 1065

```

```

SEQ ID NO: 139      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

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SEQUENCE: 139
atgcttttca atctgaggat cctgctgaac aacgtgctt ttccgaacgg tcataacttt 60
atggttcgca attttctgtg tggccagccc ctgcagaaca aggttcagct gaagggcaga 120
gatctgctga ctctgaagaa cttcactggg gaagaaatca agtatatgtt atggctgtcc 180

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-continued

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gcggatctga aatttcgaat caagcagaag ggcgaaatc ttcctctgct gcaagggaaa 240
tccttgggca tgatttttga gaagaggagc actaggacta gattgtcaac agaaacaggc 300
tttgccttgt tgggcggaaca tcctctgctt ctgacgacac aggatatcca cctcggcgta 360
aacgagtcctc tcaccgacac tgctagggtta ctgagcagca tggccgacgc tgtgctagcc 420
cgggttttaca agcagtcaga cctggacacc cttgccaaag aagcttctat tccaattatc 480
aacggcctga gtgacctgta tcaccctatt caaataactcg ccgactattt gacgcttcaa 540
gaacattaca gcagcctcaa gggcttaacc ttgagttgga taggcgacgg caacaatatc 600
ctgcattcca ttatgatgtc tgccgctaag ttggcatgc atctacaagc cgcaacaccc 660
aagggtctatg aaccgcagcg tagcgtgacc aagctggccg agcagtatgc taaggaaaat 720
ggcacaagc tccttcttac caacgatccc ctggaggctg ctacgcggcg caacgtgctg 780
attaccgata catggattag catgggccag gaggaggaga aaaagaagcg gctccaggct 840
tttcaaggct atcaggtcac catgaaaact gcaaaggctc ctgcctccga ctggactttc 900
ctgcattgtc taccocgcaa gccctgaggaa gtggacgatg aggtgttcta ctcccacgg 960
agtctggtgt tcccgggaagc agagaatcgg aagtggacca tcatggctgt catggtgtcg 1020
ctcttgactg actattctcc ccaactgcaa aaaccaagt tttag 1065

```

```

SEQ ID NO: 140      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 140
atgcttttca atctgaggat cctgctgaac aacgctgctt ttccgcaacgg tcataacttt 60
atggttcgca attttcgttg tggccagccg ctgcagaaca aggttcagct gaagggcaga 120
gatctctgta ctctgaagaa cttcactggg gaagaaatca agtatatgtt atggctgtcc 180
gcggatctga aatttcgaat caagcagaag ggcgaaatc ttcctctgct gcaagggaaa 240
tccttgggca tgatttttga gaagaggagc actaggacta gattgtcaac agaaacaggc 300
tttgccttgt tgggcggaaca tcctctgctt ctgacgacac aggatatcca cctcggcgta 360
aacgagtcctc tcaccgacac tgctagggtta ctgagcagca tggccgacgc tgtgctagcc 420
cgggttttaca agcagtcaga cctggacacc cttgccaaag aagcttctat tccaattatc 480
aacggcctga gtgacctgta tcaccctatt caaataactcg ccgactattt gacgcttcaa 540
gaacattaca gcagcctcaa gggcttaacc ttgagttgga taggcgacgg caacaatatc 600
ctgcattcca ttatgatgtc tgccgctaag ttggcatgc atctacaagc cgcaacaccc 660
aagggtctatg aaccgcagcg tagcgtgacc aagctggccg agcagtatgc taaggaaaat 720
ggcacaagc tccttcttac caacgatccc ctggaggctg ctacgcggcg caacgtgctg 780
attaccgata catggattag catgggccag gaggaggaga aaaagaagcg gctccaggct 840
tttcaaggct atcaggtcac catgaaaact gcaaaggctc ctgcctccga ctggactttc 900
ctgcattgtc taccocgcaa gccctgaggaa gtggacgatg aggtgttcta ctcccacgg 960
agtctggtgt tcccgggaagc agagaatcgg aagtggacca tcatggctgt catggtgtcg 1020
ctcttgactg actattctcc ccaactgcaa aaaccaagt tttag 1065

```

```

SEQ ID NO: 141      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 141
atgcttttca atctccgcat cctccttaac aacgcgcgt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggcgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtatatgtt atggctgtcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcggagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctggacact ctggccaagg aggcgtcaat tcccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa ggggtctgacc ctgtcatgga ttggcgatgg aaacaatatt 600
ctgcaactca tcatgatgtc cgccgcgaag ttccgaatgc atctgcaagc cgccaacgcca 660
aaaggatacg aaccggatgc gtcctgacg aagttggcgg aacagtacgc gaaggagaa 720
ggaaccaagc ttctgctgac taacgacccc ctcgaggctg cgcatggggg caacgtgctg 780
attaccgaca cctggatctc catggggcag gagggaagaga agaagaagag actgcaggca 840
tcccaggggt accaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tgccgaggaa gcccgaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tcccggagc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctag 1065

```

```

SEQ ID NO: 142      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 142
atgcttttca acctgagaat cctcttgaac aatgctgctt ttcggaatgg ccacaacttt 60
atgggttcgga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgcctca cactgaagaa ctttactgga gaagagatca agtacatgct gtggctgtcg 180
gccgacctga agttccagat caagcagaag ggagaatacc ttccgctgct tcaaggaaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagt 360
aacgaatccc tcacogatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctcgatacc ttggcaaaag aggttccat tcccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa ggggtctgacc ctgtcatgga ttggcagatg aaacaatatt 600
ctgcactcca tcgatgtgct cgccgcgaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aaccggatgc gtcctgacc aagttggcgg aacagtacgc gaaggagaac 720
ggaaaccaagc ttctgctgac taacgacccc ctcgaggctg cgcagggggg caacgtgctg 780
attaccgaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccagggtg tccaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tgccgaggaa gccggaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tcccagaggc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctag 1065

```

```

SEQ ID NO: 143      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 143
atgcttttca acctgagaat cctcttgaac aatgctgctt ttcggaatgg ccacaacttt 60
atgggttcgga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgcctca cactaaagaa ctttactgga gaagagatca agtacatgct atggctgtcg 180
gccgacctga agttccgtat caagcagaag ggagaatacc ttccgctgct tcaaggaaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagt 360
aacgaatccc tcacogatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctcgatacc ttggcaaaag aggttccat tcccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa ggggtctgacc ctgtcatgga ttggcagatg aaacaatatt 600
ctgcactcca tcgatgtgct cgccgcgaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aaccggatgc atcctgacc aagttggcgg aacagtacgc gaaggagaac 720
ggaaaccaagc ttctgctgac taacgacccc ctcgaggctg cgcagggggg taacgtgctg 780
attaccgaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccagggtg tccaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tgccgaggaa gccggaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tcccagaggc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctag 1065

```

```

SEQ ID NO: 144      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 144
atgcttttca acctgagaat cctcttgaac aatgctgctt ttcggaatgg ccacaacttt 60
atgggttcgga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgcctca cactaaagaa ctttactgga gaagagatca agtacatgct atggctgtcg 180
gccgacctga agttccgtat caagcagaag ggagaatacc ttccgctgct tcaaggaaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagt 360
aacgaatccc tcacogatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctcgatacc ttggcaaaag aggttccat tcccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa ggggtctgacc ctgtcatgga ttggcagatg aaacaatatt 600
ctgcactcca tcgatgtgct cgccgcgaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aaccggatgc gtcctgacc aagttggcgg aacagtacgc gaaggagaac 720
ggaaaccaagc ttctgctgac taacgacccc ctcgaggctg cgcagggggg caacgtgctg 780
attaccgaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccagggtg tccaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tgccgaggaa gccggaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tcccagaggc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctag 1065

```

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SEQ ID NO: 145      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct

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source          1..1065
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 145
atgctgttca acctgcgaat cctgctgaac aacgcgcgtt ttcggaacgg gcacaacttt 60
atggtgagga actttcgctg cggacagccc ctccagaata aggtccagct gaagggcagg 120
gacctgctga ccttgaaaaa ttccacaggg gaggaaatca agtatatgct gtggctgtca 180
gctgatctga agttccggat caagcagaag ggcgaatata tgctctgct ccagggcaaa 240
agcctgggga tgatcttcga aaagcgagct actcggacca gactgtcaac cgagactgga 300
ttcgctctgc tgggaggaca cccttgtttt ctgaccactc aggacattca cctgggagtg 360
aacgagtcct tgaccgacac tgctcgctgc ctgagctcta tggccgacgc tgtgctggct 420
cgagtctaca aacagtccta cctggatacc ctggccaagg aagcttctat cccaattatt 480
aacggcctgt cagacctgta tcaccccatc cagattctgg ccgattacct gaccttcag 540
gagcactatt ctagtctgaa agggctgaca ctgagttgga ttggggacgg aaacaatatc 600
ctgcaactcta ttatgatgtc agccgccaag ttggaatgc acctccaggc tgcaacccca 660
aaaggctacg aaccgatgac ctcagtgaac aagctggctg aacagtacgc caaagagaac 720
ggcactaagc tgctgctgac caacgacct ctggaggccg ctcacggagg caacgtgctg 780
atcaccgata cctggattag tatgggacag gagggaagaga agaagaagcg gctccaggcc 840
ttccagggct accaggtgac aatgaaaacc gctaaggtcg cagccagcga ttggaccttt 900
ctgcaactgcc tgcccagaaa gccgaagag gtggacgacg aggtcttcta ctctccaga 960
agcctgggtg ttcccgaagc tgagaatagg aagtggaaca ttatggcagt gatggtcagc 1020
ctgctgactg attattcacc tcagctccag aaaccaaagt tctga 1065

SEQ ID NO: 146      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 146
atgctgttca acctgcgeat cctgctgaac aacgcgcgtt ttcgcaacgg ccacaacttc 60
atggtgcgca acttcgctg cggccagccc ctgcagaaca aggtgcagct gaagggcgcg 120
gacctgctga ccttcacggc cttcaccggc gaggagatca agtatatgct gtggctgagc 180
ggcgacctga agttccgeat caagcagaag ggcgagtagc tgccctgctg gcagggcaag 240
agcctgggga tgatcttcga gaagcgagc acccgacccc ccctgagcac cgagacaggc 300
ctggccctgc tggggggcca ccctgcttc ctgaccaccc aggacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggcagcatc ccccatcatc 480
aacggcctga gcgacctgta caaccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttccgcatgc acctgcaggc cgccaccccc 660
aaaggctacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaac 720
ggcaccgaagc tgctgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgata cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac cctgaagacc gccaaagtggt cgcacagcga ctggaccttc 900
ctgcaactgcc tgcccgcgaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctgggtg tcccagagcg cgagaaccgc aagtggaaca tcatggcgtg gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaaagt tctga 1065

SEQ ID NO: 147      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 147
atgctgttca acctgcgaat cctgctgaac aacgcgcgtt ttcggaacgg gcacaacttt 60
atggtgagga actttcgctg cggacagccc ctccagaata aggtccagct gaagggcagg 120
gacctgctga ccttgaaaaa ttccacaggg gaggaaatca agtatatgct gtggctgtca 180
gctgatctga agttccggat caagcagaag ggcgaatata tgctctgct ccagggcaaa 240
agcctgggga tgatcttcga aaagcgagct actcggacca gactgtcaac cgagactgga 300
ttcgctctgc tgggaggaca cccttgtttt ctgaccactc aggacattca cctgggagtg 360
aacgagtcct tgaccgacac tgctcgctgc ctgagctcta tggccgacgc tgtgctagct 420
cgagtctaca aacagtccta cctggatacc ctggccaagg aagcttctat cccaattatt 480
aacggcctgt cagacctgta tcaccccatc cagattctgg ccgattacct gaccttcag 540
gagcactatt ctagtctgaa agggctgaca ctgagttgga ttggggacgg aaacaatatc 600
ctgcaactcta ttatgatgtc agccgccaag ttggaatgc acctccaggc tgcaacccca 660
aaaggctacg aaccgatgac ctcagtgaac aagctggctg aacagtacgc caaagagaac 720
ggcactaagc tgctgctgac caacgacct ctggaggccg ctcacggagg caacgtgctg 780
atcaccgata cctggattag tatgggacag gagggaagaga agaagaagcg gctccaggcc 840
ttccagggct accaggtgac aatgaaaacc gctaaggtcg cagccagcga ttggaccttt 900
ctgcaactgcc tgcccagaaa gccgaagag gtggacgacg aggtcttcta ctctccaga 960
agcctgggtg ttcccgaagc tgagaatagg aagtggaaca ttatggcagt gatggtcagc 1020
ctgctgactg attattcacc tcagctccag aaaccaaagt tctga 1065

SEQ ID NO: 148      moltype = DNA length = 1065
FEATURE
Location/Qualifiers

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misc_feature      1..1065
                  note = Synthetic construct
source            1..1065
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 148
atgctgttca acctgcgaat cctgtgaac aatgccgctt ttcggaacgg gcacaatttc 60
atgggtgagga actttcgtcg cggacagccc ctccagaaca aggtccagct gaagggcagg 120
gacctgtctga ccttgaaaaa ttccacaggg gaggaaatca agtacatgct gtggctgtca 180
gccgatctga agttccggat caagcagaag ggcgaatatc tgcctctgct ccaggggcaa 240
agcctgggga tgatcttcga aaagcgagc actcggacca gactgtcaac agagactgga 300
ttcgactgctc tgggaggaca cccatgtttt ctgaccacac aggacattca tctgggagt 360
aacgagtccc tgaccgacac agcacgcgtc ctgagctcca tggctgatgc agtctggct 420
cgagtctaca aacagctctga cctggatacc ctggccaagg aagcttctat cccaatcatt 480
aatggcctga gtgacctgta tccccccatc cagattcttg ccgattacct gaccctccag 540
gagcattatt ctagtctgaa agggctgaca ctgagctgga ttggggacgg aaacaatatc 600
ctgcaactca ctatgatgag cgcgcgcaag ttggaatgc acctccaggg tgcaacccca 660
aaaggctacg aacccgatgc ctccgtgaca aagctggcag aacagtagtc caaagagaac 720
ggcactaagc tctgtctgac caatgacct ctggaggccg ctccaggagg caactgtctg 780
atcaactgata ttatgatgag tatgggacag gaggaagaga agaagaagcg gctccaggcc 840
ttccagggct accagggtgac aatgaaaact gctaaggctc cagccagcga ctggaccttt 900
ctgcattgcc tggccagaaa gccctgaagag gtggacgacg aggtcttcta ctccaccaga 960
agcctgtgtg ttctctgaagc tgagaatagg aagtggacaa tcatggcagc gatggtcagc 1020
ctgctgactg attattcccc tcagctccag aaaccaagt tctga 1065

SEQ ID NO: 149      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 149
atgcttttca accttcgcat tctctcaac aacgcgcgct ttagaaacgg acacaacttc 60
atgggtccgca acttcgctcg cggacagccg ctgcagaaca aggtccagct caagggtcgg 120
gatctcctga cgctgaagaa ctttaccggc gaagagatta agtacatgct gtggctgtcc 180
gccgacctta agttccggat caagcagaag ggcgaatacc ttccctgct gcaaggaaag 240
tccctgggca tgatcttcga gaagcgagc accagaacca gactctccac tgaaccggg 300
ttcgcgctgc ttggcgccca cccgtgtttc ctcaactacg aagacatcca tcttggcgtg 360
aacgagtccc ttaccgacac gcccagggtg ctgtcaagca tggccgacgc cgtccttgcc 420
cgcgtgtaca agcagtcaga ccttgatact ctggccaagg aagcctccat cctattattc 480
aacggcctat ccgaccttta cccccgatc cagatcctcg ctgactacct gaccctgcaa 540
gaacactaca gcagcctcaa gggactgact ctgtcctgga tcggcgacgg gaacaacatc 600
ctgcaactca tcatgatgag cgcagccaag ttccgcatgc atctccaagc cgctacaccc 660
aagggttatg aaccggacgc ctctgtgacc aagtggcagc aacagtagc caaggagaac 720
ggtaactaagc tctctttaac caacgacccc ctcgaaagcag cccatggcgg gaatgtgctc 780
attaccgata cctggatttc gatgggccag gaggaggaga agaagaagcg gctgcaggcg 840
ttccagggct accaggtcac catgaaaact gccaaagtgg ccgcctcgga ttggaccttt 900
ctccactgcc tgcctcgga gctgaggag gtggacgacg aagtgttcta ctccaccag 960
tccctcgtgt tccccgagc gcaaaatagg aagtggacca tcatggccgt gatgggtgctc 1020
ctcttgaccg attacagccc gcagcttcag aagcctaaat tctag 1065

SEQ ID NO: 150      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 150
atgcttttca atcttcgcat cctgttgaac aacgcgcgct tccgcaatgg tcacaacttc 60
atgggtccgga acttcagatg tggacagcct ctccaaaaca aggtccagct gaagggaaag 120
gacctcttaa cctccaaaaa ctttactgga gaggagatca agtacatgct gtggcttagc 180
gccgacctta agttccggat caagcagaag ggagagtacc tcccgctgct gcaaggaaag 240
agtcttggaa tgatcttcga gaagcggtcc accagaactc gcctctccac tgaaccggga 300
ttcgactccc tgggtggaca cccgtgcttt ctgaccaccc aagacatcca cctcggagt 360
aacgagagcc tcacggacac cgcgagagtg ctgtcatcca tggccgacgc cgtgcttgca 420
cgggtctaca agcagtcgga tctggacact cttgccaagg aagcctccat tctatcatt 480
aacggctctg cggtctgtta cccccgatt cagatccttg cggactacct cacacttcaa 540
gaacactatt caagcctaaa gggcttgacc ctgtcctgga tcggagatgg aaacaacatt 600
ctccattcca tcatgatgag cgcctgccaag ttccgaaatgc atctccaagc agcgactcct 660
aagggttacg agccggacgc ctcaagtact aagctggccg agcagtagc caaggagaac 720
ggtaaccaac tgtgtcttac taacgacccg cttgaagcgg cccatggagg aaacgtgctg 780
attaccgaca cctggatttc catgggacag gaagaggaga agaagaagcg gctccaggcg 840
ttccagggat accaggtcac catgaaaacg gccaaagtgg ccgctagcga ttggaccttt 900
ctgcaactgc tcccgcgcaa gccctgaagaa gtggacgacg aagtgttcta ctccctcgc 960
tctcttgtgt tcccgaagc cgaacaacag aagtggacca tcatggccgt gatgggtgctc 1020
ctcctgaccg attacagccc gcagctgcag aagcctaagt tctag 1065

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SEQ ID NO: 151      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 151
atgcttttca atctccgcat cctcctcaac aacgccgcgt ttagaaacgg ccacaacttc 60
atggtcggga acttcagatg tggccagccg cttcagaaca aggtccagct caaggggccg 120
gatcttctga ccttgaagaa ctttactggc gaagaaatca agtacatgct ctggctctcc 180
gccgacttga agttccgcat taagcagaag ggggaatacc ttccgctgct gcaaggaaag 240
tcgctcgcca tgatctttga gaagcgctca acccgaccca ggctgtccac tgaaccgggg 300
ttcgcgctgc ttggtggcca cccctgcttc ctgaccaccc aagacattca cctcgagtg 360
aacgaatcgc tcaactgatac tgcccgggtg ctgtcgtcga tggccgatgc agtgcctggc 420
aggggtgtaca aacagtcgga tctggacact ctggccaagg aggcgtccat ccctattatc 480
aacggccttt ccgacctcta ccaccgatt cagatccttg ccgattacct caccctgcaa 540
gaacactact cgtcactgaa gggcttgacc ttgtcctgga tcggcgacgg caacaacatc 600
ctccattcca ttatgatgtc cgccgccaaa ttccgcatgc atcttcaagc cgcaaccctc 660
aagggttacg agccggagcg ttccgtgacc aagctcgccg agcagtagcg taaggagaaac 720
ggaaccaagc ttctgtgac taacgacccc cttagggcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggacag gaagaagaga agaagaagcg gttacaggcg 840
ttccagggtc atcagggtc catgaaaaac gccaaaggtc ctgcctcgga ctggaccttc 900
ctgcattgcc tgctcgcgaa gccgaagaa gtggacgacg aggtgttcta ctgccacgg 960
tcctctgtgt tcctgagggc cgagaataga aagtggacca ttatggccgt gatggtgtcc 1020
cttctcacgg actactcgcc gcaactgcag aaaccaagt tctag 1065

SEQ ID NO: 152      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 152
atgcttttca atcttcgcat cctcctcaac aacgccgcct tccggaacgg tcacaacttc 60
atggtcggga acttcgctg cggccagccg ctccaaaaca aagtgcagct taaggggccg 120
gatcttctga ccttgaagaa cttcaccgga gaggaatca agtacatgct gtggctctcg 180
gccgacctga agtttaggat taagcagaag ggggagatgc tgccgctgct ccaagggaag 240
tcctttggca tgatcttga aaagaggtcc acccggactc ggctcagcac cgaacacaggt 300
tttgcacttc tggggggcca cccgtgcttc ctgacgaccc aggcacatcca tctgggtgtc 360
aacgagagtt tgaccgacac tgccagagtg ctgtcatcca tggcggacgc ggtgctcgcg 420
agagtgatga agcagtcgga tcttgacacc ctggcaaaaag aggcctcaat cccgatcatt 480
aacggactct cggatctgta ccacctatc caaatcttg ccgactacct gaccctgcaa 540
gaacactaca gctcctgaa gggcctgact ctttctcgga ttggcgatgg aaacaacatt 600
ctccattcta ttatgatgtc cgccgccaaag ttccgcatgc accttcaagc cgccaccccg 660
aagggttacg aacctgacgc ctccgtgact aagctagccg aacagtacgc taaggagaaac 720
ggcactaagc ttctccttac caacgatccg ctggaggcgg cccatggcgg aaatgtgctt 780
atcaccgaca ccttccttag catggggcag gaagaagaga agaagaacg gctccaggca 840
ttccagggtc accaggtcac catgaaaact gccaaaggtc ccgctagcga ctggaccttc 900
ctccactgtc tgctcgcgaa gctgaagaa gtggacgacg aggtgttcta ctcccgcgc 960
tcctctgtgt ttctgagggc cgagaacaga aagtggacca tcatggccgt gatggtgtca 1020
ttacttacgg actacagccc gcagctgcag aagccgaagt tctag 1065

SEQ ID NO: 153      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 153
atgcttttca acttgagaat ccttctgaac aacgccgcct tccgcaacgg tcataacttc 60
atggtcggga acttcagatg tggccagccc ctccaaaaca aagtgcagct gaaggggccg 120
gaccttctta cgctgaagaa ttccaccggc gaagaaatca agtacatgct ctggctgtcc 180
gccgatctta agttccgcat taagcagaag ggggaatacc ttccgctgct gcaagggaag 240
tcgctgggca tgatttttga gaagcgggtc actcgcaccc gcctgtccac tgaacttgga 300
ttcgactgct tcggtggcca tcctgcttc ctgaccaccc aagacatcca cctcggcgtg 360
aacgagtcct cgactgacac cgcccgggtc ttatcctcga tggccgatgc tgtgcttgcg 420
aggggtgtaca agcagtcgga cctcgacaca ctgcggaagg aggcctccat ccccatcctc 480
aacggcctgt ccgaccttta ccaccaatt cagatcctcg ccgattacct gaccctgcaa 540
gagcactact cgtcgctcaa ggggcttacc ctctcgtgga ttggcgacgg caacaacatc 600
cttactcca tcctgatgtc ggcagcgaag ttccgcatgc atctgcaagc cgccacgcct 660
aagggttatg aaccgatgct ctacgtgacc aagctcgccg aacagtacgc gaaagagaat 720
ggaaccaagc tacttctgac caacgacccc ctggaggcgg ctacaggcgg caacgtcctc 780
attaccgata cttggatttc gatgggacag gaagaggaaa agaagaagag actgcaggcg 840
ttccagggat accaggtcac catgaaaact gccaaagtg cagcctccga ctggaccttc 900

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cttcactgcc tgcgcaggaa gctgaagag gtggacgacg aggtgttcta ctccccgcgc 960
tccttgggtg ttcttgaggg cgaataaccg aagtggacta tcatggccgt gatgggtgtc 1020
ctctcaccg actactcgcc gcaactgcag aagcctaagt tctag 1065

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SEQ ID NO: 154      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 154
atgttattca accttagaat tctccttaac aacgcgcgct tccggaatgg gcataacttt 60
atggtccgca atttccgctg tggacagcct ctgcaaaaaca aggtccagct caaggggccgg 120
gatctgctga ctctcaagaa cttcactggg gaagaaatca agtacatgct ctggctgagc 180
gcccagctca agttccgcat caagcagaag ggagagtacc tcccgtgctt ccaagggaag 240
tccctgggca tgatctctga gaagagatcc acccgcacca gactttccac tgagactggc 300
ttcgcttgcg tgggaggcca cccatgcttc ctgacgaccc aggcatttca ccttggcgtg 360
aacgagtgcc tgactgacac gcgaagggtg ttgtcctcga tggccgacgc cgtgcttgcc 420
cgggtgtaca agcagagcga tcttgacacc ctggctaagg aagcttccat tcccatcctc 480
aacggtctga ggcacctgta ccacccgatt cagatcctgg cggactacct aaccctgcaa 540
gagcactata gctccctgaa gggcctcaca ctttcatgga tcggcgacgg caacaacatc 600
ctgcaactta ttatgatgag cgctgccaac ttccgcatgc acctccaagc cgccacgcct 660
aaaggctacg agcccgacgc ctccgtgacc aagcttgccg agcagtacgc gaaggaaaac 720
ggcaccgaag cgttctctac caacgatcct ctggaagcgg cccatggtgg caacgtgctc 780
attaccgaca ttctggatctc catgggacag gaggaggaaa agaagaagcg gctccaggcg 840
tttcagggtt accaggtcac catgaaaacc gccaaaggtc cagcctccga ctggaccttc 900
cttcattgct ttccgcgcaa gccgaagaa gtggacgatg aagtgtttta ctacactcgg 960
tcactcgtgt tcccggaagc agagaacagg aaatggacca ttatggccgt gatgggtgtc 1020
ctgctcaccg attacagtcc gcaactgcag aagcccaagt tctag 1065

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```

SEQ ID NO: 155      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 155
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaaca aggtccagct gaaggggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcccagctta agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggcatttca cctcgagatg 360
aacgaatccc tcacgcgatac cgcccggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt caaatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaa ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgct cgcgcgcaaaa ttccgcatgc atcttcaagc cgccacgcct 660
aagggttacg aacccgacgc ttccgtgact aagctcgccg agcagtacgc taaggagaaac 720
ggaaaccaagc tctgctgac taacgacccg ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaggaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggtcac catgaaaacc gccaaaggtc ctgcctccga ctggaccttc 900
ctgcactgct tgcctcgcaa gcttgaagaa gtggacgacg aggtgttcta ctgcgacgg 960
agcctcgtgt tccccgagcg cgagaataga aagtggacca tcatggccgt gatgggtgtc 1020
cttctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

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SEQ ID NO: 156      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 156
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaaca aggtccagct taaggggccgg 120
gatctcctca ccttataaaa cttcaccggc gaagagatca agtacatgct ctggctctcc 180
gcccagctta agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgcca tgatctttga gaagcgctca accaggacca ggctttctac tgaactggg 300
ttcgcgcttc tcggcggtca tccctgcttc ctcacgaccc aagacatcca cctcgagatg 360
aacgaatccc tcacggatac tgcccgctg ctttcgagca tggcagacgc cgtgctgcgc 420
cgggtgtaca aacagtccga tctgcacact ctgcgcaagg aggcgtcaat tcctattatc 480
aacggtctta gtgaccttta ccaccgatc cagatcctcg ccgattacct cacactccaa 540
gaacactaca gctcccttaa gggctctacc ctctcctgga tcggcgacgg caacaacatt 600
ctccactcca tcatgatgct cgcgcgcaaa ttccgcatgc atcttcaagc cgccaccccg 660
aagggttacg agcctgatgc ttccgtgact aagctcgccg agcagtacgc taaggagaaac 720
ggaaaccaagc ttcttctcac taacgacca ctcgaagcag cccatggggg caacgtgctt 780

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atcactgaca cctggatctc catgggccag gaagaagaga agaagaagcg gctccaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctccga ctggaccttt 900
ctccactgcc tccctcgcaa acctgaagaa gtggacgacg aggtgttcta ctgcgcccg 960
agcctcggtg tccccgaggg cgagaataga aagtggacca ttatggccgt gatggtgtca 1020
ctcctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

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```

SEQ ID NO: 157      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

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SEQUENCE: 157
atgcttttca atctccgcat cctccttaac aacgccgctg ttagaaacgg acataacttc 60
atggtccgga acttcagatg tggacagccg cttcaaaaca aggtccagct gaagggtcgg 120
gatcttctga ccttgaagaa ctttacggga gaagagatca agtacatgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggagaatacc tcccgctgct tcaaggaaaag 240
agcctcgga tgaattttga gaagcgctca accaggaccg gcctttctac tgaactgga 300
ttcgcgctgc tgggtggaca cccctgcttc ctgacgaccc aggacatcca cctcgagtg 360
aacgaatccc tcaactgata cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcoga tctggacact ctggccaagg aggcgtcaat tctatcatc 480
aacggactta gtgacctcta ccatccgatt caaatcctgg ccgactacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggagatgg aaacaacatt 600
ctccactcca tcatgatgtc gcgcgcaaaa ttcggaatgc atctccaagc cgccacgcct 660
aagggttacg aaccgcagcg tcccgctgact aagctcgccg agcagtacgc taaggagaaac 720
ggtaccaagc ttctcctgac caacgaccca ctagaagcag cccacggtgg aaacgtgctt 780
attactgaca cttggatctc catgggacag gaggaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctccga ctggaccttc 900
ctgcaactgcc tgctcgcgaa gctgaagaa gtggacgacg aggtgttcta ctgcgcccg 960
agcctcggtg tccccgaggg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 158      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

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SEQUENCE: 158
atgcttttca acctccgcat tctcctcaac aacgctgcct tccggaatgg acataacttc 60
atggtccgga acttcagatg cggacagccg cttcagaaca aggtccagct taaggggaga 120
gatctcctta ccttcaaaaa cttcactggc gaagaaatca agtacatgct ctggcttagt 180
gcggacttca agttccgcat caagcagaag ggagaatacc tcccgctcct tcaaggaaaag 240
agcctcgga tgaattttga gaagaggtcc accagaactc gcctttcaac cgagactggg 300
ttcgccctgc ttggcggtca cccctgcttc ctcaactacc aagacatcca cctcggcgtg 360
aacgagagcc ttaccgacac cgcccgcgtg ctctcctcaa tggccgacgc tgtgtcgccc 420
cgggtgtaca agcagtcoga ccttgatact ctgcgcaagg aggcctccat cccaattatc 480
aacgggctct ctgactcteta ccaccctatc caaatcctcg cggactacct caccctccaa 540
gagcactata gctcgctcaa gggcctcacc ctttctgga ttggcgacgg caacaacatt 600
cttcaactga tcatgatgtc gcgcgccaag ttcggcattg atctccaagc cgcgaccccc 660
aagggttacg agcctgacgc atccgtgacc aagctcgccg agcagtacgc gaaggaaaat 720
ggcaccagc ttcttctcac caacgacccc cttgaggccg ctatggcgcg caacgtgctc 780
atcactgaca cttggatctc catgggcccag gaggaggaaa agaagaagcg ctttcaggca 840
ttccagggtt accaggtcac catgaaaacc gccaaagtgg ccgcctccga ctggaccttt 900
cttcaactgtc tcccgcggaa gctgaagaa gtggatgacg aagtgtttta ctcccctcgg 960
tcaactcggt tcccggaagc agaaaacagg aagtggacca ttatggccgt catggtgtcc 1020
ctcctcaccg actacagccc gcagcttcag aaaccaagt tctag 1065

```

```

SEQ ID NO: 159      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature         1..1065
                     note = Synthetic construct
source               1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 159
atgcttttca atctccgcat cctccttaac aacgcagcgt ttagaaacgg tcacaacttc 60
atggtccgga acttccgctg tggacagccg cttcaaaaca aggtccagct gaagggtcgg 120
gaaccttctga ccttgaagaa ctttactgga gaagagatca agtacatgct ttggctgtcc 180
gcggacttga agttccgcat taagcagaag ggagaatacc tcccgctgct ccaaggaaaag 240
agcctgggaa tgaattttga gaagcgctca accaggaccg gcctttctac tgaactgga 300
ttcgcgctgc tgggtggata cccctgcttc ctgacgaccc aggacattca cctcgagtg 360
aacgagtccc tcaactgata cgccagagtg ttatcgagca tggcagatgc cgtgctggct 420
aggggtgtaca aacagtccga tctggacacc ctggccaagg aggcattcaat tctattatc 480
aacggactta gtgacctcta ccatccgatt caaatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggagatgg aaacaacatt 600
ctccattcca tcatgatgtc cgcgcccaag ttcggaatgc atctccaagc cgccacgcgg 660

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```

aaaggatcacg agccggagcgc ttccgtgact aagctcgccg agcagtagcg taaggagaaac 720
ggaaaccaagc ttctgtgtgac taacgacccg ctagaagccg cccacggtgg aaacgtgtct 780
attactgaca cctggatctc catgggacag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ccgcctccga ctggaccttc 900
cttcaactgcc tgcctcggaa gectgaagaa gtggacgacg aggtgttcta ctgcgcgcg 960
agcctcgtgt tccctgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctctcaccg actacagccc gcagcttcag aagcctaagt tctag 1065

```

```

SEQ ID NO: 160      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 160
atgcttttca atctccgeat tctctcaac aacgcagcct ttagaaacgg ccacaacttc 60
atgggtccgga acttcagatg tggccagccg cttcagaaca aggtccagct caagggccgg 120
gacctctca cctcaaaaaa cttaccggc gaagagatca agtacatgct ctggctttcg 180
gccgacctta agttccgcat caagcagaag ggggaatacc ttccgtgct tcaaggaaag 240
tccctcggca tgatctttga aaagcgctcg accaggaccc gcctttccac tgaaccggcg 300
ttcgcgcttc tcggtggcca cccctgcttc ctacaccacc aagacattca cctcgagtg 360
aacgaatccc ttaccgatac cgcaagagtg ctttcgtcga tggccgatgc cgtgcttgcg 420
cggtgtgata agcagtcaga tctcgacact ctgcaccaag aggcgtccat tccattatc 480
aacggccttt cgcaccttta ccaccgatt cagatcctcg ccgattacct caccctgcaa 540
gagcactact cgtcactcaa gggctcttacc ctctcctgga tcggcgacgg aaacaacatc 600
ctccattcga tcatgatgtc cgcgcgcaaa ttccgcatgc acctccaaag cgcgaccccg 660
aagggttacg agcccgacgc ttccgtgacc aagctcgccg aacagtacgc taaggaaaac 720
ggcaccagc tcctctctac taacgacct ctcgaaagcag cccatggggg caacgtgctc 780
attactgaca cttggatctc gatgggcccag gaagaggaga aaaagaagcg gcttcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctcgga ctggaccttc 900
cttcaactgcc ttcccgcaaa gectgaagag gtggacgatg aggtgttcta ctccccacgg 960
tcccttgtgt tccccgaggc cgagaatagg aagtggacca tcatggccgt gatggtgtcg 1020
ctctcactg actactcccc gcaacttcag aagcctaagt tctag 1065

```

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SEQ ID NO: 161      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 161
atgctgttta atctgagaat acttctaaac aacgcgcct tccggaatgg ccataacttt 60
atgggtccgga atttccgctg cggccagccg ctgcagaaca aggtccagct gaagggaaga 120
gacttgctga cctcaagaa cttcaccgga gaagaaatca agtatatgct gtggctgtcc 180
gccgacctga aattccgcat caagcagaag ggcgaatgc tgccgctgtt gcaagggaag 240
tccctgggga tgatcttcga gaagaggtcc accagaacac ggctttcaac cgaaccggcg 300
tttgactgc tgggtggaca cccctgtttt ctgaccactc aagatatcca cctggcgctg 360
aacgagtcctc ttaccgacac tgctaggggtg ttgtccagca tggccgatgc cgtcctggct 420
cgctgtgata agcagtcaga cctggatacc ctggcaaaag aagcgtccat tccattatc 480
aacgggctgt ccgacctgta ccatccgatt caaatcctgg ccgactacct gactctgcaa 540
gagcattaca gcagcttgaa ggggcttact ctctcgtgga tcggcgacgg gaacaacatc 600
ctgcactcca tcatgatgtc cgcgcgcaag ttccggatgc atttgcaagc tgcgaccccg 660
aaaggttacg agcccgatgc tagcgtaact aagcttgccg aacagtacgc caaagagaat 720
ggtacaaaac tgcttctgac taacgacccg ctggaagcag cccacggcgg gaacgtgctg 780
ataaccgaca cctggatttc aatggggcag gaggaagaga agaagaagcg actgcaggcg 840
ttccaaggct atcaggttac catgaaaacc gccaaagtgg cagccagcga ttggactttc 900
ctgcactgtc tgcccgcgaa gcccgaggaa gttgatgacg aagtattcta ctaccccg 960
agcctcgtgt tccccgaggc cgaaaaccgg aagtggacta ttatggccgt gatggtgtcg 1020
ctgttgaccg actacagccc gcaactgcag aagccgaagt ttag 1065

```

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SEQ ID NO: 162      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 162
atgcttttca acctgaggat ccttttgaac aacgcgcct ttcgcaacgg ccacaacttt 60
atgggtccgca atttccgctg cgggcagccg ctgcagaaca aggtccagct gaagggccgg 120
gatctgtgta ccttgaagaa cttcaccggg gaggaatca agtacatgct ttggctctcc 180
gccgatctga agttcagaag caagcagaag ggagagtacc tccgtgtgct gcaaggaaag 240
tcaactcgaa tgattttcga aaagagaagc actaggaccc gcctctcaac tgaaccggcg 300
ttcgcgctgc tcggggcgca tccgtgtttc ctgactaccc aagacatcca cctgggagtg 360
aacgagtcgc tgaccgacac cgcacgcgtg ctgtcatcca tggcggacgc agtgcttgcc 420
cggtgtgata agcagtcgga cctggacact cttgccaagg aggcataaat cccatcatt 480
aacggactgt ccgatctcta ccaccgatt cagatcctgg ctgactacct aacctgcaa 540

```

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```

gagcactact caagcctgaa ggggctgacc ctgtcgtgga tcgggggacgg caacaacatt 600
ctgcaactcca tcatgatgtc ggcggctaag ttcgggatgc atttgcaagc ggcaactccg 660
aagggttatg aaccgcagcg ctcctgaccc aagctggccg aacagtacgc caaggaaaac 720
ggaaccaagt tgctgtgac taatgatccc ctggaggcgg ccacggggg gaacgtgtctg 780
ataaccgata cctggatctc catggggcag gaagaagaga agaaaaagcg gctgcaggca 840
ttccagggat accaggctac catgaaaacc gcaaaagtgg cagccagcga ctggactttc 900
ctccattgcc tgcgcgaaa gccggaggag gtccgatgac aggtgttcta ctcccgcgg 960
tcgctgggtg tcccggaggc ggaaccggg aagtggacca ttatggccgt gatggtgtca 1020
ctcctgactg actacagccc gcaactgcag aagccgaagt tctag 1065

```

```

SEQ ID NO: 163      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 163
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaaca aggtccagct gaagggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagtg 360
aacgaaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcgga tctggacact ctggccaagg aggcgtcaat tctattatc 480
aacggcctta gtgacctcta ccattccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgtc cgccgcaaaa ttccgcatgc atcttcaagc cgccacgccc 660
aagggttacg agccgcagcg ttccgtgact aagctcgccc agcagtacgc taaggagaac 720
ggaaccaagc tctgtgtgac taacgaccca ctagaagcag ccacgggggg caacgtgctt 780
attactgaca ccttcctctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggctac catgaaaacc gccaaaggtc ctgcctccga ctggaccttc 900
ctgcaactgcc tgcctcgcaa gcctgaagaa gtggacgacg aggtgttcta ctgcacacg 960
agcctcggtg tcccggaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctagataag tgaa 1074

```

```

SEQ ID NO: 164      moltype = DNA length = 1073
FEATURE            Location/Qualifiers
misc_feature        1..1073
                    note = Synthetic construct
source              1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 164
atgctgttca acctgcgcat cctgctgaac aacgcgcgct tccgcaacgg ccacaacttc 60
atggtgcgca acttcgctg cgccagcccc ctgcagaaca aggtgcagct gaagggcccg 120
gacctgtgta cctgaagaa cttcacccgc gaggagatca agtacatgct gtggctgagc 180
gccgaacctga agttccgcat caagcagaag ggcagtagcc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcca gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ctggccctgc tggcgcccca cccctgcttc ctgaccaccc aggacatcca cctgggctgtg 360
aacgagagacc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgtaca agcagagcga cctggacacc ctggccaagg aggcagcat ccccatcatc 480
aacggcctga gcgacctgta ccccccattc cagatcctgg ccgactacct gacctgcag 540
gagcaactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttccgcatgc acctgcaggc cgccaccccc 660
aagggttacg agccgcagcg cagcgtgacc aagctggccc agcagtacgc caaggagaac 720
ggcaccaca gctgtgtgac caacgacccc ctggaggccg ccacggcgcg caacgtgtctg 780
atcaccgaca cctggatcag catgggccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accagggtgac catgaagacc gccaaaggtg ccgcccagcga ctggaccttc 900
ctgcaactgcc tgcctcgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccgc 960
agcctgggtg tcccggaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctgaataag tga 1073

```

```

SEQ ID NO: 165      moltype = DNA length = 1073
FEATURE            Location/Qualifiers
misc_feature        1..1073
                    note = Synthetic construct
source              1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 165
atgctgttca acctgcgcat cctgctgaac aacgcgcgct tccgcaacgg ccacaacttc 60
atggtgcgca acttcgctg cgccagcccc ctgcagaaca aggtgcagct gaagggcccg 120
gacctgtgta cctgaagaa cttcacccgc gaggagatca agtacatgct gtggctgagc 180
gccgaacctga agttccgcat caagcagaag ggcagtagcc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcca gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ctggccctgc tggcgcccca cccctgcttc ctgaccaccc aggacatcca cctgggctgtg 360
aacgagagacc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420

```

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```

cgcggtgtaca agcagagcgga cctggacacc ctggccaagg aggccagcat ccccatcacc 480
aacggcctga ggcacctgta ccaccccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcctgatgag cgcgcgcaag ttccgcatgc acctgcaggg cgccaccccc 660
aagggctacg agcccgacgc cagcgtgacc aagctggccg agcagtagcg caaggagaaac 720
ggcaccgaag tgctgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accagggtgac catgaagacc gccaaaggtg ccgccagcga ctggaccttc 900
ctgcaactgc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctggtgt tccccaggcg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgacgg actacagccc ccagctgcag aagcccaagt tctgaataag tga 1073

```

```

SEQ ID NO: 166      moltype = DNA length = 1074
FEATURE             Location/Qualifiers
misc_feature         1..1074
                     note = Synthetic construct
source               1..1074
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 166
atgcttttca atctccgcat cctccttaac aacgcgcgt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcoga tctggacact ctggccaagg aggcgtcaat tctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcctgatgtc gcccgcaaaa ttccgcatgc atcttcaagc cgccacgccg 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtagcg taaggagaaac 720
ggaaccaagc tctctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggctac catgaaaaacc gccaaagtcg ctgcctccga ctggaccttc 900
ctgcaactgc tgccctgcaa gccctgaagaa gtggacgacg aggtgttcta ctgcgccagg 960
agcctcgtgt tccccaggcg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcacgg actacagccc gcagcttcag aagcccaagt tctagataag tgaa 1074

```

```

SEQ ID NO: 167      moltype = DNA length = 1073
FEATURE             Location/Qualifiers
misc_feature         1..1073
                     note = Synthetic construct
source               1..1073
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 167
atgcttttca acctgagaat cctcttgaac aatgctgctt ttccgaatgg ccacaacttt 60
atggttccga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgctca cactaaagaa ctttactgga gaagagatca agtacatgct atggctgtcg 180
gccgacctga agttccgcat caagcagaag ggaagaatacc ttccgctgct tcaaggaaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcoga tctcgatacc ttggcaaaag aggccttccat tccatcacc 480
aacggcctga ggcacctgta ccaccaatc caaatcctgg ctgactacct gaccctgcaa 540
gagcactaca gcagcctgaa gggctctgacc ctgtcatgga ttggcgatgg aaacaatatt 600
ctgcactcca tcctgatgtc cgcgcggaag ttccgaatgc atctgcaagc cgccactcca 660
aaaggatacg aacgggatgc atccgtgacc aagttggcgg aacagtagcg gaaggagaaac 720
ggaaccaagc tctctgctgac taacgacccc ctcgaggctg cgcctggggg taacgtgctg 780
attacggaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccagggtt accaggctac catgaaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcatatgc tgccgaggaa gcccgaggaa gtgcacgacg aagtgttcta ctgcctcgg 960
tccctggtgt tccccaggcg cgaaaaaccg aagtggacca tcatggccgt gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctagataag tga 1073

```

```

SEQ ID NO: 168      moltype = DNA length = 1073
FEATURE             Location/Qualifiers
misc_feature         1..1073
                     note = Synthetic construct
source               1..1073
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 168
atgctgttca acctgcgcat cctgctgaac aacgcgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgg 120
gacctgtgta ccttgaagaa cttcaccggc gaggagatca agtacatgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcagtagc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcta gaagcgacg acccgacccc gcctgagcac cgagacaggc 300

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ctggccctgc tggggcgcca cccctgcttc ctgaccaccc agggacatcca cctgggctgt 360
aacgagagcc tgacogacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgttaca agcagagcga cctggacacc ctggccaagg agggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccaccccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcggcatgc acctgcaggg cgccaccccc 660
aagggtctac agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720
ggcaccAAGC tgctgctgac caacgacccc ctggaggccg ccacggcgcg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accaggtgac catgaagacc gccAaggtgg ccgcccagcga ctggaccttc 900
ctgcaactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctgaataag tga 1073

```

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SEQ ID NO: 169      moltype = DNA length = 1074
FEATURE
misc_feature       1..1074
                    note = Synthetic construct
source             1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 169
atgcttttca atctccgeat cctccttaac aacgcccgcgt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacctgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaataacc ttccgctgct tcaaggaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc agggacatcca cctcgagtg 360
aacgaatccc tcaccgatgc cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgata aacagtccga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa ggggtctgaa ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgtc cgccgcaaaa ttcggcatgc atcttcaagc cgccacggcg 660
aagggttacc agcccgacgc ttccgtgact aagctcgccg agcagtacgc taaggagAAC 720
ggaaaccaagc ttctgtgac taacgaccca ctagaagcag ccacgggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggtc atcagggtcac catgaaaacc gccAaggtcg ctgcctccga ctggaccttc 900
ctgcaactgcc tgccccgcaa gccctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcgtgt tccccgaggg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc cgagcttcag aagcccaagt tctagataag tgaa 1074

```

```

SEQ ID NO: 170      moltype = DNA length = 1073
FEATURE
misc_feature       1..1073
                    note = Synthetic construct
source             1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 170
atgctgttca acctgcgcgt cctgctgaac aacgcccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cgccagcccc ctgcagaaca aggtgcagct gaagggccgc 120
gaactgtcga ccttgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180
gcccacctga agttccgcat caagcagaag ggcgagtacc tgccccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ctggccctgc tggggcgcca cccctgcttc ctgaccaccc agggacatcca cctgggctgt 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgttaca agcagagcga cctggacacc ctggccaagg agggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccaccccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcggcatgc acctgcaggg cgccaccccc 660
aagggtctac agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720
ggcaccAAGC tgctgctgac caacgacccc ctggaggccg ccacggcgcg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accaggtgac catgaagacc gccAaggtgg ccgcccagcga ctggaccttc 900
ctgcaactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctgaataag tga 1073

```

```

SEQ ID NO: 171      moltype = DNA length = 1073
FEATURE
misc_feature       1..1073
                    note = Synthetic construct
source             1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 171
atgcttttca acctgagaat cctcttgaac aatgctgctt ttccgaatgg ccacaacttt 60
atggttcgga acttccgctg cgccagccct ttacaaaaca aggtccagct gaagggccgg 120
gatttgctca cactaaagaa ctttactgga gaagagatca agtacctgct atggtgctgc 180

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```

gccgacctga agttccgtat caagcagaag ggagaatacc ttccgctgct tcaaggaaag 240
agcctcgga tgaattttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcggagt 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcoga tctcgatacc ttggcaaaag aggcctccat tccatcatc 480
aacggcctga gcgacctgta cccccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa gggcttgacc ctgtcatgga ttggcgatgg aaacaatatt 600
ctgcactcca tcatgatgtc cgccgcgaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aacgggatgc atcctgaccc aagttggcgg aacagtacgc gaaggagaac 720
ggaaccaagc tctgtctgac taacgacccg ctcgaggctg cgcattgggg taactgtctg 780
attacggaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccagggtt accaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tggcaggagaa gccggaggaa gtgcagcagc aagtgttcta ctgcctcgg 960
tccctgggtg tcccgaggc cgaaaaccgg aagtggacca tcatggcctg gatgggtgc 1020
tgctgactg actatagccc gcagctgcag aagcctaagt tctagataag tga 1073

```

```

SEQ ID NO: 172      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 172
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaacgg ccacaacttc 60
atgggtccga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacctgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgga tgaattttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcggagt 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtcoga tctcgatacc ttggcaaaag aggcctccat tccatcatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggcttgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgtc cgccgcaaaa ttcgcatgc atcttcaagc cgccacggcg 660
aagggttacg agcccagcgc ttccgtgact aagctcgccg agcagtacgc taaggagaac 720
ggaaccaagc tctgtctgac taacgaccca ctagaagcag cccacggggg caactgtgct 780
attactgaca cctggatctc catggggcag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggtcac catgaaaacc gccaaagtcg ctgcctcoga ctggaccttc 900
ctgcactgcc tgcctcgcaa gccctgaaga gtggacgacg aggtgttcta ctgcacacg 960
agcctcgtgt tccccaggc cgagaataga aagtggacca tcatggcctg gatgggtgca 1020
ctgctcacgg actacagccc gcagcttcag aagcccaagt tctagataag tgaa 1073

```

```

SEQ ID NO: 173      moltype = DNA length = 1073
FEATURE            Location/Qualifiers
misc_feature        1..1073
                    note = Synthetic construct
source              1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 173
atgctgttca acctcgcat cctgctgaac aacgcgcgct tccgcaacgg ccacaacttc 60
atgggtgcga acttcgctg cgccagcccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta cctgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180
gcggaactga agttccgcat caagcagaag ggcagatacc tgcctctgct gcagggcaag 240
agcctgggca tgaattttga gaagcgacgc acccgacccc gcctgagcac cgagacaggg 300
ctggccctgc tggcgcgcca cccctgcttc ctgaccaccc aggcacatcca cctgggctg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggcacgcat cccatcatc 480
aacggcctga gcgacctgta ccaaccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatt 600
ctgcacagca tcatgatgag cgccgcaaa ttcgcatgc acctgcaggc cgccaccccc 660
aagggttacg agcccagcgc cagcgtgacc aagctggccg agcagtacgc caaggagaac 720
ggcaccgaag tctgtctgac caacgacccc ctggaggccg cccacggcgg caactgtgct 780
atcaccgaca cctggatcag catggggcag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accaggtgac ctgaagacc gccaaagtcg ccgcaagcga ctggaccttc 900
ctgcactgcc tggcccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccaggc cgagaaccgc aagtggacca tcatggcctg gatgggtgag 1020
ctgctgacgg actacagccc ccagctgcag aagcccaagt tctgaataag tga 1073

```

```

SEQ ID NO: 174      moltype = DNA length = 1073
FEATURE            Location/Qualifiers
misc_feature        1..1073
                    note = Synthetic construct
source              1..1073
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 174
atgcttttca acctgagaat cctcttgaac aatgctgctt ttccgaatgg ccacaacttt 60
atggttcgga acttccgttg cggccagcct ttacaaaaca aggtccagct gaagggccgg 120
gatttgetca cactaaagaa ctttactgga gaagagatca agtacatget atggctgtcg 180
gccgacctga agttccgtat caagcagaag ggagaatacc ttccgtgct tcaaggaaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagt 360
aacgaatccc tcacogatac cgcgggggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctcgatacc ttggcaaaag aggtctccat tcccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa gggctctgac ctgtcatgga ttggcgatgg aaacaatatt 600
ctgcactcca tcgatgtgct cgcgcgaag ttccgaatgc atctgcaagc cgcactcca 660
aaaggatagc aaccggatgc atccgtgacc aagtgtggcg aacagtacgc gaaggagaa 720
ggaaaccaagc tctgctgac taacgacccg ctcgaggctg cgcagggggg taacgtgctg 780
attacggaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccaggggt accaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcatgtcc tgcgagggaa gccggaggaa gtgcgacgag aagtgttcta ctgcctcgg 960
tccctggtgt tccccgaggc cgaaaaccgg aagtggacca tcatggcctg gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt ttagataag tga 1073

```

```

SEQ ID NO: 175      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 175
atgctgttca acctgcgcat cctgctgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgttg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta cctgaagaa cttcaccggc gaggagatca agtacatget gtggctgagc 180
gccgacctga agttccgtat caagcagaag ggcgagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatcttcca gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggacatcca cctgggcgtg 360
aacgagagcc tgacogacac cgcggcgctg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccacccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tggcgacgg caacaacatc 600
ctgcacagca tcgatgtgag cgcgcgaag ttccgcatgc acctgcaggc cgccaccccc 660
aaaggctacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaa 720
ggcaccgaagc tgctgctgac caacgacccc ctggaggccg ccacggcgcg caacgtgctg 780
atcaccgaca cctggatcag catggggcag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac ctgaagacc gccaaagtgg ccgccaagcga ctggaccttc 900
ctgcaactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctggtgt tccccgaggc cgagaaccgc aagtggacca tcatggcctg gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 176      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct
source              1..1065
                     mol_type = other DNA
                     organism = synthetic construct

```

```

SEQUENCE: 176
atgctgttca acctgcgcat cctgctgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgttg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta cctgaagaa cttcaccggc gaggagatca agtacatget gtggctgagc 180
gccgacctga agttccgtat caagcagaag ggcgagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatcttcca gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggacatcca cctgggcgtg 360
aacgagagcc tgacogacac cgcggcgctg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccacccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tggcgacgg caacaacatc 600
ctgcacagca tcgatgtgag cgcgcgaag ttccgcatgc acctgcaggc cgccaccccc 660
aaaggctacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaa 720
ggcaccgaagc tgctgctgac caacgacccc ctggaggccg ccacggcgcg caacgtgctg 780
atcaccgaca cctggatcag catggggcag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac ctgaagacc gccaaagtgg ccgccaagcga ctggaccttc 900
ctgcaactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctggtgt tccccgaggc cgagaaccgc aagtggacca tcatggcctg gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 177      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                     note = Synthetic construct

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```

source          1..1065
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 177
atgctgttca acctgcgcat cctgctgaac aacgcgcgct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta ccttgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgagc acccgacccc ccctgagcac cgagacaggc 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggacatcca cctgggcgtg 360
aacgagagcc tgaccgacac cgcgcgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgttaca agcagagcga cctggacacc ctggccaagg aggcacgcat ccccatcatc 480
aacggcctga gcgacctgta ccaccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgcgcgcaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggctacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaac 720
ggcaccgaag tgctgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccaaaggtg ccgcccagcga ctggaccttc 900
ctgcaactgcc tgcccgcgaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctgggtg tccccgaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

SEQ ID NO: 178      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 178
atgctgttca acctgcgcat cctgctgaac aacgcgcgct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta ccttgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgagc acccgacccc ccctgagcac cgagacaggc 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggacatcca cctgggcgtg 360
aacgagagcc tgaccgacac cgcgcgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgttaca agcagagcga cctggacacc ctggccaagg aggcacgcat ccccatcatc 480
aacggcctga gcgacctgta ccaccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgcgcgcaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggctacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaac 720
ggcaccgaag tgctgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccaaaggtg ccgcccagcga ctggaccttc 900
ctgcaactgcc tgcccgcgaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctgggtg tccccgaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctgaataag tgaa 1074

SEQ ID NO: 179      moltype = DNA length = 1077
FEATURE            Location/Qualifiers
misc_feature        1..1077
                    note = Synthetic construct
source              1..1077
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 179
atgggcgtct tcaacctgcg gatcctgctg aacaacgcgc ccttcgggaa cggccacaac 60
ttcatggtcc gcaacttcag atcgcgccag cccctgcaga acaaggtgca gctgaagggc 120
cgggaacctg tgacctgaa gaacttcacc ggcaagaga tcaagtacat gctgtggctg 180
agcggccgac tgaagtccg gatcaagcag aagggcgagt acctgcccc gctgcaaggc 240
aagagccctg gcatgatctt cgagaagcgg agcacccgga cccggctgag caccgagaca 300
ggctttgccc tgctgggagg ccacccctgc tttctgacca ccaggacat ccacctgggc 360
gtgaacgaga gcctgaccga caccgccaga gtgctgagca gcatggccga cggcgtgctg 420
gcccgggtgt acaagcagag cgacctggac accctggcca aagaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggaaacct acagtccctt gaaggccctg acctgagct ggatcggcga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgcc aagttcggca tgcactgca gcccgccacc 660
cccaagggtc acgagcctga tgccagcgtg accaagctgg ccgagcagta cgccaagag 720
aacggcacca agctgctgct gaccaacgac cccctggaag ccgcccacgg cggcaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaagagg aaaagaagaa gcggctgcag 840
gccttccagg gctaccaggt cacaatgaag accgccaaag tggcccgacg cgaactggacc 900
ttcctgactc gctgccccg gaagcccgaa gaggtggacg acgaggtgtt ctacagcccc 960
cgggtccctg gtgtcccgga ggccgagaa cgaagtggga ccattatggc cgtgatgggtg 1020
tccctgctga ccgactactc cccccagctg cagaagccca agttctagat aagtga 1077

SEQ ID NO: 180      moltype = DNA length = 1077
FEATURE            Location/Qualifiers

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misc_feature      1..1077
                  note = Synthetic construct
source            1..1077
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 180
atgggcgtct tcaacctgcg gatcctgctg aacaacgccg ccttcgggaa cggccacaac 60
ttcatgggtcc gcaacttcag atcgcgccag cccctgcaga acaggggtgca gctgaagggc 120
cgggacctgc tgacctgaa gaacttcacc ggccaagaga tcagggtacat gctgtggctg 180
agcgccgacc tgaagtccg gatcaagcag aagggcgagt acctgcccc gctgcaaggc 240
aagagcctgg gcatgatctt cgagaagcgg agcacccgga cccggctgag caccgagaca 300
ggctttgccc tgctgggagg cccccctgc tttctgacca cccaggacat ccacctgggc 360
gtgaacgaga gctgaccga caccgccaga gtgctgagca gcatggccga cgccgtgctg 420
gcccgggtgt acaagcagag cgacctggac accctggcca aagaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggaacact acagctccct gaagggcctg acctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgcc aagttcgga tgcactgca ggccgccacc 660
cccaagggtc acgagcctga tgcacgctg accaagctgg ccgagcagta cgccaagag 720
aacggcacca agctgtgct gaccaacgac cccctggaag ccgcccacgg cggcaacgtg 780
ctgataccgg acactggat cagcatgggc caggaagagg aaaagaagaa gcggctgcag 840
gcttccagg gctaccaggt cacaatgaag accgccaagg tggcccgacg cgaactggacc 900
ttcctgactc tgctgcccc gaagcccgaa gaggtggacg acgaggtgtt ctacagcccc 960
cgggtccctg gtgtcccgga ggccgagaa cgaagtggga ccattatggc cgtgatgggtg 1020
tccctgctga ccgactactc cccccagctg cagaagccca agttctagat aagtga 1077

SEQ ID NO: 181      moltype = DNA length = 1077
FEATURE            Location/Qualifiers
misc_feature        1..1077
                  note = Synthetic construct
source              1..1077
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 181
atgctggtct tcaacctgcg gatcctgctg aacaacgccg ccttcgggaa cggccacaac 60
ttcatgggtcc gcaacttcag atcgcgccag cccctgcaga acaggggtgca gctgaagggc 120
cgggacctgc tgacctgaa gaacttcacc ggccaagaga tcagggtacat gctgtggctg 180
agcgccgacc tgaagtccg gatcaagcag aagggcgagt acctgcccc gctgcaaggc 240
aagagcctgg gcatgatctt cgagaagcgg agcacccgga cccggctgag caccgagaca 300
ggctttgccc tgctgggagg cccccctgc tttctgacca cccaggacat ccacctgggc 360
gtgaacgaga gctgaccga caccgccaga gtgctgagca gcatggccga cgccgtgctg 420
gcccgggtgt acaagcagag cgacctggac accctggcca aagaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggaacact acagctccct gaagggcctg acctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgcc aagttcgga tgcactgca ggccgccacc 660
cccaagggtc acgagcctga tgcacgctg accaagctgg ccgagcagta cgccaagag 720
aacggcacca agctgtgct gaccaacgac cccctggaag ccgcccacgg cggcaacgtg 780
ctgataccgg acactggat cagcatgggc caggaagagg aaaagaagaa gcggctgcag 840
gcttccagg gctaccaggt cacaatgaag accgccaagg tggcccgacg cgaactggacc 900
ttcctgactc tgctgcccc gaagcccgaa gaggtggacg acgaggtgtt ctacagcccc 960
cgggtccctg gtgtcccgga ggccgagaa cgaagtggga ccattatggc cgtgatgggtg 1020
tccctgctga ccgactactc cccccagctg cagaagccca agttctagat aagtga 1077

SEQ ID NO: 182      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                  note = Synthetic construct
source              1..1074
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 182
atgctgttca acctgaggat cctgctgaac aacgcagctt tcaggaacgg ccacaacttc 60
atgggtgagga acttccggtg cggccagccc ctgcagaaca aggtgcagct gaagggcagg 120
gacctgctga cctgaagaa cttcaccgga gaggagatca agtatcatgt gtggctgagc 180
gcagacctga agttcaggat caagcagaag ggagagtacc tgccccgtct gcaggggaag 240
tccctgggca tgatcttcga gaagaggagt accaggacca ggctgagcac cgaaccgggc 300
tcgccccctg tgggaggaca cccctgcttc ctgaccaccc aggacatcca cctgggctg 360
aacgagagtc tgaccgacac cgccagggtg ctgtctagca tggccgacgc cgtgctggcc 420
aggggtgata agcagtcaga cctggacacc ctggctaagg aggcacgcat ccccatcatc 480
aacggcctga gcgacctgta ccaccccatc cagatcctgg ctgactacct gacctgctg 540
gagcactaca gctctctgaa gggcctgacc ctgagctgga tcggcgacgg gaacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcggcatgc acctgcaggc cgctaccccc 660
aaggggttac agccccagc cagcgtgacc aagctggcag agcagtagc caaggagaa 720
ggcaccaaag cgtgtgtgac caacgacccc ctggaggccg cccacggagg caacgtgctg 780
atcaccgaca cctggatcac catgggacag gaggaggaga agaagaagcg gctgcaggct 840
ttccagggtt accaggtag catgaagacc gccaaaggtg ctgccagcga ctggaccttc 900
ctgcaactgc tgcccaggaa gcccgaggag gtggacgacg aggtgttcta ctctccagg 960
agcctggtgt tccccaggc cgagaacagg aagtggacca tcatggctgt gatggtgtcc 1020
ctgctgacgg actacagccc ccagctgcag aagcccaagt tctgaataag tgaa 1074

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SEQ ID NO: 183      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature       1..1074
                   note = Synthetic construct
source            1..1074
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 183
atgctgttca acctgaggat cctgctgaac aacgcagctt tcaggaacgg ccacaacttc 60
atggtgagga acttccgggt cggccagccc ctgcagaaca aggtgcagct gaagggcagg 120
gacctgtgta ccttgaagaa cttcaccgga gaggagatca agtacatgct gtggctgagc 180
gcagacctga agttcaggat caagcagaag ggagagtacc tgcccctgct gcaggggaa 240
tccctgggca tgatcttcga gaagaggagt accaggacca ggctgagcac cgaaacccgc 300
ttcgccctgc tgggaggaca cccctgcttc ctgacgaccc aggacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcaagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggctaagg aggccagcat ccccatcatc 480
aacggcctga ggcacctgta ccaccccatc cagatcctgg ctgactacct gaccctgcag 540
gagcactaca gctctctgaa gggcctgacc ctgagctgga tcggcgacgg gaacaacatc 600
ctgcactcca tcctgatgtc gcccgcgaag ttcggaatgc atctgaagc cgccacgcca 660
aaaggatagc aacccgatgc gccctgaca aagttggcgg aacagtacgc taaggagaac 720
ggaaccaagc tgctgtgac caacgacccc ctggaggccg cccacggagg caacgtgctg 780
atcacccgca cctggatcac catgggacag gaggaggaga agaagaagcg gctgcaggct 840
ttccagggtt accaggtgac catgaagacc gccaaagtggt ctgccagcga ctggaccttc 900
ctgcaactgcc tgcccaggaa gcccgaggag gtggacgacg aggtgttcta ctctccagg 960
agcctggtgt tcccaggagg cgagaacagg aagtggacca tcattgctgt gatggtgtcc 1020
ctgctgacgg actacagccc ccaagctgac aagcccaagt tctgaataag tgaa 1074

SEQ ID NO: 184      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature       1..1074
                   note = Synthetic construct
source            1..1074
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 184
atgctgttca acctgcgcat cctgctgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta ccttgaagaa cttcaccggc gaggagatca agtacatgct gtggctgagc 180
gcccacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggggggcca cccctgcttc ctgaccaccc aggacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgcggtgtaca agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga ggcacctgta ccaccccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gacgctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcctgatgag cgccgccaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggtacg agcccagcgc cagcgtgacc aagctggccc agcagtacgc caaggagaac 720
ggcaccgaagc tgctgtgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcacccgca cctggatcac catgggccaag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtt accaggtgac catgaagacc gccaaagtggt ccgcccagca ctggaccttc 900
ctgcaactgcc tgcccgcgaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctggtgt tcccaggagg cgagaacccc aagtggacca tcattgcccgt gatggtgagc 1020
ctgctgacgg actacagccc ccaagctgac aagcccaagt tctgaataag tgaa 1074

SEQ ID NO: 185      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature       1..1074
                   note = Synthetic construct
source            1..1074
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 185
atgctgttca acctgcgcat cctgctgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgta ccttgaagaa cttcaccggc gaggagatca agtacatgct gtggctgagc 180
gcccacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgacgc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggggggcca cccctgcttc ctgaccaccc aggacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgcggtgtaca agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga ggcacctgta ccaccccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gacgctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcctgatgag cgccgccaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggtacg agcccagcgc cagcgtgacc aagctggccc agcagtacgc caaggagaac 720
ggcaccgaagc tgctgtgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcacccgca cctggatcac catgggccaag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtt accaggtgac catgaagacc gccaaagtggt ccgcccagca ctggaccttc 900

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ctgctactgcc tgcctccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctgggtg tccctcgaggc cgagaaccgc aagtggacca tcatggcggt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctgaataag tgaa 1074

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SEQ ID NO: 186      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 186
atgcttttca acttgagaat cctgctgaac aacgcgcgct ttccgaacgg tcacaatttt 60
atggtcagaa acttcagatg cggacagccc ctccaaaaca aggtccagct gaagggccgc 120
gatctcctca cctcgaagaa cttcacgggg gaggagatca agtacatgct gtggctctcc 180
gctgacctga agttcaggat caagcagaag ggagaatata tgcgcgtgct gcaagggaaag 240
tccctgggga tgattttcga gaagcggagc acccggaact ggctctccac tgaaactggg 300
ttcgcccttc tgggcggtca cccctgcttc ctgaccactc aagacattca cctcggagtg 360
aacgagtcct tgactgacac gcccgggtg ctgtcgagca tggcagacgc cgtgctagcc 420
cgctgtgata agcagtcaga cctcgatacc ctggccaagg aggtctcgat cccgatcctc 480
aacgggttgt ccgacctgta ccacccgatt cagattctcg ccgactacct caccctgcaa 540
gagcattaca gctccctgaa ggggcttacc ctgtcctgga ttggcgacgg aaacaacatc 600
ctgcaactca ttatgatgtc ggcggccaag ttccgcatgc acctccaagc cgcgacccct 660
aagggttacg aaccagacgc gtcagtgaat aagctggccc aacagtacgc aaaggaaaat 720
ggcaccgaag tgcctctgac caacgatccg ttggaagccg cccatggcgg aatatgtgctc 780
atcaccgaca cctggatctc gatgggacag gaggaagaga agaagaagcg gctgcaggcg 840
ttccagggtc accaggtcac catgaaaact gccaaaggtg ccgcccagca ctggaccttc 900
ctgcaactgcc ttccgcgcaa gctgaggag gtggacgacg aagtgttcta ctctccacg 960
tccctgggtg tccctcgaggc ggagaaccgc aagtggacca tcatggcggt gatggtcagc 1020
ctgctgaccg attacagccc tcagttgcaa aagccgaagt tttga 1065

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SEQ ID NO: 187      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 187
atgctgttca acctccgcat cctcctcaac aacgcgcgct tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaaca aggtccagct caagggggcg 120
gacctcctga cctcgaagaa cttcacgggc gaagagatca agtacatgct gtggctctcc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaaag 240
tcgctgggga tgatctctga gaagcgggtc accagaaccc ggctgtcaac cgaaccggcg 300
ttcgcactgc tggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagtg 360
aacgaatcgc tgaccgacac gcccgcgctg ctgagctcaa tggcggaacg cgtgctggcc 420
cgctgtgata agcagtcaga cctggacacc ctggccaagg aagctccatc cccgatcctc 480
aacggactgt ccgacctgta ccacccgatc cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tgggggaacg gaacaacatc 600
ctgcaactca taatgatgtc agccgccaag ttccggaatg acctccaagc cgcacacccg 660
aagggttacg aaccggacgc atcagtgaac aaactggccc agcagtacgc caaggaaaac 720
ggcaccgaag tccctcctga caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcaccgaca cctggatctc catgggacag gaggaaggaa agaagaagcg gctgcaggcg 840
ttccagggtc accaggtcac catgaaaacc gccaaaggtg ccgcatcaga ctggaccttc 900
ctgcaactgcc ttccctcgaa gccggaagag gtggacgacg aggtgttcta ctcccgcg 960
tcgctgggtg tccctcgaggc ggagaacagg aagtggacca tcatggcggt gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

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SEQ ID NO: 188      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 188
atgcttttca atctccgcat cctccttaac aacgcgcgct ttagaaacgg ccacaacttc 60
atggtcaggga acttcagatg tggccagccg cttcaaaaaca aggtccagct gaagggccgg 120
gatcttctga cctcgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
goggacttga agttccgcat taagcagaag ggggaaatacc ttccgctgct tcaaggaaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggaatccca cctcggagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgata aacagtcoga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgtc cgccgcaaaa ttccgcatgc atcttcaagc cgccacgcgc 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtacgc taaggagaac 720
ggaaccaaac ttctgctgac taacgacca ctagaagcag cccacggggg caacgtgctt 780

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attactgaca cctggatctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggtcg ctgcctccga ctggacottc 900
ctgcactgcc tgccctcgcaa gccctgaagaa gtggacgacg aggtgttcta ctgccacagg 960
agcctcgtgt tccccgaggg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctgaataag taga 1074

```

```

SEQ ID NO: 189      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 189
atgcttttca acctgagaat cctcttgaac aatgctgctt ttcggaatgg ccacaacttt 60
atggttcgga acttccgctt cgccagcctt ttacaaaaca aggtccagct gaagggccgg 120
gatttgctca cactaaagaa ctttactgga gaagagatca agtacatgct atggctgtcg 180
gccgacctga agttccgctt caagcagaag ggagaatacc ttccgctgct tcaaggaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctcgatacc ttggcaagg aggttccat tccatcatc 480
aacggcctga gcgacctgta ccaccaatc caaatcctgg ctgactacct gacctgcaa 540
gagcactaca gcagcctgaa gggctctgacc ctgtcatgga ttggcgatgg aaacaatat 600
ctgcaactcca tcatgatgtc cgcgcggaag ttcggaatgc atctgcaagc cgccactcca 660
aaaggatacg aacgcgatgc atccgtgacc aagttggcgg aacagtacgc gaaggagaa 720
ggaaccaagc tcctgctgac taacgacccg ctcgaggctg cgcatggggg taacgtgctg 780
attacggaca cctggatctc catggggcag gaggaagaga agaagaagag actgcaggca 840
ttccaggggt accaggtcac catgaaaacc gcaaaagtgg cagcttcgga ctggactttc 900
ctgcattgcc tgccgaggaa gccggaggaa gtcgacgacg aagtgttcta ctgcctcgg 960
tcctggtgtg tccccgaggg cgaaaaaccg aagtggacca tcatggccgt gatggtgtcc 1020
ttgctgactg actatagccc gcagctgcag aagcctaagt tctgaataag taga 1074

```

```

SEQ ID NO: 190      moltype = DNA length = 1074
FEATURE            Location/Qualifiers
misc_feature        1..1074
                    note = Synthetic construct
source              1..1074
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 190
atgctgttca acctgcgcat cctgtgaac aacgccgctt tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cgccagcctt ctgcagaaca aggtgcagct gaagggccgc 120
gacctgtgca cctgaagaa cttcacccgc gaggagatca agtacatgct gtggctgagc 180
gccgacctga agttccgctt caagcagaag ggcgagtacc tgccctgctg gcagggcaa 240
agcctgggca tgatcttcca gaagcgagc acccgacccc gctgagcac cgagacaggc 300
ctggccctgc tggggcgcca cccctgcttc ctgaccaccc aggacatcca cctgggctg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgcggtgtaca agcagagcga cctggacacc ctggccaagg aggcagcat cccatcatc 480
aacggcctga gcgacctgta ccacccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcggtatgc acctgcaggc cgccaccccc 660
aagggtctac agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaa 720
ggcaccagc tctgtctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccaaaggtg ccgcccagca ctggaccttc 900
ctgcactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgacgg actacagccc ccagctgcag aagcccaagt tctgaataag taga 1074

```

```

SEQ ID NO: 191      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 191
atgcttttca atctccgcat cctccttaac aacgccgctt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gctgacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgcca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggccaagg aggcgtcaat tctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggcttgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgtc cgccgcaaaa ttcggtatgc atcttcaagc cgccacggcg 660

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-continued

```

aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtagcg taaggagAAC 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctccga ctggaccttc 900
ctgcactgcc tgccctcgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 192      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 192
atgcttttca atctccgcat cctccttaac aacgcccgtg ttagaaacgg ccacaacttc 60
atgggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaaggggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaataacc ttccgctgct tcaaggaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttccgctgctc tgggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg cgcattacct caccctgcaa 540
gaacactaca gctccctgaa ggggtctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcgatgtgct cgccgcaaaa ttccgcatgc atcttcaagc cgccacggcg 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtagcg taaggagAAC 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctccga ctggaccttc 900
ctgcactgcc tgccctcgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 193      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 193
atgcttttca atctccgcat cctccttaac aacgcccgtg ttagaaacgg ccacaacttc 60
atgggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaaggggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaataacc ttccgctgct tcaaggaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttccgctgctc tgggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg cgcattacct caccctgcaa 540
gaacactaca gctccctgaa ggggtctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcgatgtgct cgccgcaaaa ttccgcatgc atcttcaagc cgccacggcg 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtagcg taaggagAAC 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggcccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcagggtcac catgaaaacc gccaaaggctg ctgcctccga ctggaccttc 900
ctgcactgcc tgccctcgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 194      moltype = DNA length = 1065
FEATURE
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 194
atgcttttca atctccgcat cctccttaac aacgcccgtg ttagaaacgg ccacaacttc 60
atgggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaaggggccgg 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacatgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaataacc ttccgctgct tcaaggaaag 240
agcctcggca tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttccgctgctc tgggtggcca cccctgcttc ctgacgaccc aggcacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgtaca aacagtccga tctggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg cgcattacct caccctgcaa 540

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-continued

```

gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca gctcatgtgc cgccgcaaaa ttccgcatgc atcttcaagc cgccacgccc 660
aagggttacg agcccgacgc ttccgtgact aagctcgccc agcagtagcg taaggagaac 720
ggaaccaagc ttctgtgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggctac catgaaaacc gccaaaggtcg ctgcctccga ctggaccttc 900
ctgcactgcc tgccctgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcggtg tccccgagcg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 195      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 195
atgcttttca atctccgcat cctccttaac aacgcccgtg ttagaaacgg ccacaacttc 60
atggctccga acttcagatg tggccagccg cttcaaaaac aggtccagct gaaggggccc 120
gatcttctga cctgaagaa ctttactggc gaagagatca agtacctgct ctggctctcc 180
gcggaactga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcggca tgcctcttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
tcgcgctgct tcggtggcca cccctgcttc ctgacgaccc aggacatcca cctcggagt 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgata aacagtcgca tctggacact ctggccaagg aggcgtcaat tctattatc 480
aacggcctta gtgacctcta ccattccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa gggctctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca gctcatgtgc cgccgcaaaa ttccgcatgc atcttcaagc cgccacgccc 660
aagggttacg agcccgacgc ttccgtgact aagctcgccc agcagtagcg taaggagaac 720
ggaaccaagc ttctgtgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggctac catgaaaacc gccaaaggtcg ctgcctccga ctggaccttc 900
ctgcactgcc tgccctgcaa gctgaagaa gtggacgacg aggtgttcta ctgccacgg 960
agcctcggtg tccccgagcg cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 196      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 196
atgggtgttca acctccgcat cctcctcaac aacgcccgtg tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaaca aggtccagct caagggggcg 120
gacctcctga cctgaagaa cttcaccggc gaagagatca agtacctgct gtggctctcc 180
gcccgaactga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaag 240
tcgctgggga tgcattctga gaagcgggtc accagaaccc ggctgtcaac cgaaccggg 300
tcgcactgct tggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagt 360
aacgaatcgc tgaccgacac cgcccgcgtg ctgagctcaa tggcggacgc cgtgctggcc 420
cgctgtgata agcagtcgca cctggacacc ctggccaagg aagcgtccat cccgatcatc 480
aacggactgt ccgacctgta ccacccgatc cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcactcca taatgatgtc agccgccaaag ttccggaatgc acctccaagc cgcaacccc 660
aagggttacg aaccggacgc atcagtgacc aaactggccc agcagtagcg caaggaaaac 720
ggcaccgaag tcctgtgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcaccgaca cctggatctc catgggacag gagggagaaa agaagaagcg gctgcaggcg 840
ttccaggggg accaggctac catgaaaacc gcgaaggtcg cggcacatga ctggaccttc 900
ctgcactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcccgcg 960
tcgctgggtg tccccgagcg ggagaacagg aagtggacca tcatggccgt gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 197      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 197
atgggtgttca acctccgcat cctcctcaac aacgcccgtg tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaacc ggtccagct caagggggcg 120
gacctcctga cctgaagaa cttcaccggc gaagagatca agtacctgct gtggctctcc 180
gcccgaactga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaag 240
tcgctgggga tgcattctga gaagcgggtc accagaaccc ggctgtcaac cgaaccggg 300
tcgcactgct tggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagt 360
aacgaatcgc tgaccgacac cgcccgcgtg ctgagctcaa tggcggacgc cgtgctggcc 420

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cgcggtgtaca agcagtcgga cctggacacc ctggccaagg aagcgtccat cccgatcacc 480
aacggactgt ccgacctgta ccacccgac cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcaactcca taatgatgtc agccgccaag ttcggaatgc acctccaagc cgcaaccccg 660
aagggctacg aaccggacgc atcagtgacc aaactggccg agcagtagcg caaggaaaac 720
ggcaccgaag tcctgctgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcacccgaca cctggatctc catgggacag gaggagggaaa agaagaagcg gctgcaggcg 840
ttccaggggt accaggtcac catgaaaacc gcgaagggtcg cggcatcaga ctggaccttc 900
ctgcaactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcgcgcgc 960
tcgctggtgt tccccgaggg ggagaacagg aagtggacca tcattggcggg gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 198      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 198
atggtgttca acctccgcat cctcctcaac aacgcgcgat tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaacc gggtccagct caagggggcg 120
gacctcctga cctgaagaa cttcaccggc gaagagatcc ggtacatgct gtggtctctc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaag 240
tcgctgggga tgatcttcga gaagcgggtc accagaaccc ggctgtcaac cgaaacccgg 300
ttcgcaactgc tggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagtg 360
aacgaatcgc tgaccgacac cgcccgcgtg ctgagctcaa tggcggacgc cgtgctggcc 420
cgcggtgtaca agcagtcgga cctggacacc ctggccaagg aagcgtccat cccgatcacc 480
aacggactgt ccgacctgta ccacccgatc cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcaactcca taatgatgtc agccgccaag ttcggaatgc acctccaagc cgcaaccccg 660
aagggctacg aaccggacgc atcagtgacc aaactggccg agcagtagcg caaggaaaac 720
ggcaccgaag tcctgctgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcacccgaca cctggatctc catgggacag gaggagggaaa agaagaagcg gctgcaggcg 840
ttccaggggt accaggtcac catgaaaacc gcgaagggtcg cggcatcaga ctggaccttc 900
ctgcaactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcgcgcgc 960
tcgctggtgt tccccgaggg ggagaacagg aagtggacca tcattggcggg gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 199      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 199
atgctgggtca acctccgcat cctcctcaac aacgcgcgat tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaacc aggtccagct caagggggcg 120
gacctcctga cctgaagaa cttcaccggc gaagagatca agtacatgct gtggtctctc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaag 240
tcgctgggga tgatcttcga gaagcgggtc accagaaccc ggctgtcaac cgaaacccgg 300
ttcgcaactgc tggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagtg 360
aacgaatcgc tgaccgacac cgcccgcgtg ctgagctcaa tggcggacgc cgtgctggcc 420
cgcggtgtaca agcagtcgga cctggacacc ctggccaagg aagcgtccat cccgatcacc 480
aacggactgt ccgacctgta ccacccgatc cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcaactcca taatgatgtc agccgccaag ttcggaatgc acctccaagc cgcaaccccg 660
aagggctacg aaccggacgc atcagtgacc aaactggccg agcagtagcg caaggaaaac 720
ggcaccgaag tcctgctgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcacccgaca cctggatctc catgggacag gaggagggaaa agaagaagcg gctgcaggcg 840
ttccaggggt accaggtcac catgaaaacc gcgaagggtcg cggcatcaga ctggaccttc 900
ctgcaactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcgcgcgc 960
tcgctggtgt tccccgaggg ggagaacagg aagtggacca tcattggcggg gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 200      moltype = DNA length = 1065
FEATURE             Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 200
atgctgggtca acctccgcat cctcctcaac aacgcgcgat tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaacc gggtccagct caagggggcg 120
gacctcctga cctgaagaa cttcaccggc gaagagatca agtacatgct gtggtctctc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tcccgctgct gcaagggaag 240
tcgctgggga tgatcttcga gaagcgggtc accagaaccc ggctgtcaac cgaaacccgg 300

```


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```

ttcgcaactgc tgggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagtg 360
aacgaatcgc tgaccgacac cgcgccgctg ctgagctcaa tggcggaacg cgtgctggcc 420
cgctgtgaca agcagtcgca cctggacacc ctggccaagg aagcgtccat cccgatcatc 480
aacggactgt ccgacctgta ccaccgacg cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcaactcca taatgatgtc agccgccaag ttcggaatgc acctccaagc cgcaaccccg 660
aagggctacg aaccggacgc atcagtgacc aaactggccg agcagtacgc caaggaaaac 720
ggcaccgaag tcctgtgtgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcaccgaca cctggatctc catgggacag gaggaggaaa agaagaagcg gctgcaggcg 840
ttccaggggg accaggtcac catgaaaacc gcgaaggctc cggcatcaga ctggaccttc 900
ctgcaactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcgcgcg 960
tcgctggtgt tccccgagcg ggagaacagg aagtggacca tcatggcggt gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 201      moltype = DNA length = 1065
FEATURE
misc_feature       1..1065
                    note = Synthetic construct
source             1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 201
atgctgggtca acctccgcat cctctcaaac aagcccgcat tcagaaacgg gcacaacttc 60
atggtcagaa acttccgctg cgggcaaccc ctacaaaacc gggtcacagt caagggcgcg 120
gacctctga cctgaagaa cttcaccggc gaagagatcc ggtacatgct gtggctctcc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tcccgtgctg gcaagggaag 240
tcgctgggga tgatctctga gaagcgggtc accagaaccc ggctgtcaac cgaacccggg 300
ttcgcaactgc tgggggggaca cccgtgcttc ctgaccaccc aagacatcca cctgggagtg 360
aacgaatcgc tgaccgacac cgcgccgctg ctgagctcaa tggcggaacg cgtgctggcc 420
cgctgtgaca agcagtcgca cctggacacc ctggccaagg aagcgtccat cccgatcatc 480
aacggactgt ccgacctgta ccaccgacg cagatcctgg cagactacct gaccctgcaa 540
gaacactaca gctccctgaa gggcctgacc ctgtcatgga tcggggacgg gaacaacatc 600
ctgcaactcca taatgatgtc agccgccaag ttcggaatgc acctccaagc cgcaaccccg 660
aagggctacg aaccggacgc atcagtgacc aaactggccg agcagtacgc caaggaaaac 720
ggcaccgaag tcctgtgtgac caacgacccg ctggaggccg cacacggggg gaacgtgctg 780
atcaccgaca cctggatctc catgggacag gaggaggaaa agaagaagcg gctgcaggcg 840
ttccaggggg accaggtcac catgaaaacc gcgaaggctc cggcatcaga ctggaccttc 900
ctgcaactgcc tgccccggaa gccggaagag gtggacgacg aggtgttcta ctgcgcgcg 960
tcgctggtgt tccccgagcg ggagaacagg aagtggacca tcatggcggt gatggtcagc 1020
ctcctgaccg actactcgcc gcagctgcag aagccgaagt tctga 1065

```

```

SEQ ID NO: 202      moltype = DNA length = 1065
FEATURE
misc_feature       1..1065
                    note = Synthetic construct
source             1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 202
atgctgggtca acctgcgcat cctgctgaac aagcccgct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggcgcg 120
gacctgtga cctgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggagagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatctctga gaagcgcagc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tgggggggaca cccgtgcttc ctgaccaccc agacatcca cctgggctgt 360
aacgagagcc tgaccgacac cgcgccgctg ctgagcagca tggcggaacg cgtgctggcc 420
cgctgtgaca agcagagcga cctggacacc ctggccaagg aggcacagat ccccatcatc 480
aacggctga ggcacctgta ccacccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgcgcccaag ttcgctatgc acctgcagcg cgccaccccc 660
aagggctacg agcccgacg cagcgtgacc aagctggccg agcagtacgc caaggagaa 720
ggcaccgaag tgctgtgtgac caacgacccc ctggaggccg ccacggcgcg caacgtgctg 780
atcaccgaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccaaaggtg ccgccaagcg ctggaccttc 900
ctgcaactgcc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccgagcg cgagaacccg aagtggacca tcatggcggt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

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SEQ ID NO: 203      moltype = DNA length = 1065
FEATURE
misc_feature       1..1065
                    note = Synthetic construct
source             1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 203
atgctgggtca acctgcgcat cctgctgaac aagcccgct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaacc ggggtcagct gaagggcgcg 120
gacctgtga cctgaagaa cttcaccggc gaggagatca agtacctgct gtggctgagc 180

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-continued

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gccgacctga agttccgcat caagcagaag ggcgagtacc tgccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgagc acccgacccc gctgagcac cgagacaggc 300
ttcgccctgc tggcgcgcca cccctgcttc ctgaccacccc aggcacatcca cctggcgctg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgcggtgtaca agcagagcga cctggacacc ctggccaagg aggcagcat ccccatcatc 480
aacggcctga gcgacctgta ccccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcgcatgc acctgcaggc cgccaccccc 660
aagggtctac agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720
ggcaccAagc tgcgtctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accaggtgac catgaagacc gccAaggtg ccgccagcga ctggaccttc 900
ctgcactgcc tgcctcgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgc 960
agcctgtgtg tccccgaggc cgagaacccg aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

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SEQ ID NO: 204      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 204
atgcttgta atctccgcat cctccttaac aacgcgcgt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaaca aggtccagct gaagggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacctgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgga tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgacccc aggcacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgata aacagtcga ctgggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa ggggtctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgct cgccgcaaaa ttcgcatgc atcttcaagc cgccacggcg 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtacgc taaggagAAC 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggtcac catgaaaacc gccAaggtcg ctgcttcga ctggaccttc 900
ctgcactgcc tgcttcgcaa gccctgaagaa gtggacgacg aggtgttcta ctgcacacg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

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SEQ ID NO: 205      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 205
atgcttgta atctccgcat cctccttaac aacgcgcgt ttagaaacgg ccacaacttc 60
atggtccgga acttcagatg tggccagccg cttcaaaacc gggtccagct gaagggccgg 120
gatcttctga ccttgaagaa ctttactggc gaagagatca agtacctgct ctggctctcc 180
gcggacttga agttccgcat taagcagaag ggggaatacc ttccgctgct tcaaggaaag 240
agcctcgga tgatctttga gaagcgctca accaggaccc gcctttctac tgaactggg 300
ttcgcgctgc tcggtggcca cccctgcttc ctgacgacccc aggcacatcca cctcgagtg 360
aacgaatccc tcaccgatac cgcccgggtg ttatcgagca tggcagatgc cgtgctggcc 420
aggggtgata aacagtcga ctgggacact ctggccaagg aggcgtcaat tcctattatc 480
aacggcctta gtgacctcta ccatccgatt cagatcctgg ccgattacct caccctgcaa 540
gaacactaca gctccctgaa ggggtctgaca ttgtcctgga tcggcgacgg caacaacatt 600
ctccattcca tcatgatgct cgccgcaaaa ttcgcatgc atcttcaagc cgccacggcg 660
aagggttacg agcccgacgc ttccgtgact aagctcgccg agcagtacgc taaggagAAC 720
ggaaccaagc ttctgctgac taacgaccca ctagaagcag cccacggggg caacgtgctt 780
attactgaca cctggatctc catgggccag gaagaagaga aaaagaagcg gctgcaggcg 840
ttccagggat atcaggtcac catgaaaacc gccAaggtcg ctgcttcga ctggaccttc 900
ctgcactgcc tgcttcgcaa gccctgaagaa gtggacgacg aggtgttcta ctgcacacg 960
agcctcgtgt tccccgaggc cgagaataga aagtggacca tcatggccgt gatggtgtca 1020
ctgctcaccg actacagccc gcagcttcag aagcccaagt tctag 1065

```

```

SEQ ID NO: 206      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 206
atgggccttg tcaatctccg catcctcctt aacaacgcgc cgtttagaaa cgccacaaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaaa acaagggtcca gctgaagggc 120
cgggatcttc tgaccttgaa gaactttact ggccgaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagttccg cattaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcattgatctt tgagaagcgc tcaaccaggga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggttg ccaccctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat cctcaccga taccgcccgc gtgttatcga gcatggcaga tgcctgctg 420
gccaggggtg acaaacagtc cgatctggac actctggcca aggaggctc aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcgccga cggcaacaac 600
attctccatt ccatcatgat gtccgcccga aaattcgcca tgcattctca agccgccacg 660
ccgaagggtt acgagcccca cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttcacgg gatattcagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttccccga ggccgagaat agaaagtggg ccatcatggc cgtgatgggt 1020
tactgtctca ccgactacag cccgcagctt cagaagccca agttctag 1068

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SEQ ID NO: 207      moltype = DNA length = 1068
FEATURE
misc_feature        1..1068
                     note = Synthetic construct
source              1..1068
                     mol_type = other DNA
                     organism = synthetic construct

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SEQUENCE: 207
atgggccttg tcaatctccg catcctcctt aacaacgcgc cgtttagaaa cgccacaaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaaa accgggtcca gctgaagggc 120
cgggatcttc tgaccttgaa gaactttact ggccgaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagttccg cattaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcattgatctt tgagaagcgc tcaaccaggga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggttg ccaccctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat cctcaccga taccgcccgc gtgttatcga gcatggcaga tgcctgctg 420
gccaggggtg acaaacagtc cgatctggac actctggcca aggaggctc aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcgccga cggcaacaac 600
attctccatt ccatcatgat gtccgcccga aaattcgcca tgcattctca agccgccacg 660
ccgaagggtt acgagcccca cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttcacgg gatattcagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttccccga ggccgagaat agaaagtggg ccatcatggc cgtgatgggt 1020
tactgtctca ccgactacag cccgcagctt cagaagccca agttctag 1068

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SEQ ID NO: 208      moltype = DNA length = 1071
FEATURE
misc_feature        1..1071
                     note = Synthetic construct
source              1..1071
                     mol_type = other DNA
                     organism = synthetic construct

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SEQUENCE: 208
atgggcggac ttgtcaatct ccgcacctc cttaacaacg ccgcgtttag aaacggccac 60
aacttcattg tccggaactt cagatgtggc cagccgcttc aaaacaaggt ccagctgaag 120
ggccggggtc ttctgacctt gaagaacttt actggcgaag agatcaagta catgctctgg 180
ctctccgcgg acttgaagtt ccgcattaag cagaaggggg aataccttcc gctgcttcaa 240
ggaaagagcc tcggcatgat ctttgagaag cgctcaacca ggaccgcctt ttctactgaa 300
actgggttcg cgctgctcgg tggccacccc tgcttctga cgaccaggga catccacctc 360
ggagtgaacg aatccctcac cgataccgcc cgggtgttat cgagcatggc agatgcctg 420
ctggccaggg tgtacaaaca gtccgatctg gacactctgg ccaaggaggc gtcaattcct 480
attatcaacg gccttagtga cctctaccat ccgattcaga tctggccga ttacctcacc 540
ctgcaagaac actacagctc cctgaagggt ctgacattgt cctggatcgg cgacggcaac 600
aacattctcc attccatcat gatgtccgcc gcaaaattcg gcatgcatct tcaagccgcc 660
acgccgaagg gttacagacc cgacgcttcc gtgactaagc tcgcccagca gtacgctaag 720
gagaacggaa ccaagcttct gctgactaac gaccacttag aagcagccca cgggggcaac 780
gtgcttatta ctgacacctg gatctccatg ggccaggaag aagagaaaaa gaagcggctg 840
caggcggttc agggatatca ggtcacccatg aaaaccgcca aggtcgctgc ctccgactgg 900
accttctcgc actgctgccc tcgcaagcct gaagaagtgg acgacgaggt gttctactcg 960
ccacggagcc tcgtgttccc cgaggccgag aatagaaagt ggaccatcat ggccgtgatg 1020
gtgtcactgc tcaccgacta cagcccgcag cttcagaagc ccaagttcta g 1071

```

```

SEQ ID NO: 209      moltype = DNA length = 1077
FEATURE
misc_feature        1..1077
                     note = Synthetic construct

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source          1..1077
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 209
atggcccttt tcaatctccg catcctcctt aacaacgccg cgtttagaaa cgccacaaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag gcaagggtcca gctgaagggc 120
cgggatcttc tgacctgaa gaactttact ggccaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagtccg cattaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccaggga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg ccacctctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat ccttcaccga taccgcccgg gtgttatcga gcatggcaga tgcctgtctg 420
gccaggggtg acaaacagtc cgaatctggac actctggcca aggaggcgct aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccatcatgat gtccgcccga aaattcggca tgcattctca agccgccacg 660
ccgaagggtt acgagcccgga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttccagg gatcatcagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttcccoga ggccgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tactgtctca ccgactacag cccgcagctt cagaagccca agttctagat aagtga 1077

SEQ ID NO: 210      moltype = DNA length = 1077
FEATURE            Location/Qualifiers
misc_feature        1..1077
                    note = Synthetic construct
source              1..1077
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 210
atggcccttt tcaatctccg catcctcctt aacaacgccg cgtttagaaa cgccacaaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag gccgggtcca gctgaagggc 120
cgggatcttc tgacctgaa gaactttact ggccaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagtccg cattaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccaggga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg ccacctctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat ccttcaccga taccgcccgg gtgttatcga gcatggcaga tgcctgtctg 420
gccaggggtg acaaacagtc cgaatctggac actctggcca aggaggcgct aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccatcatgat gtccgcccga aaattcggca tgcattctca agccgccacg 660
ccgaagggtt acgagcccgga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttccagg gatcatcagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttcccoga ggccgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tactgtctca ccgactacag cccgcagctt cagaagccca agttctagat aagtga 1077

SEQ ID NO: 211      moltype = DNA length = 1077
FEATURE            Location/Qualifiers
misc_feature        1..1077
                    note = Synthetic construct
source              1..1077
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 211
atggcccttt tcaatctccg catcctcctt aacaacgccg cgtttagaaa cgccacaaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag gccgggtcca gctgaagggc 120
cgggatcttc tgacctgaa gaactttact ggccaagaga tcaggtagat gctctggctc 180
tccgcggact tgaagtccg cattaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccaggga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg ccacctctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat ccttcaccga taccgcccgg gtgttatcga gcatggcaga tgcctgtctg 420
gccaggggtg acaaacagtc cgaatctggac actctggcca aggaggcgct aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccatcatgat gtccgcccga aaattcggca tgcattctca agccgccacg 660
ccgaagggtt acgagcccgga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttccagg gatcatcagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttcccoga ggccgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tactgtctca ccgactacag cccgcagctt cagaagccca agttctagat aagtga 1077

SEQ ID NO: 212      moltype = DNA length = 1077
FEATURE            Location/Qualifiers

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misc_feature      1..1077
                  note = Synthetic construct
source            1..1077
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 212
atggcccttg tcaatctccg cactctcctt aacaacgccg cgtttagaaa cggccacaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag gcagggtcca gctgaagggc 120
cgggatcttc tgaccctgaa gaactttact ggcgaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagtccg cattaaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccagga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg cccccctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat cccctaccga taccgcccgg gtgttatcga gcatggcaga tgccgtgctg 420
gccaggggtg acaaacagtc cgatctggac actctggcca aggaggcgtc aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccactcatgat gtccgcccga aaattcgcca tgcattctca agccgccacg 660
ccgaagggtt acgagcccga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttcacag gatatacagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttccccga ggcgcgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tcactgctca ccgactacag cccgcagctt cagaagccca agttctagat gaataga 1077

SEQ ID NO: 213      moltype = DNA length = 1077
FEATURE             Location/Qualifiers
misc_feature         1..1077
                    note = Synthetic construct
source               1..1077
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 213
atggcccttt tcaatctccg cactctcctt aacaacgccg cgtttagaaa cggccacaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag tcaagggtcca gctgaagggc 120
cgggatcttc tgaccctgaa gaactttact ggcgaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagtccg cattaaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccagga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg cccccctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat cccctaccga taccgcccgg gtgttatcga gcatggcaga tgccgtgctg 420
gccaggggtg acaaacagtc cgatctggac actctggcca aggaggcgtc aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccactcatgat gtccgcccga aaattcgcca tgcattctca agccgccacg 660
ccgaagggtt acgagcccga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttcacag gatatacagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttccccga ggcgcgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tcactgctca ccgactacag cccgcagctt cagaagccca agttctagat aagtga 1077

SEQ ID NO: 214      moltype = DNA length = 1077
FEATURE             Location/Qualifiers
misc_feature         1..1077
                    note = Synthetic construct
source               1..1077
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 214
atggcccttt tcaatctccg cactctcctt aacaacgccg cgtttagaaa cggccacaac 60
ttcatgggtcc ggaacttcag atgtggccag ccgcttcaag tcaagggtcca gctgaagggc 120
cgggatcttc tgaccctgaa gaactttact ggcgaagaga tcaagtacat gctctggctc 180
tccgcggact tgaagtccg cattaaagcag aagggggaat accttccgct gcttcaagga 240
aagagcctcg gcatgatctt tgagaagcgc tcaaccagga cccgccttcc tactgaaact 300
gggttcgcgc tgctcggtgg cccccctgc ttctgacga cccaggacat ccacctcgga 360
gtgaacgaat cccctaccga taccgcccgg gtgttatcga gcatggcaga tgccgtgctg 420
gccaggggtg acaaacagtc cgatctggac actctggcca aggaggcgtc aattcctatt 480
atcaacggcc ttagtgacct ctaccatccg attcagatcc tggccgatta cctcaccctg 540
caagaacact acagctccct gaagggtctg acattgtcct ggatcggcga cggcaacaac 600
attctccatt ccactcatgat gtccgcccga aaattcgcca tgcattctca agccgccacg 660
ccgaagggtt acgagcccga cgcttccgtg actaagctcg ccgagcagta cgctaaggag 720
aacggaacca agcttctgct gactaacgac ccactagaag cagcccacgg gggcaacgtg 780
cttattactg acacctggat ctccatgggc caggaagaag agaaaaagaa gcggctgcag 840
gcgttcacag gatatacagg caccatgaaa accgccaagg tcgctgcctc cgactggacc 900
ttcctgcact gcctgcctcg caagcctgaa gaagtggacg acgaggtgtt ctactcgcca 960
cggagcctcg tgttccccga ggcgcgagaat agaaagtgga ccatcatggc cgtgatgggtg 1020
tcactgctca ccgactacag cccgcagctt cagaagccca agttctagat aagtga 1077

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SEQ ID NO: 215      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 215
atgctgttca acctgcecat cctgetgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttccgctg cggccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gaactgtctga ccttgaagaa cttcacggcg gaggagatca agtacatgct gtggtgagc 180
gcccacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgagc acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggcgggcca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgttaca agcagagcga cctggacacc ctggccaagg agggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccacccatc cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcgcatgc acctgcaggc cgccaccccc 660
aagggtacg agcccgacgc cagcgtgacc aagctggccg agcagtacgc caaggagaa 720
ggcaccgaag tgctgtgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcacggaca cctggatgag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggtc accaggtgac catgaagacc gccaaagtg cgcacagcga ctggaccttc 900
ctgcaactgc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccccg 960
agcctggtgt tccccgaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgacgg actacagccc ccaagtgcag aagcccaagt tctga 1065

SEQ ID NO: 216      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 216
atgggagtat tcaacctgcg catcctgctg aacaacgccg ccttccgcaa cggccacaac 60
ttcatggtgc gcaacttccg ctgcggccag cccctgcaga acaaggtgca gctgaagggc 120
cgcgacctgc tgaccctgaa gaacttcacc ggcgaggaga tcaagtacat gctgtggctg 180
agcggccgacc tgaagtccg catcaagcag aaggggcagct acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcacccgca cccgctgag caccgagaca 300
ggcttcgccc tgcctggggcgg ccacccctgc ttcttgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgcccgc gtgctgagca gcatggccga cgcctgtgctg 420
gcccgctgtg acaagcagag cgacctggac accctggcca aggaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctgt 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcctcatgat gagcgccgcc aagttcgcca tgcacctgca ggccgccacc 660
cccaagggtc acgagccgga cgcagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgctgct gaccaacgac cccctggagg ccgcccacgg cggcaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840
gccttcaggg gctaccaggt gaccatgaag accgccaagg tggccgccag cgactggacc 900
ttcctgcact gccctgcccc caagcccag gaggtggacg acgaggtgtt ctacagcccc 960
cgcagcctgg tgttccccga ggccgagaa cgcgaagtga ccatcatggc cgtgatggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

SEQ ID NO: 217      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 217
atgggagtat tcaacctgcg catcctgctg aacaacgccg ccttccgcaa cggccacaac 60
ttcatggtgc gcaacttccg ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgaccctgaa gaacttcacc ggcgaggaga tccggtacat gctgtggctg 180
agcggccgacc tgaagtccg catcaagcag aaggggcagct acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcacccgca cccgctgag caccgagaca 300
ggcttcgccc tgcctggggcgg ccacccctgc ttcttgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgcccgc gtgctgagca gcatggccga cgcctgtgctg 420
gcccgctgtg acaagcagag cgacctggac accctggcca aggaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctgt 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcctcatgat gagcgccgcc aagttcgcca tgcacctgca ggccgccacc 660
cccaagggtc acgagccgga cgcagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgctgct gaccaacgac cccctggagg ccgcccacgg cggcaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840
gccttcaggg gctaccaggt gaccatgaag accgccaagg tggccgccag cgactggacc 900
ttcctgcact gccctgcccc caagcccag gaggtggacg acgaggtgtt ctacagcccc 960

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cgcagcctgg tgttccccga ggccgagaa cgcgaagtga ccatcatggc cgtgatgggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

```

```

SEQ ID NO: 218      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 218
atgctgggtat tcaacctgcg catcctgctg aacaacgccc ccttcgcgaa cggccacaac 60
ttcatgggtgc gcaacttcgc ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgaccttgaa gaacttcacc ggccaggaga tccggtacat gctgtggctg 180
agcggccgacc tgaagtccg catcaagcag aaggcgaggt acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcaccgcga cccgcctgag caccgagaca 300
ggcttcgccc tgctggggcg ccacctctgc ttctgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgcccgc gtgctgagca gcatggccga cgcctgtgctg 420
gcccgcgtgt acaagcagag cgacctggac accctggcca aggaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgccc aagttcgcca tgcacctgca ggccgccacc 660
cccaagggtc acgagcccga cgccagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgtgctt gaccaacgac cccctggagg ccgcccacgg cgccaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840
gcttccagg gctaccaggt gacctgaag accgccaagg tggccgccag cgactggacc 900
ttctgcact gccctgcccc caagcccag gaggtggacg acgaggtgt ctacagcccc 960
cgcagcctgg tgttccccga ggccgagaa cgcgaagtga ccatcatggc cgtgatgggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

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```

SEQ ID NO: 219      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 219
atgggagtat tcaacctgcg catcctgctg aacaacgccc ccttcgcgaa cggccacaac 60
ttcatgggtgc gcaacttcgc ctgcggccag cccctgcaga acaaggtgca gctgaagggc 120
cgcgacctgc tgaccttgaa gaacttcacc ggccaggaga tcaagtacat gctgtggctg 180
agcggccgacc tgaagtccg catcaagcag aaggcgaggt acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcaccgcga cccgcctgag caccgagaca 300
ggcttcgccc tgctggggcg ccacctctgc ttctgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgcccgc gtgctgagca gcatggccga cgcctgtgctg 420
gcccgcgtgt acaagcagag cgacctggac accctggcca aggaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgccc aagttcgcca tgcacctgca ggccgccacc 660
cccaagggtc acgagcccga cgccagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgtgctt gaccaacgac cccctggagg ccgcccacgg cgccaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840
gcttccagg gctaccaggt gacctgaag accgccaagg tggccgccag cgactggacc 900
ttctgcact gccctgcccc caagcccag gaggtggacg acgaggtgt ctacagcccc 960
cgcagcctgg tgttccccga ggccgagaa cgcgaagtga ccatcatggc cgtgatgggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

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SEQ ID NO: 220      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature        1..1068
                    note = Synthetic construct
source              1..1068
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 220
atgggagtat tcaacctgcg catcctgctg aacaacgccc ccttcgcgaa cggccacaac 60
ttcatgggtgc gcaacttcgc ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgaccttgaa gaacttcacc ggccaggaga tccggtacat gctgtggctg 180
agcggccgacc tgaagtccg catcaagcag aaggcgaggt acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcaccgcga cccgcctgag caccgagaca 300
ggcttcgccc tgctggggcg ccacctctgc ttctgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgcccgc gtgctgagca gcatggccga cgcctgtgctg 420
gcccgcgtgt acaagcagag cgacctggac accctggcca aggaggccag catccccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgccc aagttcgcca tgcacctgca ggccgccacc 660
cccaagggtc acgagcccga cgccagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgtgctt gaccaacgac cccctggagg ccgcccacgg cgccaacgtg 780
ctgatcaccg acacctggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840

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gccttcagg gctaccaggt gaccatgaag accgccaagg tggccgcccag cgaactggacc 900
ttcctgcact gcctgccccg caagcccagag gaggtggacg acgaggtgtt ctacagcccc 960
cgcagccttg tggtccccga ggccgagaac cgcaagtgga ccatcatggc cgtgatgggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

```

```

SEQ ID NO: 221      moltype = DNA length = 1068
FEATURE
misc_feature       1..1068
                    note = Synthetic construct
source             1..1068
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 221
atgctgggtat tcaacctgcg catcctgctg aacaacgccc ccttcgcgaa cgccacaaac 60
ttcatgggtgc gcaacttcgg ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgacctgaa gaacttcacc ggcgaggaga tccggtacat gctgtggctg 180
agcgccgacc tgaagtccc catcaagcag aaggcgaggt acctgcccct gctgcagggc 240
aagagccttg gcatgatctt cgagaagcgc agcaccgcga cccgcctgag caccgagaca 300
ggcttcgccc tgctggggcg ccacccctgc ttctgacca ccaggacat ccacctgggc 360
gtgaacgaga gcttgaccga caccgcccgc gtgctgagca gcatggccga cgccgtgctg 420
gccccgctgt acaagcagag cgacctggac accctggcca aggaggccag catcccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggagcact acagcagcct gaaggcgctg accctgagct ggatcgccga cggcaacaac 600
atcctgcaca gcatcatgat gagcgccgccc aagttcgga tgcacctgca ggccgccacc 660
cccaagggct acgagccgga cgccagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgtgctt gaccaacgac cccctggagg ccgcccacgc cggcaacgtg 780
ctgatccacc acactggat cagcatgggc caggaggagg agaagaagaa ggcctgcag 840
gccttcagg gctaccaggt gaccatgaag accgccaagg tggccgcccag cgaactggacc 900
ttcctgcact gcctgccccg caagcccagag gaggtggacg acgaggtgtt ctacagcccc 960
cgcagccttg tggtccccga ggccgagaac cgcaagtgga ccatcatggc cgtgatgggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

```

```

SEQ ID NO: 222      moltype = DNA length = 1073
FEATURE
misc_feature       1..1073
                    note = Synthetic construct
source             1..1073
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 222
atgttgttca acttgaggat cttgttgaac aacgcccgcct tcaggaaagg acacaacttc 60
atggtaaggga acttcagggt cggacagccc ttgcagaaca aagtacagtt gaaagggaagg 120
gaacttgtga cattgaaaaa cttcacagga gaagaaatca aatacatgtt gtggtgtgctg 180
gcccacttga aattcaggat caaacagaaa ggagaatact tgcccttgtt gcagggaaaa 240
tcgttgggaa tgatcttcga aaaaaggctc acaaggacaa ggttgtcgac agaaccagga 300
ttcgcttgtt tgggaggaca cccctgcttc ttgacaacac aggcacatca cttgggagta 360
aacgaatcgt tgacagacac agccagggtg ttgtcgtcga tggccgacgc cgtattggcc 420
agggatataca aacagtcgga ctgggacaca ttggccaaag aagcctcgat ccccatcatc 480
aacggattgt cggacttgta ccaccccatc cagatcttgg ccgactactt gacattgcag 540
gaacactact cgtcgttgaa aggattgaca ttgtcgtgga tcggagacgg aaacaacatc 600
ttgactcga tcatgatgtc ggccgcccaga ttccgaatgc acttgacggc cgccacaccc 660
aaaggatagc aaccgcagc ctccgttaaca aaattggccg aacagtacgc caaagaaaac 720
ggaacaaaat tgttgttgac aaacgacccc ttggaagccg cccacggagg aaacgtattg 780
atcacagaca catggatctc gatgggacag gaagaagaaa aaaaaaaag gttgcaggcc 840
ttccagggat accaggtaac aatgaaaaa gccaagtag ccgctcgga cttggacattc 900
ttgactgct tgcccaggaa acccgaagaa gtgacgacg aagtattcta ctgcccagg 960
tcgttggtat tccccgaagc cgaacacagg aaatggacaa tcatggccgt aatgggatcg 1020
ttgttgacag actactcgcc ccagttgcag aaacccaaat tctgaatagt gaa 1073

```

```

SEQ ID NO: 223      moltype = DNA length = 1065
FEATURE
misc_feature       1..1065
                    note = Synthetic construct
source             1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 223
atgctgttca acctgcgcat cctgctgaac aacgcccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acttcgcgtg cggccagccc ctgcagggca aggtgcagct gaagggccgc 120
gacctgctga ccctgaagaa cttcaccggc gaggagatca agtatcatgt gtggctgagc 180
gcccacctga agttccgcat caagcagaag ggcgagtacc tgcccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgcagc acccgcaccc gcctgagcac cgagacaggc 300
ttcgccttgc tggcgccgca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaaag aggccagcat ccccatcatc 480
aacggcctga ggcacctgta ccccccacat cagatcctgg ccgactacct gaccctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcgcatgc acctgcaggc cgccaccccc 660
aagggtacg agcccagcgc cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720

```


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```

ggcaccgaagc tgctgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgcaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accagggtgac catgaagacc gccaaaggtg cgcacagcga ctggaccttc 900
ctgcaactgcc tgcctccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctgggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 224      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 224
atgctgttca acctgcgcat cctgtgaac aacgcccgcct tccgcaacgg ccacaacttc 60
atgggtgcgca acttccgctg cggccagccc ctgcagggcc ggggtgcagct gaagggccgc 120
gacctgtctga cctgaagaa cttcaccggc gaggagatca agtacatgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatctctga gaagcgacgc acccgacccc gctgagcac cgagacaggg 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cccccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgcgcgcaag ttccgcatgc acctgcaggc cgccaccccc 660
aagggctacg agcccgacgc cagcgtgacc aagctggccc agcagtacgc caaggagaac 720
ggcaccgaagc tctgtctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgcaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accagggtgac catgaagacc gccaaaggtg cgcacagcga ctggaccttc 900
ctgcaactgcc tgcctccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctgggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 225      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 225
atgctgttca acctgcgcat cctgtgaac aacgcccgcct tccgcaacgg ccacaacttc 60
atgggtgcgca acttccgctg cggccagccc ctgcagggcc ggggtgcagct gaagggccgc 120
gacctgtctga cctgaagaa cttcaccggc gaggagatcc ggtacatgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatctctga gaagcgacgc acccgacccc gctgagcac cgagacaggg 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cccccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgcgcgcaag ttccgcatgc acctgcaggc cgccaccccc 660
aagggctacg agcccgacgc cagcgtgacc aagctggccc agcagtacgc caaggagaac 720
ggcaccgaagc tctgtctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgcaca cctggatcag catgggcccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accagggtgac catgaagacc gccaaaggtg cgcacagcga ctggaccttc 900
ctgcaactgcc tgcctccgcaa gcccgaggag gtggacgacg aggtgttcta cagccccgcg 960
agcctgggtgt tccccgaggg cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 226      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature        1..1065
                    note = Synthetic construct
source              1..1065
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 226
atgctgttca acctgcgcat cctgtgaac aacgcccgcct tccgcaacgg ccacaacttc 60
atgggtgcgca acttccgctg cggccagccc ctgcagggcc aggtgacagct gaagggccgc 120
gacctgtctga cctgaagaa cttcaccggc gaggagatca agtacatgct gtggctgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgccccctgt gcagggcaag 240
agcctgggca tgatctctga gaagcgacgc acccgacccc gctgagcac cgagacaggg 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cccccgcgtg ctgagcagca tggccgacgc cgtgctggcc 420
cgctgtgata agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggcctga gcgacctgta ccccccatc cagatcctgg ccgactacct gacctgcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600

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ctgcacagca tcatgatgag cgccgccaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggtctacg agcccgagcg cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720
ggcaccAagc tgcTgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccAaggtgg ccgcccagcga ctggacottc 900
ctgcaactgc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccgaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 227      moltype = DNA length = 1065
FEATURE            Location/Qualifiers
misc_feature       1..1065
                   note = Synthetic construct
source             1..1065
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 227
atgctgttca acctgcgcat cctgtgaac aacgccgcct tccgcaacgg ccacaacttc 60
atggtgcgca acctccgctg cgccagccc ctgcagaaca aggtgcagct gaagggccgc 120
gaactgtcga ccttcagaa cttcacggc gaggagatca agtacctgct gtggtgagc 180
gccgacctga agttccgcat caagcagaag ggcgagtacc tgccccctgct gcagggcaag 240
agcctgggca tgatcttcga gaagcgacg acccgacccc gcctgagcac cgagacaggc 300
ttcgccctgc tggggcgcca cccctgcttc ctgaccaccc aggcacatcca cctgggctgtg 360
aacgagagcc tgaccgacac cgcccccgctg ctgagcagca tggccgacgc cgtgctggcc 420
cgcggtgtaca agcagagcga cctggacacc ctggccaagg aggccagcat ccccatcatc 480
aacggctcga gcgacctgta ccacccatc cagatcctgg ccgactacct gacctgtcag 540
gagcactaca gcagcctgaa gggcctgacc ctgagctgga tcggcgacgg caacaacatc 600
ctgcacagca tcatgatgag cgccgccaag ttcggcatgc acctgcaggc cgccaccccc 660
aagggtctacg agcccgagcg cagcgtgacc aagctggccg agcagtacgc caaggagAAC 720
ggcaccAagc tgcTgctgac caacgacccc ctggaggccg cccacggcgg caacgtgctg 780
atcaccgaca cctggatcag catgggccag gaggaggaga agaagaagcg cctgcaggcc 840
ttccagggct accaggtgac catgaagacc gccAaggtgg ccgcccagcga ctggacottc 900
ctgcaactgc tgccccgcaa gcccgaggag gtggacgacg aggtgttcta cagcccccg 960
agcctggtgt tccccgaggc cgagaaccgc aagtggacca tcatggccgt gatggtgagc 1020
ctgctgaccg actacagccc ccagctgcag aagcccaagt tctga 1065

```

```

SEQ ID NO: 228      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature       1..1068
                   note = Synthetic construct
source             1..1068
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 228
atgctggtat tcaacctgcg catcctgctg aacaacgccg ccttcgcga cgccacaaac 60
ttcatggtgc gcaacttcgc ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgacctgaa gaacttcacc ggcgaggaga tccggtacat gctgtggctg 180
agcggcgacc tgaagtccg catcaagcag aagggcgagt acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcaccgcga cccgctgag caccagagaca 300
ggcttcgccc tgcTggggcg ccacctctgc ttcctgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgccccg gtgctgagca gcatggccga cgccgtgctg 420
gccccgctgt acaagcagag gcacctggac accctggcca aggaggccag catcccatc 480
atcaacggcc tgagcgacct gtaccacccc atccagatcc tggccgacta cctgacctg 540
caggagcact acagcagcct gaagggcctg accctgagct ggatcgccga cggcaacaac 600
atcctgcata gcatcatgat gagcgccgcc aagttcgga tgcacctgca ggcgcgccacc 660
cccaagggtc acgagcccca cgccagcgtg accaagctgg ccgagcagta cgccaaggag 720
aacggcacca agctgctgct gaccaacgac cccctggagg ccgcccacgg cggcaacgtg 780
ctgatacccg acactggat cagcatgggc caggaggagg agaagaagaa gcgcctgcag 840
gcttccagg gctaccaggt gacctgaag accgccaagg tggccgccag cgactggacc 900
ttcctgcact gcctgccccg caagcccgag gagggtgacg acgaggtgt ctacagcccc 960
cgcagcctgg tgttccccga ggcgagaac cgcaagtgga ccatcatggc cgtgatggtg 1020
agcctgctga ccgactacag cccccagctg cagaagccca agttctga 1068

```

```

SEQ ID NO: 229      moltype = DNA length = 1068
FEATURE            Location/Qualifiers
misc_feature       1..1068
                   note = Synthetic construct
source             1..1068
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 229
atgctggtat tcaacctgcg catcctgctg aacaacgccg ccttcgcga cgccacaaac 60
ttcatggtgc gcaacttcgc ctgcggccag cccctgcaga accgggtgca gctgaagggc 120
cgcgacctgc tgacctgaa gaacttcacc ggcgaggaga tccggtacat gctgtggctg 180
agcggcgacc tgaagtccg catcaagcag aagggcgagt acctgccccct gctgcagggc 240
aagagcctgg gcatgatctt cgagaagcgc agcaccgcga cccgctgag caccagagaca 300
ggcttcgccc tgcTggggcg ccacctctgc ttcctgacca cccaggacat ccacctgggc 360
gtgaacgaga gccctgaccga caccgccccg gtgctgagca gcatggccga cgccgtgctg 420
gccccgctgt acaagcagag gcacctggac accctggcca aggaggccag catcccatc 480

```

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atcaacggcc  tgagcgacct  gtaccacccc  atccagatcc  tggccgacta  cctgaccctg  540
caggagcact  acagcagcct  gaagggcctg  accctgagct  ggatcggcga  cggcaacaac  600
atcctgcaca  gcatcatgat  gagcgccgcc  aagttcggca  tgcacctgca  ggccgccacc  660
cccaagggct  acgagcccca  cgccagcgtg  accaagctgg  ccgagcagta  cgccaaggag  720
aacggcacca  agctgctgct  gaccaacgac  cccttgaggg  ccgccacgg  cggcaacgtg  780
ctgatcaccg  acacctggat  cagcatgggc  caggaggagg  agaagaagaa  gcgcctgcag  840
gccttcagg  gctaccaggt  gaccatgaag  accgccaagg  tggccgccag  cgactggacc  900
ttcctgcact  gcctgccccg  caagcccgag  gaggtggacg  acgaggtgtt  ctacagcccc  960
cgcagcctgg  tgttccccga  ggccgagaa  cgcgaagtga  ccatcatggc  cgtgatgggt  1020
agcctgctga  ccgactacag  cccccagctg  cagaagccca  agttctga  1068

```

```

SEQ ID NO: 230      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 230
aaccaatcga aagaaccaa a 21

```

```

SEQ ID NO: 231      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 231
ctctaatacac caggagtaaa a 21

```

```

SEQ ID NO: 232      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 232
gagagagatc ttaacaaaaa a 21

```

```

SEQ ID NO: 233      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 233
tgtgtaacaa caacaacaac a 21

```

```

SEQ ID NO: 234      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 234
ccgcagtagg aagagaaagc c 21

```

```

SEQ ID NO: 235      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 235
aaaaaaaaaa gaaatcataa a 21

```

```

SEQ ID NO: 236      moltype = RNA  length = 21
FEATURE
misc_feature       1..21
                    note = Synthetic construct
source             1..21
                    mol_type = other RNA
                    organism = synthetic construct

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SEQUENCE: 236		
gagagaagaa agaagaagac g		21
SEQ ID NO: 237	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 237		
caattaaaaa tacttaccaa a		21
SEQ ID NO: 238	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 238		
gcaaacagag taagcgaaac g		21
SEQ ID NO: 239	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 239		
gcgaagaaga cgaacgcaaa g		21
SEQ ID NO: 240	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 240		
ttaggactgt attgactggc c		21
SEQ ID NO: 241	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 241		
atcatcgaa ttcggaaaaa g		21
SEQ ID NO: 242	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 242		
aaaacaaaag ttaaagcaga c		21
SEQ ID NO: 243	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	
source	1..21	
	mol_type = other RNA	
	organism = synthetic construct	
SEQUENCE: 243		
tttatctcaa ataagaaggc a		21
SEQ ID NO: 244	moltype = RNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Synthetic construct	

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source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 244
ggtggggagg tgagatttct t                                21

SEQ ID NO: 245        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 245
tgattaggaa actacaaagc c                                21

SEQ ID NO: 246        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 246
catttttcaa ttccataaaa c                                21

SEQ ID NO: 247        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 247
ttacttttaa gcccaacaaa a                                21

SEQ ID NO: 248        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 248
ggcgtgtgtg tgtgttgttg a                                21

SEQ ID NO: 249        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 249
gtggtgaagg ggaaggttta g                                21

SEQ ID NO: 250        moltype = RNA length = 21
FEATURE              Location/Qualifiers
misc_feature          1..21
                      note = Synthetic construct
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 250
ttgttttttt ttggtttggt t                                21

SEQ ID NO: 251        moltype = RNA length = 1331
FEATURE              Location/Qualifiers
misc_feature          1..1331
                      note = Synthetic construct
source                1..1331
                      mol_type = other RNA
                      organism = synthetic construct

SEQUENCE: 251
aggattatta catcaaaaca aaaagccgcc accatgctgg tattcaacct ggcgcatcctg 60
ctgaacaacg ccgccttcgc caacggccac aacttcatgg tgcgcaactt ccgctgcggc 120
cagccctgcg agaaccgggt gcagctgaag ggccgcgacc tgctgacctt gaagaacttc 180
accggcgagg agatccggta catgctgtgg ctgagcgccg acctgaagtt ccgcatcaag 240
cagaagggcg agtacctgcc cctgctgcag ggcaagagcc tgggcatgat cttcagagaag 300

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```

cgcagcacc cgcacccgct gagcaccgag acaggtctcg cctgtgtggg cggccacccc 360
tgcttctga ccacccagga catccacctg ggcgtgaacg agagcctgac cgacaccgcc 420
cgctgtctga gcagcatggc cgacgcccgt ctggcccgcg tgtacaagca gagcgacctg 480
gacacctgg ccaaggaggc cagcatcccc atcatcaacg gcctgagcga cctgtaccac 540
cccatccaga tcttgccgca ctacctgacc ctgcaggagc actacagcag cctgaagggc 600
ctgacctga gctggatcgg cgacggcaac aacatcctgc acagcatcat gatgagcgcc 660
gccaagtctg gcatgcacct gcaggccgcc acccccaagg gctacgagcc cgacgccagc 720
gtgaccaagc tggccgagca gtacgccaag gagaacggca ccaagctgct gctgaccaac 780
gaccccttg aggccgcca cgccggcaac gtgtgatca ccgacacctg gatcagcatg 840
ggccaggagg aggagaagaa gaagcgctg caggccttcc agggctacca ggtgacctg 900
aagaccgcca aggtggccgc cagcgactgg accttctgc actgcctgcc ccgcaagccc 960
gaggaggttg acgacgaggt gttctacagc ccccgagcc ttgtgttccc cgaggccgag 1020
aaccgcaagt ggcacatcat ggcgtgatg gtgagcctgc tgaccgacta cagccccag 1080
ctgcagaagc ccaagttctg aggtctctag taatgagctg gagcctcggt agcgttcc 1140
cctgcccgt gggcctccca acgggcccct cccccctct tgcaccggcc cttcctggtc 1200
tttgaataaa gtctgagtgg gcagcatcta gaaaaaaaa aaaaaaaaa aaaaaaaaa 1260
aaaaaaaaa aaaaaaaaa aaaaaaaaa aaaaaaaaa aaaaaaaaa 1320
aaaaaaaaa a 1331

```

```

SEQ ID NO: 252      moltype = RNA length = 1334
FEATURE            Location/Qualifiers
misc_feature        1..1334
                    note = Synthetic construct
source              1..1334
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 252
aggattatta catcaaaaca aaaagccgcc accatgctgg tattcaacct gcgcacctg 60
ctgaacaacg ccgcttccg caacggccac aactcatgg tgcgaacct ccgctgcggc 120
cagcccttgc agaaccgggt gcagctgaag ggcgcgacc tgctgacct gaagaacttc 180
accggcgagg agatccggtg catgtgtgg ctgagcgccg acctgaagt ccgcataag 240
cagaaggggc agtacctgcc cctgctgcag ggcaagagcc tgggcatgat cttcgagaag 300
cgcagcacc cgcacccgct gagcaccgag acaggtctcg cctgtgtggg cggccacccc 360
tgcttctga ccaccagga catccacctg ggcgtgaacg agagcctgac cgacaccgcc 420
cgctgtctga gcagcatggc cgacgcccgt ctggcccgcg tgtacaagca gagcgacctg 480
gacaccttg ccaaggaggc cagcatcccc atcatcaacg gcctgagcga cctgtaccac 540
cccatccaga tcttgccgca ctacctgacc ctgcaggagc actacagcag cctgaagggc 600
ctgacctga gctggatcgg cgacggcaac aacatcctgc acagcatcat gatgagcgcc 660
gccaagtctg gcatgcacct gcaggccgcc acccccaagg gctacgagcc cgacgccagc 720
gtgaccaagc tggccgagca gtacgccaag gagaacggca ccaagctgct gctgaccaac 780
gaccccttg aggccgcca cgccggcaac gtgtgatca ccgacacctg gatcagcatg 840
ggccaggagg aggagaagaa gaagcgctg caggccttcc agggctacca ggtgacctg 900
aagaccgcca aggtggccgc cagcgactgg accttctgc actgcctgcc ccgcaagccc 960
gaggaggttg acgacgaggt gttctacagc ccccgagcc ttgtgttccc cgaggccgag 1020
aaccgcaagt ggcacatcat ggcgtgatg gtgagcctgc tgaccgacta cagccccag 1080
ctgcagaagc ccaagttctg aggtctctag taatgagctg gagcctcggt agcgttcc 1140
cctgcccgt gggcctccca acgggcccct cccccctct tgcaccggcc cttcctggtc 1200
tttgaataaa gtctgagtgg gcagcatcta gaaaaaaaa aaaaaaaaa aaaaaaaaa 1260
aaaaaaaaa aaaaaaaaa aaaaaaaaa aaaaaaaaa aaaaaaaaa 1320
aaaaaaaaa agct 1334

```

```

SEQ ID NO: 253      moltype = RNA length = 1328
FEATURE            Location/Qualifiers
misc_feature        1..1328
                    note = Synthetic construct
source              1..1328
                    mol_type = other RNA
                    organism = synthetic construct

```

```

SEQUENCE: 253
aggattatta catcaaaaca aaaagccgcc accatgctgt tcaacctgag catcctgctg 60
aacaacggcg ctttccgcaa cgccacaac ttcattggtg gcaacttccg ctgcggccag 120
ccctgcaga acaaggtgca gctgaagggc cgcgacctgc tgacctgaa gaacttacc 180
ggcgaggaga tcaagtacat gctgtggctg agcgcgacc tgaagttccg catcaagcag 240
aaggcgaggt acctgcccct gctgcagggc aagagcctgg gcatgatctt cgagaagcgc 300
agcaccgcga cccgctgag caccgagaca ggcttcgccc tgctgggagg ccacctctgc 360
ttcctgacca cccaggacat ccacctgggc gtgaacgaga gcctgaccga caccgcccgc 420
gtgctgagca gcatggccga gcgcgtgtg gcccgctgt acaagcagag cgacctggac 480
acctggcca aggagggcag catccccatc atcaacggcc tgagcgacct gtaccacccc 540
atccagatcc tggccgacta cctgaccttg caggagcact acagcagcct gaagggcctg 600
accttgagct ggatcggcga cggcaacaac atcctgcaca gcatcatgat gagcgccgcc 660
aagttcgcca tgcacctgca ggcggccacc cccaagggct acgagcccga cgccagcgtg 720
accaagctgg ccgagcagta cgccaaggag aacggcacca agctgctgct gaccaacgac 780
cccttgagg ccgcccacgg cggcaacgtg ctgatcaccg acacctggat cagcatgggc 840
caggaggagg agaagaagaa gcgcctgcag gccttccagg gctaccaggt gacctgaag 900
accgccaagg tggccgccc cgactggacc ttcctgcact gcctgcccgc caagcccag 960
gagggtgagc acgaggtgtt ctacagcccc cgcagcctgg tgttcccga ggccgagaac 1020
cgcaagtcca ccatcatggc cgtgatggtg agcctgtga ccgactacag cccccagctg 1080
cagaagccga agttctgagg tctctagtaa tgagctggag cctcggtagc cgttctcct 1140
gcccgtggg cctcccaacg ggcctcctc cctccttgc accggccctt cctggtttt 1200

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gaataaagtc tgagtgggca gcatctagaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1260
aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa 1320
aaaaaaaaaa                                     1328

```

What is claimed is:

1. A polynucleotide encoding an ornithine transcarbamylase (OTC) protein comprising the amino acid sequence of SEQ ID NO:4 and having OTC enzymatic activity, wherein

(i) the polynucleotide comprises a sequence of SEQ ID NO:221; or

(ii) the polynucleotide comprises a sequence of SEQ ID NO:221, wherein T is substituted with U.

2. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA comprising a 3' poly A tail.

3. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA comprising a 5' untranslated region (5'UTR).

4. The polynucleotide of claim 3, wherein the 5'UTR is derived from a gene expressed by *Arabidopsis thaliana*.

5. The polynucleotide of claim 4, wherein the 5' UTR comprises a sequence selected from SEQ ID NO: 6, SEQ ID NOS: 125-127, and SEQ ID NOS: 230-250.

6. The polynucleotide of claim 5, wherein the 5'UTR comprises a sequence of SEQ ID NO: 6.

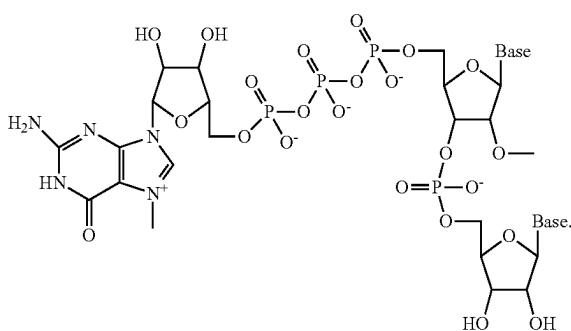
7. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA comprising a 3' UTR.

8. The polynucleotide of claim 7, wherein the 3' UTR comprises a sequence selected from SEQ ID NOS: 16-22.

9. The polynucleotide of claim 1, comprising a Kozak sequence of SEQ ID NO: 23 or a partial Kozak sequence of SEQ ID NO: 24.

10. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA comprising a 5' cap.

11. The polynucleotide of claim 10, wherein the 5' cap is m7GpppGm having the following structure:



12. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA comprising

(a) a sequence of SEQ ID NO: 21; or

(b) a sequence of SEQ ID NO: 6 and a sequence of SEQ ID NO: 21.

13. The polynucleotide of claim 1, wherein the polynucleotide is an mRNA, and wherein

(a) the percentage of uracil nucleobases in the coding region of the polynucleotide is reduced with respect to the percentage of uracil nucleobases in the wild-type OTC nucleic acid sequence; or

(b) 1-100% of the uridines are modified uridine analogs; or

(c) 1-100% of the uridine nucleotides are each independently a modified uridine analog selected from the group consisting of 5-methoxyuridine, N¹-methylpseudouridine, 5-hydroxyuridine, 5-methyluridine, 5-hydroxymethyluridine, 5-carboxyuridine, 5-carboxymethylesteruridine, 5-formyluridine, 5-propynyluridine, 5-bromouridine, 5-fluorouridine, 5-iodouridine, 2-thiouridine, 6-methyluridine, N¹-ethylpseudouridine, N¹-propylpseudouridine, N¹-cyclopropylpseudouridine, N¹-phenylpseudouridine, N¹-aminomethylpseudouridine, N³-methylpseudouridine, N¹-hydroxypseudouridine, N¹-hydroxymethylpseudouridine,

5-methoxycarbonylmethyl-2-thiouridine, 5-methylaminomethyl-2-thiouridine, 5-carbamoylmethyluridine, 5-carbamoylmethyl-2'-O-methyluridine, 1-methyl-3-(3-amino-3-carboxypropyl) pseudouridine, 5-methylaminomethyl-2-selenouridine, 5-carboxymethyluridine, 5-methyldihydrouridine, 5-taurinomethyluridine, 5-taurinomethyl-2-thiouridine, 5-(isopentenylaminomethyl) uridine, 2'-O-methylpseudouridine, 2-thio-2'-O-methyluridine, and 3,2'-O-dimethyluridine; or

(d) 100% of the uridine nucleotides are 5-methoxyuridine; or

(e) 100% of the uridine nucleotides are N¹-methylpseudouridine.

14. A pharmaceutical composition comprising the polynucleotide of claim 1 and pharmaceutically acceptable carrier, wherein the pharmaceutically acceptable carrier is a lipid formulation.

15. A pharmaceutical composition comprising the polynucleotide of claim 13 and pharmaceutically acceptable carrier, wherein the pharmaceutically acceptable carrier is a lipid formulation.

16. A method of treating OTC deficiency in a patient identified as suffering from OTC deficiency, comprising administering to the patient a pharmaceutical composition of claim 15, wherein upon administration of the pharmaceutical composition to the patient, the protein of SEQ ID NO: 4 is expressed in the patient.

17. A method of treating OTC deficiency in a patient identified as suffering from OTC deficiency, comprising administering to the patient a polynucleotide of claim 1, wherein the polynucleotide expresses the protein of SEQ ID NO: 4 in the patient.

18. A vector comprising the polynucleotide of claim 1.

19. An mRNA encoding an OTC protein having the amino acid sequence of SEQ ID NO: 4, wherein the mRNA comprises a polynucleotide sequence of SEQ ID NO: 221, wherein T is substituted with U, and wherein the encoded protein has OTC enzymatic activity.

20. The mRNA of claim 19, wherein

(a) 1-100% of the uridines are modified uridine analogs; or

(b) 1-100% of the uridine nucleotides are each independently a modified uridine analog selected from the group consisting of 5-methoxyuridine, N¹-methylpseudouridine, 5-hydroxyuridine, 5-methyluridine, 5-hydroxymethyluridine, 5-carboxyuridine, 5-car-

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boxymethylesteruridine, 5-formyluridine, 5-propyny-
 luridine, 5-bromouridine, 5-fluorouridine, 5-iodouri-
 dine, 2-thiouridine, 6-methyluridine,
 N1-ethylpseudouridine, N1-propylpseudouridine,
 N¹-cyclopropylpseudouridine, N¹-phenylpseudouri- 5
 dine, N¹-aminomethylpseudouridine, N³-methylp-
 pseudouridine, N¹-hydroxypseudouridine, N¹-hy-
 droxymethylpseudouridine,
 5-methoxycarbonylmethyl-2-thiouridine, 5-methyl-
 aminomethyl-2-thiouridine, 5-carbamoylmethyluri- 10
 dine, 5-carbamoylmethyl-2'-O-methyluridine,
 1-methyl-3-(3-amino-3-carboxypropy) pseudouridine,
 5-methylaminomethyl-2-selenouridine, 5-carboxym-
 ethyluridine, 5-methyldihydrouridine, 5-taurinomethy-
 luridine, 5-taurinomethyl-2-thiouridine, 5-(isopente- 15
 nylaminomethyl) uridine, 2'-O-methylpseudouridine,
 2-thio-2'O-methyluridine, and 3,2'-O-dimethyluridine;
 or
 (c) 100% of the uridine nucleotides are 5-methoxyuridine;
 or 20
 (d) 100% of the uridine nucleotides are N¹-methylp-
 pseudouridine.

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